Supplemental Material: Finite spectral triples as towerless intervals

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Part 03

Part A. Statement.

For a constant finite operator commuting with the gauge connection, the Yang–Mills term is

$$\boxed{S_{\mathrm{YM}} = -\frac{1}{24\pi^2}\int_M \sum_j f(m_j^2/\Lambda^2)\ \mathrm{tr}_{H_j}(F_{\mu\nu}F^{\mu\nu})\,dV.}$$

Here H_j is an eigenspace of D_F^2. The trace includes its entire finite multiplicity; the spin trace has already been taken.

Hypotheses.

Use Hermitian Euclidean gamma matrices, $\{\gamma^\mu,\gamma^\nu\} = 2g^{\mu\nu}$, and skew-Hermitian gauge curvature

$$F_{\mu\nu} = [\nabla_\mu + A_\mu, \nabla_\nu + A_\nu]_{\mathrm{gauge}}.$$

The spectral action is the literal trace on $L^2(M,S)\otimes H_F$, without a subsequent chirality or reality projection.

To distinguish the Laplace measure from the mass ratios, write

$$f(z) = \int_{(0,\infty)} e^{-sz}\,d\nu(s), \qquad 0 < \nu((0,\infty)) < \infty, \qquad \int (s^{-2}+s^2)\,d\nu(s) < \infty.$$

This class includes $f(z) = e^{-z}$. In particular, $f(0) < \infty$, $f(\infty) = 0$, and the spectral action is trace class.

Assume the finite-sector decomposition is fixed as Λ varies, A is smooth and independent of Λ, and

$$\partial_\mu D_F = 0, \qquad [A_\mu, D_F] = 0.$$

Proof.

Put $\mathcal{D} = D_{M,A}$, suppressing identity factors. Chirality satisfies

$$\{\mathcal{D},\gamma_5\} = 0, \qquad \gamma_5^2 = 1.$$

Consequently,

$$(\mathcal{D}+\gamma_5 D_F)^2 = \mathcal{D}^2 + D_F^2 + \gamma_5[D_F,\mathcal{D}] = \mathcal{D}^2 + D_F^2.$$

The two summands strongly commute. Since A preserves every eigenspace H_j of D_F^2,

$$\boxed{\operatorname{Tr} e^{-tD_A^2} = \sum_j e^{-tm_j^2} \operatorname{Tr} e^{-tD_{M,A_j}^2}.}$$

Let $P_j = D_{M,A_j}^2$, and use integrated heat coefficients:

$$\operatorname{Tr} e^{-tP_j} = t^{-2}a_0(P_j) + t^{-1}a_2(P_j) + a_4(P_j) + R_j(t).$$

For fixed smooth backgrounds, $|R_j(t)| \le Ct$ for all $t > 0$: the small-t estimate is the heat expansion, and the large-t estimate follows from $P_j \ge 0$.

Define

$$x_j = m_j^2/\Lambda^2, \qquad \Phi_k(x) = \int s^{-k} e^{-sx}\, d\nu(s).$$

Integration of the exact factorization gives

$$S = \sum_j \left[\Lambda^4 \Phi_2(x_j) a_0(P_j) + \Lambda^2 \Phi_1(x_j) a_2(P_j) + f(x_j) a_4(P_j)\right] + O(\Lambda^{-2}).$$

The remainder is bounded by

$$C\Lambda^{-2} \int s\, d\nu(s),$$

uniformly for all $x_j \ge 0$. It is also uniform over compact smooth families of connections. Thus fixed mass ratios require no expansion in m_j/Λ.

It remains to compute the gauge part of a_4. With $\gamma^{\mu\nu} = \frac{1}{2}[\gamma^\mu, \gamma^\nu]$, the gauge-dependent endomorphism in the convention $P = -(\nabla^2 + E)$ is

$$E_F = \tfrac{1}{2}\gamma^{\mu\nu} F_{\mu\nu}.$$

The relevant terms are

$$a_4(P)\big|_{F^2} = (4\pi)^{-2} \int_M \operatorname{tr}_{S,F} \left(\frac{1}{12} F_{\mu\nu}F^{\mu\nu} + \frac{1}{2} E_F^2\right) dV.$$

Using

$$\operatorname{tr}_S 1 = 4, \qquad \operatorname{tr}_S(\gamma^{\mu\nu}\gamma^{\rho\sigma}) = 4(g^{\mu\sigma}g^{\nu\rho} - g^{\mu\rho}g^{\nu\sigma}),$$

the two contributions are $+\frac{1}{3}\operatorname{tr}_F F^2$ and $-\operatorname{tr}_F F^2$. Hence

$$a_4(P)\big|_{F^2} = -\frac{1}{24\pi^2} \int_M \operatorname{tr}_F F^2\, dV.$$

This proves the statement.

For $F_{\mu\nu} = F_{\mu\nu}^a T_a$, with skew-Hermitian T_a, the coefficient matrix is therefore

$$-\frac{1}{24\pi^2} \sum_j f(x_j) \operatorname{tr}_{H_j}(T_a T_b).$$

It is positive semidefinite. The normalization is per full four-component spin Dirac operator, with the finite representation traced explicitly.

The three regimes are:

Heavy mass ratio	Weight	Consequence for the kinetic term
$m_h/\Lambda \to 0$	$f(0)$	Full inherited finite trace
$m_h/\Lambda \to c \in (0,\infty)$	$f(c^2)$	Partial suppression
$m_h/\Lambda \to \infty$	0	Trace over the light sectors

Absolute divergence $m_h \to \infty$ does not distinguish these regimes.

Part B. Statement.

Write

$$\ell = f(0), \qquad h = f(\mu^2).$$

For N generations with exactly one heavy up-type quark, the coefficients on the full finite space are

$$\boxed{\begin{aligned}\kappa_{\mathrm{em}}^{(N)} &= \frac{1}{24\pi^2}\frac{16}{3}[(2N-1)\ell + h],\\ \kappa_s^{(N)} &= \frac{1}{24\pi^2}2[(2N-1)\ell + h].\end{aligned}}$$

Here $S_{\mathrm{YM}} = \int(\kappa_{\mathrm{em}}F_{\mathrm{em}}^2 + \kappa_s\sum_a G_a^2)\,dV$; there is no $1/4$ in this definition.

Thus

$$\boxed{\frac{\kappa_{\mathrm{em}}^{(N)}(\mu)}{\kappa_{\mathrm{em}}^{(N)}(0)} = \frac{\kappa_s^{(N)}(\mu)}{\kappa_s^{(N)}(0)} = \frac{(2N-1)f(0) + f(\mu^2)}{2Nf(0)}.}$$

Hypotheses.

Use $Q = T_3 + Y$ and the following particle states:

Multiplet	Number of states	Y	T_3	Q
(u_L, d_L)	6	$1/6$	$+1/2, -1/2$	$2/3, -1/3$
u_R	3	$2/3$	0	$2/3$
d_R	3	$-1/3$	0	$-1/3$
(ν_L, e_L)	2	$-1/2$	$+1/2, -1/2$	$0, -1$
ν_R	1	0	0	0
e_R	1	-1	0	-1

Charge conjugates double this to the standard 32-dimensional finite space. For color, use Hermitian generators with

$$\mathrm{tr}_{\mathbf{3}}(t_a t_b) = \tfrac{1}{2}\delta_{ab}.$$

All three colors of the selected up-type quark have the same Dirac mass. Photon and gluon generators commute with these masses.

Proof.

On the particle space, both chiralities contribute

	u	d	e
$\mathrm{tr}\,Q^2$	8/3	2/3	2
color index	1	1	0.

Therefore, for one generation,

$$K_{\mathrm{em}}^{\mathrm{particle}} = \frac{8}{3}(h+\ell), \qquad K_s^{\mathrm{particle}} = h+\ell.$$

Charge conjugation doubles both expressions. Each additional light generation adds $32\ell/3$ and 4ℓ, respectively, on the full finite space. Part A now gives the stated coefficients.

In particular:

Heavy sector	Both kinetic-coefficient ratios	$\mu \to \infty$
Up quark in one generation	$\dfrac{f(0)+f(\mu^2)}{2f(0)}$	1/2
Top only in three generations	$\dfrac{5f(0)+f(\mu^2)}{6f(0)}$	5/6

The common suppression implies

$$\boxed{\kappa_{\mathrm{em}}/\kappa_s = 8/3}$$

throughout this particular family. The absolute kinetic form changes, while its photon-to-gluon ratio remains unchanged.

The same charge table gives, on the particle space,

$$K_W = 3\cdot\tfrac{1}{2}+\tfrac{1}{2} = 2,$$

$$K_Y = 6(1/6)^2 + 3(2/3)^2 + 3(1/3)^2 + 2(1/2)^2 + 1 = \frac{10}{3},$$

reproducing the conventions in section 14.3.

Part C. Statement.

For broken generators, a mass-dependent "F^2 coefficient" requires a prescription for separating gauge and Higgs derivative terms. I use the **integrated quadratic transverse kinetic coefficient**: expand about zero gauge field with D_F held constant, remove the gauge-field mass term, and extract the coefficient of $q^2|a_T(q)|^2$. Since

$$F_{\mu\nu}(q)^* F_{\mu\nu}(q) = 2q^2|a_T(q)|^2,$$

this specifies the quadratic F^2 coefficient, including all Higgs contributions to it.

For a vectorial doublet with nonnegative masses m_1, m_2, put

$$x_i = m_i^2/\Lambda^2, \quad z(u) = (1-u)x_1 + ux_2, \quad b(u) = u(1-u).$$

The normalized weight is

$$\boxed{w_f(x_1,x_2) = \frac{3}{2}\int_0^1 \left[(1-2b)f(z) + b(\sqrt{x_1}-\sqrt{x_2})^2 f'(z)\right] du.}$$

It is symmetric and satisfies $w_f(x,x) = f(x)$.

For the heat cutoff,

$$\boxed{w_{\exp}(x_1,x_2) = \frac{3}{2}\int_0^1 e^{-z(u)}\left[1-(2+(\sqrt{x_1}-\sqrt{x_2})^2)b(u)\right] du.}$$

This is generally more complicated than a divided difference of an antiderivative.

Hypotheses and scope.

Compute the local, gravity-independent coefficient in flat coordinates, with fixed m_i/Λ and $q/\Lambda \to 0$. The result concerns the quadratic action; longitudinal derivative terms and the full covariant Higgs-operator expansion remain distinct.

For an odd finite operator, the vectorial model can be realized as

$$D_F = \begin{pmatrix} 0 & M \\ M & 0 \end{pmatrix}, \qquad M = \mathrm{diag}(m_1, m_2), \qquad T = \begin{pmatrix} t & 0 \\ 0 & t \end{pmatrix}.$$

The Standard Model instead has a left doublet and right singlets. That representation changes the answer, as calculated below.

This distinction is necessary: prescribing D_F^2 alone does not prescribe $[T, D_F]$. Moreover,

$$[\nabla_\mu, \nabla_\nu] D_F = [F_{\mu\nu}, D_F],$$

so a noncommuting curvature cannot coexist with a covariantly constant finite operator. A separate "pure curvature" coefficient obtained by discarding Higgs derivative operators would require an additional operator-basis convention.

Proof by Duhamel expansion.

Without the commuting assumption,

$$\boxed{D_A^2 = D_{M,A}^2 + D_F^2 + i\gamma^\mu\gamma_5[A_\mu, D_F].}$$

The last term is essential at fixed D_F/Λ.

First take $f(z) = e^{-z}$, put $\tau = \Lambda^{-2}$, and diagonalize

$$D_F = \mathrm{diag}(\lambda_i).$$

Use $A_\mu = i\, a_\mu T$, with T Hermitian. The free squared operator is

$$P_0 = -\partial^2 + D_F^2.$$

The linear vertex between internal states j and i, carrying momentum q, is

$$(V_1)_{ij} = T_{ij}\left[(2p+q)\cdot a + \gamma^{\mu\nu} q_\mu a_\nu + (\lambda_i - \lambda_j)\gamma^\mu\gamma_5 a_\mu\right].$$

The quadratic vertex has no external momentum and therefore makes no contribution to the q^2 coefficient.

The second-order heat trace is

$$\frac{\tau^2}{2}\int_0^1 \mathrm{Tr}\left(e^{-(1-u)\tau P_0} V_1 e^{-u\tau P_0} V_1\right) du,$$

together with the quadratic-vertex term. Completing the Gaussian square gives the factor

$$(4\pi\tau)^{-2} e^{-\tau[(1-u)\lambda_i^2 + u\lambda_j^2 + bq^2]}.$$

For $q\cdot a = 0$, the Gaussian-averaged spin trace of the vertex product is

$$4\left[\frac{2}{\tau} + (\lambda_i - \lambda_j)^2 + q^2\right]|a|^2.$$

Expanding to order q^2, and writing $r_i = \lambda_i/\Lambda$, yields

$$\frac{q^2|a|^2}{8\pi^2}\sum_{ij}|T_{ij}|^2 \int_0^1 e^{-[(1-u)r_i^2 + u r_j^2]}\left[1 - 2b - b(r_i - r_j)^2\right] du.$$

Comparison with $F^2 = 2q^2|a|^2$ and Part A's normalization proves the heat-cutoff formula.

Integration over the Laplace measure replaces

$$e^{-z} \mapsto f(z), \qquad -se^{-sz} \mapsto f'(z),$$

giving the stated general formula. The moment assumptions justify these operations. The small-momentum expansion is uniform for mass ratios in compact sets.

For a vectorial positive-mass pair, the two relevant signed-mass branches both have $(r_i - r_j)^2 = (\sqrt{x_1} - \sqrt{x_2})^2$. At equal masses, that term vanishes and

$$\frac{3}{2}\int_0^1 (1 - 2u(1-u))\, du = 1,$$

which recovers Part A.

The divided difference does occur as

$$\int_0^1 f(z(u))\, du = \frac{\mathcal{F}(x_1) - \mathcal{F}(x_2)}{x_1 - x_2}, \qquad \mathcal{F}' = f.$$

The orbital vertices and finite-operator commutator supply the additional terms in w_f.

Standard Model doublet.

For the left-handed weak generator,

$$T = \begin{pmatrix} t & 0 \\ 0 & 0 \end{pmatrix}, \qquad [T, D_F] = \begin{pmatrix} 0 & tM \\ -Mt & 0 \end{pmatrix}.$$

For an off-diagonal t, summing these matrix elements replaces

$$(\sqrt{x_1} - \sqrt{x_2})^2 \quad \text{by} \quad x_1 + x_2.$$

Thus

$$w_L(x_1, x_2) = \frac{3}{2}\int_0^1 e^{-z(u)}[1 - (2 + x_1 + x_2)b(u)]\, du.$$

In particular,

$$w_L(x, x) = e^{-x}(1 - x/2).$$

Equal masses do not make a left-only weak generator commute with D_F, because its right partners are weak singlets. The conjectured diagonal identity therefore applies to the vectorial model, but not to this chiral representation.

For a heavy up quark and massless down quark, both models give

$$\boxed{w(x) := w_L(x, 0) = 3\int_0^1 u^2 e^{-xu}\, du = \frac{3[2 - (x^2 + 2x + 2)e^{-x}]}{x^3}, \qquad x = \mu^2.}$$

The value at zero is defined continuously. In particular,

$$w(x) = 1 - \frac{3}{4}x + \frac{3}{10}x^2 + O(x^3), \qquad w(x) \sim \frac{6}{x^3} \quad (x \to \infty).$$

The divided-difference candidate $(1 - e^{-x})/x$ already disagrees at first order.

Weak-angle consequences.

Use particle traces below; charge conjugation doubles the entire kinetic matrix. Let $N = 1$ or 3, with one heavy up-type quark, $x = \mu^2$, and $r = e^{-x}$. The charged and neutral transverse coefficients are

$$\begin{aligned} K_W &= 2N - \frac{3}{2} + \frac{3}{2}w(x), \\ K_{33} &= 2N - \frac{3}{4} + \left(\frac{3}{4} - \frac{3x}{8}\right) r, \\ K_{YY} &= \frac{10N}{3} - \frac{17}{12} + \left(\frac{17}{12} - \frac{3x}{8}\right) r, \\ K_{3Y} &= -\frac{1}{4} + \left(\frac{1}{4} + \frac{3x}{8}\right) r. \end{aligned}$$

The neutral action contains $K_{33}F_3^2 + 2K_{3Y}F_3F_Y + K_{YY}F_Y^2$.

For completeness, a Dirac species of mass ratio squared x, with neutral charges q_L^a, q_R^a, contributes

$$e^{-x}\left[q_L^a q_L^b + q_R^a q_R^b - \frac{x}{2}(q_L^a - q_R^a)(q_L^b - q_R^b)\right].$$

Applying this formula to the three up colors proves the neutral entries above. The commutator term is responsible for the terms proportional to xr.

At $x = 0$,

$$K_W = K_{33} = 2N, \qquad K_{YY} = 10N/3, \qquad K_{3Y} = 0,$$

so the usual matching angle is 5/8.

For $x > 0$, generally $K_W \neq K_{33}$ and $K_{3Y} \neq 0$. Consequently, the usual electroweak relations no longer define one common weak angle. Two precise continuations are:

1. **Continue the charged-W/hypercharge matching prescription.** Define

$$\boxed{\cos^2\theta_{\text{match}}(x) := \frac{K_{YY}}{K_W + K_{YY}} = \frac{80N - 34 + (34 - 9x)e^{-x}}{128N - 70 + (34 - 9x)e^{-x} + 36w(x)}.}$$

   Then

$$\cos^2\theta_{\text{match}}(0) = \frac{5}{8}, \qquad \cos^2\theta_{\text{match}}(\infty) = \begin{cases} 23/29, & N = 1, \\ 103/157, & N = 3. \end{cases}$$

   This prescription alone does not diagonalize the neutral kinetic terms.

2. **Define the angle from the neutral current.** When the neutral mass term has the Standard Model direction $(W_3 - B)^2$, the photon direction is $(1, 1)$. Its kinetic-orthogonal massive direction is proportional to

$$(K_{YY} + K_{3Y},\, -K_{33} - K_{3Y}).$$

   Writing its current as $T_3 - \sin^2\theta_{\text{NC}} Q$ gives

$$\cos^2\theta_{\text{NC}} = \frac{K_{YY} + K_{3Y}}{K_{33} + 2K_{3Y} + K_{YY}}.$$

   Here

$$K_{33} + K_{3Y} = 2N - 1 + r, \qquad K_{YY} + K_{3Y} = \frac{5}{3}(2N - 1 + r),$$

   and therefore

$$\boxed{\cos^2\theta_{\text{NC}}(x) = 5/8.}$$

   This equality is special to the chosen heavy-up family. It does not restore a common $SU(2)$ kinetic coefficient or the usual relations among charged and neutral couplings.

As a consistency check,

$$K_{33} + 2K_{3Y} + K_{YY} = \frac{8}{3}(2N - 1 + e^{-x}),$$

exactly reproducing Part B's photon coefficient on the particle space.

Verdict on corpus claim.

Item 1 states: "The centralizer and compressed resolvent do not determine the surviving gauge kinetic terms." The calculation confirms that qualification.

- **The full inherited kinetic form survives** the joint limit $m_h \to \infty$, $\Lambda \to \infty$, $m_h/\Lambda \to 0$.
- **Finite nonzero ratios produce threshold weights.** When $m_h/\Lambda \to \infty$, the residual kinetic form is the light-sector trace. For the specified up-quark removals, photon and gluon coefficients retain their ratio but acquire the common reductions 1/2 or 5/6.
-

  The corpus's $5/8$ is a valid electroweak matching condition when the full module is light relative to the matching cutoff, or when that condition is imposed before taking the heavy limit. The broken action does not retain one universal weak angle. Its neutral-current angle happens to remain $5/8$ in this family, whereas the continued charged-W/hypercharge matching ratio changes as above. At the strict $SU(3)\times U(1)_{\mathrm{em}}$ endpoint, a weak angle requires reference to such an earlier matching prescription.

Independent checks (Claude, 2 Nov 2026)

- Part A: (1/12) tr_S 1 + (1/2)(1/4) tr_S(gamma^{mu nu} gamma^{rho sigma}) F F gives (1/3 - 1) tr F^2 = -(2/3) tr F^2. Dividing by 16 pi^2 gives -1/(24 pi^2), the standard spectral-action Yang-Mills coefficient.
- Part B: the per-generation traces tr Q^2 = 8/3 (u), 2/3 (d), 2 (e) and color index 1, 1, 0 were rechecked by hand.
- Part C: w_L(x, 0) = 3 int_0^1 u^2 e^{-xu} du = 3[2 - (x^2+2x+2) e^{-x}]/x^3 agrees by quadrature to 1e-10 at x = 0, 0.3, 1, 4, 12. The limits cos^2 theta_match(infinity) = (80N-34)/(128N-70) give 23/29 (N=1) and 103/157 (N=3). The identity (K_YY + K_3Y) = (5/3)(K_33 + K_3Y), hence cos^2 theta_NC = 5/8 for every x, was verified algebraically.

Part 05

The geometric identification exists: the minimal grading-preserving spectral truncation of a suitable chiral interval is exactly Connes–Lott, with

$$|m| = \frac{\pi}{2l}, \qquad d_{\mathrm{trunc}} = \frac{2l}{\pi}.$$

The truncated endpoint states are smeared states on the original interval. Moreover, this interval has no zero mode, which limits the physical interpretation of both extremes.

Part 1. Deconstructed chain

Statement. There is a natural family with opposite endpoint chiralities, an exact Connes–Lott member at $N = 2$, and endpoint distance converging to l. The answer depends on the representation of the site algebra; the familiar lattice-distance anomaly is not universal.

Hypotheses. We consider complex even spectral triples, without imposing additional real-structure or KO-dimension axioms. Let $N \geq 2$, $n = N - 1$, and

$$A_N = \mathbb{C}^N, \qquad H_N = \mathrm{span}\{L_1, \dots, L_{N-1}, R_2, \dots, R_N\}.$$

Thus interior sites carry both components, site 1 only L, and site N only R. Define

$$\pi_N(a) = \mathrm{diag}(a_1, \dots, a_{N-1}, a_2, \dots, a_N), \qquad \Gamma_N = \mathrm{diag}(I_n, -I_n).$$

If S_n is the lower shift, set

$$B_N = h_N(I_n - S_n), \qquad D_N = \begin{pmatrix} 0 & B_N \\ B_N^* & 0 \end{pmatrix}.$$

These are forward/backward differences:

$$(D_N\psi)_{L_i} = h_N(R_{i+1} - R_i), \qquad (D_N\psi)_{R_j} = h_N(L_{j-1} - L_j),$$

with the missing components interpreted as

$$R_1 = 0, \qquad L_N = 0.$$

There are nearest-neighbour couplings and onsite L-R terms at interior sites.

For compatibility with the spectral truncation in Part 3, choose

$$\theta_N = \frac{\pi}{4N-2}, \qquad h_N = \frac{\pi}{4l\sin\theta_N} \sim \frac{N}{l}. \tag{1}$$

This fixes the smallest positive eigenvalue to $\pi/(2l)$ at every N.

Proof.

The representation is faithful, $[\Gamma_N, \pi_N(a)] = 0$, and $\Gamma_N D_N = -D_N\Gamma_N$. At $N = 2$,

$$A_2 = \mathbb{C}^2, \quad H_2 = \mathbb{C}^2, \quad D_2 = \begin{pmatrix} 0 & h_2 \\ h_2 & 0 \end{pmatrix}, \quad h_2 = \frac{\pi}{2l}.$$

This is exactly the stated two-point triple; a diagonal unitary supplies any desired phase of m.

Ordering the basis as

$$L_1, R_2, L_2, R_3, \dots, L_{N-1}, R_N$$

makes D_N a path adjacency matrix with alternating signs. A diagonal unitary removes those signs. The usual sine eigenvectors of a path therefore give

$$\mathrm{spec}(D_N) = \left\{ 2h_N \cos\frac{j\pi}{2N-1} : 1 \leq j \leq 2N-2 \right\} = \{\pm\lambda_{N,r} : 1 \leq r \leq N-1\},$$

where

$$\boxed{\lambda_{N,r} = 2h_N \sin((2r-1)\theta_N).} \tag{2}$$

Consequently,

$$\boxed{K_N(t) = \mathrm{Tr}\, e^{-tD_N^2} = 2\sum_{r=1}^{N-1} e^{-4th_N^2 \sin^2((2r-1)\theta_N)}.} \tag{3}$$

For a real site function a, the onsite terms commute with $\pi_N(a)$, while

$$B_N\pi_R(a) - \pi_L(a)B_N = h_N\,\mathrm{diag}(a_2 - a_1, \dots, a_N - a_{N-1}).$$

Hence

$$\|[D_N, \pi_N(a)]\| = h_N \max_i |a_{i+1} - a_i|.$$

The triangle inequality bounds $|a_j - a_i|$ by $|j-i|/h_N$; the function $a_i = (i-1)/h_N$ attains this bound. Thus

$$\boxed{d_N(i,j) = \frac{|i-j|}{h_N}.} \tag{4}$$

In particular,

$$\boxed{d_N(1,N) = \frac{4l(N-1)}{\pi} \sin\frac{\pi}{4N-2} \longrightarrow l.} \tag{5}$$

One can instead choose the geometric normalization

$$h_N = \frac{N-1}{l}. \tag{6}$$

Then $d_N(1,N) = l$ exactly for every N, with the same continuum limit. However, its $N = 2$ mass is $1/l$, rather than the continuum lowest eigenvalue $\pi/(2l)$. Equations (1) and (6) distinguish spectral calibration from metric calibration at finite resolution.

Two other natural choices clarify the literature:

- **One-component staggered path.** Take $A = \mathbb{C}^M$, $H = \mathbb{C}^M$, alternating grading, and $D = h$ times path adjacency. Opposite endpoint chiralities require even M. Its spectrum is $2h\cos(j\pi/(M+1))$. For $k = |i-j|$,

$$\boxed{d(i,j) = \begin{cases} (k+1)/(2h), & k \text{ odd}, \\ \sqrt{k(k+2)}/(2h), & k \text{ even}. \end{cases}} \tag{7}$$

  In particular $d(1,M) = M/(2h)$. Thus $h = M/(2l)$ gives endpoint distance l, whereas $h = M/l$ gives $l/2$.

  Here is a direct proof of (7). Restrict the commutator to the connecting subpath and extend the function constantly outside it. After removing phases, its norm is that of weighted adjacency, with dimensionless edge weights $w_i = h|\Delta a_i|$. If its two bipartition sizes are p, q, the off-diagonal block C satisfies

$$\sum_i w_i = \mathbf{1}_p^T C \mathbf{1}_q \le \sqrt{pq}\,\|C\|.$$

  For $p = q$, alternating weights $1, 0, 1, \ldots, 1$ attain the bound. For $p = q+1$, the weights

$$\frac{(q, 1, q-1, 2, \ldots, 1, q)}{\sqrt{q(q+1)}}$$

  attain it: the normalized constant vectors on the two bipartitions are singular vectors with singular value 1, and positivity gives the norm 1. This proves (7).

  These are the nonadditive distances studied by Bimonte–Lizzi–Sparano, §5 and given explicitly by Atzmon, Eq. (30). Reduction to the connecting path also agrees with Iochum–Krajewski–Martinetti, Proposition 4.

- **Vertex-edge incidence Dirac.** Let $\partial : \mathbb{C}^N \to \mathbb{C}^{N-1}$ be the path difference matrix and use

$$D_{\text{inc}} = h \begin{pmatrix} 0 & \partial^* \\ \partial & 0 \end{pmatrix}.$$

  Its spectrum and heat trace are

$$\{0\} \cup \left\{ \pm 2h \sin\frac{k\pi}{2N} : 1 \le k < N \right\},$$

$$K_{\mathrm{inc}}(t) = 1 + 2\sum_{k=1}^{N-1} e^{-4th^2 \sin^2(k\pi/(2N))}.$$

These follow by substituting the cosine eigenvectors into the path Laplacian $\partial^*\partial$. Representing a site function on each edge by its value at the tail gives distance $|i-j|/h$. But H has dimension $2N-1$, has a zero mode, and both endpoint vertex components have the same grading. At $N = 2$, its spectrum is $0, \pm\sqrt{2}h$, so it is not Connes–Lott.

Thus the quoted integer deconstruction tower belongs to the incidence construction. Opposite chiral boundary conditions produce the shifted tower (2).

Status: PROVED

Part 2. Continuum limit

Statement. The chain above converges to an interval Dirac operator in three precise senses: ordered low eigenvalues, heat traces locally uniformly for $t > 0$, and pure-state metric spaces in Gromov–Hausdorff distance. It also has a controlled one-dimensional heat window.

Hypotheses. On $H_l = L^2([0,l], \mathbb{C}^2)$, define

$$D_l = \begin{pmatrix} 0 & \partial_x \\ -\partial_x & 0 \end{pmatrix}, \qquad \mathrm{Dom}\, D_l = \{(L,R) \in H^1 : R(0) = 0,\ L(l) = 0\}, \tag{8}$$

with scalar multiplication by $C^\infty([0,l])$ and grading $\Gamma = \mathrm{diag}(1,-1)$.

These are opposite local chiral conditions. Reflection parity is Γ at 0 and $-\Gamma$ at l; their composition gives antiperiodicity on the doubled interval.

Proof.

Integration by parts gives the boundary form

$$[\overline{L}\,\widetilde{R} - \overline{R}\,\widetilde{L}]_0^l.$$

The conditions in (8) form a maximal isotropic boundary subspace, so D_l is self-adjoint. Its resolvent is compact by the compact embedding $H^1 \hookrightarrow L^2$.

Solving $D_l\psi = \lambda\psi$ gives

$$\psi_\lambda(x) = c(\cos\lambda x, \sin\lambda x), \qquad \cos(\lambda l) = 0.$$

Therefore

$$\boxed{\mathrm{spec}(D_l) = \left\{ \pm\frac{(r-\frac{1}{2})\pi}{l} : r = 1, 2, ... \right\}.} \tag{9}$$

The kernel is zero. Squaring gives mixed Neumann–Dirichlet conditions on L, and Dirichlet–Neumann conditions on R.

For scalar f,

$$\|[D_l, f]\| = \|f'\|_\infty, \qquad d_l(x,y) = |x-y|. \tag{10}$$

Under normalization (1), with $m_1 = \pi/(2l)$,

$$\lambda_{N,r} = m_1 \frac{\sin((2r-1)\theta_N)}{\sin\theta_N} \longrightarrow (2r-1)m_1. \tag{11}$$

This proves low-eigenvalue convergence. Since

$$\sin y \geq \frac{2y}{\pi} \quad (0 \leq y \leq \pi/2),$$

we also have

$$\lambda_{N,r} \geq \frac{2r-1}{l}. \tag{12}$$

A summable Gaussian majorant now proves, uniformly on every $t \in [\varepsilon, T]$,

$$K_N(t) \longrightarrow K_l(t) = 2\sum_{r=1}^{\infty} e^{-t((r-\frac{1}{2})\pi/l)^2}. \tag{13}$$

Poisson summation gives

$$\boxed{K_l(t) = \frac{l}{\sqrt{\pi t}}\left(1 + 2\sum_{q=1}^{\infty}(-1)^q e^{-q^2 l^2/t}\right).} \tag{14}$$

In particular, $K_l(t) \sim l/\sqrt{\pi t}$ as $t/l^2 \to 0$.

The finite-chain heat window has the following rigorous meaning. For every sequence $t_N > 0$ satisfying

$$\frac{t_N}{l^2} \to 0, \qquad \frac{N^2 t_N}{l^2} \to \infty, \tag{15}$$

one has

$$\boxed{K_N(t_N) \sim \frac{l}{\sqrt{\pi t_N}}, \qquad -2t_N \frac{K_N'(t_N)}{K_N(t_N)} \to 1.} \tag{16}$$

To prove this, put $u_r = \pi\sqrt{t_N}(r-\frac{1}{2})/l$. On bounded u_r-intervals, the sine argument in (11) tends uniformly to zero, so $\sqrt{t_N}\lambda_{N,r}/u_r \to 1$. The mesh tends to zero and its upper endpoint tends to infinity. Bound (12) controls the tails. Gaussian Riemann sums, and then the same sums weighted by $t_N\lambda_{N,r}^2$, prove both assertions.

Finally, identify site i with $x_i = (i-1)l/(N-1)$. Equation (4) gives

$$d_N(i,j) = \rho_N |x_i - x_j|, \qquad \rho_N = \frac{N-1}{l h_N} = 1 + O(N^{-1}).$$

The metric distortion and the grid mesh are both $O(l/N)$, proving Gromov–Hausdorff convergence to $[0, l]$.

The endpoint chirality here concerns allowed boundary components. It does not supply localized chiral zero modes.

Status: PROVED

Part 3. Minimal spectral truncation

Statement. The minimal nonzero grading-preserving spectral truncation of (8) is unitarily equivalent to Connes–Lott, with

$$\boxed{|m| = m_1 = \frac{\pi}{2l}, \qquad c = \frac{\pi}{2}.}$$

Its natural truncated endpoint states have distance $1/m_1 = 2l/\pi$. Literal endpoint evaluations do not descend through compression.

Hypotheses. Let

$$P = \mathbf{1}_{[0,E]}(|D_l|), \qquad m_1 \leq E < 3m_1,$$

and give $S = P\,C([0,l])\,P$ its inherited operator-system structure. This is the compression construction described in Connes–van Suijlekom, §1.

Proof.

An orthonormal chiral basis of PH_l is

$$u_L(x) = \sqrt{\frac{2}{l}} \begin{pmatrix} \cos(\pi x/(2l)) \\ 0 \end{pmatrix}, \qquad u_R(x) = \sqrt{\frac{2}{l}} \begin{pmatrix} 0 \\ \sin(\pi x/(2l)) \end{pmatrix}.$$

In this basis,

$$PD_lP = \begin{pmatrix} 0 & m_1 \\ m_1 & 0 \end{pmatrix}, \qquad P\Gamma P = \mathrm{diag}(1,-1). \tag{17}$$

Compression of multiplication is

$$Q(f) = \begin{pmatrix} \frac{2}{l}\int_0^l f(x)\cos^2\frac{\pi x}{2l}\,dx & 0 \\ 0 & \frac{2}{l}\int_0^l f(x)\sin^2\frac{\pi x}{2l}\,dx \end{pmatrix}. \tag{18}$$

Since

$$Q(1) = I, \qquad Q(\cos(\pi x/l)) = \tfrac{1}{2}\,\mathrm{diag}(1,-1),$$

its range is exactly the full diagonal algebra:

$$\boxed{S = \mathbb{C} \oplus \mathbb{C}.} \tag{19}$$

Here the compressed operator system happens to be an algebra, although Q is not multiplicative.

Evaluation on the finite-dimensional space PH_l is bounded. At 0, only u_L has a nonzero boundary value; at l, only u_R does. The normalized evaluation vectors therefore define

$$\omega_0(\mathrm{diag}(a,b)) = a, \qquad \omega_l(\mathrm{diag}(a,b)) = b.$$

Using (17),

$$\|[PD_lP, \mathrm{diag}(a,b)]\| = m_1|a-b|,$$

and hence

$$\boxed{d_{\mathrm{trunc}}(\omega_0,\omega_l) = \frac{1}{m_1} = \frac{2l}{\pi} < l.} \tag{20}$$

Minimality follows because there is no zero mode and the grading exchanges the eigenspaces at $+m_1$ and $-m_1$. A nonzero grading-preserving spectral subspace must retain both.

There is an essential distinction between these states and literal evaluations. For

$$g(x) = \cos(3\pi x/l),$$

equation (18) gives $Q(g) = 0$, while $g(0) = 1$ and $g(l) = -1$. Thus neither endpoint evaluation factors through Q.

The pullbacks of the truncated endpoint states are instead the probability measures

$$d\mu_L = \frac{2}{l}\cos^2\frac{\pi x}{2l}\,dx, \qquad d\mu_R = \frac{2}{l}\sin^2\frac{\pi x}{2l}\,dx.$$

Their distance using the **original** continuum seminorm is

$$d_{D_l}(\mu_L,\mu_R) = \int_0^l \left|\int_0^x (d\mu_L - d\mu_R)\right| dx = \frac{4l}{\pi^2}. \tag{21}$$

The equality follows by integration by parts and is attained by $f = -x$. Equations (10), (20), and (21) concern three different state/metric comparisons.

Status: PROVED

Part 4. The two limits

Amendment (Terra 6.1, 3 Nov 2026; report). The statements of this part are correct for the stated bare chain, with one precision: the condition N/l -> infinity gives uniform convergence of the dispersion on bounded energy windows and of normalized heat traces and counting densities, and does not by itself give convergence of the unnormalized difference of heat traces (for N = floor(l^(5/4)) it diverges). Escapes for the empty light sector: a tuned bulk mass or a scaled boundary coupling.

Statement. At fixed finite rank, $m \to 0$ produces infinitely separated finite geometry, not an additional heat dimension. A resolved interval requires a growing tower. As $m \to \infty$, the metric collapses, but the light Hilbert space is zero.

Hypotheses. Identify

$$m = m_1 = \frac{\pi}{2l},$$

use normalization (1), and hold physical energies or heat times fixed unless stated otherwise.

Proof.

For fixed N, $D_N = mT_N$ with T_N independent of m. Therefore, as $m \to 0$,

$$D_N \to 0, \qquad K_N(t) \to 2(N-1), \qquad d_N(i,j) \to \infty \quad (i \neq j).$$

For the minimal truncation,

$$K_2(t) = 2e^{-tm^2}.$$

Every finite N has ultraviolet heat dimension zero. Making its finitely many modes light does not change that conclusion.

For the full interval, a fixed energy E is in the resolved one-dimensional regime when

$$\frac{1}{l} \ll E.$$

For its finite-chain approximation, one additionally needs

$$\boxed{\frac{1}{l} \ll E \ll \frac{N}{l}.} \tag{22}$$

Thus, as $l \to \infty$, an exact continuum limit throughout a fixed positive energy window follows if $N/l \to \infty$. Equivalently, as $m \to 0$, one needs $Nm \to \infty$.

The continuum counting function and heat density make the emergence explicit:

$$\#\{|\lambda| \le E\} = 2\left\lfloor \frac{lE}{\pi} + \frac{1}{2} \right\rfloor, \qquad \frac{K_l(t)}{l} \longrightarrow \frac{1}{\sqrt{\pi t}} \quad (l \to \infty). \tag{23}$$

Based at an endpoint, the metric limit is a half-line; the opposite endpoint recedes to infinity.

The order of limits matters even for the normalized heat trace:

$$\lim_{l\to\infty}\lim_{N\to\infty} \frac{K_N(t)}{l} = \frac{1}{\sqrt{\pi t}}, \qquad \lim_{N\to\infty}\lim_{l\to\infty} \frac{K_N(t)}{l} = 0. \tag{24}$$

Likewise, taking $t \to 0$ at fixed N gives dimension zero, whereas taking $N \to \infty$ first gives the interval's dimension one.

As $m \to \infty$, all distances tend to zero. At fixed N, the algebra elements with uniformly bounded commutators reduce to constants. But every eigenvalue has magnitude at least m, so

$$\|(D_N - i)^{-1}\| \le \frac{1}{\sqrt{m^2+1}} \longrightarrow 0. \tag{25}$$

The same holds for D_l, after identifying the varying intervals by dilation. Its heat trace tends to zero at every fixed $t > 0$.

Thus the metric limit is a point, while the embedded resolvent limit has

$$H_{\text{light}} = \ker T_N = 0.$$

There is no finite operator limit on the original nonzero Hilbert space, and no surviving fermion pair at fixed energy.

Finally, the two-state spectral truncation exists for

$$m_1 \leq E < 3m_1.$$

A genuine spectral cutoff $E \ll 1/l$ retains no modes. The ratio between the first two positive levels is always 3, so the minimal pair has no parametrically large separation from the next level.

Status: PROVED

Verdict on hypothesis H (theorem T1)

Supported for T1's geometric assertion. Connes–Lott is exactly the minimal nonzero grading-preserving spectral compression of the opposite-chirality interval above, with

$$l = \frac{\pi}{2|m|}.$$

The chain recovers its spectrum, heat window, and metric.

The precise replacement for the stronger interpretation of H is:

> The two sheets are truncated endpoint evaluation states, whose pullbacks are smeared over the interval. Refinement restores the interval. Decompactification produces a one-dimensional heat regime only when the number of retained modes grows sufficiently rapidly. Collapse identifies the metric points but removes all fermionic modes of this particular interval at fixed energy.

Accordingly, a nonempty two-state sector at $E \ll 1/l$, or surviving light fermions after collapse, is disproved by this model. The construction supplies no KO-6 or Standard Model identification.

Independent numerical check (Claude, 2 Nov 2026)

For l = 1 and N = 2, 3, 5, 9 we checked the following. The spectrum of D_N agrees with +/- 2 h_N sin((2r-1) theta_N) to 1e-14. The lowest positive eigenvalue equals pi/(2l) for every N. For random real site functions, ||[D_N, pi_N(a)]|| equals h_N max abs(a_{i+1} - a_i), which is the identity behind d_N(i,j) = abs(i-j)/h_N. The endpoint distance d_N(1,N) = 0.6366, 0.7869, 0.8844, 0.9398 tends to l. For the minimal truncation, Q(cos(3 pi x/l)) = 0, and the smeared-state distance 0.405290 matches 4l/pi^2. The truncated distance is 2l/pi = 0.6366, identical to d_2(1,2).

Literature (../LITERATURE.md, section A). The Kaluza-Klein reading of the two-point space is in Connes-Lott (sections I and III: "two copies of Z separated by a distance L"); the two points as points of a fifth dimension are in Viet-Wali 1996; the two-point model as an approximation of a manifold, to be related to dimensional reduction, is in Dimakis-Muller-Hoissen 1994 (section III, Example 4, and section VII). Connes-van Suijlekom (section 1) note that compressions P A P are operator systems, of which the chiral interval is an exception. The truncation theorem with |m| = pi/(2l) was not found in print.

Part 06

The proposed uniqueness is false: **the same KO-6 triple can admit both reductions**. Moreover, under the stated definition, the strict heavy-limit triple is a doubling of KO-7, although it is not a doubling of KO-5. The distinction between the full real Hilbert space and its physical particle subspace is essential.

1. Lemma — Product convention and necessary correction of the "Clifford point."

Under ordinary spectral-triple conventions, a graded triple on `C^2` is not itself an odd KO-1 or KO-7 triple. The appropriate interpretation is an odd point

$$P_k = (\mathbb{C}, \mathbb{C}, 0, K), \qquad k = 1 \text{ or } 7,$$

together with the auxiliary `C^2` introduced by the odd-times-odd product. For real algebras, use the point algebra `R`, so that the product algebra remains `A`.

I use the following products, with the standard antilinear tensor product of real structures.

For even × even,

$$D = D_1 \otimes 1 + \gamma_1 \otimes D_2, \qquad \gamma = \gamma_1 \otimes \gamma_2,$$
$$J = J_1 \otimes (J_2 \gamma_2^b), \qquad b = (1 - \epsilon_1'')/2.$$

For even × odd, use the same formula for `D`, no grading, and

$$J = (J_1 \gamma_1^a) \otimes J_2, \qquad (-1)^a = \epsilon_1'' \epsilon_2'.$$

Odd × even is obtained by transporting the even-first construction through the tensor flip.

For odd × odd,

$$H = H_1 \otimes H_2 \otimes \mathbb{C}^2, \quad D = D_1 \otimes 1 \otimes \sigma_1 + 1 \otimes D_2 \otimes \sigma_2, \quad \gamma = 1 \otimes 1 \otimes \sigma_3,$$
$$J = J_1 \otimes J_2 \otimes (BK),$$

where

$(\epsilon_1', \epsilon_2')$	$(+,+)$	$(+,-)$	$(-,+)$	$(-,-)$
B	σ_1	1	σ_3	$-i\sigma_2$.

This agrees, up to harmless phases, with Dąbrowski–Dossena, equation (25), Table 6.

Proof. In the odd-odd case,

$$B\sigma_1 = \epsilon_1' \sigma_1 B, \qquad B\sigma_2 = -\epsilon_2' \sigma_2 B.$$

The second minus sign compensates for complex conjugation of `sigma_2`. Consequently `JD=DJ`; computing `J^2` and `J gamma` gives the signs of the sum degree. The grading twists in the other two formulas follow by the same calculation. Order zero and first order pass to these products.

Thus the intended doubling has Hilbert space `H_0 ⊗ C^2`, with the displayed grading on the auxiliary factor. All subsequent statements use this corrected interpretation and the zero-Dirac point. ■

2. Theorem — Exact doubling criterion in KO-dimension six.

Let `(A,H,D,J,gamma)` be a finite KO-6 triple. For `s=±1`, the following are equivalent:

1. It is a product `T_n × P_k`, where

$$(s, n, k) = (+1, 7, 7) \quad \text{or} \quad (-1, 5, 1).$$

2. There is a complex-linear unitary `U` satisfying

$$U^2 = 1, \quad U\gamma = -\gamma U, \quad [U, a] = [U, a^o] = [U, D] = 0, \quad UJ = sJU,$$

where `a^o=Ja^*J^{-1}`.

Write `H_L=H_{\gamma=+1}`, `H_R=H_{\gamma=-1}`, and

$$D = \begin{pmatrix} 0 & M^* \\ M & 0 \end{pmatrix}.$$

Equivalently, there is an `A`-bimodule unitary `V:H_L→H_R` such that

$$V^*M = (V^*M)^*, \qquad (V^*C)^2 = s, \quad C = J|_{H_L} : H_L \longrightarrow H_R.$$

Here `C` and `V^*C` are antilinear.

Proof. Given `U`, identify

$$H_L \oplus H_L \longrightarrow H, \qquad (x, y) \longmapsto x + Uy.$$

Put

$$H_0 = H_L, \qquad d = (UD)|_{H_L}, \qquad j = (UJ)|_{H_L}.$$

The assumptions give

$$d = d^*, \qquad j^2 = s, \qquad jd = s\, dj.$$

Hence `(j^2,jd/dj)` is `(+,+)` for KO-7 and `(-,-)` for KO-5. In these coordinates,

$$a = \begin{pmatrix} a_0 & 0 \\ 0 & a_0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & d \\ d & 0 \end{pmatrix}, \quad \gamma = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$J(x, y) = (sjy, jx).$$

These are precisely the products in Result 1: `B=σ_1` for `(7,7)` and `B=-iσ_2` for `(5,1)`.

The descended opposite action is

$$ja_0^*j^{-1} = a^o|_{H_L}.$$

Since `[d,a_0]=U[D,a]|_{H_L}`, order zero and first order descend. Conversely, the product has `U=1⊗σ_1`, with all the required properties.

Finally,

$$U = \begin{pmatrix} 0 & V^* \\ V & 0 \end{pmatrix}.$$

Its commutation with `D` is equivalent to self-adjointness of `V^*M`, while `UJ=sJU` is equivalent to `(V^*C)^2=s`.

If one prefers an anticommuting convention, replacing `U` by `i gamma U` exchanges `[U,D]=0` with `{U,D}=0` and preserves `s`. ■

In particular, ordinary equality of left and right module multiplicities is only part of the criterion; compatibility with `D` and `J` must also be checked.

3. Counterexample — J does not select a unique reduction; electrodynamics admits both.

Consider the electrodynamics triple in the basis `(e_L,e_R,bar e_L,bar e_R)`. After a phase change compatible with `A,J,gamma`, its matrices are

$$\rho(a, b) = \mathrm{diag}(a, a, b, b), \qquad \gamma = \mathrm{diag}(1, -1, -1, 1),$$

$$J = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix} K, \qquad D = m \begin{pmatrix} \sigma_1 & 0 \\ 0 & \sigma_1 \end{pmatrix}, \quad m \geq 0.$$

These are the input data of van den Dungen–van Suijlekom, §4.1, equations (4.1)–(4.2).

Define

$$U_+ = \mathrm{diag}(\sigma_1, \sigma_1), \qquad U_- = \mathrm{diag}(\sigma_1, -\sigma_1).$$

Proof. Both involutions reverse the grading and commute with the algebra, opposite algebra, and `D`. Direct multiplication gives

$$U_+J = JU_+, \qquad U_-J = -JU_-.$$

Thus the same triple, with the same J, admits both factorizations.

On H_0=span(e_L,bar e_R), the two descended triples have

$$d_+ = mI_2, \qquad j_+ = \sigma_1 K$$

and

$$d_- = \mathrm{diag}(m, -m), \qquad j_- = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} K.$$

They have KO-degrees seven and five, respectively. This also holds at m=0. ■

4. Lemma — Exact neutral-sector criterion, including Majorana masses.

Suppose a KO-6 sector has

$$H = H_+ \oplus H_-, \qquad \dim H_+ = \dim H_- = r,$$

and both algebra actions are the same scalar character throughout this sector. Then:

- a KO-7 reduction always exists;
- a KO-5 reduction exists exactly when **every singular value of M, including zero, has even multiplicity**.

Proof. Choose an orthonormal basis of H_+ and its J-image as the basis of H_-. Then

$$J = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} K.$$

The equation JD=DJ says M^T=M. Takagi factorization, implemented by a basis change preserving this form of J, reduces M to

$$\Sigma = \mathrm{diag}(\mu_1, \ldots, \mu_r), \qquad \mu_i \geq 0.$$

Taking V=I satisfies Result 2 with s=+1.

For s=-1, the descended antiunitary j squares to -1 and commutes with d^2=D^2|_{H_+}. Every eigenspace of d^2 must therefore have even complex dimension. These dimensions are exactly the singular-value multiplicities.

Conversely, pair equal entries of Sigma and choose V=σ_2 on each pair. Then V is a skew-symmetric Hermitian unitary commuting with Sigma; it satisfies the criterion with s=-1. ■

A single neutral Weyl state and its conjugate therefore **are** a KO-7 doubling, even without a Dirac partner. They cannot be a KO-5 doubling. The same statement holds for a single Majorana pair, whether its mass vanishes or not.

5. Proposition — The strict one-generation heavy limit is a KO-7 doubling, not a KO-5 doubling.

Here is an explicit convention for the Standard Model representation. Set

$$F = \mathbb{C} \oplus \mathbb{C}^3, \qquad H_p = (\mathbb{C}^2 \otimes F)_L \oplus (\mathbb{C}^2 \otimes F)_R, \qquad H = H_p \oplus H_{\bar{p}}.$$

For a=(lambda,q,m), put

$$Q_\lambda = \mathrm{diag}(\lambda, \bar{\lambda}), \qquad B_{\lambda,m} = \mathrm{diag}(\lambda, m),$$
$$\rho_p(a) = (q \otimes I_4) \oplus (Q_\lambda \otimes I_4),$$
$$\rho_{\bar{p}}(a) = (I_2 \otimes B_{\lambda,m}) \oplus (I_2 \otimes B_{\lambda,m}).$$

Use

$$J(p, \bar{p}) = (K\bar{p}, Kp), \qquad \gamma = \mathrm{diag}(I_8, -I_8, -I_8, I_8).$$

On particles, the Dirac block is

$$S_p = \begin{pmatrix} 0 & Y^* \\ Y & 0 \end{pmatrix},$$

where `Y` has entries `y_nu,y_e,y_u I_3,y_d I_3` on the corresponding species. The antiparticle block is `bar S_p`; the additional Majorana block connects `nu_R` with `J nu_R`. This realizes the bimodule and mass-block structure in CCM, Proposition 2.4 and Theorem 2.21.

Assume precisely the strict projection described in canonical §3: remove `nu_R,J nu_R` and the complete up-type Dirac sector, and restrict to

$$B = \{(\lambda, Q_\lambda, m)\} \simeq \mathbb{C} \oplus M_3(\mathbb{C}).$$

The surviving space consists of `d_L,d_R,e_L,e_R,nu_L` and their conjugates, with `D_light=PSP`.

Proof. The surviving down-quark and electron masses pair identical `B`-bimodules. After making their masses nonnegative, each sector has both involutions from Result 3, with the identity on color.

On

$$N = \mathrm{span}(\nu_L, J\nu_L),$$

both algebra actions are the scalar character `lambda`, and `PSP|_N=0`. Thus

$$\gamma_N = \sigma_3, \qquad J_N = \sigma_1 K, \qquad U_N = \sigma_1$$

gives a KO-7 reduction. Combining this with `U_+` on the charged sectors gives a global KO-7 reduction.

A KO-5 reduction is impossible: the full space has dimension eighteen, so its descended space would have dimension nine, which cannot carry an antiunitary squaring to `-1`. More specifically, the neutral bimodule is inequivalent to the charged bimodules, so any permitted `U` preserves it; Result 4 already obstructs KO-5 there. ■

Thus the anticipated claim "the neutral sector prevents any doubling" is false under the definition in Question 1.

6. Proposition — Modifications and the distinction from physical vectorlikeness.

For the reduced algebra `B` above, the following conclusions hold.

Surviving neutral content	Odd reductions of the full light triple
No neutral sector	Both KO-5 and KO-7
`nu_L,nu_R` and conjugates, pure Dirac mass	Both KO-5 and KO-7
Only `nu_L,J nu_L`	KO-7 only
A single isolated Majorana pair	KO-7 only
General neutral Dirac–Majorana block	Always KO-7; KO-5 exactly under Result 4

Proof. The charged sectors admit both reductions by Result 3. Pure Dirac neutrinos have the same four-state construction, so adding or retaining `nu_R` restores both possibilities, including when its Dirac mass is zero.

For one generation with Dirac mass `m` and right Majorana mass `R`, the neutral symmetric matrix can be written

$$M_N = \begin{pmatrix} 0 & m \\ m & R \end{pmatrix}.$$

Phase changes make `m,R` nonnegative. Its singular values are

$$\mu_\pm = \frac{\sqrt{|R|^2 + 4|m|^2} \pm |R|}{2}.$$

They coincide exactly when `R=0`. Hence any nonzero right Majorana mass obstructs KO-5 in this one-generation neutral sector, but never KO-7. Treating that sector separately leaves a charged KO-5 reduction; it does not turn the complete direct sum into a KO-5 reduction. ■

There is an additional physically meaningful condition absent from Question 1. Let `Q` be `+1` on particles and `-1` on antiparticles.

No-go theorem under the extra hypothesis `[U,Q]=0`. An unpaired `nu_L` prevents doubling.

Proof. Such a `U` must map particle states of positive grading to particle states of negative grading within the same bimodule. No particle `nu_R` exists. The neutral involution used in Result 5 instead maps `nu_L` to its conjugate and violates `[U,Q]=0`. ■

This is why physical chirality and doubling of the complete real triple are different tests.

7. Proposition — Three generations and arbitrary flavor mixing.

Assume the reduced algebra acts trivially on generation indices.

If all three right-neutrino Majorana pairs are removed and only the top Dirac sector is made heavy, the light space has

$$96 - 6 - 12 = 78$$

complex dimensions. If all three up-type sectors are removed, it has dimension fifty-four. In either case, with three surviving left neutrinos and `D_light=PSP`, the light triple admits KO-7 and excludes KO-5.

Dropping the remaining neutral sector, or retaining three right neutrinos with purely Dirac neutrino masses, permits both reductions.

Proof. Each surviving charged species has equal left/right flavor multiplicities. Singular-value decomposition of its mass matrix uses bimodule unitaries; conjugate transformations on antiparticles preserve `J`. Consequently every charged sector reduces to copies of Result 3, irrespective of mass rank or flavor mixing.

The three unpaired neutral states give `r=3`, `M_N=0` in Result 4, excluding KO-5 and admitting KO-7. Equivalently, the proposed descended spaces have odd dimensions thirty-nine and twenty-seven.

With three right neutrinos retained, the general neutral matrix is

$$M_N = \begin{pmatrix} 0 & Y_\nu^T \\ Y_\nu & R \end{pmatrix}, \qquad R^T = R.$$

For `R=0`, every singular value of `Y_nu` appears twice, so both reductions exist. For general `R`, KO-7 always exists, while KO-5 exists exactly when every singular value of this six-by-six matrix has even multiplicity. Thus nonzero Majorana masses are not an unconditional obstruction in several generations; the exact obstruction is the stated multiplicity condition. ■

8. No-go theorem — The massless Standard Model triple is not a doubling.

Let the algebra remain the full

$$A = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$$

with the representation of Result 5, for `g≥1` generations. Then no involution satisfying Result 2 exists, for any admissible `D`. In particular this applies at `v=0`, when only the Majorana block remains.

Proof. Consider the central projection

$$e_{\mathbb{H}} = (0, 1_{\mathbb{H}}, 0).$$

Its represented range lies entirely in the particle left-handed sector:

$$\dim \rho(e_{\mathbb{H}}) H_+ = 8g, \qquad \dim \rho(e_{\mathbb{H}}) H_- = 0.$$

An algebra-linear grading-reversing unitary would identify these spaces, which is impossible.

In physical representation-ring notation,

$$V_L = (3,2)_{1/6} \oplus (1,2)_{-1/2},$$

$$V_R = (3,1)_{2/3} \oplus (3,1)_{-1/3} \oplus (1,1)_{-1} \oplus (1,1)_0.$$

Thus `I_G=[V_L]-[V_R]≠0`. For the complete real graded Hilbert space, the relevant gauge character is

$$[H_+] - [H_-] = I_G - \overline{I_G},$$

not merely `I_G`. It too is nonzero: the coefficient of `(3,2)_{1/6}` is `g`. Either this full-character calculation or the central-projection argument proves the obstruction. ■

Consequence for the 10 -> 9 claim

9. Proposition — What dimension reduction is actually established.

The heavy projection itself preserves KO-degree six, as canonical §3 explicitly states. Hence its spacetime product still has degree `4+6=10 mod 8`.

A subsequent choice of a KO-5 reduction removes a degree-one Clifford factor and gives the formal representative `4+5=9`. Such a reduction exists for the charged truncation or the pure-Dirac completion, but **both also admit KO-7 reductions**. Their fixed `J` does not select nine.

For the strict light spectrum with one or three unpaired left neutrinos, **KO-5 is obstructed; only KO-7 is available among the proposed reductions**. This gives the formal representative eleven, not nine. If particle preservation is additionally required, that strict light spectrum admits neither reduction.

Proof. These are Results 2–7. For completeness, the spacetime real structures are

$$J_{4+5} = (J_M \gamma_M) \otimes j_5, \qquad J_{4+7} = J_M \otimes j_7.$$

Thus even the odd reductions require different product conventions. No finite-dimensional operation here changes heat dimension: a finite internal factor remains spectrally zero-dimensional. Consequently the claimed loss of one dimension is, at most, a chosen Clifford-factor removal—not a proved metric collapse, and not a consequence of vectorlike charged matter alone. ■ —

Independent numerical check (Claude, 2 Nov 2026)

We used the electrodynamics data of Result 3, with m = 0.7, J = [[0, I_2],[I_2, 0]] K and gamma = diag(1,-1,-1,1). The triple has J^2 = 1, JD = DJ and J gamma = -gamma J, so its KO-dimension is 6. Both U_+ = diag(sigma_1, sigma_1) and U_- = diag(sigma_1, -sigma_1) satisfy U^2 = 1 and U gamma = -gamma U. Both commute with random algebra elements, their opposites and D; and U_+ J = J U_+, U_- J = -J U_-. On H_0 = span(e_L, bar e_R), U_+ descends to j^2 = +1 with jd = dj (KO-7) and U_- to j^2 = -1 with jd = -dj (KO-5). For No-go 8 we also rechecked by hand the dimension count dim rho(e_H) H_+ = 8g, dim rho(e_H) H_- = 0 in the representation of Result 5.

Terminology. Here "doubling" means the factorization of an even triple as an odd triple times a Clifford point (the doubling of an odd triple in Dabrowski-Dossena's product construction). It is unrelated to the "fermion doubling" of the noncommutative Standard Model (Lizzi-Mangano-Miele-Sparano). Prior work on products and on J versus J gamma: ../LITERATURE.md, section D; the factorization criterion and the 9/11 results were not found in print.

Referee note

Terra 6.1 (3 Nov 2026; report) checked Result 3, Lemma 4, Proposition 5, Result 7, No-go theorem 8 and the no-go under [U, Q] = 0: all correct. One amendment to Result 7: "arbitrary flavor mixing" means arbitrary matrices within the displayed species blocks, with the heavy projection chosen onto the specified top Dirac sector; mixing between inequivalent species is outside the displayed Dirac operator. Escapes: for KO-5, neutral-sector pairing or a symmetry enforcing even multiplicities; for No-go 8, a change of algebra or representation, the even description, and boundary or defect realizations of chirality; for the no-go under [U, Q] = 0, a matching particle bimodule of opposite grading.

Part 07

Does the chiral interval select 9 or 11?

Questions 1-3 of prompts/07. The Astra job stopped on the codex usage limit before producing output, so these are Claude Opus 5.5's proofs, checked by hand. Astra job 08 confirmed Lemmas 1-3 under the pointwise reality hypothesis of Lemma 1 and answered Question 4: see lemma 08 and the amendments below.

Let I_l = [0, l], H = L^2(I_l, C^2) with psi = (L, R), D = E d/dx with E = [[0, 1], [-1, 0]], Dom D = {psi in H^1 : R(0) = 0, L(l) = 0}, Gamma = diag(1, -1) (job 05). D and Gamma are real.

Lemma 1 (real structures on the chiral interval). Let J be antiunitary with J f J^{-1} = conj(f) for f in C^infty(I_l), J Dom D = Dom D, J^2 = eps, JD = eps' DJ, J Gamma = eps'' Gamma J. Then J = W K with W = diag(alpha, beta) constant, abs(alpha) = abs(beta) = 1, and

```
eps = +1,   eps'' = +1,   alpha = eps' beta,   eps' in {+1, -1}.
```

Both values of eps' occur, and they are exchanged by J -> J Gamma. The chiral boundary conditions exclude every real structure anticommuting with Gamma; the bare line allows such ones, e.g. W = E with signs (-1, +1, -1).

Proof. Antilinearity together with J f = conj(f) J gives J = W K for a unitary matrix function W(x). Since E is real, D J psi = E W' conj(psi) + E W conj(psi'), while J D psi = W E conj(psi'). Hence eps' D J = J D forces E W' = 0, so W is constant, and W E = eps' E W. Preservation of R(0) = 0 for arbitrary L(0) forces W_21 = 0, and preservation of L(l) = 0 forces W_12 = 0. So W is diagonal and unitary. Then W conj(W) = 1 gives eps = +1, and W commutes with Gamma, so eps'' = +1. Finally diag(alpha, beta) E = [[0, alpha], [-beta, 0]] and E diag(alpha, beta) = [[0, beta], [-alpha, 0]], so alpha = eps' beta. Replacing J by J Gamma keeps eps = eps'' = +1 and flips eps'. ■

With Connes' even convention (eps' = +1) the chiral interval therefore has KO-dimension 0 mod 8, although its metric dimension is 1.

Lemma 2 (compression). Every J of Lemma 1 commutes with P = 1_{[0,E]}(abs(D)). On the truncation span(u_L, u_R) of job 05 it induces J_2 = diag(alpha, beta) K, with D_2 = m_1 sigma_1, Gamma_2 = sigma_3 and the signs (+1, eps', +1).

Proof. JD = eps' DJ gives J D^2 = D^2 J, so J commutes with abs(D) and with its spectral projections. The basis functions u_L = sqrt(2/l)(cos(pi x/2l), 0) and u_R = sqrt(2/l)(0, sin(pi x/2l)) are real, so J u_L = alpha u_L and J u_R = beta u_R. ■

Lemma 3 (the reflection fixes s = eps'). Let (sigma psi)(x) = a (R(l - x), L(l - x)). Then sigma preserves Dom D, anticommutes with Gamma and commutes with constants; [sigma, D] = 0 holds exactly when the phases on the two components agree, and sigma^2 = 1 forces a = +1 or -1. On the truncation, sigma = a sigma_1. For every J of Lemma 1,

```
sigma J = s J sigma   with   s = beta/alpha = eps'.
```

Proof. The swap with x -> l - x exchanges the conditions R(0) = 0 and L(l) = 0. With general phases a, b on the two components, (D sigma psi)_L = -b L'(l - x) and (sigma D psi)_L = -a L'(l - x), so commutation forces a = b. One checks sigma u_L = u_R and sigma u_R = u_L. Then sigma J psi = a (beta conj R(l - x), alpha conj L(l - x)) and J sigma psi = a (alpha conj R(l - x), beta conj L(l - x)), which gives s = beta/alpha = eps'. ■

Consequence. As the interval collapses, only constants keep bounded commutators, and the reflection becomes a canonical L <-> R involution. Its sign against J is s = eps', so the interval fixes s exactly as far as the real structure on it is fixed, and no further. The 9 versus 11 ambiguity of job 06 is the same as the standard ambiguity J versus J gamma for even real spectral triples (Dabrowski-Dossena). This matches job 06, Proposition 9: the two odd reductions use J_M gamma_M (for 4 + 5) and J_M (for 4 + 7) on spacetime. Under Connes' convention eps' = +1 the reflection gives s = +1, i.e. KO-7, formally 11.

Conjecture (Question 4; false as stated, see amendment 4). For F = F_0 x (collapsing chiral interval) with the product conventions of job 06, Lemma 1, the induced U = 1 (x) sigma satisfies UJ = sJU with s = eps' (interval) times a sign determined by the product grading twist. In particular, which of 9 and 11 is selected is fixed by choosing which spacetime real structure (J_M or J_M gamma_5) represents physical charge conjugation. A KO-0 interval does not change the KO-degree of a product, so the "Clifford point" of job 06 must come from the odd x odd product structure rather than from the interval alone; the open point is how the two descriptions match. Attempted by: Claude Opus 5.5 (Questions 1-3 above). The gpt-6-astra job was cut off by the usage limit before attempting it.

Amendments from job 08

1. Lemma 1 needs its hypothesis J f J^-1 = conj(f). Without it, J_ref = sigma_1 R_l K (R_l the reflection x -> l - x) preserves the chiral boundary conditions and has the KO-6 signs (Counterexample 08.4), so the chiral boundary conditions alone do not force KO-dimension 0.
2. Lemma 2 holds for the compressed operators. With the full compressed diagonal algebra and the inherited J_2, the two-point data fail the first-order condition: [[D_2, e], e°] = D_2 for e = diag(1, 0) (Counterexample 08.4).
3. The sign s = eps' in the Consequence above is the sign of the reflection against the interval's own J. In the standard internal-first KO-6 product the required grading twist changes it: the bare reflection 1 (x) sigma always has s = -1, formally 4 + 5 = 9, for either eps' (Theorem 08.5).
4. Question 4 (Conjecture 07.4) is false as stated: a fixed spacetime J_M and particle-number conservation admit both signs (Proposition 08.9, Counterexample 08.11, Theorem 08.12).

Part 08

Product real structures and the choice between 9 and 11

Prompt: prompts/08 (Task A: check Lemmas 07.1-07.3; Task B: Conjecture 07.4). Attempted by gpt-6-astra (codex 0.164, xhigh, 3 Nov 2026), working alone. Checked by Claude Opus 5.5: by hand (results 1-3, 4 second part, 8, 12) and numerically in Python (results 5, 6, 7, 11 and 8; section at the end).

Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): all 12 numbered statements are correct. Two points of scope: the case t = -1 in Result 1 uses the extended convention 0_-, since the standard KO-0 signs have t = +1; and Theorem 7 concerns the scalar-algebra truncation, since the full compressed two-sheet algebra with its inherited real structure fails first order (Counterexample 4).

Outcome. Lemmas 07.1-07.3 hold under the pointwise reality hypothesis. Conjecture 07.4 is false as stated: Counterexample 11 fixes the physical D_F, J_F, Gamma_F and particle number Q and still admits admissible involutions of both signs. What holds instead: in the standard internal-first KO-6 product, the bare interval reflection always has s = -1 (Theorem 5, formally 4+5 = 9); the product is exactly T_5(m) x P_1 with d_5 = D_0 + m Gamma_0 (Theorem 7); particle preservation pairs the two signs whenever [Q, D] = 0 (Theorem 12); and the literal collapse of the massless interval leaves no light sector (No-go theorem 8).

We use the sign convention

$$J^2 = \epsilon, \qquad JD = \epsilon' DJ, \qquad J\Gamma = \epsilon'' \Gamma J.$$

Unless stated otherwise, KO-6 means the standard signs $(+,+,-)$. We require the order-zero and first-order conditions when calling data a real spectral triple; we impose no orientability or Poincare-duality assumption.

1. Lemma (07.1: classification under pointwise reality).

Let

$$H_I = L^2([0,l],\mathbb{C}^2), \quad D_I = E\partial_x, \quad E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \Gamma_I = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

with

$$\operatorname{Dom} D_I = \{(L,R) \in H^1 : R(0) = 0,\ L(l) = 0\}.$$

Suppose J_I is antiunitary, preserves this domain, and satisfies

$$J_I f J_I^{-1} = \overline{f}, \qquad J_I D_I = t D_I J_I, \quad t \in \{+1, -1\}.$$

Then

$$J_I = \mathrm{diag}(\alpha, \beta) K, \qquad |\alpha| = |\beta| = 1, \qquad \alpha = t\beta.$$

Consequently,

$$(\epsilon_I, \epsilon'_I, \epsilon''_I) = (+, t, +).$$

Proof. The linear unitary $W = J_I K$ commutes with every scalar multiplication operator. Hence W is multiplication by a measurable unitary matrix $W(x)$.

Domain preservation, applied to compactly supported smooth sections, gives $W \in H^1_{\mathrm{loc}}$. Comparing the coefficients of derivatives in

$$WE\overline{\psi'} = tE(W'\overline{\psi} + W\overline{\psi'})$$

gives

$$WE = tEW, \qquad EW' = 0.$$

Thus W is constant. Preservation of the boundary conditions gives $W_{21} = W_{12} = 0$, because the remaining endpoint traces are arbitrary.

Writing $W = \mathrm{diag}(\alpha, \beta)$, the equation $WE = tEW$ becomes $\alpha = t\beta$. Finally,

$$W\overline{W} = 1, \qquad W\Gamma_I = \Gamma_I W.$$

These imply the asserted signs. Conversely, every displayed J_I satisfies all the hypotheses. □

Thus Lemma 07.1 is correct under its stated pointwise reality hypothesis. Up to phase, the two possibilities are K and $\Gamma_I K$. They are 0_+ and 0_- in the extended convention; Connes' even convention selects $t = +1$. See Dąbrowski–Dossena, arXiv v1, Table 1, p. 5.

2. Lemma (07.2: spectral compression).

Every J_I in Result 1 commutes with the spectral projections of $|D_I|$. On the minimal nonzero grading-preserving spectral subspace,

$$D_2 = m\sigma_1, \qquad \Gamma_2 = \sigma_3, \qquad J_2 = \mathrm{diag}(\alpha, \beta) K, \qquad m = \frac{\pi}{2l}.$$

Proof. The relation $J_I D_I = t D_I J_I$ implies that J_I preserves every eigenspace of D_I^2, hence every spectral subspace of $|D_I|$.

Solving $D_I \psi = \lambda \psi$ with the boundary conditions gives $\lambda = \pm(r - \frac{1}{2})\pi/l$. The lowest pair has the real orthonormal chiral basis

$$u_L = \sqrt{\frac{2}{l}} \begin{pmatrix} \cos(mx) \\ 0 \end{pmatrix}, \qquad u_R = \sqrt{\frac{2}{l}} \begin{pmatrix} 0 \\ \sin(mx) \end{pmatrix}.$$

Direct differentiation gives $D_I u_L = m u_R$, $D_I u_R = m u_L$. Also $J_I u_L = \alpha u_L$, $J_I u_R = \beta u_R$. □

This proves Lemma 07.2 as a statement about the compressed operators. Result 4 below addresses the first-order condition.

3. Lemma (07.3: reflection sign).

Let $R_l \psi(x) = \psi(l - x)$. A component-exchanging reflection

$$\sigma = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix} R_l, \qquad |a| = |b| = 1,$$

commutes with D_I exactly when $a = b$. Requiring additionally $\sigma^2 = 1$ gives $a = b = \pm 1$. For these reflections,

$$\sigma \Gamma_I = -\Gamma_I \sigma, \qquad \sigma J_I = t J_I \sigma.$$

On the minimal truncation, $\sigma = \pm\sigma_1$.

Proof. Reflection exchanges the two endpoint conditions. Since $\partial_x R_l = -R_l \partial_x$, commutation with D_I requires

$$-E \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix} = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix} E,$$

which is equivalent to $a = b$. Its square is then a^2, so $a = \pm 1$.

For real a, multiplication of the matrices representing σ and J_I gives

$$\sigma J_I = t J_I \sigma$$

using $\alpha = t\beta$. The identities $\cos(m(l-x)) = \sin(mx)$ and $\sin(m(l-x)) = \cos(mx)$ give the compressed reflection. □

Thus all three lemmas agree with the direct calculation. Notice, however, that

$$\sigma f \sigma^{-1} = f(l - \cdot).$$

The reflection commutes with constants, but generally does not commute with the interval algebra.

4. Counterexample (two stronger interpretations of the interval lemmas fail).

First, chiral boundary conditions alone do not force KO-dimension zero. Without pointwise reality, the antiunitary

$$J_{\text{ref}} = \sigma K, \qquad \sigma = \sigma_1 R_l,$$

has the standard KO-6 signs.

Second, the minimal two-point compression with its full diagonal algebra and inherited J_2 is not a real spectral triple satisfying the first-order condition when $m > 0$.

Proof. For the first assertion,

$$J_{\text{ref}}^2 = 1, \qquad J_{\text{ref}} D_I = D_I J_{\text{ref}}, \qquad J_{\text{ref}} \Gamma_I = -\Gamma_I J_{\text{ref}}.$$

It implements

$$J_{\text{ref}} f J_{\text{ref}}^{-1} = \overline{f}(l - \cdot).$$

The opposite action is still scalar multiplication, so order zero and first order hold. This example violates precisely the pointwise reality hypothesis of Result 1.

For the second assertion, the compressed algebra is

$$A_2 = \{\text{diag}(a, b)\}.$$

Indeed, compression sends 1 to 1 and $\cos(\pi x/l)$ to $\frac{1}{2}\sigma_3$. Let $e = \text{diag}(1, 0)$. The inherited diagonal J_2 gives $e^o = e$, whereas

$$[[D_2, e], e^o] = D_2 \neq 0.$$

Moreover, $\sigma_1 e \sigma_1 = 1 - e$, so the reflection is not an A_2-module map. □

For the finite products below, we therefore explicitly restrict the interval algebra to scalar constants. This restriction makes the compressed interval a real triple and makes its reflection algebra-linear. It is an additional assumption about the proposed collapsed model.

5. Theorem (B1–B2: the internal-first KO-6 product).

Let

$$F_0 = (A_0, H_0, D_0, J_0, \Gamma_0)$$

be a finite real KO-6 triple:

$$J_0^2 = 1, \qquad J_0 D_0 = D_0 J_0, \qquad J_0 \Gamma_0 = -\Gamma_0 J_0.$$

Take either the interval or its scalar-algebra truncation, and use the internal-first product

$$D_F = D_0 \otimes 1 + \Gamma_0 \otimes D_I, \qquad \Gamma_F = \Gamma_0 \otimes \Gamma_I.$$

Among the grading-twisted tensor real structures

$$J_{a,b} = (J_0\Gamma_0^a) \otimes (J_I\Gamma_I^b), \qquad a, b \in \{0, 1\},$$

the standard KO-6 conditions force

$$a = 0, \qquad t(-1)^b = -1.$$

For the induced reflection $U = 1 \otimes \sigma$,

$$\boxed{UJ_F = -J_F U.}$$

Proof. Direct calculation gives

$$J_{a,b}^2 = (-1)^a, \qquad J_{a,b}\Gamma_F = -\Gamma_F J_{a,b}.$$

The two Dirac summands satisfy

$$J_{a,b}(D_0 \otimes 1) = (-1)^a (D_0 \otimes 1) J_{a,b},$$

$$J_{a,b}(\Gamma_0 \otimes D_I) = -t(-1)^b (\Gamma_0 \otimes D_I) J_{a,b}.$$

Thus $J_F^2 = 1$ requires $a = 0$, and $J_F D_F = D_F J_F$ requires $t(-1)^b = -1$.

Since $\sigma J_I = tJ_I\sigma$ and $\sigma\Gamma_I = -\Gamma_I\sigma$,

$$UJ_{a,b} = t(-1)^b J_{a,b} U.$$

The required product sign is therefore -1. □

Explicitly:

Interval $t = \epsilon'_I$	Required b	Product reflection sign s
$+1$	1	-1
-1	0	-1

For $t = +1$, this is exactly job 06's formula

$$J_F = J_0 \otimes (J_I\Gamma_I).$$

Changing J_I to $J_I\Gamma_I$ while adjusting the required twist leaves the resulting J_F unchanged, up to phase. The general formula $s = t(-1)^b$ is proved, but t and b are not independent choices when the product must have standard KO-6 signs. The relevant product constructions are Dąbrowski–Dossena, §4.1, equation (18) and Tables 2–3.

A KO-0 factor leaves the KO-degree unchanged. Hence F_0 must already have degree six in this ordinary even-even construction. An odd internal triple multiplied by an even interval remains odd under the standard product rules.

6. Proposition (B1–B2: the grading and the lift of reflection).

The preceding construction has three necessary distinctions.

1. The interval grading alone cannot be the total KO-6 grading for a tensor real structure of the displayed form.
2. Twisting the reflection by the internal grading can change s, but generally destroys commutation with D_F.
3. Transporting all data between equivalent product conventions preserves s.

Proof. If one sets $\Gamma_F = 1 \otimes \Gamma_I$, then every $J_{a,b}$ commutes with Γ_F, contradicting the required KO-6 relation. The product grading must be $\Gamma_0 \otimes \Gamma_I$.

For

$$U_c = \Gamma_0^c \otimes \sigma, \qquad c \in \{0, 1\},$$

we obtain

$$U_c J_F = (-1)^c \, t(-1)^b J_F U_c,$$

and

$$[U_c, D_F] = [\Gamma_0^c, D_0] \otimes \sigma.$$

Thus U_0 always commutes with D_F. For $c = 1$,

$$[U_1, D_F] = 0 \quad \Longleftrightarrow \quad D_0 = 0,$$

because $\{\Gamma_0, D_0\} = 0$. When $D_0 = 0$, both lifts are admissible and have opposite signs.

To compare product conventions, let

$$W = P_+^0 \otimes 1 + P_-^0 \otimes \Gamma_I, \qquad P_\pm^0 = \frac{1 \pm \Gamma_0}{2}.$$

Then

$$W D_F W = D_0 \otimes \Gamma_I + 1 \otimes D_I,$$

while

$$W(1 \otimes \sigma)W = \Gamma_0 \otimes \sigma.$$

Unitary transport preserves the relation $UJ = sJU$. Keeping the bare symbol $1 \otimes \sigma$ after changing convention can therefore mean choosing a different involution. □

The interval can still carry the physical L/R label. For example, if $\Gamma_0 = Q_0$ distinguishes particles from antiparticles, then

$$\Gamma_F = Q_0 \otimes \Gamma_I$$

has the usual reversal of finite grading between a particle and its conjugate.

Multiplying the total real structure by Γ_F also flips s, but changes its signs to $(-, -, -)$. It therefore leaves the standard KO-6 convention required by job 06.

7. Theorem (B1: exact reconciliation with the odd Clifford point).

For the scalar-algebra minimal truncation, choose the harmless overall phase so that

$$D_F = D_0 \otimes 1 + m\Gamma_0 \otimes \sigma_1, \quad \Gamma_F = \Gamma_0 \otimes \sigma_3, \quad J_F = J_0 \otimes (\sigma_3 K).$$

With $U = 1 \otimes \sigma_1$, this triple is exactly the odd-odd product

$$\boxed{F \cong T_5(m) \times P_1,}$$

where

$$T_5(m) = (A_0, H_0, \; d_5 = D_0 + m\Gamma_0, \; j_5 = J_0\Gamma_0),$$

and

$$P_1 = (\mathbb{C}, \mathbb{C}, 0, K).$$

For real algebras, use the scalar point algebra $\mathbb{R}$.

Proof. The operator d_5 is self-adjoint, and

$$j_5^2 = -1, \qquad j_5 d_5 = -d_5 j_5.$$

Its opposite action is unchanged, and $[d_5, a] = [D_0, a]$, so order zero and first order descend.

Let $e_\pm$ be the eigenvectors of σ_3, and define

$$R : H_0 \longrightarrow H_{F,+}, \qquad Rx = P^0_+ x \otimes e_+ + P^0_- x \otimes e_-.$$

This is unitary onto the positive total-grading subspace. Direct calculation gives

$$R^* U D_F R = D_0 + m\Gamma_0, \qquad R^* U J_F R = J_0 \Gamma_0.$$

Under the unitary identification

$$(x, y) \longmapsto Rx + URy,$$

the operators become

$$D_F = \begin{pmatrix} 0 & d_5 \\ d_5 & 0 \end{pmatrix}, \quad \Gamma_F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J_F(x, y) = (-j_5 y, j_5 x).$$

These are precisely the odd-odd product matrices with auxiliary matrix $B = -i\sigma_2$. □

More generally, any admissible involution in a standard KO-6 triple gives

$$d = (UD)|_{H_+}, \qquad j = (UJ)|_{H_+},$$

with

$$j^2 = s, \qquad jd = s\, dj.$$

Thus

$$s = -1 : \quad F \cong T_5 \times P_1, \qquad s = +1 : \quad F \cong T_7 \times P_7.$$

The auxiliary matrices are respectively $-i\sigma_2$ and σ_1, in agreement with Dąbrowski–Dossena, equation (25), Table 6, p. 12.

The even KO-0 interval is therefore not identified with an odd point. Its two-dimensional space becomes the auxiliary doubling space after reorganizing the complete product, while its mass enters d_5. The point itself has one-dimensional Hilbert space and zero Dirac operator. Because its Dirac operator is zero, the distinction between P_1 and P_7 includes a choice of KO label.

8. No-go theorem (literal collapse produces no light finite sector).

Assume the heavy-limit model uses exactly the product Dirac operator of Result 5, with $m = \pi/(2l)$. At every fixed energy cutoff Λ, its light subspace is zero for sufficiently small l.

Proof. Anticommutation of D_0 and Γ_0 gives

$$D_F^2 = D_0^2 \otimes 1 + 1 \otimes D_I^2.$$

The chiral interval and its minimal truncation satisfy

$$D_I^2 \geq \frac{\pi^2}{4l^2}.$$

Hence

$$|D_F| \geq \frac{\pi}{2l}, \qquad \mathbf{1}_{[0,\Lambda]}(|D_F|) = 0 \quad \text{if } l < \frac{\pi}{2\Lambda}.$$

The same conclusion holds for the alternative product convention by unitary equivalence. □

Consequently, the exact product supplies an algebraic model at finite l, but cannot yield a nonempty light vectorlike sector by literal collapse at fixed energy. Retaining such a sector requires additional dynamics or a modified limiting prescription.

9. Proposition (B3: spacetime grading twists do not select the finite involution).

Let the four-dimensional Euclidean spacetime triple satisfy

$$J_M^2 = -1, \qquad J_M D_M = D_M J_M, \qquad J_M \gamma_M = \gamma_M J_M.$$

For an odd reduction (d, j_s) with

$$j_s^2 = s, \qquad j_s d = s\, d j_s,$$

the job-06 even-odd convention requires

$$J_{M\times T_s} = (J_M \gamma_M^{a_s}) \otimes j_s, \qquad (-1)^{a_s} = s.$$

Therefore

$$J_{4+5} = (J_M \gamma_M) \otimes j_5, \qquad J_{4+7} = J_M \otimes j_7.$$

These formulas do not imply that choosing physical spacetime charge conjugation selects s.

Proof. For

$$D_{M\times T_s} = D_M \otimes 1 + \gamma_M \otimes d,$$

the proposed real structure has commutation signs $(-1)^{a_s}$ and s against the two summands. They must agree. Its square is $-s$, giving KO-1 for $s = -1$ and KO-3 for $s = +1$.

We can also retain the removed point and compare the complete products. Write

$$F = T_s \times P_k, \qquad J_F = j_s \otimes (B_s K),$$

where $B_- = -i\sigma_2$, $B_+ = \sigma_1$. The unitary

$$V = P_+^M \otimes 1 \otimes 1 + P_-^M \otimes 1 \otimes \sigma_1$$

identifies $(M \times T_s) \times P_k$ with $M \times F$. In particular,

$$V((J_M \gamma_M^{a_s}) \otimes j_s \otimes B_s K) V^* = J_M \otimes J_F.$$

Here the extra factor $\gamma_M^{a_s}$ cancels because $\sigma_1 B_s = s B_s \sigma_1$. □

Thus both reconstructions can use the same original J_M. The different spacetime factors in the reduced products are required by the product rules.

10. Proposition (B3: Euclidean and Lorentzian charge conjugation).

In the conventions used by Chamseddine–Connes–Marcolli and van den Dungen–van Suijlekom, the Euclidean spacetime charge-conjugation operator is J_M, with

$$(J_M^2, \epsilon'_M, \epsilon''_M) = (-, +, +).$$

The operator $J_M\gamma_5$ is its other even real-structure convention. Lorentzian charge conjugation requires changing signature and is not obtained merely by choosing between these two Euclidean operators.

Proof and source identification. CCM state, "The operator J is the charge conjugation operator," in §3.1, and use $J = J_M \otimes J_F$ in §3.2. Their calculation in §3.5.2 explicitly uses $J_M D_M = D_M J_M$. See CCM, arXiv v1, §§3.1–3.2, p. 20, and §3.5.2, p. 27.

Van den Dungen–van Suijlekom likewise identify J_M as "the charge conjugation operator on the spinors" in Example 2.1 and use $J = J_M \otimes J_F$ in Definition 2.3. Their fermionic action uses this J_M. See the published paper, pp. 435–436 and Proposition 4.3, p. 452.

For an explicit signature comparison, take Lorentzian signature $(+---)$, chiral gamma matrices, and

$$\gamma_E^0 = \gamma_L^0, \qquad \gamma_E^j = -i\gamma_L^j.$$

Up to overall phases, standard conjugations are

$$C_E = i\gamma_E^0\gamma_E^2 K, \qquad C_L = -i\gamma_L^2 K.$$

These are the conventions in D'Andrea–Kurkov–Lizzi, arXiv v2, equations (3.4), (3.13)–(3.15).

The Clifford relations give

$$C_E^2 = -1, \quad [C_E, \gamma_5] = 0, \quad [C_E, D_E] = 0, \qquad D_E = -i\gamma_E^\mu \nabla_\mu,$$

whereas

$$C_L^2 = 1, \qquad C_L \gamma_5 = -\gamma_5 C_L.$$

In the displayed common matrix basis,

$$C_L = -i\gamma_E^0 C_E.$$

Both C_E and $C_E\gamma_5$ square to -1 and preserve Euclidean chirality; neither is already C_L.

Finally, for real m, Lorentzian charge conjugation carries

$$(i\gamma_L^\mu(\partial_\mu + ieA_\mu) - m)\psi = 0$$

to the same equation with charge $-e$. Composing C_L with γ_5 instead reverses the scalar mass sign. Thus the usual massive Dirac-field identification distinguishes charge conjugation once the signature and Dirac-operator conventions are fixed. □

This identification determines the starting spacetime convention. By Result 9, it does not determine the finite involution.

11. Counterexample (fixed physical real structure permits both 9 and 11).

There is a single scalar-interval product with fixed D_F, J_F, Γ_F, Q admitting particle-preserving involutions of both signs.

Proof. Let τ_i act on an internal particle/antiparticle space and σ_i on the interval truncation. Set

$$\rho(a,b) = \mathrm{diag}(a,b) \otimes 1, \qquad D_F = m\tau_3 \otimes \sigma_1,$$

$$\Gamma_F = \tau_3 \otimes \sigma_3, \qquad J_F = (\tau_1 \otimes \sigma_3)K, \qquad Q = \tau_3 \otimes 1, \quad m > 0.$$

This is Result 5 with

$$D_0 = 0, \quad \Gamma_0 = \tau_3, \quad J_0 = \tau_1 K.$$

It has standard KO-6 signs. Its Dirac operator commutes with the algebra, so the first-order condition holds.

Define

$$U_- = 1 \otimes \sigma_1, \qquad U_+ = \tau_3 \otimes \sigma_1 = QU_-.$$

Both are grading-reversing unitary involutions commuting with D_F, the two algebra actions, and Q. Nevertheless,

$$U_- J_F = -J_F U_-, \qquad U_+ J_F = J_F U_+.$$

The unitary

$$W = \mathrm{diag}(1, \sigma_3)$$

on the particle/antiparticle blocks transforms the data into

$$D_F' = m\, 1 \otimes \sigma_1, \qquad J_F' = (\tau_1 \otimes 1)K,$$

while preserving Γ_F, Q, ρ. The transformed involutions are

$$U_-' = \mathrm{diag}(\sigma_1, -\sigma_1), \qquad U_+' = \mathrm{diag}(\sigma_1, \sigma_1).$$

Thus this is also the electrodynamics counterexample in job 06, expressed as an interval product. Tensoring with a fixed spacetime J_M leaves both possibilities available. □

This disproves the assertion in Conjecture 07.4 that physical spacetime charge conjugation alone selects the reduction. Selecting specifically the bare reflection $1 \otimes \sigma$ is an additional restriction and gives $s = -1$ in the internal-first convention.

12. Theorem (B4: particle preservation and sign pairing).

Let Q be a self-adjoint involution equal to $+1$ on particles and -1 on antiparticles.

For an interval product with $Q = Q_0 \otimes 1$, the bare reflection automatically satisfies

$$[1 \otimes \sigma, Q] = 0.$$

More generally, assume a standard KO-6 triple satisfies

$$[Q, D] = [Q, \Gamma] = [Q, a] = [Q, a^o] = 0, \qquad QJ = -JQ.$$

If an admissible doubling involution U preserves particle number, then QU is another particle-preserving doubling involution with the opposite sign against J.

Proof. The tensor-product assertion is immediate. For the general assertion, $[Q, U] = 0$ gives

$$(QU)^2 = 1.$$

The remaining commutation and grading properties follow from those of Q and U. If $UJ = sJU$, then

$$(QU)J = sQJU = -sJ(QU).$$

Also $[QU, Q] = 0$. □

Thus particle preservation does not select a sign in a particle-number-conserving sector. The additional assumption $[Q, D] = 0$ is essential to the sign-pairing argument; Majorana terms can violate it.

Conversely, particle-preserving doubling requires the particle spaces of opposite grading to be isomorphic as bimodules: U restricts to such an isomorphism. An unpaired left-handed neutrino with no matching right-handed particle in its bimodule therefore obstructs particle-preserving doubling. The formal interval product builds in the missing partner and cannot represent that strict light spectrum without changing its particle content.

Consequence for 9 versus 11

Lemmas 07.1–07.3 hold under pointwise reality, but their proposed physical consequence does not follow. For the internal-first standard KO-6 product, the required grading twist gives the bare interval reflection $s = -1$, hence a KO-5 reduction and formal $4+5 = 9$. Changing the interval's even real-structure convention consistently does not change this result. A fixed spacetime charge conjugation J_M, and even particle-number preservation, can coexist with both reductions; choosing the lift of L/R reflection remains additional data. Finally, literal collapse of this chiral interval leaves no light fermions, so neither 9 nor 11 has been established as the dimension of a surviving heavy-W/Z limit.

Independent numerical check (Claude, 3 Nov 2026)

Python (numpy), antiunitaries represented as J = M K, with J X J^-1 = M conj(X) M^-1, J^2 = M conj(M), and U J = s J U iff U M = s M conj(U). F_0: H_0 = C^2 (+) C^2 with Gamma_0 = diag(g, -g), D_0 = diag(d, conj(d)) (d off-diagonal, random complex entry), J_0 = block swap times K; KO-6 signs verified. Interval truncation D_I = m sigma_1 (m = 0.7), Gamma_I = sigma_3, J_I = diag(t, 1) K.

- **Theorem 5.** Of the eight choices (t, a, b), exactly (t, a, b) = (+1, 0, 1) and (-1, 0, 0) give the KO-6 signs (+, +, -); both give s = -1 for U = 1 (x) sigma_1, and U commutes with D_F.
- **Theorem 7.** With R x = P_+ x (x) e_+ + P_- x (x) e_-: R* U D_F R = D_0 + m Gamma_0 and R* U J_F R = J_0 Gamma_0, with j_5^2 = -1 and j_5 d_5 = -d_5 j_5 (KO-5 signs).
- **Proposition 6.** Gamma_0 (x) sigma_1 commutes with D_F for D_0 = 0 and fails to commute for D_0 != 0.
- **Counterexample 11.** KO-6 signs, order zero and first order hold for random diagonal algebra elements; U_- and U_+ are both admissible (involutive, grading-reversing, commuting with D_F, Q, the algebra and its opposite), with s = -1 and s = +1.

- **No-go theorem 8.** D_F^2 = D_0^2 (x) 1 + 1 (x) D_I^2 holds exactly.
- **Counterexample 4 (first order), by hand.** For e = diag(1, 0) and the inherited diagonal J_2, e° = e and [[m sigma_1, e], e] = m sigma_1.

Terminology. Here "doubling" means the factorization of an even triple as an odd triple times a Clifford point (the doubling of an odd triple in Dabrowski-Dossena's product construction). It is unrelated to the "fermion doubling" of the noncommutative Standard Model (Lizzi-Mangano-Miele-Sparano). Prior work on products and on J versus J gamma: ../LITERATURE.md, section D; the factorization criterion and the 9/11 results were not found in print.

Part 09

The chiral interval with a bulk mass: chirality, light pairs and both limits

Theorem T3 of ../STRATEGY.md, core part. Proved by Claude Opus 5.5 without Astra or Fable; checked numerically against an independent first-order discretization. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026): of 11 statements, 4 correct, 3 needed amendments, 2 had gaps (closed by the referee) and 2 were wrong as first stated (Corollary 9.10 and Lemma 9.11). All corrections are incorporated below; the referee report is appended.

Setting

Let I = [0, l], H = L^2(I; C^2) with psi = (L, R), and Gamma = diag(1, -1). For M in R let A = -d/dx + M, so that A^dagger = d/dx + M, and

```
D_M psi = (A^dagger R, A L),   i.e.   (D psi)_L = R' + M R,   (D psi)_R = -L' + M L.
```

At M = 0 this is the operator of job 05. We consider three boundary conditions:

- (a) R(0) = R(l) = 0;
- (b) L(0) = L(l) = 0;
- (c) R(0) = 0 and L(l) = 0 (the chiral interval of job 05).

Lemma 9.1. For each of (a), (b) and (c), D_M is self-adjoint on {psi in H^1 : boundary condition} and has compact resolvent.

Proof. For psi, phi in H^1 the mass terms cancel (M is real) and integration by parts gives <D psi, phi> - <psi, D phi> = [conj(R) phi_L - conj(L) phi_R]_0^l. Each condition is a Lagrangian subspace of the four-dimensional space of boundary values: two independent conditions on which the form vanishes. Distributional differentiation shows Dom D_M^* is contained in H^1, so the boundary form forces the same conditions and D_M is self-adjoint. The estimate norm(psi') <= norm(D_M psi) + abs(M) norm(psi) makes the graph norm control the H^1 norm, so the resolvent maps L^2 boundedly into H^1, which embeds compactly in L^2. ■

Chirality lives in the boundary conditions

Theorem 9.2. For every M in R and l > 0:

- (a) ker D = C (e^{Mx}, 0), so ind = dim ker_{Gamma=+1} - dim ker_{Gamma=-1} = +1;
- (b) ker D = C (0, e^{-Mx}), so ind = -1;
- (c) ker D = 0, so ind = 0.

Proof. D psi = 0 iff A L = 0 and A^dagger R = 0, i.e. L = alpha e^{Mx} and R = beta e^{-Mx}. Exponentials never vanish, so a condition at either end forces the corresponding coefficient to 0. ■

Corollary 9.3 (equivariant index). Let a compact group G act on H_G = (+)_rho V_rho (x) L^2(I; C^2), a sum of finitely many finite-dimensional representations V_rho, with boundary type tau(rho) in {a, b, c} and mass M_rho on each summand. (With infinitely many summands the operator need not be Fredholm: the referee report gives a counterexample.) Then the G-index of D in R(G) is

sum over tau(rho) = a of [V_rho] minus sum over tau(rho) = b of [V_rho],

independent of every M_rho and of l. Assigning (a) to the summands of V_L and (b) to those of V_R realizes I_G = [V_L] - [V_R] of research_note 7. Vectorlike pairs of type (c) contribute 0. ■

Spectrum of the mixed condition with a bulk mass

Theorem 9.4. Take condition (c). The spectrum is symmetric under lambda -> -lambda, all eigenvalues are simple, and 0 is not an eigenvalue. Write c = mu l.

1. M = -mu with mu > 0 and c > 1: there is exactly one positive eigenvalue below mu,

   lambda_0 = mu sech(kappa_0 l), where kappa_0 in (0, mu) is the unique positive root of tanh(kappa

   and the remaining positive eigenvalues are lambda_n = sqrt(mu^2 + k_n^2), n >= 1, with k_n the unique root of tan(k l) = k/mu in (n pi/l, (n + 1/2) pi/l).

2. M = -mu and c < 1: no eigenvalue in (0, mu]. The positive eigenvalues are sqrt(mu^2 + k_n^2), n >= 0, with k_0 in (0, pi/(2l)) and k_n in (n pi/l, (n + 1/2) pi/l) roots of tan(k l) = k/mu. At c = 1 the lowest eigenvalue is exactly mu.

3. M = +mu > 0: every positive eigenvalue is sqrt(mu^2 + k_n^2) with k_n in ((n - 1/2) pi/l, n pi/l), n >= 1, a root of tan(k l) = -k/mu, so all exceed sqrt(mu^2 + (pi/2l)^2).

For M = 0 the spectrum is +/-(n + 1/2) pi/l, n >= 0 (job 05).

Proof. If D psi = lambda psi with lambda > 0, then R = A L / lambda and A^dagger A L = lambda^2 L, where A^dagger A = $-d^{2/dx}2$ + M^2. The conditions become L(l) = 0 and L'(0) = M L(0). Conversely, every solution of this Sturm-Liouville problem with lambda^2 > 0 yields the eigenvector (L, A L/lambda), whose R-component vanishes at 0. The Sturm-Liouville problem has simple eigenvalues, and the condition L(l) = 0 forces each eigenfunction into one of three forms:

- E = M^2 + k^2: L = sin(k(l - x)), giving tan(k l) = -k/M;
- E = M^2 - kappa^2 with 0 < kappa < abs(M): L = sinh(kappa(l - x)), giving tanh(kappa l) = -kappa/M;
- E = M^2: L = l - x, giving M l = -1.

Counting roots for M = -mu, with t = k l or t = kappa l:

- g(t) = tanh t - t/c is concave on t > 0, with g(0) = 0, g'(0) = 1 - 1/c and g -> -infinity. It has a positive root iff c > 1, and that root is unique.
- h(t) = tan t - t/c is negative on (n pi - pi/2, n pi). On (n pi, n pi + pi/2) we have h' = sec^2 t - 1/c, which changes sign at most once from - to +. Since h(n pi) <= 0 and h tends to +infinity, there is exactly one root there for n >= 1. For n = 0 there is one iff c < 1.

For M = +mu, the function tan t + t/c is strictly increasing on (n pi - pi/2, n pi), from -infinity to n pi/c > 0, so it has exactly one root there. The hyperbolic equation and M l = -1 have no solutions. Finally, s = kappa_0 l satisfies s = c tanh s, which gives lambda_0^2 = mu^2 - kappa_0^2 = mu^2 sech^2 s. ■

Lemma 9.5 (size and shape of the light pair). For c > 1,

mu sech(mu l) <= lambda_0 <= 2 mu e^{-kappa_0 l}, so lambda_0 = 2 mu e^{-mu l} (1 + o(1)) as mu l

The pair is spanned by L_0 = sinh(kappa_0 (l - x)), concentrated at x = 0, and R_0 = A L_0 / lambda_0 = (kappa_0 cosh(kappa_0(l - x)) - mu sinh(kappa_0(l - x)))/lambda_0, which vanishes at 0 and is concentrated at x = l. Moreover ||R_0|| = ||L_0||.

Proof. The lower bound holds because sech is decreasing and kappa_0 < mu. For the upper bound, lambda_0^2 = (mu - kappa_0)(mu + kappa_0) <= 2 mu (mu - kappa_0) = 2 mu^2 (1 - tanh(kappa_0 l)) <= 4 mu^2 e^{-2 kappa_0 l}. The equality of norms follows from ||A L_0||^2 = <A^dagger A L_0, L_0> = lambda_0^2 ||L_0||^2. For the asymptotic, put s = kappa_0 l, so that c = s/tanh s and 0 < c - s = c(1 - tanh s) <= 2c e^{-2s} -> 0; hence lambda_0 = mu sech s = 2 mu e^{-c} e^{c-s}/(1 + e^{-2s}) = 2 mu

e^{-c}(1 + o(1)). Finally, mu = kappa_0 coth(kappa_0 l) and lambda_0 = kappa_0/sinh(kappa_0 l) give the exact identity R_0(x) = sinh(kappa_0 x) = L_0(l - x): the two profiles are mirror images, decreasing away from their walls, with localization on the scale 1/kappa_0 as s -> infinity (and approaching multiples of l - x and x as c -> 1). ■

Theorem 9.6 (the light pair is Connes-Lott with m = lambda_0). Let P_0 project onto the eigenvalues +/- lambda_0 (c > 1), and use the basis u_L = (L_0, 0)/||L_0|| and u_R = (0, R_0)/||R_0||. Then:

- P_0 D P_0 = lambda_0 sigma_1 and P_0 Gamma P_0 = sigma_3;
- the compression of C(I) is Q(f) = diag(int f abs(u_L)^2, int f abs(u_R)^2), whose range is C + C;
- the truncated end states are at Connes distance 1/lambda_0.

Proof. We have D u_L = A L_0/||L_0|| = lambda_0 u_R. Pointwise orthogonality of the L and R components makes Q(f) diagonal. Q(1) = I and Q(x) = diag(_L, _R) with _L < _R, so the range is the full diagonal algebra. Finally, |||[lambda_0 sigma_1, diag(a, b)]||| = lambda_0 abs(a - b). ■

The two-sheet distance is therefore 1/(fermion mass), as in the Standard Model finite geometry; it depends on l only through lambda_0. At M = 0 (job 05) it was 2l/pi.

The two limits

Lemma 9.7. For c > 1 let s(c) be the positive root of tanh s = s/c, and set F(c) = (lambda_0 l)^2 = c^2 sech^2 s(c). Then F is strictly decreasing on (1, infinity), F(1+) = 1 and F(infinity) = 0.

Proof. Differentiating tanh s = s/c gives s'(1/c - sech^2 s) = s/c^2. Here 1/c - sech^2 s > 0, since 1/c = tanh(s)/s and sinh 2s > 2s. Using c^2 - s^2 = c^2 sech^2 s, the condition F'(c) = 2c - 2 s s' < 0 is equivalent to c^2 sech^2 s < c^3 sech^2 s, i.e. to c > 1. The limits follow from s -> 0 as c -> 1 and s/c -> 1 as c -> infinity. ■

Theorem 9.8 (decompactification, l -> infinity at fixed mu, M = -mu). We have lambda_0 <= 2 mu e^{-kappa_0 l} -> 0 with kappa_0 -> mu. The normalized profiles of L_0 near x = 0 and of R_0 near x = l converge, uniformly on compact sets of y and in L^2([0, infinity)) after extension by zero, to sqrt(2 mu) e^{-mu y}, where y is the distance to the respective wall: these are exact chiral zero modes of the two half-line operators (proof: norm(L_0)^2 = sinh(2s)/(4 kappa_0) - l/2 ~ e^{2s}/(8 kappa_0), kappa_0 -> mu, L_0(y) = R_0(l - y), and the same formula gives a uniform exponential bound). Since k_n lies in (n pi/l, (n + 1/2) pi/l), the rest of the spectrum obeys

```
#{n : lambda_n <= E} = (l/pi) sqrt(E^2 - mu^2) + O(1)   for E > mu,
```

a one-dimensional Weyl law (for the positive spectrum; both signs double the leading term). One dimension emerges above the bulk gap mu, and the chirality sits on the two walls. ■

Theorem 9.9 (collapse at fixed light mass). Fix m_f > 0. For each l in (0, 1/m_f) there is a unique mu(l) > 1/l with lambda_0 = m_f. As l -> 0:

- c(l) = mu(l) l -> infinity, with c = log(2c/(m_f l)) + o(1); equivalently c = -W_{-1}(-m_f l/2) + o(1) (Lambert W, lower branch), so c = L + log L + o(1) with L = log(2/(m_f l)): the leading scaling log(1/(m_f l))/l carries a log log correction, and it is specific to this localization model;
- the next eigenvalue exceeds sqrt(mu(l)^2 + (pi/l)^2) -> infinity;
- the light pair stays Connes-Lott with mass m_f (Theorem 9.6), while the interval collapses to a point in Gromov-Hausdorff distance. This is a statement about the compression at each l. As whole triples in the spectral propinquity, the family converges to (C, C^2, m_f sigma_1), a point carrying a massive two-dimensional Dirac block, at distance at least min(sqrt(2)/2, (1/m_f - l)/2) from the Connes-Lott triple (Corollary 15.5 of lemma 15). Both statements hold: the metric compares the whole algebra, and the compression is a different object.

Proof. Existence and uniqueness follow from Lemma 9.7 applied to F(c) = (m_f l)^2 in (0, 1). As l -> 0 we have F -> 0, so c -> infinity, and Lemma 9.5 gives m_f = 2 (c/l) e^{-c}(1 + o(1)). The bound on the next eigenvalue is Theorem 9.4.1. ■

Corollary 9.10 (collapse at fixed bulk mass). If instead mu is fixed and l -> 0, then eventually c < 1, every eigenvalue has absolute value greater than mu, and the smallest absolute value is (pi/(2l))(1 + o(1)) (the lowest root t_0 = k_0 l of tan t/t = 1/(mu l) tends to pi/2). No light fermion survives. This extends Proposition 05-4 to M != 0.

Gauge fields on the interval (scale lemma for T4)

Lemma 9.11. For -$d^{2/dx}2$ on [0, l]:

- Neumann-Neumann conditions give eigenvalues (n pi/l)^2, n >= 0, including a massless mode;
- Neumann-Dirichlet (in either order) gives ((n - 1/2) pi/l)^2, n >= 1;
- Dirichlet-Dirichlet gives (n pi/l)^2, n >= 1.

A gauge field with a Dirichlet condition at one end and a Neumann condition at the other therefore has lightest mass pi/(2l), which equals the Connes-Lott mass of the massless chiral interval (job 05); with Dirichlet conditions at both ends the lightest mass is pi/l. ■

Remark: fermion mass as internal momentum

In this family, masses are internal momenta, with an explicit rule for their quantization. At M = 0 the Connes-Lott mass is the lowest allowed momentum (r - 1/2) pi/l, quantized by the chiral boundary conditions. With a bulk mass, lambda_n^2 = mu^2 + k_n^2 adds the bulk mass and the quantized momentum in quadrature. The light pair's mass, mu sech(kappa_0 l), is exponentially sensitive to mu l. Order-one differences of mu_i l between generations therefore give hierarchical masses of about 2 mu_i e^{-mu_i l}. This is the localization mechanism for Yukawa hierarchies (Arkani-Hamed-Schmaltz 2000; Gherghetta-Pomarol 2000; Grossman-Neubert 2000). In noncommutative-geometry terms, the finite operator D_F of the two-sheet space is the towerless remnant of internal momentum, and its texture becomes localization and overlap data.

Numerical check (Claude)

The secular equations of Theorem 9.4 were compared with the eigenvalues of an independent first-order discretization: the chain of job 05 with mass term (M/2)(R_{i+1} + R_i), N = 4000.

mu (l = 1)	secular	chain
1e-4	1.57073, 4.71237, 7.85397	1.57054, 4.71178, 7.85299
0.5	1.26828, 4.63129, 7.80591	1.26809, 4.63070, 7.80493
3	0.30255, 5.06274, 8.05194	0.30244, 5.06227, 8.05103
6	0.02975, 7.04564, 9.33873	0.02972, 7.04536, 9.33799

At mu = 6, 2 mu e^{-mu l} = 0.029745, against lambda_0 = 0.029747. Lemma 9.7 was confirmed on (1, 40] using the cancellation-free form c^2 sech^2 s. For the collapse at m_f = 1:

l	mu l	mu	lambda_0	next eigenvalue
0.5	2.234	4.47	1.000000	9.56
0.1	4.501	45.0	1.000000	59.2
0.01	7.284	728	1.000000	812
0.001	9.893	9893	1.000000	10487

Referee report (gpt-6.1-terra, effort max, 3 Nov 2026)

Prompt: prompts/terra_check_09.md. Reproduced as received; the corrections above implement it. Claude checked the two errors, the identity R_0(x) = L_0(l - x) and the asymptotic of 9.5 by hand.

We take $\ell > 0$ throughout. All checks below are analytic; we used no numerical computation.

Lemma 9.1 Verdict: Correct with amendment. The compact-resolvent argument needs the graph-norm estimate

$$\|\psi'\|_{L^2} \le \|D_M\psi\|_{L^2} + |M|\,\|\psi\|_{L^2}.$$

The boundary form in the proof is correct. Each proposed boundary subspace is maximal isotropic. To apply this to the adjoint, distributional differentiation shows that every vector in $\operatorname{Dom} D_M^*$ lies in H^1; the boundary form then forces it to satisfy the same boundary conditions. Thus D_M is self-adjoint.

The displayed estimate makes the graph norm control the H^1-norm. Consequently, the resolvent maps L^2 boundedly into H^1, and compactness of $H^1 \hookrightarrow L^2$ gives compact resolvent.

Theorem 9.2 Verdict: Correct.

The zero-mode equations give

$$L(x) = \alpha e^{Mx}, \qquad R(x) = \beta e^{-Mx}.$$

The boundary conditions force $\beta = 0$ in case (a), $\alpha = 0$ in case (b), and both coefficients to vanish in case (c). The grading preserves each domain and anticommutes with D_M, so the stated chiral indices follow.

Corollary 9.3 Verdict: Correct with amendment. Assume that the displayed direct sum has finitely many finite-dimensional G-representations. Under that hypothesis, Theorem 9.2 gives the stated element of $R(G)$.

Without such a hypothesis, the direct-sum operator need not be Fredholm, even if every summand has boundary type (c). For a counterexample, take $\ell = 1$, $G = S^1$, the distinct one-dimensional representations $V_j(z) = z^j$, $j \ge 2$, and $M_j = -j$. Choose a smooth χ equal to 1 on $[0, 1/2]$ and equal to 0 at 1, and put

$$f_j(x) = \chi(x)e^{-jx}.$$

Then $(f_j, 0)$ satisfies condition (c), while

$$A_j f_j = -\chi'(x)e^{-jx}, \qquad \frac{\|A_j f_j\|}{\|f_j\|} = O\big(\sqrt{j}\, e^{-j/2}\big) \longrightarrow 0.$$

After normalization, these vectors lie in distinct orthogonal representation summands. The direct sum has no kernel, but it is not bounded below and hence is not Fredholm. Its ordinary $R(G)$-valued Fredholm index is therefore undefined.

Theorem 9.4 Verdict: Correct with amendment. Specify $\mu > 0$ in the negative-mass cases and $n \ge 1$ in case 3. For an exhaustive list, add $M = 0$, where

$$\operatorname{spec} D_0 = \left\{ \pm \frac{(n + \frac{1}{2})\pi}{\ell} : n \ge 0 \right\}.$$

At $c = 1$, the threshold eigenvalue is μ, and the remaining positive roots have $n \ge 1$ in the intervals stated in case 1.

For $\lambda \ne 0$, eliminating R gives

$$-L'' + M^2 L = \lambda^2 L, \qquad L(\ell) = 0, \quad L'(0) = ML(0).$$

Conversely, every solution gives $R = AL/\lambda$. The squared operator is nonnegative, and Theorem 9.2 excludes $\lambda = 0$. Separated boundary conditions make each squared eigenvalue simple; $\Gamma D_M = -D_M\Gamma$ gives the symmetric spectrum.

For $M = -\mu$, put $c = \mu\ell$. In the subthreshold case, the equation is

$$\frac{\tanh t}{t} = \frac{1}{c}, \qquad t = \kappa\ell > 0.$$

The left side decreases strictly from 1 to 0, so there is exactly one positive root if $c > 1$, and none if $c \le 1$. It satisfies $\kappa < \mu$.

For the oscillatory roots, let $h(t) = \tan t - t/c$. It is negative on each positive interval $(n\pi - \pi/2, n\pi)$. On $(n\pi, n\pi + \pi/2)$, its derivative increases strictly. For $n \ge 1$, it starts negative in value and tends to $+\infty$, so it has exactly one root. On $(0, \pi/2)$, it has exactly one positive root when $c < 1$, and none when $c \ge 1$. At $c = 1$, the separate linear solution $L = \ell - x$ gives $\lambda = \mu$.

For $M = +\mu$, $\tan t + t/c$ increases strictly from $-\infty$ to a positive value on each $((n - \frac{1}{2})\pi, n\pi)$, $n \ge 1$, giving exactly one root there. It has no positive root on the other half-periods, and there is no subthreshold or threshold solution. These counts establish all three stated cases.

Lemma 9.5 Verdict: Gap, closed here. The displayed proof does not justify the relative asymptotic or make "concentrated" precise.

Write $s = \kappa_0 \ell$ and $c = \mu\ell$. The root equation gives $c = s/\tanh s$. As $c \to \infty$, $s \to \infty$ and $s/c \to 1$; moreover,

$$0 < c - s = c(1 - \tanh s) \le 2ce^{-2s} \longrightarrow 0.$$

Thus

$$\lambda_0 = \mu \operatorname{sech} s = 2\mu e^{-c} \frac{e^{c-s}}{1 + e^{-2s}} = 2\mu e^{-c}(1 + o(1)).$$

The two stated inequalities also follow directly from $s < c$ and $\operatorname{sech} s \le 2e^{-s}$.

The root equation simplifies the right profile exactly:

$$R_0(x) = \sinh(\kappa_0 x) = L_0(\ell - x).$$

This proves the equality of norms and shows that the two profiles decrease away from their respective walls. Exponential localization on the scale $1/\kappa_0$ holds when $s \to \infty$. It is not uniform throughout $c > 1$: at fixed ℓ, as $c \downarrow 1$, the normalized profiles approach multiples of $\ell - x$ and x.

Theorem 9.6 Verdict: Correct.

The two chiral basis vectors span the $\pm\lambda_0$ eigenspaces together, and the first two matrix identities follow from

$$D_M u_L = \lambda_0 u_R, \qquad D_M u_R = \lambda_0 u_L.$$

By the reflection identity in Lemma 9.5, the normalized left density exceeds the right density on the left half of the interval, with the reverse inequality on the right half. Hence

$$\langle x \rangle_L < \ell/2 < \langle x \rangle_R.$$

Thus $Q(1)$ and $Q(x)$ span the diagonal algebra, and the commutator norm is $\lambda_0 |a - b|$. The distance between its two pure states is $1/\lambda_0$.

The pullbacks of these states through Q are the two smeared probability measures, not literal endpoint evaluations. Also, the unnumbered sentence immediately after the proof needs correction: the distance is **not determined by ℓ alone**, but it does depend on ℓ. In fact, Lemma 9.7 gives $1/\lambda_0 > \ell$ for $c > 1$.

Lemma 9.7 Verdict: Correct.

Set $q = \operatorname{sech}^2 s$. Since

$$\frac{1}{c} = \frac{\tanh s}{s} > q,$$

implicit differentiation is valid and gives

$$s' = \frac{s}{c(1 - cq)}, \qquad F'(c) = \frac{2cq(1 - c)}{1 - cq} < 0.$$

As $c \downarrow 1$, the inverse relation $c = s/\tanh s$ gives $s \to 0$, so $F \to 1$. As $c \to \infty$, $s \to \infty$ and $c/s \to 1$, so $c^2 \operatorname{sech}^2 s \to 0$.

Theorem 9.8 Verdict: Gap, closed here. The statement asserts convergence of the normalized profiles without proving it or specifying a mode of convergence. The convergence holds uniformly on compact sets of wall distance and in $L^2([0,\infty))$, after extension by zero.

Let $B_\ell = \|L_0\| = \|R_0\|$. With $s = \kappa_0 \ell$,

$$B_\ell^2 = \frac{\sinh(2s)}{4\kappa_0} - \frac{\ell}{2} \sim \frac{e^{2s}}{8\kappa_0}.$$

At fixed μ, $s/c \to 1$, so $\kappa_0 \to \mu$. For fixed $y \geq 0$,

$$\frac{L_0(y)}{B_\ell} = \frac{R_0(\ell - y)}{B_\ell} \longrightarrow \sqrt{2\mu}\, e^{-\mu y}.$$

The same formula supplies a uniform exponential bound, proving the stated L^2 convergence. The limits are exact zero modes of the two separately oriented half-line operators, with left and right chirality respectively. They are not zero modes at finite ℓ.

For $E > \mu$, set $K = \sqrt{E^2 - \mu^2}$. The root bounds

$$n\pi < k_n \ell < (n + \tfrac{1}{2})\pi$$

give

$$\#\{n \geq 1 : k_n \leq K\} = \frac{\ell K}{\pi} + O(1).$$

This is the displayed one-sided counting law. Counting both signs of the Dirac spectrum doubles its leading term.

Theorem 9.9 Verdict: Correct.

Lemma 9.7 makes F a continuous strictly decreasing map from $(1,\infty)$ onto $(0,1)$. Since $m_f \ell \in (0,1)$, it gives the unique $c = \mu\ell > 1$. As $\ell \to 0$, $c \to \infty$, and Lemma 9.5 yields

$$m_f \ell = 2ce^{-c}(1 + o(1)), \qquad c = \log \frac{2c}{m_f \ell} + o(1).$$

The first root above the light pair has $k_1 > \pi/\ell$, proving the stated lower bound and divergence.

The original interval metric is $|x - y|$: the mass term commutes with scalar functions, so it does not change the Dirac commutator seminorm. Therefore

$$d_{\mathrm{GH}}([0,\ell], \{\text{point}\}) = \ell/2 \longrightarrow 0.$$

The compressed two-state geometry uses a different seminorm and retains distance $1/m_f$.

Corollary 9.10 Verdict: Error. The assertion that "every eigenvalue exceeds μ" is false: the spectrum is symmetric and contains negative eigenvalues.

The corrected statement is that, for fixed bulk mass and sufficiently small ℓ, **every eigenvalue has absolute value greater than** μ, and the smallest absolute eigenvalue is

$$\min_{\lambda \in \operatorname{spec} D_M} |\lambda| = \frac{\pi}{2\ell}(1 + o(1)).$$

For $M = -\mu < 0$, the lowest root $t_0 = k_0 \ell \in (0, \pi/2)$ satisfies

$$\frac{\tan t_0}{t_0} = \frac{1}{\mu\ell} \longrightarrow \infty,$$

so $t_0 \to \pi/2$, proving the formula. For $M = 0$ it is exact; the positive-mass case follows similarly from the root just above $\pi/2$. Thus the conclusion that no light fermion survives is correct.

Lemma 9.11 Verdict: Error. The three eigenvalue lists are correct, but the gauge-field conclusion lacks a boundary condition at the other end.

For a Dirichlet condition at **both** ends, which in particular includes a Dirichlet condition at one end, the lightest mass is π/ℓ, not $\pi/(2\ell)$. The conclusion is correct for a Dirichlet condition at one end **and a Neumann condition at the other**, for the stated operator $-d^2/dx^2$.

Even if "at one end" is read as exactly one Dirichlet condition, another boundary condition need not be Neumann: $u(0) = 0$, $u'(\ell) = u(\ell)/\ell$ admits the zero mode $u(x) = x$.

Amendments and errors

- **9.1:** Add the graph-norm estimate for compact resolvent.
- **9.3:** Require a finite direct sum, or impose other hypotheses ensuring an equivariant Fredholm operator with index in $R(G)$.
- **9.4:** Specify the root indices and $\mu > 0$, and include $M = 0$ for completeness.
- **9.5:** Supply the relative-asymptotic argument and state localization precisely; the gap is closed above.
- **9.6:** Correct the adjacent unnumbered claim that the distance is unrelated to ℓ.
- **9.8:** Prove and specify the profile convergence; the gap is closed above.
- **9.10:** Replace "every eigenvalue exceeds μ" by the corresponding statement about absolute values.
- **9.11:** Require Neumann at the opposite endpoint for the claimed lightest mass.

Overall: 11 statements checked; 4 correct, 3 correct with amendment, 2 gaps closed above, and 2 errors.

Literature (../LITERATURE.md, section B). Opposite-chirality boundary modes with exponentially small mixing are in Jackiw-Rebbi 1976, Kaplan 1992 and Shamir 1993 (section 3); collapse of spectral triples to a point is treated by Farsi-Latremoliere (J. London Math. Soc., 2026). The tuned collapse mu(l) of Theorem 9.9 was not found in print.

Part 10

Toeplitz compression of the twistor coordinates on CP^3

Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026): the three identities are correct; the displayed fourth moment had an index error (pairing two barred indices), corrected below. This answers Part A of parked job 04 (research_note 4 and canonical 4, which labelled the result "Checked calculation", although no script computes it). Proved by Claude Opus 5.5 without Astra or Fable; Monte Carlo check with 2 x 10^6 samples.

Let Gamma_1, ..., Gamma_5 be Hermitian generators of Cl(5) on C^4, so that Gamma_a Gamma_b + Gamma_b Gamma_a = 2 delta_ab. For [w] in CP^3 put x_a([w]) = w* Gamma_a w / w* w, and define the level-one Berezin-Toeplitz compression

```
Q(f) = 4 int_{CP^3} f([w]) P_w dmu([w]),
```

where P_w is the rank-one projector onto w and mu is the Fubini-Study probability measure, normalized so that Q(1) = I.

Proposition 10.1.

```
Q(x_a) = Gamma_a / 5,   Q(x_a)^2 = I/25,   Q(x_a^2) = I/5.
```

In particular Q is not multiplicative.

Proof. Lift to w uniform on S^7 in C^4. For n = 4, writing w = g/abs(g) with g a standard complex Gaussian (abs(g) and g/abs(g) are independent), Wick's theorem gives

```
E[w_j w_k conj(w_i) conj(w_l)] = (delta_ij delta_kl + delta_jl delta_ki) / (n(n+1)),
```

and the sixth moment is the sum over the six matchings of unbarred with barred indices, divided by n(n+1)(n+2) = 120.

First, Q(x_a)_kl = 4 sum_ij (Gamma_a)_ij E[conj(w_i) w_j w_k conj(w_l)] = (4/20)(tr(Gamma_a) delta_kl + (Gamma_a)_kl) = Gamma_a/5, because tr Gamma_a = 0.

Second, Q(x_a^2)_kl = (4/120) sum over matchings of {j, q, k} with {i, p, l} of the corresponding contraction of (Gamma_a)_ij (Gamma_a)_pq. Matchings that contain tr Gamma_a vanish. The remaining three give tr(Gamma_a^2) delta_kl = 4 delta_kl, (Gamma_a^2)_kl = delta_kl and (Gamma_a^2)_kl = delta_kl. The total is 6 delta_kl, so Q(x_a^2) = (24/120) I = I/5.

The middle identity follows from Gamma_a^2 = I. ■

Numerical check. With Gamma_a built as Pauli tensor products (Clifford relations verified), the Monte Carlo deviations are max abs(Q(x_a) - Gamma_a/5) <= 0.0009 and max abs(Q(x_a^2) - I/5) <= 0.0006, at a sampling error of about 10^-3.

Part 11

Heavy W in the spectral-action family forces a heavy fermion

Theorem T5 of ../STRATEGY.md. Proved by Claude Opus 5.5 without Astra or Fable. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026): Proposition 11.1 correct; Corollaries 11.2 and 11.3 and the regime amendment overstated their conclusions as first written; corrected below, report appended. The result is conditional on the Chamseddine-Connes-Marcolli normalization at the matching (unification) scale.

Hypothesis (CCM relation). In the spectral action of the almost-commutative Standard Model, the gauge normalization g^2 f_0 / (2 pi^2) = 1/4 and the Higgs normalization by a = Tr(Y_nu* Y_nu + Y_e* Y_e + 3(Y_u* Y_u + Y_d* Y_d)) give, at the matching scale,

```
8 M_W^2 = sum over generations of (m_nu^2 + m_e^2 + 3 m_u^2 + 3 m_d^2),
```

where the m are Dirac masses (Chamseddine-Connes-Marcolli, Adv. Theor. Math. Phys. 11 (2007), section 5). In outline (H the unrescaled Higgs field of CCM; the normalization is the three-generation one): the Dirac masses satisfy sum N_f m_f^2 = a abs()^2, while M_W^2 = (g^2/4)(a f_0/pi^2) abs()^2 with f_0 = pi^2/(2 g^2). Majorana masses do not enter a.

Proposition 11.1. Assume the CCM relation at fixed gauge coupling, with N_g generations, and let m_max be the largest Dirac mass. Then

```
m_max / (2 sqrt 2) <= M_W <= sqrt(N_g) m_max.
```

In particular M_W -> infinity if and only if m_max -> infinity.

Proof. Every term has weight at least 1, so 8 M_W^2 >= m_max^2. Each generation has weights 1 + 1 + 3 + 3 = 8, so 8 M_W^2 <= 8 N_g m_max^2. ■

Corollary 11.2 (no-go for the fixed-mass prescription). The heavy-W prescription of canonical 14.1 and research_state 9 (v -> infinity with y_f v fixed for every fermion and the gauge couplings fixed, so that the Yukawa couplings tend to zero) is incompatible with the CCM relation: the right-hand side stays bounded while M_W -> infinity. In the spectral-action family a heavy-W/Z limit forces m_max -> infinity, with m_max >= M_W/sqrt(N_g); some species is then unbounded along a subsequence, though no single fixed species need diverge along the whole family (the referee report gives a family with m_e^2 + m_nu^2 = 8 t^2 in which each mass returns to zero infinitely often). ■

Corollary 11.3 (only the top heavy). Assume that the up-type Dirac mass of one generation (three colors) diverges and every other Dirac mass stays bounded. Then

```
m_t^2 = (8/3) M_W^2 + O(1),   so   m_t / M_W -> sqrt(8/3), approximately 1.633.
```

Top dominance is a hypothesis here: the relation alone also allows, for example, y_e = 2g with every other Yukawa zero, which gives m_e = 2 sqrt(2) M_W and m_t = 0. Under the hypothesis, the surviving SU(3) x U(1)_em theory is vectorlike and anomaly-free, and job 03 gives the residual kinetic weights. ■

Remark (interval comparison). In the massless chiral interval of job 05, the fermion truncation mass and the Neumann-Dirichlet gauge mass both equal pi/(2l) (Lemma 9.11). Under top dominance (Corollary 11.3), the CCM relation instead fixes m_t / M_W = sqrt(8/3) at matching. The interval model and the spectral action therefore assign the top and the W to the same scale with different order-one ratios. Matching them is part of T4.

Amendment (regime of validity; review of 3 Nov 2026; corrected after the referee report). The CCM relation is a relation in the truncated asymptotic (heat-kernel) action at the matching scale Lambda; small Dirac masses relative to Lambda alone do not make it exact for the full finite-cutoff action, which also needs control of momenta and of the Majorana sector. With fixed Yukawa couplings, M_W and every mass with nonzero Yukawa diverge together as v -> infinity, with ratios fixed by all the Yukawas through the matching constraint (for example y_t = y_nu = g gives m_t/M_W = sqrt 2); Corollary 11.3's ratio needs top dominance. At fixed Lambda a heavy-W limit leaves every fixed small-mass hierarchy (m_max <= eps Lambda implies M_W <= sqrt(N_g) eps Lambda), so it needs Lambda -> infinity jointly or a calculation of the finite-cutoff action. The weights f(m$^{2/\mathrm{Lambda}}$2) of lemma 03, Part A, hold under its commuting finite-operator hypothesis and do not apply directly to the broken gauge and Higgs kinetic terms (Part C's w_L(x, 0) = 1 - 3x/4 + O(x^2) differs from f); near the cutoff the coefficients need their own calculation (STRATEGY T5b).

Literature (../LITERATURE.md, section E). The mass relation and the top ratio are in CCM 2007: equation (1.5), Lemma 4.6 equation (4.16) and section 5.4 equation (5.27). Corollary 11.3 restates that result; Proposition 11.1 and Corollary 11.2 apply it to the heavy-W prescription of the corpus.

Referee report on lemmas 10 and 11 (gpt-6.1-terra, effort max, 3 Nov 2026)

Prompt: prompts/terra_check_10_11.md. Reproduced as received; the corrections above (and in lemma 10) implement it. Claude checked the counterexamples (the rotating e/nu family; y_e = 2g; y_t = y_nu = g) and the corrected fourth moment by hand.

We count Proposition 10.1, Proposition 11.1, the two corollaries, and the regime-of-validity amendment as five statements. The checks below are analytic; we did not rerun the Monte Carlo calculation.

Proposition 10.1 **Verdict: Correct with amendment.** All three identities and the normalization of Q are correct. The displayed fourth-moment formula in the proof has an index error, and its subsequent contraction is consequently incorrect as written.

For w uniformly distributed on $S^7 \subset \mathbb{C}^4$, the moments are

$$\mathbb{E}\left[\prod_{r=1}^{p} w_{j_r}\overline{w_{i_r}}\right] = \frac{\displaystyle\sum_{\sigma\in S_p}\prod_{r=1}^{p}\delta_{j_r i_{\sigma(r)}}}{4\cdot 5\cdots(3+p)}.$$

Thus the fourth moment required in the proof is

$$\mathbb{E}[w_j w_k \overline{w_i w_l}] = \frac{\delta_{ij}\delta_{kl}+\delta_{jl}\delta_{ki}}{20}.$$

The file instead has $\delta_{il}\delta_{kj}$ as its second term. For example, with $i = l = 1$ and $j = k = 2$, its formula gives $1/20$, whereas the expectation is zero by phase invariance. Contracting the erroneous formula with A_{ij} gives A_{lk}, not the claimed A_{kl}.

The sixth-moment statement is correct: its denominator is $4 \cdot 5 \cdot 6 = 120$, and its numerator is the sum over all six matchings. Also,

$$Q(1) = 4\mathbb{E}[ww^*] = I.$$

For any 4×4 matrix A, the corrected moments give

$$Q(w^*Aw) = \frac{\operatorname{tr}(A)I + A}{5}$$

and

$$Q((w^*Aw)^2) = \frac{((\operatorname{tr} A)^2 + \operatorname{tr}(A^2))I + 2\operatorname{tr}(A)A + 2A^2}{30}.$$

For $A = \Gamma_a$, conjugation by a different Clifford generator shows $\operatorname{tr}\Gamma_a = 0$, while $\Gamma_a^2 = I$ and $\operatorname{tr}(\Gamma_a^2) = 4$. Consequently,

$$Q(x_a) = \frac{\Gamma_a}{5}, \qquad Q(x_a)^2 = \frac{I}{25}, \qquad Q(x_a^2) = \frac{I}{5}.$$

Proposition 11.1 **Verdict: Correct.** For a fixed, finite N_g, the proof follows from the assumed relation. If S denotes its weighted mass sum, then

$$m_{\max}^2 \le S \le 8N_g m_{\max}^2.$$

Substituting $S = 8M_W^2$ proves both bounds and the equivalence of the two divergence statements.

The CCM relation is quoted correctly for the three-generation model. In Chamseddine–Connes–Marcolli, arXiv:hep-th/0610241v1, equations (1.5), (4.10), (4.15)–(4.17), and (5.25)–(5.26), the weights are $1, 1, 3, 3$; g is the weak coupling at matching; and the neutrino terms are **Dirac Yukawa mass parameters**, not the eigenmasses of the full seesaw matrix. The Majorana matrix does not enter the trace a, although it enters other terms in the action. CCM states that the mass relation is valid "at unification scale."

The outline preceding the proposition should identify its $\langle H \rangle$ as the **unrescaled** spectral Higgs field. CCM's canonically normalized field is $\mathbf{H} = (\sqrt{af_0}/\pi)\varphi$. Equivalently, using the standard normalized Yukawas and v,

$$m_f = \frac{y_f v}{\sqrt{2}}, \qquad M_W = \frac{gv}{2}, \qquad \sum_f N_f y_f^2 = 4g^2,$$

which gives the quoted mass relation.

There is also a scope distinction for $N_g \neq 3$. Proposition 11.1 remains true under its explicitly assumed relation, but CCM derives the coefficient 8 for three generations. If we replicate the same gauge representations N_g times and normalize their gauge kinetic term in the same way, the gauge trace scales by $N_g/3$, giving instead

$$\sum_f N_f m_f^2 = \frac{8N_g}{3} M_W^2.$$

Thus the N_g-generation version with coefficient 8 is an additional assumption, not the automatic extension of the cited normalization.

Corollary 11.2 **Verdict: Error.** The fixed-mass no-go and the displayed bound are correct. The subsequent assertion that a particular Dirac mass must *tend to infinity* is stronger than the relation proves for a general family.

If all finitely many Dirac masses are bounded, Proposition 11.1 bounds M_W. Moreover, it gives exactly

$$m_{\max} \geq \frac{M_W}{\sqrt{N_g}}.$$

Thus $M_W \to \infty$ forces $m_{\max} \to \infty$, and at least one fixed species is unbounded along a subsequence. It need not force one fixed species to tend to infinity along the whole family.

For a counterexample with fixed gauge coupling, let $t \to \infty$, put $M_W = t$, and take all Dirac masses except one charged-lepton mass and one neutrino Dirac mass to be zero. Set

$$\theta(t) = \frac{\pi}{4}(1 + \sin t), \quad m_e = 2\sqrt{2}\, t \cos\theta(t), \quad m_\nu = 2\sqrt{2}\, t \sin\theta(t).$$

Then $m_e^2 + m_\nu^2 = 8t^2$, so the assumed CCM relation holds. Nevertheless, each of m_e and m_ν vanishes on an unbounded sequence. The corresponding Yukawa couplings vary; the gauge coupling can remain fixed.

The description of the fixed-mass prescription should also say **gauge couplings fixed**: nonzero fixed fermion masses with $v \to \infty$ require their Yukawa couplings to tend to zero.

Corollary 11.3 **Verdict: Error.** The displayed asymptotic formula is correct under the stated "only the top heavy" hypothesis. The claim that a heavy top follows from the spectral action, without that choice or another hypothesis selecting it, is false.

Let R be the weighted sum of all the other Dirac masses squared. Under the hypothesis, $R = O(1)$, and

$$8M_W^2 = 3m_t^2 + R \quad \Longrightarrow \quad m_t^2 = \frac{8}{3}M_W^2 - \frac{1}{3}R.$$

This proves the stated $O(1)$ formula and $m_t/M_W \to \sqrt{8/3}$.

For a counterexample to the unconditional heavy-top claim, take $y_e = 2g$ and every other normalized Yukawa coupling, including y_t, equal to zero. The matching constraint is satisfied, $m_e = 2\sqrt{2}\, M_W$, and $m_t = 0$. The relation therefore permits a heavy charged lepton with no heavy top. CCM's equation (5.27) likewise obtains the top ratio by neglecting the other Yukawa couplings; its equation (5.29) considers a comparable neutrino Yukawa instead. The interval-comparison remark needs the same top-dominance qualification.

The assertion about the surviving $SU(3) \times U(1)_{\mathrm{em}}$ representation is correct for the stated Standard Model field content: each surviving charged Dirac fermion has identical left- and right-handed representations of the unbroken group. Their anomaly contributions cancel pairwise, and neutral Majorana states do not contribute to these gauge anomalies.

Amendment on the regime of validity **Verdict: Error.** The warning about a fixed cutoff is justified, but the amendment overstates both the fixed-Yukawa conclusion and the form of the threshold weights.

First, fixed Yukawa couplings do not by themselves imply a heavy top or the ratio in Corollary 11.3. A fixed nonzero y_t gives

$$\frac{m_t}{M_W} = \frac{\sqrt{2}\, y_t}{g},$$

but its value depends on the other Yukawas through the matching constraint. For example, $y_t = y_\nu = g$, with all others zero, satisfies that constraint and gives $m_t/M_W = \sqrt{2}$; both masses diverge as $v \to \infty$. If $y_t = 0$, the top does not diverge. Corollary 11.3 applies only when the other Dirac masses remain bounded.

Second, the displayed CCM relation is a relation in the **truncated asymptotic action**. Small Dirac masses relative to Λ alone do not establish it as an exact relation for the full finite-cutoff action: the expansion also needs control of the other relevant scales, including momenta and the Majorana sector, or a suitable uniform remainder estimate. CCM's own discussion places a Majorana scale of order Λ in one scenario; see its equations (5.16)–(5.21).

At fixed Λ, the amendment correctly observes that a heavy-W limit cannot remain within a fixed small-mass hierarchy. Indeed, if $m_{\max} \le \varepsilon\Lambda$, Proposition 11.1 gives $M_W \le \sqrt{N_g}\varepsilon\Lambda$. A limit within that hierarchy requires a joint family with $\Lambda \to \infty$, together with the necessary control of the remaining scales.

Finally, the simple weights $f(m^2/\Lambda^2)$ in lemma 03, Part A have a commuting finite-operator hypothesis. They cannot be assigned generally to broken gauge and Higgs kinetic terms. In particular, that same file's Part C gives, for a broken weak generator with one heavy and one massless partner and the squared-operator cutoff $f(x) = e^{-x}$,

$$w_L(x,0) = 3\int_0^1 u^2 e^{-xu}\,du = 1 - \frac{3}{4}x + O(x^2),$$

which differs from $f(x) = 1 - x + O(x^2)$. Near the cutoff, the appropriate coefficients and any replacement for the CCM relation require a calculation under specified hypotheses.

Amendments and errors found

- Correct the fourth-moment indices in Proposition 10.1's proof; the three proposition identities are unchanged.
- Specify the unrescaled Higgs field in the CCM outline and retain the three-generation scope of the cited normalization.
- In Corollary 11.2, assert divergence of $m_{\max}$, or add a hypothesis ensuring that one fixed species tends to infinity.
- In Corollary 11.3 and the interval remark, make top dominance a hypothesis; it is not forced by the mass relation.
- In the regime amendment, qualify fixed-Yukawa claims, distinguish the truncated action from the finite-cutoff action, and restrict the simple threshold weights to their applicable commuting setting.

Overall: 5 statements checked; 1 correct, 1 amended, 0 gaps, 3 errors.

Part 12

Boundary Yukawa matrices on the interval: from chiral to vectorlike boundary conditions, and towerless textures

Theorems T3 (remaining part) and T8 of ../STRATEGY.md. Proved by Claude Opus 5.5 without Astra or Fable. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026): of 9 statements, 4 correct, 2 needed amendments and 3 were wrong as first stated (Theorem 12.4 item 2, Theorem 12.7, Corollary 12.9); all corrections are incorporated below and the referee report is appended. The secular equation was checked against direct numerical integration of the ODE; the bounds of Theorem 12.6 and Lemma 12.8 were checked on random samples (section "Numerical check").

Setting

Let I = [0, 1]. A doublet-type field Q in L^2(I; C^2) (x) C^{N_Q} and a singlet-type field U in L^2(I; C^2) (x) C^{N_U} have components (L, R), with grading Gamma = diag(1, -1) on each. As in lemma 09,

the bulk operator is (D psi)_L = R' + M R, (D psi)_R = -L' + M L, with diagonal bulk masses M_Q = diag(m_i^Q) and M_U = diag(m_j^U). For a complex N_Q x N_U matrix X, let D_X be this operator on the domain of H^1 functions with

```
Q_R(0) = 0,   U_L(0) = 0,   Q_R(1) = X U_R(1),   U_L(1) = -X^dagger Q_L(1).
```

At X = 0, Q has boundary type (a) and U type (b) of lemma 09. The matrix X is a boundary Yukawa coupling: it couples the two fields only at the wall x = l. We write c_i^Q = -m_i^Q l and c_j^U = m_j^U l, so that c > 0 means that the zero mode of the field at X = 0 is localized near x = 0, away from the Yukawa wall.

Lemma 12.1. D_X is self-adjoint with compact resolvent, and Gamma D_X = -D_X Gamma.

Proof. The boundary form of lemma 9.1, summed over both fields, is [conj(Q_R) phi_{Q,L} - conj(Q_L) phi_{Q,R} + conj(U_R) phi_{U,L} - conj(U_L) phi_{U,R}] evaluated between 0 and l. At x = l the conditions leave Q_L and U_R free, which is half of the boundary values, and on them the form becomes conj(U_R) X^dagger phi_{Q,L} - conj(Q_L) X phi_{U,R} - conj(U_R) X^dagger phi_{Q,L} + conj(Q_L) X phi_{U,R} = 0. At x = 0 the conditions are those of types (a) and (b). Hence the conditions define a Lagrangian subspace, and the argument of lemma 9.1 applies. Gamma flips the signs of Q_R and U_R together, so it preserves each condition. ■

Kernel, index and symmetry breaking

Theorem 12.2. For every X,

```
ker D_X = {(e^{M_Q x} a, 0; 0, 0) : X^dagger e^{M_Q l} a = 0}  (+)  {(0, 0; 0, e^{-M_U x} b) : X e^{-M_U
```

so dim ker_{Gamma=+1} = N_Q - rank X, dim ker_{Gamma=-1} = N_U - rank X, and ind D_X = N_Q - N_U.

Proof. As in Theorem 9.2, D psi = 0 forces L = e^{Mx} a and R = e^{-Mx} b on each field. The conditions at 0 remove Q_R and U_L, and the conditions at l reduce to X U_R(l) = 0 and X^dagger Q_L(l) = 0. ■

Let a compact group G act unitarily on C^{N_Q} and C^{N_U} by rho_Q and rho_U, commuting with the bulk masses. The domain of D_X is G-invariant exactly for g in the stabilizer H_X = {g : rho_Q(g) X = X rho_U(g)}.

Corollary 12.3 (equivariant index and a no-go). In R(H_X),

```
ind_{H_X} D_X = [ker X^dagger] - [ker X] = [rho_Q] - [rho_U]   (restricted to H_X),
```

because X maps (ker X)^perp isomorphically and H_X-equivariantly onto ran X = (ker X^{dagger})perp. At X = 0 we have H_X = G, and the index is the chiral class [V_L] - [V_R] of Corollary 9.3. D_X has no zero modes iff X is invertible. This requires rho_Q and rho_U to be isomorphic as H_X-representations, i.e. the residual theory must be vectorlike. For the Standard Model with H = SU(3) x U(1)_em, the quark sector satisfies this. A lepton sector without nu_R does not: a massless left-handed neutrino is forced for every X preserving electromagnetism (H contained in H_X), in agreement with the unpaired nu_L of job 06. ■

Exact secular equation

For a field of mass m and lambda > 0 let omega = (m^2 - $\mathrm{lambda}^{2)}\{1/2\}$, and read omega coth(omega l) as k cot(k l) when omega = ik and as 1/l when omega = 0. Define

```
alpha_i(lambda) = omega coth(omega l) + m_i^Q,     beta_j(lambda) = omega coth(omega l) - m_j^U,
```

A = diag(alpha_i) and B = diag(beta_j). Let W be the set of lambda > 0 with lambda^2 < m^2 + $\mathrm{pi}^{2/l}2$ for every bulk mass m of the system.

Theorem 12.4. For lambda in W, lambda is an eigenvalue of D_X iff det(lambda^2 - A(lambda) X B(lambda) X^dagger) = 0, and its multiplicity is dim ker(lambda^2 - A X B X^dagger). Moreover:

1. alpha_i(0) = 2m/(1 - e^{-2ml}) with m = m_i^Q, which equals f_{Q_i}^2, the value at x = l of the normalized zero mode e^{mx}; likewise beta_j(0) = 2m/(e^{2ml} - 1) = f_{U_j}^2 for e^{-mx}. In terms of c, f^2 = phi(c)^2/l with phi(c)^2 = 2c/(e^{2c} - 1) and phi(0) = 1.
2. On its own window 0 < lambda^2 < m^2 + pi$^{2/l}$2, each alpha_i (and each beta_j) is a strictly decreasing, concave function of lambda^2, positive at 0 and tending to -infinity at the edge of that window, so it vanishes at exactly one point: the lowest positive eigenvalue Lambda_i^Q of Q_i alone with R(0) = 0 and L(l) = 0, respectively Lambda_j^U of U_j alone with L(0) = 0 and R(l) = 0 (the type-(c) problems of lemma 09). On the common window W it is still decreasing and concave but may stay positive (e.g. l = 1, M_Q = diag(0, 4): alpha_2 > 0 on all of W = (0, pi)).

Proof. With R(0) = 0, the Q-field solution is R = a sinh(omega x)/omega and L = a(cosh(omega x) + m sinh(omega x)/omega)/lambda. With L(0) = 0, the U-field solution is L = b sinh(omega x)/omega and R = b(m sinh(omega x)/omega - cosh(omega x))/lambda. On W, every S = sinh(omega l)/omega is nonzero. Inserting these into the conditions at l and eliminating b gives lambda^2 c = X B X^dagger A c for c = diag(S_Q) a. The map c -> eigenfunction is linear and injective. Finally, lambda^2 - X B X^dagger A and lambda^2 - A X B X^dagger have kernels of equal dimension for lambda != 0. For item 2, the expansion x coth x = 1 + sum_{n >= 1} 2x$^{2/(x}$2 + n^2 pi^2) gives

```
z^{1/2} coth(z^{1/2} l) = (1/l) [1 + sum_{n >= 1} 2 z l^2 / (z l^2 + n^2 pi^2)],
```

which is strictly increasing and concave in z > -pi$^{2/l}$2. Substituting z = m^2 - lambda^2 gives the claim. The function tends to -infinity at the edge of its own window, and alpha_i = 0 means Q_L(l) = 0. ■

Example 12.5 (one flavor, no bulk mass: the interpolation). For N_Q = N_U = 1 and m = 0 we get alpha = beta = lambda cot(lambda l), and the equation reduces to tan^2(lambda l) = abs(X)^2. With theta = arctan abs(X) in [0, pi/2], the positive spectrum is

```
theta/l,   (n pi - theta)/l,   (n pi + theta)/l   (n >= 1).
```

At X = 0 the light pair is the pair of chiral zero modes of types (a) and (b): zero then has multiplicity two and the positive eigenvalues are n pi/l, each of multiplicity two; theta = pi/2 is reached only as a limit. As abs(X) -> infinity the conditions tend to Q_L(l) = 0 and U_R(l) = 0, so each field becomes type (c), and the spectrum tends to two copies of (r - 1/2) pi/l. The light pair theta/l therefore interpolates continuously between the chiral zero modes and the Connes-Lott mass pi/(2l) of job 05. Its eigenvectors span a Gamma-graded two-dimensional space on which D acts as (theta/l) sigma_1, with the Gamma = +1 vector carried by (Q_L, U_L) and the Gamma = -1 vector by (Q_R, U_R). ■

Light spectrum and towerless limit

Let F_Q = diag(f_{Q_i}), F_U = diag(f_{U_j}) and Y_0 = F_Q X F_U. Let Lambda_Q = min_i Lambda_i^Q, Lambda_U = min_j Lambda_j^U and lambda_* = min(Lambda_Q, Lambda_U). For 0 < lambda < lambda_* the matrices A and B are positive, and we put Y(lambda) = A(lambda)^{1/2} X B(lambda)^{1/2}. Singular values sigma_k are listed in decreasing order.

Theorem 12.6.

1. lambda in (0, lambda_*) is an eigenvalue of multiplicity m iff exactly m indices k satisfy sigma_k(Y(lambda)) = lambda.
2. For each k, the equation lambda = sigma_k(Y(lambda)) has at most one solution lambda_k in (0, lambda_). *It has one whenever 0 < sigma_k(Y_0) < lambda_.*
3. Every such solution satisfies

   ```
   sigma_k(Y_0) (1 - lambda_k^2/Lambda_Q^2 - lambda_k^2/Lambda_U^2) <= lambda_k <= sigma_k(Y_0).
   ```

Proof. Each Lambda lies in the window of its own field, so (0, lambda_) *is contained in W. (1) Conjugating by A^{1/2} turns A X B X^dagger into Y Y^dagger, and dim ker(lambda^2 - Y Y^dagger) = #{k : sigma_k(Y) = lambda}. Then apply Theorem 12.4. (2) A and B decrease entrywise in lambda, and sigma_k(P Z Q) <= norm(P) norm(Q) sigma_k(Z). Hence sigma_k(Y(lambda)) is continuous and nonincreasing, so lambda*

- *sigma_k(Y(lambda)) is strictly increasing. It is negative near 0 and at least lambda_* - sigma_k(Y_0) > 0 near lambda_*. (3) The upper bound is (2). For the lower bound, concavity places alpha_i above its chord: alpha_i(lambda) >= alpha_i(0)(1 - lambda^{2/Lambda_i Q)^2), and likewise for beta_j. Therefore Y(lambda) = D_1 Y_0 D_2 with diagonal D_1, D_2 whose entries are at least (1 - t_Q)^{1/2} and (1 - t_U)^{1/2}, where t_Q = lambda^{2/Lambda_Q}2 and t_U = lambda^{2/Lambda_U}2. Hence sigma_k(Y) >= ((1 - t_Q)(1 - t_U))^{1/2} sigma_k(Y_0) >= (1 - t_Q - t_U) sigma_k(Y_0). The last step uses a^2 + ab + b^2 <= a + b when a + b <= 1; for a + b > 1 the bound is trivial. ■

Theorem 12.7 (towerless limit of the texture). Fix the localization parameters c_i^Q and c_j^U and the matrix Y_hat = X/l, and let l -> 0. Put Phi_Q = diag(phi(c_i^Q)), Phi_U = diag(phi(c_j^U)) and M_F = Phi_Q Y_hat Phi_U. For every E > 0 and all small l, the spectrum of D_X in [-E, E] consists of:

- 0 with multiplicity N_Q + N_U - 2 rank Y_hat;
- plus/minus lambda_k(l) for those k <= rank Y_hat with sigma_k(M_F) <= E (since lambda_k(l) < sigma_k(M_F) and lambda_k(l) -> sigma_k(M_F)), where

  ```
  abs(lambda_k(l) - sigma_k(M_F)) <= sigma_k(M_F)^3 l^2 (1/L_Q^2 + 1/L_U^2),
  ```

  and L_Q = l Lambda_Q and L_U = l Lambda_U depend only on the c's.

All eigenvalues outside zero and the light set {+/- lambda_k} have absolute value greater than lambda_* = min(L_Q, L_U)/l, which tends to infinity. For E > norm(M_F) the light space has dimension N_Q + N_U and, after grading-preserving unitary identifications, its compressed Dirac matrices converge to D_F below. Calling the limit a spectral triple requires specifying an algebra and its representation; with that understood, the towerless remnant is the finite even triple with H_F = C^{N_Q} (+) C^{N_U}, Gamma_F = diag(1, -1) and

```
D_F = [[0, M_F], [M_F^dagger, 0]],   M_F = Phi_Q Y_hat Phi_U,
```

up to unitaries on each chirality, which is the Yukawa block of the Chamseddine-Connes finite Dirac operator.

Proof. By item 1 of Theorem 12.4, Y_0 = (1/l) Phi_Q X Phi_U = M_F. Rescaling x -> x/l shows that l Lambda_i depends on c_i alone. Apply Theorem 12.6 with sigma_k(Y_0) fixed and lambda_* -> infinity, and use lambda_k <= sigma_k(M_F) in the error term. Theorem 12.2 counts the zero modes. ■

At fixed fermion masses, the towerless limit therefore requires X = l Y_hat -> 0: the boundary condition tends to the chiral type (a) (+) (b), while the normalized profiles at the wall grow like l^{-1/2}.

Hierarchies from localization

Lemma 12.8 (graded singular values). Let M = F X G with F = diag(f_1 <= … <= f_N) and G = diag(g_1 <= … <= g_N) positive, and X in M_N(C). For p = 1, …, N let T_p = {N-p+1, …, N}, Delta_p = abs(det X_{T_p T_p}), D_p = max over p-subsets I, J of abs(det X_{IJ}), eps_p^F = f_{N-p}/f_{N-p+1}, eps_p^G = g_{N-p}/g_{N-p+1} (both 0 for p = N), and K_p = binom(N, p) - 1. Then

```
f_{T_p} g_{T_p} Delta_p  <=  sigma_1 ... sigma_p  <=  f_{T_p} g_{T_p} [Delta_p + D_p ((1 + K_p eps_p^F)
```

where f_I is the product of f_i over i in I.

Proof. By Cauchy-Binet, the p-th compound is Lambda^p M = Lambda^p F Lambda^p X Lambda^p G, with entries f_I g_J det X_{IJ}, and sigma_1 … sigma_p = norm(Lambda^p M). The lower bound is the (T_p, T_p) entry. For the upper bound, norm(A) <= sum abs(A_{IJ}). Any p-subset I != T_p trades at least one index of T_p for an index <= N - p, so f_I/f_{T_p} <= eps_p^F. Hence sum_I f_I <= f_{T_p}(1 + K_p eps_p^F), and similarly for g. ■

Corollary 12.9 (masses from localization). Assume every trailing minor Delta_p is nonzero. Put E_0 = E_N = 0 and E_p = (D_p/Delta_p)((1 + K_p eps_p^F)(1 + K_p eps_p^G) - 1) for 1 <= p < N. Then

```
1/(1 + E_{p-1}) <= sigma_p / (f_{N-p+1} g_{N-p+1} Delta_p/Delta_{p-1}) <= 1 + E_p,
```

by dividing the bounds of Lemma 12.8 for p and p - 1. Indexing generations by k = N - p + 1, m_k = f_k g_k (Delta_{N-k+1}/Delta_{N-k})(1 + O(eps)) as eps = max_p (eps_p^F + eps_p^G) -> 0, for fixed X and N (uniform statements need uniform bounds on D_p/Delta_p). Without the minor condition the formula fails: F = G = diag(t, 1) and X = sigma_1 give Delta_1 = 0 while both singular values equal t. With M_F = Phi_Q Y_hat Phi_U and Y_hat anarchic (all minors of order one), the masses follow phi(c_k^Q) phi(c_k^U). Since phi(c) = (2c)^{1/2} e^{-c} (1 + O(e^{-2c})) for large c, order-one differences of the c's between generations give exponential hierarchies (the ratio also carries the factor sqrt(c_k^Q c_k$^{U/(c_\{k+1\}Q}$ c_{k+1}^U)), bounded when adjacent c's differ by bounded amounts):

```
m_k / m_{k+1} ~ e^{-(c_k^Q - c_{k+1}^Q) - (c_k^U - c_{k+1}^U)}   up to factors of order one.
```

■

Remark 12.10 (mixing; numerical, not proved here). For an up sector and a down sector sharing Phi_Q, we drew 500 anarchic complex samples with c^Q = (3, 2, 0.5). The medians of abs(V_ij)/(phi(c_i$^{Q)/phi(c_j}$Q)) were 1.02, 0.92 and 1.09 for (ij) = (12), (23) and (13). So the left mixing follows the doublet localization alone, abs(V_ij) ~ phi(c_i$^{Q)/phi(c_j}$Q), as in the split-fermion and warped-flavor literature (Huber-Shafi 2001; Agashe-Perez-Soni 2005; Casagrande et al. 2008). An upper bound with explicit constants, |V_ij| <= 3 K_u K_d f_i/f_j for i < j, is proved in the referee report appended below, which also notes that there is no universal lower bound; Astra 6 wrote out the cofactor step that the report names (validation, item 10).

Consequences for STRATEGY

- **T3 (remaining part).** The boundary Yukawa X deforms the chiral types (a) (+) (b) (X = 0) continuously into two copies of the vectorlike type (c) (abs(X) -> infinity). The index N_Q - N_U is constant along the way. What changes is the symmetry: X reduces G to H_X, and the zero modes lift exactly as far as the H_X-index allows (Corollary 12.3).
- **T8.** In the towerless limit, the Yukawa block of D_F is Phi_Q Y_hat Phi_U. The texture of the finite geometry is localization data (the c's) times an order-one boundary matrix, the masses are internal momenta and overlaps, and the hierarchies are exponential in the c's (Corollary 12.9).

Numerical check (Claude)

Method: roots of the secular equation are bracketed on fine grids and refined by Brent's method; the shooting check integrates the ODE from x = 0 with scipy `solve_ivp` (rtol 1e-12) and finds the zeros of the boundary-condition determinant at x = l.

- **Secular equation against ODE shooting** (scipy `solve_ivp`, rtol 1e-12, real X). For one massless flavor and X = 0.3, 1 and 5, the shooting roots agree with (n pi +/- arctan X)/l to 6 digits. For three flavors with masses (-3, -1.5, 0.2) and (2.5, 1, -0.3), l = 1 and random X, both methods give 0.024689, 0.048653, 0.534569 and 2.03374.
- **Theorem 12.6** (c^Q = (2, 1, 0), c^U = (1.5, 0.5, -0.5), X = 0.05 x Gaussian, lambda_* = 0.577). The eigenvalues in the window are 0.000576, 0.035602 and 0.061096, against sigma_k(Y_0) = 0.000576, 0.035632 and 0.061188. All three two-sided bounds hold.
- **Theorem 12.7** (c^Q = (4, 2, 0.3), c^U = (3, 1, -0.2), Gaussian Y_hat, sigma(M_F) = 0.000735, 0.072042, 0.786929). The relative errors of the three light eigenvalues are:

l	relative errors	lambda_*
1	1.5e-5, 1.1e-2, 2.5e-1	0.147
0.1	below 1e-6, 1.1e-4, 4.2e-3	1.47
0.03	below 1e-6, 1.0e-5, 3.9e-4	4.90

  The errors decrease as l^2.

- **Lemma 12.8:** no violation in 2000 random complex 3 x 3 cases with log-uniform gradings over 8 e-folds (6000 inequalities).

Referee report (gpt-6.1-terra, effort max, 3 Nov 2026)

Prompt: prompts/terra_check_12.md. Reproduced as received; the corrections above implement it. Claude checked the three counterexamples (M_Q = diag(0, 4); arctan(l)/l -> 1; F = G = diag(t, 1), X = sigma_1) by hand.

We take $\ell > 0$ and the bulk masses to be real, as in Lemma 09. All arguments below are analytic; we used no numerical computation.

Lemma 12.1 Verdict: Correct.

At $x = \ell$, the free boundary values are $Q_L(\ell)$ and $U_R(\ell)$. Substitution of the boundary conditions into the boundary form gives zero, and the allowed boundary values have half the total boundary dimension. The conditions at $x = 0$ are likewise maximal isotropic. The adjoint boundary conditions therefore coincide with the stated ones, so D_X is self-adjoint.

The graph norm controls the H^1-norm, and the embedding $H^1([0,\ell]) \hookrightarrow L^2([0,\ell])$ is compact. Thus the resolvent is compact. Finally, Γ preserves the boundary conditions and anticommutes with the differential expression.

Theorem 12.2 Verdict: Correct.

The zero-mode equations give

$$L(x) = e^{Mx}a, \qquad R(x) = e^{-Mx}b.$$

The conditions at 0 remove Q_R and U_L, and those at ℓ impose exactly the two displayed kernel conditions.

The endpoint exponentials are invertible, so the two chiral kernel dimensions are $N_Q - \operatorname{rank} X$ and $N_U - \operatorname{rank} X$. Their difference is $N_Q - N_U$.

Corollary 12.3 Verdict: Correct with amendment. The neutrino conclusion holds for every X that preserves the stated electromagnetic symmetry, meaning $H \subseteq H_X$. The phrase "for every X" needs that qualification.

The endpoint exponentials commute with the group actions. They identify the actual chiral kernels equivariantly with $\ker X^\dagger$ and $\ker X$. The orthogonal complements are equivariantly isomorphic through X, proving the displayed index identity. Both kernels vanish exactly when X is invertible, in which case X is an equivariant isomorphism.

If electromagnetic symmetry is not required, the neutrino assertion can fail. For one lepton generation with zero bulk masses, take

$$Q_L = (\nu_L, e_L), \qquad U_R = e_R, \qquad X = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The left kernel is then spanned by e_L, while ν_L is coupled to e_R. This X breaks electromagnetic charge. If X intertwines the electromagnetic representations and there is no neutral right-handed lepton, the neutrino subspace lies in $\ker X^\dagger$, as claimed.

Theorem 12.4 Verdict: Error. The secular equation, multiplicity formula, endpoint values, monotonicity and concavity are correct. The assertion that *each* function has a zero **on the common window** W is false.

For example, take $\ell = 1$, $M_Q = \operatorname{diag}(0, 4)$, and $M_U = (0)$. Then $W = (0, \pi)$, but throughout W,

$$\alpha_2(\lambda) = \sqrt{16 - \lambda^2}\,\coth\sqrt{16 - \lambda^2} + 4 > 0.$$

Its individual lowest mixed-boundary eigenvalue is greater than 4, hence lies outside W. The proof's assertion that every function tends to $-\infty$ at the edge of W fails for the same reason.

The corrected zero statement uses each mass's **individual** first-pole window

$$0 < \lambda < \sqrt{m^2 + \pi^2/\ell^2}.$$

On that interval the corresponding α or β has exactly one zero, at the lowest positive eigenvalue of its mixed-boundary problem. The U problem is equivalent to the type-(c) problem after exchanging components and changing the mass sign.

For the monotonicity and concavity, write $s = \lambda^2$ and $a_n = (n\pi/\ell)^2$. Differentiating the displayed partial-fraction expansion on the individual window gives

$$\frac{d}{ds}(\omega \coth(\omega\ell)) = -\frac{2}{\ell}\sum_{n\geq 1}\frac{a_n}{(m^2 - s + a_n)^2} < 0,$$

$$\frac{d^2}{ds^2}(\omega \coth(\omega\ell)) = -\frac{4}{\ell}\sum_{n\geq 1}\frac{a_n}{(m^2 - s + a_n)^3} < 0.$$

The function is positive at $s = 0$ after the appropriate $\pm m$ is added, and tends to $-\infty$ at its own first pole. This proves the corrected zero claim.

For the secular equation, the proof's boundary variables c, d satisfy

$$\lambda c = -XBd, \qquad \lambda d = -X^\dagger Ac.$$

Eliminating d gives $\lambda^2 c = XBX^\dagger Ac$. For $\lambda \neq 0$, the kernels of $\lambda^2 I - CA$ and $\lambda^2 I - AC$, with $C = XBX^\dagger$, are isomorphic via A, with inverse C/λ^2. This also proves the stated multiplicity.

Example 12.5 Verdict: Correct with amendment. At $X = 0$, omit $\theta/\ell = 0$ from the **positive** spectrum; instead, zero has multiplicity two. The positive eigenvalues are then $n\pi/\ell$, each with multiplicity two. For finite X, $\theta < \pi/2$; the endpoint $\theta = \pi/2$ is a limit.

For all positive λ, direct substitution of the massless solutions into the boundary conditions gives

$$\sin^2(\lambda\ell) - |X|^2\cos^2(\lambda\ell) = 0.$$

For $X \neq 0$, its roots are exactly the listed values. This derivation also includes the roots at $n\pi/\ell$ when $X = 0$, even though the secular formula of Theorem 12.4 was derived only before the first pole. The negative spectrum follows by grading symmetry. The stated two-dimensional graded block and the limiting mixed-boundary spectrum follow from these roots and their eigenvectors.

Theorem 12.6 Verdict: Correct.

Although Theorem 12.4 requires the correction above, it still supplies what is needed here. The smallest individual first pole belongs to some field; that field's first zero lies before its pole. Consequently, λ_* lies before the smallest first pole, and $(0, \lambda_*) \subset W$.

On this interval A, B are positive. Conjugation reduces the secular equation to the eigenvalue equation for $Y(\lambda)Y(\lambda)^\dagger$, proving the multiplicity claim. If $\lambda_2 > \lambda_1$, then

$$Y(\lambda_2) = P\,Y(\lambda_1)\,Q$$

for diagonal contractions P, Q. Thus every singular value is continuous and nonincreasing, so it meets the strictly increasing function λ at most once. The stated condition at $\lambda = 0$ gives a crossing before λ_*.

Concavity gives, with $t_Q = \lambda^2/\Lambda_Q^2$ and $t_U = \lambda^2/\Lambda_U^2$,

$$\sigma_k(Y(\lambda)) \geq \sqrt{(1-t_Q)(1-t_U)}\,\sigma_k(Y_0) \geq (1 - t_Q - t_U)\sigma_k(Y_0).$$

Together with $\sigma_k(Y(\lambda)) \leq \sigma_k(Y_0)$, this proves both bounds.

Theorem 12.7 Verdict: Error. The fixed-energy spectrum cannot contain every listed light eigenvalue for every $E > 0$.

Take one massless flavor, $\widehat{Y} = 1$, and $E = 1/2$. Then $X = \ell$, and Example 12.5 gives the positive light eigenvalue

$$\lambda(\ell) = \frac{\arctan \ell}{\ell} \longrightarrow 1.$$

For sufficiently small ℓ, it is outside $[-1/2, 1/2]$, contrary to the stated list.

The corrected fixed-energy statement includes only those $\pm\lambda_k(\ell)$ with $\lambda_k(\ell) \le E$. Equivalently, for each fixed $E > 0$ and all sufficiently small ℓ, include exactly those indices for which

$$\sigma_k(M_F) \le E.$$

The equality case is included because the strict decrease of A and B gives $\lambda_k(\ell) < \sigma_k(M_F)$ for every positive light eigenvalue.

The displayed error estimate follows from Theorem 12.6:

$$0 < \sigma_k(M_F) - \lambda_k(\ell) \le \sigma_k(M_F)^3 \ell^2 \left(L_Q^{-2} + L_U^{-2} \right).$$

Zero has the stated multiplicity. Apart from zero and the **entire** light set $\{\pm\lambda_k\}$, the eigenvalues have **absolute value** greater than λ_* for small ℓ. At λ_*, the limiting matrices A, B are nonnegative and the relevant singular values are at most $\|M_F\|$, whereas $\lambda_* \to \infty$; hence equality at the boundary is excluded for small ℓ.

For $E > \|M_F\|$, the full low-energy space has dimension $N_Q + N_U$. After grading-preserving unitary identifications, its compressed Dirac matrices converge to the displayed D_F. Calling the limit a spectral triple additionally requires an algebra and its representation to be specified.

Lemma 12.8 Verdict: Correct.

The entries of $\bigwedge^p M$ are $f_I g_J \det X_{IJ}$, and its operator norm is $\sigma_1 \cdots \sigma_p$. Its (T_p, T_p) entry gives the lower bound.

For every $I \neq T_p$, at least one factor from the top p indices is replaced by one no larger than $N - p$, so $f_I / f_{T_p} \le \epsilon_p^F$; the analogous estimate holds for g. Bounding the operator norm by the sum of absolute entries, while keeping the (T_p, T_p) entry separate, gives the upper bound. For $p = N$, both bounds reduce to the determinant identity.

Corollary 12.9 Verdict: Error. The displayed mass formula is not valid under the hypotheses of Lemma 12.8: its denominator can vanish, and its leading term can be zero when the singular value is positive.

For $0 < t < 1$, take

$$F = G = \operatorname{diag}(t, 1), \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then FXG has both singular values equal to t, but $\Delta_1 = 0$ and $\Delta_2 = 1$. The formula for m_2 gives zero, and the formula for m_1 divides by zero.

A sufficient correction is to require every trailing minor Δ_p to be nonzero and to define the small parameter. More precisely, put $E_0 = E_N = 0$ and, for $1 \le p < N$,

$$E_p = \frac{D_p}{\Delta_p} \left[(1 + K_p \epsilon_p^F)(1 + K_p \epsilon_p^G) - 1 \right].$$

Lemma 12.8 then gives the explicit bounds

$$\frac{1}{1 + E_{p-1}} \le \frac{\sigma_p}{f_{N-p+1} g_{N-p+1} \Delta_p / \Delta_{p-1}} \le 1 + E_p.$$

Indeed, apply the lemma to the products of the first p and first $p-1$ singular values and divide their bounds. For fixed X, N, if

$$\epsilon = \max_{p<N}(\epsilon_p^F + \epsilon_p^G) \longrightarrow 0,$$

these inequalities give the stated $1 + O(\epsilon)$ formula. Uniform estimates require uniform bounds on D_p/Δ_p.

The localization conclusion follows under quantitative order-one upper and lower bounds on the relevant minors. For large positive c, the mass ratio also contains the factor

$$\sqrt{\frac{c_k^Q c_k^U}{c_{k+1}^Q c_{k+1}^U}}.$$

It is bounded when the adjacent large c's differ by bounded amounts; without such control it cannot be suppressed uniformly as an order-one factor.

Remark 12.10 Bound. We can prove an upper bound independent of the right-hand localization profiles. It does not prove the reported sample medians or a universal lower bound.

For each sector $s = u, d$, let

$$M_s = FX_sG_s, \qquad F = \mathrm{diag}(f_1 \le f_2 \le f_3), \qquad G_s = \mathrm{diag}(g_1^{(s)} \le g_2^{(s)} \le g_3^{(s)}),$$

with all diagonal entries positive. Assume X_s is invertible and its trailing minors $\Delta_1^{(s)}, \Delta_2^{(s)}$ are nonzero. With $D_p^{(s)}$ as in Lemma 12.8, define

$$K_s = \left(1 + \frac{\sqrt{3}D_1^{(s)}}{\Delta_1^{(s)}}\right)\left(1 + \frac{\sqrt{3}D_2^{(s)}}{\Delta_2^{(s)}}\right).$$

Choose the columns of the left singular-vector matrices U_s in increasing order of singular value, and put $V = U_u^\dagger U_d$. Then, for $i < j$,

$$\boxed{|V_{ij}| \le 3K_uK_d\,\frac{f_i}{f_j}.}$$

To prove this, set $H_s = M_sM_s^\dagger$, and temporarily suppress s. Write

$$a = \frac{\sqrt{3}D_1}{\Delta_1}, \qquad b = \frac{\sqrt{3}D_2}{\Delta_2}.$$

For the largest eigenvalue λ_3 of H,

$$\lambda_3 \ge f_3^2 g_3^2 \Delta_1^2, \qquad H_{ii} \le 3D_1^2 f_i^2 g_3^2.$$

Cauchy–Schwarz therefore gives $|U_{i3}| \le af_i/f_3$. Applying the same argument to H^{-1}, and using $(X^{-1})_{11} = \det X_{\{2,3\},\{2,3\}}/\det X$, gives $|U_{i1}| \le bf_1/f_i$. Cofactor identities for the unitary U bound the two remaining off-diagonal entries. Together,

$$|U_{ij}| \le (1+a)(1+b)\min\left(\frac{f_i}{f_j}, \frac{f_j}{f_i}\right).$$

Summing the three terms in $V_{ij} = \sum_k \overline{(U_u)_{ki}}(U_d)_{kj}$ proves the boxed inequality.

There is no corresponding positive universal lower bound: identical up and down matrices have $V = I$, including when their minors are nonzero. The numerical medians also require the sampling distribution and sampled matrices, neither of which is specified in the remark.

Amendments and errors

- **12.3:** Restrict the neutrino assertion to couplings preserving electromagnetic symmetry.
- **12.4:** State the unique-zero claim on each field's individual first-pole window; it can fail on the common window.
- **12.5:** Treat zero separately at $X = 0$, and treat $\theta = \pi/2$ as a limit.
- **12.7:** Filter the light eigenvalues by E; state the gap in absolute value; specify an algebra and representation for a spectral-triple claim.
- **12.9:** Require nonzero trailing minors and define the separation parameter and its dependence. Retain the square-root localization factors unless they are uniformly bounded.

Overall: 9 statements checked: 4 correct, 2 correct with amendment, 0 gaps, and 3 errors.

Literature (../LITERATURE.md, section C). The one-flavour spectrum of Example 12.5 is equation (5.6) of Csaki-Grojean-Hubisz-Shirman-Terning, Fermions on an interval (2004); the warped multi-flavour analogue of the light spectrum and of the localization textures is in Casagrande et al. 2008 (equations 65, 87, 90, 94-97). No NCG paper deriving D_F from localization was found.

Part 13

T4: gauge fluctuations of the chiral interval

Prompt: prompts/13 (STRATEGY T4). Attempted by gpt-6-astra (codex 0.164, xhigh, 3 Nov 2026), working alone. Checked by Claude Opus 5.5 by hand: Results 2, 3, 4, 5 and 6 (the compression integrals, the counterexample B = -2a/(3 pi) with trivial Wilson line, the Neumann-Dirichlet mass, the 1 : 3 ratio of the a_4 gauge and scalar terms with the canonical normalization, and the eigenvalues 0 and 6m^2); Results 7 and 8 reproduce the standard CCM relations (8 M_W^2 = sum N_f m_f^2, M_W/M_Z = sqrt(5/8) with Q_2 = 12, Q_Y = 20).

Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): of 8 statements, 7 are correct and Proposition 8 needs one amendment, recorded there.

We prove an exact compression formula, but it does not identify the continuum gauge theory with the full Connes–Lott gauge theory. The fermion boundary conditions impose no boundary conditions on the gauge potentials, and fixed spectral compression produces only links of the form `phi = m I + i B`, with `B` Hermitian. The proposed ratio `M_W/m = sqrt(3/8)` is correct under the three-generation Standard Model CCM normalization, with one massive colour triplet and all other Dirac masses zero. It is not a consequence of the interval alone: the minimal two-sheet spectral action has a different gauge representation and gives a massive relative gauge field of mass `sqrt(6) m`.

1. Theorem (inner fluctuations and boundary conditions).

Let M be a closed Riemannian spin four-manifold. We use Hermitian Euclidean gamma matrices, $D_M = -i\gamma^\mu \nabla^S_\mu$, and

$$E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad D = D_M \otimes 1 + \gamma_5 \otimes E\partial_x.$$

Let

$$\mathcal{H} = L^2(S_M) \otimes L^2([0,l], \mathbb{C}^2) \otimes \mathbb{C}^n,$$

with the algebra acting by the same matrix multiplication on L and R. The interval boundary conditions are

$$R(y,0) = 0, \qquad L(y,l) = 0.$$

We consider complex inner fluctuations $D + \omega$, where $\omega = \sum_j a_j[D, b_j]$ and $\omega^* = \omega$. No real structure or additional bimodule is assumed.

Then the fluctuated operators are exactly

$$\boxed{D_A = -i\gamma^\mu(\nabla^S_\mu + iA_\mu) + \gamma_5 E(\partial_x + iA_5),}$$

where $A_\mu(y,x)$ and $A_5(y,x)$ are arbitrary smooth Hermitian $n \times n$ matrices. They are connections on the trivial $U(n)$ bundle over $M \times I_l$. Restriction to traceless potentials gives the $SU(n)$ case.

The fermion conditions induce **no Dirichlet, Neumann, or endpoint-value conditions on A_μ or A_5.**

Proof. For a matrix-valued function b,

$$[D, b] = -i\gamma^\mu \partial_\mu b + \gamma_5 E \partial_x b.$$

Consequently, represented one-forms have precisely these five Clifford components. Self-adjointness makes their connection coefficients anti-Hermitian, hence iA_α with A_α Hermitian. Conversely, smooth matrix-valued one-forms are finite sums $a\,db$, using local coordinates and a partition of unity.

Every such fluctuation is a bounded self-adjoint perturbation of D, so D_A is self-adjoint on the original domain. A bounded perturbation need not map that domain into itself; requiring this would impose an additional, unjustified condition.

Every smooth $g : M \times I_l \to U(n)$ preserves the fermion boundary conditions:

$$(gR)(y,0) = 0, \qquad (gL)(y,l) = 0.$$

The usual transformation

$$A_\alpha \longmapsto gA_\alpha g^* + i(\partial_\alpha g)g^*$$

therefore allows arbitrary endpoint values of g.

For smooth sections in $\mathrm{Dom}(D_A^2)$, the additional conditions are

$$(\partial_x + iA_5)L(y,0) = 0, \qquad (\partial_x + iA_5)R(y,l) = 0.$$

These are conditions on the fermions, not on A_5. They follow by applying the original boundary conditions to $D_A\psi$.

An orbifold extension would add different hypotheses. Requiring A_μ to be even and A_5 odd under endpoint reflections gives $\partial_x A_\mu = 0$ and $A_5 = 0$ there. Those parity requirements do not follow from the stated fermion domain. □

2. Theorem (exact fixed-projection compression and its image).

Let

$$m = \frac{\pi}{2l}, \qquad u_L = \sqrt{\frac{2}{l}}\begin{pmatrix}\cos(mx)\\ 0\end{pmatrix}, \qquad u_R = \sqrt{\frac{2}{l}}\begin{pmatrix}0\\ \sin(mx)\end{pmatrix},$$

and let P project onto their span, tensored with the spacetime and gauge factors. This is the unfluctuated lowest-pair projection of Lemma 05, Part 3.

Define

$$w_L(x) = \frac{2}{l}\cos^2(mx), \qquad w_R(x) = \frac{2}{l}\sin^2(mx), \qquad k(x) = \frac{1}{l}\sin\frac{\pi x}{l}.$$

Then, under the identification $P\mathcal{H} \simeq L^2(S_M) \otimes (\mathbb{C}^n_L \oplus \mathbb{C}^n_R)$,

$$\boxed{PD_AP = -i\gamma^\mu\left(\nabla^S_\mu + i\begin{pmatrix}A_{L,\mu} & 0\\ 0 & A_{R,\mu}\end{pmatrix}\right) + \gamma_5\begin{pmatrix}0 & \phi\\ \phi^* & 0\end{pmatrix},}$$

where

$$\boxed{A_{L,\mu} = \int_0^l w_L A_\mu\,dx, \qquad A_{R,\mu} = \int_0^l w_R A_\mu\,dx,}$$

and

$$\boxed{\phi = mI_n + iB, \qquad B = \int_0^l k(x) A_5(y,x)\, dx = B^*.}$$

Here ϕ denotes the total link; its additive fluctuation is $\phi - mI_n$.

The exact image consists of arbitrary Hermitian sheet connections together with links satisfying

$$\boxed{\phi + \phi^* = 2mI_n.}$$

In particular, not every Connes–Lott fluctuation arises by this compression.

Proof. The spacetime term is diagonal in interval chirality, giving the two weighted averages. The unfluctuated interval operator satisfies

$$D_I u_R = m u_L, \qquad D_I u_L = m u_R.$$

The upper-right matrix element of iEA_5 is

$$i\frac{2}{l}\int_0^l \cos(mx)\sin(mx) A_5\, dx = iB;$$

the lower-left element is $-iB$. This proves the operator identity without approximation.

The elementary integrals

$$\int_0^l w_L\, dx = \int_0^l w_R\, dx = 1, \quad \int_0^l w_L \cos\frac{\pi x}{l}\, dx = \frac{1}{2}, \quad \int_0^l w_R \cos\frac{\pi x}{l}\, dx = -\frac{1}{2}$$

show that any prescribed A_L, A_R have the lift

$$A_\mu(y,x) = \frac{A_{L,\mu} + A_{R,\mu}}{2} + (A_{L,\mu} - A_{R,\mu})\cos\frac{\pi x}{l}.$$

Since $\int_0^l k\, dx = 2/\pi$, any Hermitian B has the lift

$$A_5(y,x) = \frac{\pi}{2} B(y).$$

Thus the stated image is exact.

By contrast, the finite algebra

$$M_n(\mathbb{C}) \oplus M_n(\mathbb{C})$$

with $D_F = m\left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)$ admits an arbitrary complex matrix as its total link. Indeed, writing p_L, p_R for its central projections, $p_L[D_F, p_R]$ supplies an arbitrary upper-right block after multiplication by the left algebra; adding its adjoint supplies the lower-left block. The allowed finite fluctuation $\phi = 0$, for example, has no continuum lift under the displayed compression.

For $SU(2)$, B is additionally traceless. The continuum A_5 is adjoint-valued; it is not automatically an electroweak Higgs doublet.

If an additional finite factor is included in the internal-first convention, and the gauge algebra commutes with D_0, Γ_0, its compressed internal operator is

$$D_0 \otimes 1 + \Gamma_0 \otimes \begin{pmatrix} 0 & \phi \\ \phi^* & 0 \end{pmatrix}.$$

There is no extra overlap factor. If $[D_0, a] \neq 0$, additional finite scalar fluctuations occur. Moreover, the cited Theorem 5 explicitly restricts the interval algebra to constants for its real-triple assertion; it does not supply a Standard Model real structure for the full compressed algebra. □

3. No-go theorem (fixed compression is not a gauge-covariant Wilson-link construction).

Let P be the fixed projection of Result 2. Its link $\phi[A_5]$ is not a functional of the Wilson line alone, and fixed compression does not descend to continuum gauge-equivalence classes.

Proof. It suffices to consider $U(1)$ potentials independent of y. Define parallel transport by

$$W'(x) = -iA_5(x)W(x), \qquad W(0) = 1.$$

Then

$$D_{I,A} = WD_IW^*.$$

Multiplication by W preserves both fermion boundary conditions, so the complete interval spectrum is unchanged.

Take

$$A_5(x) = a\cos\frac{2\pi x}{l}.$$

Its Wilson line is $W(l) = 1$, but

$$B = \frac{a}{\pi}\int_0^\pi \sin t\cos(2t)\,dt = -\frac{2a}{3\pi}.$$

Thus

$$\phi = m - \frac{2ia}{3\pi}, \qquad |\phi| = \sqrt{m^2 + \frac{4a^2}{9\pi^2}}.$$

The zero connection and this connection are gauge equivalent even through a transformation equal to 1 at both endpoints, yet their fixed compressed operators have different internal eigenvalues.

The obstruction is that multiplication by an x-dependent gauge transformation generally fails to preserve $P\mathcal{H}$. Compression of multiplication is not multiplicative, and PgP generally fails to be unitary.

The normalization difference is already visible for constant $A_5 = a$:

$$\frac{\phi}{m} = 1 + i\frac{4l}{\pi^2}a, \qquad W(l)^* = e^{ial}.$$

Neither equality with the Wilson link nor a universal correction to the mass follows from these expressions.

A different, gauge-adapted construction is possible. Use the transported interval frames

$$u_L(x)W(x), \qquad u_R(x)W(x)W(l)^*.$$

Their vertical Dirac matrix is exactly

$$\begin{pmatrix} 0 & mW(l)^* \\ mW(l) & 0 \end{pmatrix}.$$

Its link obeys $\phi^*\phi = m^2I$, so it also omits general radial Higgs fluctuations. If W depends on y, derivatives of these frames additionally enter the sheet connections.

Finally, compression differs from eliminating the omitted fermions. With $Q = 1 - P$, block elimination adds

$$-PD_AQ\,[Q(D_A - z)Q]^{-1}QD_AP$$

whenever the inverse exists. This follows by solving the Q-component of $(D_A - z)\psi = 0$ and substituting into the P-component. It is the distinction relevant to the effective-operator discussion in research state §6.10. □

4. Lemma (the continuum vector mass requires specified gauge boundary conditions).

Let the quadratic transverse-vector operator in the interval direction be $-\partial_x^2$, with

$$f'(0) = 0, \qquad f(l) = 0.$$

Assume no boundary kinetic terms or other modifications of this operator. Then

$$\boxed{M_{\mathrm{KK}} = \frac{\pi}{2l} = m.}$$

Proof. The Neumann condition selects $f(x) = C\cos(kx)$. The Dirichlet condition gives

$$kl = \left(r + \frac{1}{2}\right)\pi, \qquad r = 0, 1, \dots .$$

The lowest eigenvalue is therefore m^2.

A Dirichlet condition at one endpoint alone does not determine this answer. Dirichlet conditions at both endpoints give lowest mass π/l, whereas two Neumann conditions give a zero mode. Thus the comparison uses specifically the mixed boundary problem listed in Lemma 9.11, and Result 1 does not derive that problem from the fermion boundary conditions. □

5. Proposition (general spectral-action vector-mass formula).

Let H_F be a specified finite fermion space with Hermitian represented gauge generators T_a. Let

$$\mathcal{D} = -i\gamma^\mu(\nabla^S_\mu + iA_\mu) + \gamma_5 V,$$

where $V = V^*$ is the total finite scalar matrix. Use the ordinary spectral trace, an even cutoff function with $f_0 = f(0) > 0$, and the terms through order Λ^0. Assume no additional kinetic terms.

Let V_0 be the chosen constant vacuum value. Define

$$Q_{ab} = \mathrm{Re}\,\mathrm{Tr}_{H_F}(T_a T_b), \qquad K_{ab} = \mathrm{Re}\,\mathrm{Tr}_{H_F}([T_a, V_0]^*[T_b, V_0]).$$

On the faithfully represented gauge algebra, the canonically normalized vector mass-squared matrix is

$$\boxed{\mathcal{M}^2 = \frac{3}{2}\,Q^{-1/2}KQ^{-1/2}.}$$

Consequently, relative to the mixed-boundary continuum mass,

$$\boxed{\frac{M_j^2}{M_{\mathrm{KK}}^2} = \frac{3}{2m^2}\,\lambda_j\!\left(Q^{-1/2}KQ^{-1/2}\right).}$$

Proof. We use the heat coefficient

$$a_4 \supset \frac{1}{16\pi^2}\,\mathrm{Tr}\left(\frac{1}{2}\mathcal{E}^2 + \frac{1}{12}\Omega_{\mu\nu}\Omega^{\mu\nu}\right),$$

as recorded in CCM, Theorem 6.1, equation (6.10). Here $\mathcal{D}^2 = \nabla^*\nabla - \mathcal{E}$.

Write

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu], \qquad D_\mu V = \partial_\mu V + i[A_\mu, V].$$

The relevant endomorphism terms are

$$\mathcal{E}_F = \frac{i}{2}\gamma^\mu\gamma^\nu F_{\mu\nu}, \qquad \mathcal{E}_V = i\gamma^\mu\gamma_5 D_\mu V.$$

The Clifford traces give

$$\mathrm{tr}_S(\mathcal{E}_F^2) = 2F_{\mu\nu}F^{\mu\nu}, \quad \mathrm{tr}_S(\mathcal{E}_V^2) = 4D_\mu V D^\mu V, \quad \mathrm{tr}_S(\Omega_F^2) = -4F_{\mu\nu}F^{\mu\nu}.$$

Hence the relevant Lagrangian is

$$\boxed{\mathcal{L} = c\ \mathrm{Tr}_{H_F}(F_{\mu\nu}F^{\mu\nu}) + 3c\ \mathrm{Tr}_{H_F}(D_\mu V D^\mu V), \qquad c = \frac{f_0}{24\pi^2}.}$$

At $V = V_0$, the quadratic gauge terms are

$$cQ_{ab}F^a_{\mu\nu}F^{b,\mu\nu} + 3cK_{ab}A^a_\mu A^{b,\mu}.$$

The change of variables $A_{\mathrm{can}} = 2\sqrt{c}\,Q^{1/2}A$ gives kinetic normalization $F^2_{\mathrm{can}}/4$ and mass normalization $A_{\mathrm{can}}\mathcal{M}^2 A_{\mathrm{can}}/2$, proving the formula.

Thus the necessary data are the represented gauge generators, their full trace multiplicities, and the vacuum matrix V_0. The common coefficient f_0 cancels. However, the bare finite Dirac matrix need not itself be the vacuum: determining V_0 requires the scalar potential, including its cutoff moments and curvature dependence. All masses below are conditional on the stated vacuum and refer to this leading spectral-action approximation. □

6. Counterexample (the minimal two-sheet action does not give the CCM triplet ratio).

Let

$$H_F = (\mathbb{C}^n_L \oplus \mathbb{C}^n_R) \otimes \mathbb{C}^N, \qquad V_0 = m \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix} \otimes I_N.$$

Gauge both sheet groups, $U(1)_L \times U(1)_R$ for $n = 1$, or $SU(2)_L \times SU(2)_R$ for $n = 2$. Use the action of Result 5.

Then the diagonal gauge combination is massless and the relative combination has

$$\boxed{M_{\mathrm{relative}} = \sqrt{6}\,m.}$$

This remains true for colour multiplicity $N = 3$.

Proof. For one normalized generator t, set

$$T_L = \begin{pmatrix} t & 0 \\ 0 & 0 \end{pmatrix}, \qquad T_R = \begin{pmatrix} 0 & 0 \\ 0 & t \end{pmatrix}, \qquad \kappa = N\,\mathrm{tr}(t^2).$$

Direct multiplication gives

$$Q = \kappa I_2, \qquad K = 2\kappa m^2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

Therefore

$$\mathcal{M}^2 = 3m^2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

whose eigenvalues are 0 and $6m^2$. The factor N cancels between scalar and gauge normalization. For $SU(2)$, this calculation applies to each $t = \sigma_a/2$.

If only the diagonal group is gauged, every generator commutes with V_0, and all these vector masses vanish. Neither choice realizes the assumed continuum mixed-boundary breaking automatically.

The dependence on the full fermion trace can also be exhibited explicitly. In the $U(1)$ example, add S otherwise identical charged pairs with zero finite Dirac operator. They contribute to Q but not K, giving

$$M^2_{\mathrm{relative}} = 6\frac{N}{N+S}m^2.$$

The mass and colour multiplicity of the original pair have not changed.

There is another distinction if one restricts the continuum field to its lowest mixed-boundary profile before evaluating the finite action:

$$A_\mu(y,x) = a_\mu(y)\cos(mx).$$

Result 2 then gives

$$A_{L,\mu} = \frac{8}{3\pi}a_\mu, \qquad A_{R,\mu} = \frac{4}{3\pi}a_\mu.$$

For a restriction $A_L = c_L a,\ A_R = c_R a$, Result 5 gives the one-field quadratic mass

$$M^2_{\text{restricted}} = 3m^2\frac{(c_L - c_R)^2}{c_L^2 + c_R^2}.$$

Thus this particular restriction gives

$$M^2_{\text{restricted}} = \frac{3}{5}m^2.$$

It is an additional restriction of the action, not a mass eigenvector of the unrestricted two-sheet theory. It demonstrates that retaining a continuum gauge profile and fluctuating the full compressed algebra are different prescriptions. □

7. Corollary (the precise conditional CCM comparison).

Let the finite model instead have the full three-generation Standard Model representation and its usual Higgs doublet. Retain all charged fermion states in the spectral trace, including those whose Yukawa masses vanish. Identify one colour-triplet Dirac mass at the vacuum with

$$m = \frac{\pi}{2l},$$

and suppose every other Dirac mass is zero.

Then, at the matching scale,

$$\boxed{M_W^2 = \frac{3}{8}m^2, \qquad \frac{M_W}{M_{\text{KK}}} = \sqrt{\frac{3}{8}}.}$$

No additional factor from the interval wavefunctions or from Wilson-line normalization multiplies this ratio.

Proof. We can recover the normalization directly from Result 5. For $T_1 = \sigma_1/2$, one Standard Model generation of particles contributes

$$\text{Tr}(T_1^2) = \frac{1}{2} + 3\frac{1}{2} = 2.$$

Antiparticles double this, and three generations give

$$Q_{11} = 12.$$

For a left doublet and its right singlets with mass matrix $Y = \text{diag}(m_u, m_d)$, the particle contribution to K_{11} is

$$2\,\text{tr}(Y^* T_1^2 Y) = \frac{1}{2}(m_u^2 + m_d^2).$$

Including antiparticles and all multiplicities gives

$$K_{11} = \sum_f N_f m_f^2.$$

Therefore

$$M_W^2 = \frac{3}{2}\frac{K_{11}}{12} = \frac{1}{8}\sum_f N_f m_f^2.$$

This is the relation in CCM, Lemma 4.6, equation (4.16), assumed in Lemma 11. Substitution of $K_{11} = 3m^2$ proves the claimed ratio.

The normalization identity is equivalently

$$\frac{f_0 g_2^2}{2\pi^2} = \frac{1}{4}.$$

Its coefficient comes from $Q_{11} = 12$. Applying that coefficient after deleting the other charged states would no longer canonically normalize the gauge kinetic term.

The interval basis already satisfies $\|u_L\| = \|u_R\| = 1$, and $\langle u_L, D_I u_R\rangle = m$. Thus its unfluctuated fermion mass is exactly m. The factor $2/\pi$ belongs to the coupling of a constant A_5 to the fixed compressed link; it does not rescale the background mass. A change of scalar coordinates likewise cannot change a mass ratio when both kinetic and Yukawa terms are transformed consistently.

If other Dirac masses are nonzero, the exact conditional comparison becomes

$$\frac{M_W^2}{M_{\mathrm{KK}}^2} = \frac{1}{8}\left(3 + \sum_{\text{other } f} N_f \frac{m_f^2}{m^2}\right).$$

If "a coloured doublet" means two massive coloured Dirac species, both of mass m, their contribution is $6m^2$, giving $M_W/m = \sqrt{3}/2$ when all other masses vanish. □

8. Proposition (electroweak doublet and the neutral-vector ratio).

Amendment (referee). In the last paragraph of this result, the law phi -> g_L phi g_R^* holds for the link of the independently gauged finite two-sheet completion. The fixed compression of the continuum field does not obey it in general: Result 3 gives an endpoint-fixed gauge transformation that changes phi. The decomposition 1 + 3 under the diagonal SU(2) stays correct.

Let an additional electroweak assignment supply one complex Higgs doublet h, with

$$T_a = \frac{\sigma_a}{2}, \qquad Y_h = \frac{1}{2}, \qquad \langle h\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v\end{pmatrix},$$

where $Q_{\mathrm{em}} = T_3 + Y$. Let the gauge fields have canonical kinetic terms and couplings g_2, g_Y.

Then

$$\boxed{M_W^2 = \frac{g_2^2 v^2}{4}, \qquad M_Z^2 = \frac{(g_2^2 + g_Y^2)v^2}{4},}$$

and consequently

$$\boxed{\frac{M_W}{M_Z} = \cos\theta_W, \qquad \rho = \frac{M_W^2}{M_Z^2\cos^2\theta_W} = 1, \quad \tan\theta_W = \frac{g_Y}{g_2}.}$$

Proof. At the vacuum,

$$|D_\mu h|^2 = \frac{v^2}{8}\left[g_2^2((W_\mu^1)^2 + (W_\mu^2)^2) + (g_2 W_\mu^3 - g_Y B_\mu)^2\right].$$

Diagonalizing the neutral quadratic form gives one massless photon and the displayed Z mass.

Under a common spectral-action coefficient, canonical normalization additionally gives

$$g_2^2 Q_2 = g_Y^2 Q_Y.$$

For the full three-generation Standard Model trace, including antiparticles,

$$Q_2 = 12, \qquad Q_Y = 20,$$

so

$$\boxed{\frac{M_W}{M_Z} = \sqrt{\frac{5}{8}}}$$

at that matching scale. These trace weights agree with the per-generation calculation in research state §6.10.

The original interval multiplication representation acts identically on both chiralities. Its matrix link transforms as

$$\phi \longmapsto g_L \phi g_R^*,$$

and under the diagonal $SU(2)$ it decomposes as $1 \oplus 3$. It therefore does not specify the electroweak doublet assignment used above. That assignment, its hypercharges, and the charged fermion trace must be supplied separately. Likewise, the continuum value of ρ requires the neutral-sector boundary conditions and kinetic normalization; the single mixed-boundary eigenvalue in Result 4 cannot determine it. □

Consequence for T4 and hypothesis H

The minimal fermion compression supports the geometric identification $m = \pi/(2l)$, but it does not establish an equivalence of gauge fluctuations, gauge boundary conditions, or bosonic actions. The ratio $\sqrt{3/8}$ is correct for the specified CCM Standard Model completion and receives no interval-overlap correction. Hypothesis H therefore requires an additional gauge-covariant reduction and a matching of representations, vacuum data, and kinetic forms before it can identify the finite vector spectrum with the interval Kaluza–Klein spectrum.

Part 14

T4b: gauge-covariant projection and conditional bosonic matching

Prompt: prompts/14 (STRATEGY T4b). Attempted by gpt-6-astra (codex 0.164, xhigh, 3 Nov 2026), working alone. Checked by Claude Opus 5.5 by hand: Theorem 5 (both mass matrices as quadratic forms), Counterexample 6 (the ratio 3/5 agrees with the restricted computation of lemma 13, Result 6: c_L = 8/(3 pi), c_R = 4/(3 pi)), Proposition 7 (t = 3 pi/4: M = 3 pi/(4l), lambda^2/m2 = 9/2, normalization l(1/2 + 1/(3 pi)), ratio 27/26, then the intermediate value theorem) and No-go theorem 8 (M_spec^2 <= 6 norm(V_0)^2). Reading: the spectral-action vector mass matches the Neumann-Dirichlet tower only through a fermion whose mass is at the tower scale (the matching bulk mass M_* > 0 leaves no light pair, Theorem 9.4 case 3), and with bounded light fermions no matching is possible as l -> 0.

Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): of 9 statements, 5 are correct, Theorems 1 and 5 and Proposition 4 need amendments, and the global form of Theorem 3 is wrong as first written. Each is recorded at its statement.

We prove that a gapped spectral projection defines a gauge-covariant light-fermion bundle with its induced Grassmannian connection. For the mixed interval with a gauge-invariant bulk mass, its finite link has fixed singular values; it does not supply the full scalar fluctuation space of a two-sheet finite triple. At the leading four-dimensional spectral-action order, equality with a specified Kaluza–Klein vector mass is equivalent to an explicit trace-and-overlap sum rule. Covariance does not enforce that rule: the massless-bulk example gives $M_{\mathrm{spec}}^2 = \frac{3}{5} M_{\mathrm{KK}}^2$, whereas a suitable positive bulk mass gives exact equality. We also prove that bounded retained fermion masses cannot match a Neumann–Dirichlet vector mass diverging as $1/l$. Matching one vector mass does not, by itself, determine relative gauge couplings or all scalar kinetic coefficients.

1. Theorem (covariant spectral projection and the data controlling its spectrum).

Amendment (referee). The double-coset statement needs the endpoint transformations to act independently through H_0 and H_l, with extendability understood, and it is a local pointwise classification unless the global topology is specified. Periodic conditions require g(l) = g(0), and then the holonomy is classified by conjugation.

Let $I = [0, l]$, and let

$$E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \Gamma = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Let the vertical operator be

$$D_{A_5,\mathsf{M},\mathcal{B}} = E(\partial_x + iA_5) + \begin{pmatrix} 0 & \mathsf{M} \\ \mathsf{M} & 0 \end{pmatrix},$$

where A_5 and M are Hermitian matrix-valued functions, and $\mathcal{B}$ specifies a self-adjoint boundary domain. Gauge transformations act in the same representation on both interval chiralities.

Let the operators form a smooth family in y, in the norm-resolvent sense after local identification of their domains. Assume that the selected light cluster is separated from the rest of the spectrum and contains complete eigenspaces, including any gauge degeneracy. Let $P_A(y)$ be its spectral projection.

Then

$$\boxed{P_{A^g} = G_g P_A G_g^\dagger,}$$

provided the transformation includes

$$A_5^g = gA_5g^\dagger + i(\partial_x g)g^\dagger, \qquad \mathsf{M}^g = g\mathsf{M}g^\dagger, \qquad \mathcal{B}^g = G_g\mathcal{B}.$$

For fixed $\mathsf{M}, \mathcal{B}$, this assertion applies exactly to the transformations preserving those data.

Define parallel transport by

$$\partial_x W = -iA_5 W, \qquad W(0) = 1, \qquad W_l = W(l).$$

The vertical spectrum is determined by the axial-gauge data

$$\boxed{\widetilde{\mathsf{M}}(x) = W(x)^\dagger \mathsf{M}(x) W(x), \qquad \widetilde{\mathcal{B}} = W^\dagger \mathcal{B}.}$$

Consequently:

- For a gauge-invariant bulk mass and the mixed conditions

  $$R(0) = 0, \qquad L(l) = 0,$$

  the entire spectrum is independent of A_5.
- With nontrivial endpoint data and gauge-invariant bulk coefficients, the relevant holonomy information is the Wilson line relative to those endpoint data. If the endpoint groups are H_0, H_l, its gauge class is represented by

  $$H_l \backslash G / H_0.$$
- A bulk matrix not commuting with parallel transport can introduce dependence on the whole transported profile $\widetilde{\mathsf{M}}(x)$, beyond W_l.

An open Wilson line is not generally gauge invariant:

$$W_l^g = g(l) W_l g(0)^\dagger.$$

It is invariant under transformations fixed at both endpoints; it transforms by conjugation when the endpoint transformations are identified. With independently unrestricted endpoints, its double-coset class is trivial.

Proof. Direct differentiation, including the boundary domain, gives

$$D_{A^g,\mathsf{M}^g,\mathcal{B}^g} = G_g D_{A,\mathsf{M},\mathcal{B}} G_g^\dagger.$$

For a contour enclosing the selected cluster,

$$P_A = \frac{1}{2\pi i} \oint (z - D_A)^{-1}\, dz.$$

Conjugation of the resolvent proves covariance. Smoothness follows from the same formula and the assumed gap.

Multiplication by W removes the connection:

$$W^\dagger D_{A_5,\mathsf{M},\mathcal{B}} W = D_{0,\widetilde{\mathsf{M}},\widetilde{\mathcal{B}}}.$$

This proves the assertion about spectral data. For a commuting bulk mass and the mixed boundary conditions, both transformed data equal their unfluctuated values.

Finally, the transport equation gives $W^g(x) = g(x)W(x)g(0)^\dagger$. The endpoint transformation law and the stated quotients follow. If the gap closes, no fixed-rank smooth light bundle is asserted. □

2. Theorem (the induced connection and finite operator).

Let M_4 be a closed Riemannian spin four-manifold. Use Hermitian Euclidean gamma matrices and

$$\mathcal{D}_A = -i\gamma^\mu(\nabla^S_\mu + \partial_\mu + iA_\mu) + \gamma_5 D_{I,A_5}.$$

Assume the hypotheses of Result 1, and let

$$\mathcal{E}_y = \operatorname{Ran} P_A(y).$$

Then the induced connection and vertical endomorphism are

$$\boxed{\nabla^{\mathcal{E}}_\mu = P_A(\partial_\mu + iA_\mu)P_A, \qquad V = P_A D_{I,A_5} P_A\big|_{\mathcal{E}}.}$$

The compressed first-order operator is exactly

$$\boxed{P_A \mathcal{D}_A P_A\big|_{\mathcal{E}} = -i\gamma^\mu(\nabla^S_\mu + \nabla^{\mathcal{E}}_\mu) + \gamma_5 V.}$$

In a local orthonormal frame $U(y) : \mathbb{C}^r \to \mathcal{E}_y$,

$$\boxed{\mathcal{A}_\mu = U^\dagger A_\mu U - iU^\dagger \partial_\mu U, \qquad V_F = U^\dagger D_{I,A_5} U.}$$

The inner products here include integration over the interval. Its curvature is

$$[\nabla^{\mathcal{E}}_\mu, \nabla^{\mathcal{E}}_\nu] = P_A(iF_{\mu\nu})P_A + P_A[\nabla_\mu P_A, \nabla_\nu P_A]P_A,$$

where $\nabla_\mu P_A = \partial_\mu P_A + i[A_\mu, P_A]$.

Proof. Write a light section as $U\chi$. Applying $\partial_\mu + iA_\mu$ and multiplying by $U^\dagger$ gives the displayed connection. The spectral projection commutes with the vertical operator, giving the displayed finite endomorphism. Expanding the commutator of the projected covariant derivatives and using $P_A^2 = P_A$ gives the curvature formula.

If

$$U^g = G_g U h^\dagger,$$

then

$$\mathcal{A}^g_\mu = h\mathcal{A}_\mu h^\dagger + i(\partial_\mu h)h^\dagger, \qquad V^g_F = hV_F h^\dagger.$$

Thus the construction is gauge covariant.

This is an exact compression. It is not generally the operator obtained by eliminating the other fermions. With $P^\perp = 1 - P_A$, elimination additionally produces

$$-P_A \mathcal{D}_A P^\perp \big[P^\perp(\mathcal{D}_A - z)P^\perp\big]^{-1} P^\perp \mathcal{D}_A P_A,$$

when the inverse exists. Horizontal derivatives and $A_\mu(x)$ generally make these off-diagonal blocks nonzero. □

3. Theorem (the finite algebra and fluctuation image for the mixed interval).

Correction (referee; the global statement as first written is false). The statement that the continuum link image consists of all links with Phi^* Phi = lambda^2 I holds locally in spacetime. Globally the unitary link must admit a path from the identity in the chosen trivialization. For M_4 = S^1 x S^3, n = 1 and Phi = lambda exp(i theta), every globally smooth real A_5 gives Phi = lambda exp(i int A_5 dx), with zero winding around S^1, so this Phi is not attained. The displayed formulas, the compression range and the reducing-subalgebra criterion are correct.

Let

$$D_{I,A_5} = \begin{pmatrix} 0 & \partial_x + iA_5 + M \\ -\partial_x - iA_5 + M & 0 \end{pmatrix}, \qquad M \in \mathbb{R},$$

with $R(0) = 0,\ L(l) = 0$, acting on $\mathbb{C}^n$-valued fields. Let the parent algebra be $C^\infty(I, M_n(\mathbb{C}))$, acting identically on both interval chiralities.

Let p_L, p_R be real normalized profiles for the lowest pair:

$$(-\partial_x + M)p_L = \lambda p_R, \qquad (\partial_x + M)p_R = \lambda p_L, \qquad \|p_L\| = \|p_R\| = 1.$$

Use the endpoint-adapted light frame

$$U_L = p_L W, \qquad U_R = p_R W W_l^\dagger.$$

Then

$$\boxed{V_F = \begin{pmatrix} 0 & \lambda W_l^\dagger \\ \lambda W_l & 0 \end{pmatrix}.}$$

Writing $w_L = p_L^2,\ w_R = p_R^2$ and

$$B_\mu = W^\dagger A_\mu W - iW^\dagger \partial_\mu W,$$

the induced connections are

$$\boxed{\begin{aligned} \mathcal{A}_{L,\mu} &= \int_0^l w_L B_\mu \, dx, \\ \mathcal{A}_{R,\mu} &= W_l \left(\int_0^l w_R B_\mu \, dx \right) W_l^\dagger - iW_l \partial_\mu W_l^\dagger. \end{aligned}}$$

The range of algebra compression is

$$A_F = M_n(\mathbb{C}) \oplus M_n(\mathbb{C})$$

on

$$H_F = \mathbb{C}_L^n \oplus \mathbb{C}_R^n, \qquad \Gamma_F = \mathrm{diag}(1, -1).$$

Taking this range with its matrix product gives a complex even finite triple. Its unitary group is $U(n)_L \times U(n)_R$, and its unrestricted inner fluctuations have an arbitrary complex matrix as their total link.

The continuum construction, however, produces exactly

$$\boxed{\Phi^\dagger \Phi = \Phi \Phi^\dagger = \lambda^2 I_n}$$

for the link. With unrestricted A_μ, it produces arbitrary pairs of Hermitian sheet connections locally in spacetime. Thus its field space is the fixed-singular-value restriction of that finite triple's fluctuation space.

The inherited local gauge action is

$$h = \mathrm{diag}(g(0), g(l)).$$

Additional gauge boundary conditions restrict this endpoint group. Globally, only endpoint transformations extending across $M_4 \times I$ are inherited.

Proof. Gauge removal and the two profile equations give the formula for V_F. Applying Result 2 to U_L, U_R gives the connection formulas. Under a continuum transformation,

$$U_L^g = gU_L g(0)^\dagger, \qquad U_R^g = gU_R g(l)^\dagger,$$

which proves the endpoint action.

The two functions w_L, w_R are linearly independent: one vanishes at $x = 0$, and the other does not. Therefore the map

$$b(x) \longmapsto \left(\int w_L b, \int w_R b \right)$$

is onto $M_n(\mathbb{C}) \oplus M_n(\mathbb{C})$. This proves both the algebra-range assertion and the connection-surjectivity assertion. Every unitary W_l has a local smooth path lift, proving the stated link image.

For the finite triple with

$$D_F = \lambda \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix},$$

multiplication of $p_L[D_F, p_R]$ by the left algebra generates arbitrary upper-right matrices; adjoining the adjoint generates arbitrary self-adjoint odd fluctuations. The full finite fluctuation space therefore includes radial changes absent from the continuum image.

There is a further distinction concerning algebra multiplication. For any compression $C(a) = PaP|_{PH}$,

$$C(a^\dagger a) - C(a)^\dagger C(a) = Pa^\dagger(1 - P)aP.$$

Hence C is a representation on a unital $*$-subalgebra precisely when that subalgebra reduces P. In the present model, multiplication by $a(x)$ reduces the light pair precisely when

$$a(x) = W(x) b W(x)^\dagger$$

for a constant matrix b. Its restriction is a single diagonal $M_n(\mathbb{C})$, commuting with V_F, and its finite scalar one-forms vanish.

Thus the two-sheet algebra is obtained by equipping the compression range with its matrix product; it is not a multiplicative quotient of the interval multiplication algebra. No real structure, real gauge-group prescription, or Standard Model representation follows from this construction. □

4. Proposition (boundary Yukawa data determine a finite operator, but not a unique finite triple).

Amendment (referee). To obtain the entire limiting block, retain the full zero-mode-descended cluster of dimension N_Q + N_U: for two massless flavours and X = l diag(1, 2), the cluster containing only the lowest pair is gapped and has a 1 x 1 limiting block. The nonzero singular values of Y are the retained positive eigenvalues; zero singular values and unmatched chiral zero modes are included separately.

Let D_X have the boundary conditions of Lemma 12:

$$Q_R(0) = 0, \quad U_L(0) = 0, \qquad Q_R(l) = XU_R(l), \quad U_L(l) = -X^\dagger Q_L(l).$$

Let the bulk masses be M_Q, M_U, and let the gauge representations commute with these masses.

Gauge covariance requires

$$X^g = g_Q(l) X g_U(l)^\dagger.$$

If X is fixed, the allowed endpoint transformations satisfy

$$g_Q(l) X = X g_U(l).$$

After axial gauge fixing,

$$\widetilde{X} = W_Q(l)^\dagger X W_U(l).$$

Let P_A retain a gapped grading-invariant light cluster. Then

$$H_F = H_{F,+} \oplus H_{F,-}, \qquad V_F = \begin{pmatrix} 0 & Y \\ Y^\dagger & 0 \end{pmatrix},$$

where the singular values of Y are the retained positive vertical eigenvalues. These eigenvalues alone do not determine the represented gauge generators or their trace data.

In the towerless scaling of Theorem 12.7,

$$X = l\widehat{X}, \qquad M_Q l,\ M_U l \text{ fixed},$$

the limiting block is, up to changes of chiral orthonormal frames,

$$\boxed{Y = \Phi_Q \widehat{X} \Phi_U,} \qquad \phi(c)^2 = \frac{2c}{e^{2c}-1}, \quad \phi(0) = 1.$$

Every rectangular complex matrix Y is attainable as an operator block by choosing

$$\widehat{X} = \Phi_Q^{-1} Y \Phi_U^{-1}.$$

This does not establish that every finite algebra, real structure, gauge representation, or inner-fluctuation space is attained.

Proof. Transforming the boundary equations gives the transformation law for X. Applying $W_Q^\dagger, W_U^\dagger$ gives $\widetilde{X}$. Since the domain is grading invariant, its symmetric spectral projection commutes with the grading; the restricted operator is therefore odd and has the displayed block form.

For an explicit spectral determination, define

$$\alpha_i(\lambda) = \omega_i \coth(\omega_i l) + m_i^Q, \qquad \beta_j(\lambda) = \omega_j \coth(\omega_j l) - m_j^U, \qquad \omega^2 = m^2 - \lambda^2.$$

In the low window where the transfer denominators are nonzero, the boundary equations give

$$\det\left[\lambda^2 I - \operatorname{diag}(\alpha_i)\widetilde{X}\operatorname{diag}(\beta_j)\widetilde{X}^\dagger\right] = 0.$$

This follows by solving the constant-coefficient first-order equations and imposing the two conditions at l, as in Theorem 12.4.

At zero spectral parameter, the normalized wall values give

$$F_Q = l^{-1/2}\Phi_Q, \qquad F_U = l^{-1/2}\Phi_U.$$

Consequently

$$F_Q X F_U = \Phi_Q \widehat{X} \Phi_U.$$

The bounds of Theorem 12.6 give convergence of the retained positive eigenvalues to its singular values, with error $O(l^2)$; the remaining rescaled eigenvalues stay separated from zero. Positivity of the diagonal entries of Φ_Q, Φ_U proves attainability of every operator block.

To obtain a finite triple, one must additionally specify the algebra to be represented on the retained space. A multiplicative restriction must satisfy the reducing-subalgebra criterion in Result 3; taking the algebra generated by compressed operators is a different prescription. Once an algebra A_F is specified, its complex gauge group is the represented $U(A_F)$, and its scalar one-forms are

$$\Omega^1_{V_F}(A_F) = \operatorname{span}\{a[V_F, b] : a, b \in A_F\}.$$

Neither this algebra nor this space is determined by the light eigenvalues. □

5. Theorem (the exact quadratic matching condition at leading spectral-action order).

Amendment (referee). Both quadratic forms must be compared on one common profile space on which G^(5) and Q are positive, because a Kaluza-Klein profile can be invisible to compression. For the massless mixed pair, f = cos(2 pi x/l) satisfies Neumann conditions at both ends and has positive norm, while int w_L f = int w_R f = 0, so Q = K = 0 for this mode.

Let a five-dimensional Yang–Mills theory have no boundary kinetic or mass terms. Let $\langle\ ,\ \rangle_5$ be its positive invariant gauge form, including its coupling constants. Expand the retained transverse vectors about $A_5 = 0$ as

$$A_\mu(y,x) = \sum_a a_\mu^a(y) F_a(x),$$

where $F_a(x)$ are fixed Hermitian Lie-algebra-valued profiles satisfying the chosen gauge boundary conditions. Define

$$G_{ab}^{(5)} = \int_0^l \langle F_a, F_b\rangle_5\, dx, \qquad H_{ab}^{(5)} = \int_0^l \langle F_a', F_b'\rangle_5\, dx.$$

Then

$$\mathcal{L}_{\mathrm{KK}}^{(2)} = \frac{1}{4} G_{ab}^{(5)} f_{\mu\nu}^a f^{b,\mu\nu} + \frac{1}{2} H_{ab}^{(5)} a_\mu^a a^{b,\mu},$$

and

$$\boxed{\mathcal{M}_{\mathrm{KK}}^2 = (G^{(5)})^{-1/2} H^{(5)} (G^{(5)})^{-1/2}.}$$

Let U_0 be the light frame at this background, and set

$$T_a = U_0^\dagger F_a U_0, \qquad V_0 = U_0^\dagger D_I U_0.$$

For the literal spectral trace, retain the gauge and scalar derivative terms through a_4, assume $f_0 > 0$, and add no independent kinetic terms. Define

$$Q_{ab} = \mathrm{Re}\,\mathrm{Tr}(T_a T_b), \qquad K_{ab} = \mathrm{Re}\,\mathrm{Tr}([T_a, V_0]^\dagger [T_b, V_0]).$$

On the subspace where $Q > 0$,

$$\boxed{\mathcal{M}_{\mathrm{spec}}^2 = \frac{3}{2} Q^{-1/2} K Q^{-1/2}.}$$

Thus equality of the vector mass spectra is equivalent to equality of the generalized spectra

$$\boxed{\mathrm{spec}\big(\tfrac{3}{2}K, Q\big) = \mathrm{spec}\big(H^{(5)}, G^{(5)}\big)\,.}$$

For a retained sector whose KK mass matrix is $\kappa^2 I$, this is equivalent to the sum rule

$$\boxed{K = \frac{2}{3}\kappa^2 Q.}$$

Equality of the quadratic actions in the same field coordinates is stronger:

$$\boxed{G^{(5)} = 4\eta Q, \qquad H^{(5)} = 6\eta K, \qquad \eta = \frac{f_0}{24\pi^2}.}$$

For a chiral finite block

$$V_0 = \begin{pmatrix} 0 & Y \\ Y^\dagger & 0 \end{pmatrix}, \qquad T_a = \mathrm{diag}(t_{L,a}, t_{R,a}),$$

put

$$C_a = t_{L,a} Y - Y t_{R,a}.$$

The trace data are explicitly

$$\boxed{Q_{ab} = \operatorname{Re}\operatorname{Tr}(t_{L,a}t_{L,b} + t_{R,a}t_{R,b}), \qquad K_{ab} = 2\operatorname{Re}\operatorname{Tr}(C_a^\dagger C_b).}$$

Here Y and the $t_{L/R,a}$ must both be computed from the interval eigenfunctions and the retained gauge profiles.

Proof. Expanding the five-dimensional action gives the two integral forms $G^{(5)}, H^{(5)}$. Canonical normalization by $(G^{(5)})^{1/2}$ gives its mass matrix.

For

$$\mathcal{D} = -i\gamma^\mu(\nabla_\mu^S + i\mathcal{A}_\mu) + \gamma_5 V,$$

the relevant heat-coefficient traces are

$$\operatorname{tr}_S(\mathcal{E}_F^2) = 2F_{\mu\nu}F^{\mu\nu}, \quad \operatorname{tr}_S(\mathcal{E}_V^2) = 4D_\mu V D^\mu V, \quad \operatorname{tr}_S(\Omega_F^2) = -4F_{\mu\nu}F^{\mu\nu}.$$

Their coefficients $1/2, 1/2, 1/12$ give

$$\mathcal{L}_{\text{spec}} = \eta \operatorname{Tr} F_{\mu\nu}F^{\mu\nu} + 3\eta \operatorname{Tr} D_\mu V D^\mu V.$$

At V_0, this becomes

$$\eta Q_{ab} f^a_{\mu\nu} f^{b,\mu\nu} + 3\eta K_{ab} a^a_\mu a^{b,\mu}.$$

The normalization $a_{\text{can}} = 2\sqrt{\eta}\, Q^{1/2} a$ proves the coefficient $3/2$, agreeing with Proposition 13.5.

Equality with a scalar matrix $\kappa^2 I$ is equivalent to

$$Q^{-1/2} K Q^{-1/2} = \frac{2}{3}\kappa^2 I.$$

The block formulas follow by multiplying

$$[T_a, V_0] = \begin{pmatrix} 0 & C_a \\ -C_a^\dagger & 0 \end{pmatrix}.$$

This proves all the matching conditions.

For mixed fermion pairs labelled by r, let their masses be λ_r, and let a single gauge profile f have overlaps

$$c_{L,r} = \int |p_{L,r}|^2 f, \qquad c_{R,r} = \int |p_{R,r}|^2 f.$$

If $\tau_r = \operatorname{tr}_r(t^2)$ includes every multiplicity, the one-vector condition becomes

$$\boxed{3\sum_r \tau_r \lambda_r^2 (c_{L,r} - c_{R,r})^2 = \kappa^2 \sum_r \tau_r (c_{L,r}^2 + c_{R,r}^2).}$$

This is the trace-and-overlap sum rule for that gauge reduction.

If instead both sheet connections are fluctuated independently, with the same representation on each pair, their relative mass is

$$M^2_{\text{relative}} = 6\frac{\sum_r \tau_r \lambda_r^2}{\sum_r \tau_r}.$$

Equating this single eigenvalue to κ^2 requires

$$6\sum_r \tau_r \lambda_r^2 = \kappa^2 \sum_r \tau_r.$$

That comparison has an additional massless diagonal vector and is not the same restriction as retaining one interval gauge profile.

All these spectral-action equalities concern the stated a_4 approximation and chosen vacuum. For a full finite scalar fluctuation space, identifying V_0 with the compressed bare operator additionally requires the scalar potential to select that vacuum. At masses comparable to the cutoff, Lemma 03, Part C shows why weighting Q alone does not give the broken-generator answer.

Finally, a nonabelian truncation to nonconstant lowest profiles need not be closed under gauge transformations: $[ft_a, ft_b] = f^2[t_a, t_b]$ generally leaves the retained span. The theorem compares quadratic transverse actions; it does not assert equality of full interacting theories. □

6. Counterexample (the covariant lowest-mode reduction still gives 3/5).

Let the gauge group be $U(1)$, and impose

$$\partial_x A_\mu(0) = 0, \qquad A_\mu(l) = 0,$$

with complementary conditions

$$A_5(0) = 0, \qquad \partial_x A_5(l) = 0.$$

Retain

$$A_\mu = a_\mu(y)\cos(\kappa x), \qquad A_5 = b(y)\sin(\kappa x), \qquad \kappa = \frac{\pi}{2l}.$$

Let one charge-one fermion have the mixed boundary conditions and scalar bulk mass M. Let λ be its lowest positive vertical eigenvalue, and define

$$c_L = \int_0^l p_L^2 \cos(\kappa x)\, dx, \qquad c_R = \int_0^l p_R^2 \cos(\kappa x)\, dx.$$

The gauge-covariant light projection gives

$$\boxed{M_{\text{spec}}^2 = 3\lambda^2 \frac{(c_L - c_R)^2}{c_L^2 + c_R^2}, \qquad M_{\text{KK}}^2 = \kappa^2.}$$

For $M = 0$,

$$\boxed{M_{\text{spec}}^2 = \frac{3}{5} M_{\text{KK}}^2.}$$

Proof. The vector boundary problem selects $f(x) = \cos(\kappa x)$. Direct integration gives

$$\frac{\int (f')^2}{\int f^2} = \kappa^2,$$

which is the Neumann–Dirichlet result of Lemma 9.11.

The retained abelian gauge transformations have parameter

$$\epsilon(y, x) = \alpha(y)\cos(\kappa x).$$

With the conventions of Result 1,

$$a_\mu \mapsto a_\mu - \partial_\mu \alpha, \qquad b \mapsto b + \kappa\alpha.$$

Thus this vector-scalar restriction is gauge closed.

Parallel transport is

$$W(x) = \exp\left[-\frac{ib}{\kappa}(1 - \cos\kappa x)\right].$$

Result 3 gives

$$\Phi = \lambda e^{ib/\kappa},$$

and

$$\mathcal{A}_L = c_L a - \frac{1-c_L}{\kappa}\,db, \qquad \mathcal{A}_R = c_R a + \frac{c_R}{\kappa}\,db.$$

Consequently,

$$D\Phi = i\Phi(c_L - c_R)\left(a + \frac{db}{\kappa}\right), \qquad F_L = c_L\,da, \quad F_R = c_R\,da.$$

The quadratic spectral action is therefore

$$\eta(c_L^2 + c_R^2)(da)^2 + 6\eta\lambda^2(c_L - c_R)^2\left(a + \frac{db}{\kappa}\right)^2.$$

Canonical vector normalization proves the mass formula. In particular, the Berry terms have been included; the calculation does not use a fixed projection.

At $M = 0$,

$$\lambda = \kappa, \qquad p_L = \sqrt{\frac{2}{l}}\cos(\kappa x), \qquad p_R = \sqrt{\frac{2}{l}}\sin(\kappa x).$$

Elementary integration gives

$$c_L = \frac{8}{3\pi}, \qquad c_R = \frac{4}{3\pi}.$$

Hence

$$\frac{M_{\text{spec}}^2}{M_{\text{KK}}^2} = 3\frac{(8-4)^2}{8^2+4^2} = \frac{3}{5}.$$

The matrices $\operatorname{diag}(c_L, c_R)$ describe the coefficient of the retained vector in the induced connection; they are not the endpoint $U(1)$ charge assignment. The explicit transformation formulas above establish covariance. □

7. Proposition (exact matching is possible after imposing an additional bulk-mass condition).

Keep the gauge and fermion boundary conditions of Result 6. At fixed l, let the fermion bulk mass vary over

$$0 \le M \le \frac{3\pi}{4l}.$$

Then at least one value

$$\boxed{0 < M_* < \frac{3\pi}{4l}}$$

satisfies

$$M_{\text{spec}}(M_*) = M_{\text{KK}}.$$

An exact defining condition is

$$\boxed{3\lambda(M_*)^2(c_L(M_*) - c_R(M_*))^2 = \frac{\pi^2}{4l^2}(c_L(M_*)^2 + c_R(M_*)^2).}$$

No uniqueness is asserted.

Proof. For $M \ge 0$, write

$$t = kl \in [\pi/2, \pi), \qquad M = -k\cot t, \qquad \lambda = \frac{k}{\sin t}.$$

The lowest chiral profiles are proportional to

$$p_L(x) \propto \sin(k(l-x)), \qquad p_R(x) \propto \sin(kx).$$

The relation $M = -k\cot(kl)$ follows by imposing $p_L'(0) = Mp_L(0)$; substitution in the first-order equation gives the displayed p_R and λ.

At

$$t = \frac{3\pi}{4}, \qquad M = \frac{3\pi}{4l},$$

we have

$$\lambda^2 = \frac{9\pi^2}{8l^2}.$$

The common squared normalization integral is

$$\int_0^l \sin^2(kx)\,dx = l\left(\frac{1}{2} + \frac{1}{3\pi}\right).$$

Integrating against $\cos(\pi x/(2l))$ gives

$$c_L = \frac{9}{3\pi + 2}, \qquad c_R = \frac{6}{3\pi + 2}.$$

Therefore

$$\frac{M_{\text{spec}}^2}{M_{\text{KK}}^2} = 3\frac{9}{2}\frac{(9-6)^2}{9^2+6^2} = \frac{27}{26} > 1.$$

At $M = 0$, Result 6 gives $3/5 < 1$. The lowest eigenvalue is simple, and the displayed root and normalized profiles depend continuously on M. The intermediate value theorem proves the existence of M_*. Thus universal impossibility of mass matching is false, while automatic matching is also false. □

8. No-go theorem (bounded light fermions cannot match the collapsing Neumann–Dirichlet tower).

Let a family of induced finite operators satisfy

$$\|V_0(l)\| \le C < \infty.$$

Use the leading spectral action of Result 5, with no additional vector-mass or kinetic terms. The retained gauge couplings may depend on l, and any finite multiplicities may be included.

Then every canonically normalized spectral-action vector mass satisfies

$$\boxed{M_{\text{spec}}^2 \le 6C^2.}$$

Consequently it cannot equal

$$M_{\text{KK}}^2 = \frac{\pi^2}{4l^2}$$

for arbitrarily small l.

This applies to the fixed-light-mass collapse of Theorem 9.9 and to the bounded-texture limit of Result 4.

Proof. For every Hermitian retained coupling matrix T,

$$\|[T, V_0]\|_{\text{HS}} \le \|TV_0\|_{\text{HS}} + \|V_0T\|_{\text{HS}} \le 2C\|T\|_{\text{HS}}.$$

Hence

$$K \le 4C^2Q$$

as quadratic forms. Result 5 gives

$$\mathcal{M}_{\text{spec}}^2 \le 6C^2 I.$$

The same inequality holds after restricting to any gauge-profile subspace. Multiplicities cannot evade it because they enter both Hilbert–Schmidt norms.

The Neumann–Dirichlet mass diverges as l^{-1}, proving the contradiction. More quantitatively, matching any such vector requires

$$\boxed{\|V_0(l)\| \ge \frac{\pi}{2\sqrt{6}\,l}.}$$

Thus some retained fermion scale must grow at least as $1/l$, unless the bosonic action or the stated approximation is changed. □

9. Proposition (gauge-factor and scalar normalizations require additional conditions).

Let the finite gauge kinetic form be diagonal on two specified factors, with fixed generator normalizations,

$$Q|_{\mathfrak{g}_s} = q_s I, \qquad s = 1, 2.$$

Let their five-dimensional kinetic terms have independent couplings $g_{5,s}$, and let

$$J_s = \int_0^l f_s(x)^2 \, dx.$$

Then the leading finite spectral action predicts

$$\frac{1}{g_{4,s}^2} = 4\eta q_s, \qquad \frac{g_{4,1}^2}{g_{4,2}^2} = \frac{q_2}{q_1},$$

whereas the KK action gives

$$\frac{1}{g_{4,s}^2} = \frac{J_s}{g_{5,s}^2}.$$

Vector-mass matching does not imply equality of these relative normalizations.

For scalar coordinates φ^i, let the induced connection at vanishing ambient A_μ be

$$\mathcal{A}_\mu = \mathcal{B}_i(\varphi)\partial_\mu\varphi^i,$$

and define

$$H_i = \partial_i V + i[\mathcal{B}_i, V].$$

The spectral-action scalar metric, in the convention $\frac{1}{2}Z_{ij}\partial_\mu\varphi^i\partial^\mu\varphi^j$, is

$$\boxed{Z_{ij}^{\text{spec}} = 6\eta \, \operatorname{Re}\operatorname{Tr}(H_i H_j).}$$

Mass matching constrains this metric on the corresponding gauge-orbit directions. It does not generally determine the metric on additional physical scalar directions.

Proof. Canonical normalization of $\eta q_s F_s^2$ gives the finite coupling formula. Integrating the five-dimensional kinetic term gives the KK formula. Rescaling $g_{5,s}^{-2}$ multiplies both $G_s^{(5)}$ and $H_s^{(5)}$ by the same factor and leaves every KK mass unchanged. Therefore mass matching cannot determine the independent ratio $g_{5,1}/g_{5,2}$. A common simple parent group with a specified invariant form could remove this freedom, but that is additional geometric input.

For the scalar metric,

$$D_\mu V = H_i \partial_\mu \varphi^i.$$

Substitution into $3\eta \operatorname{Tr}(D_\mu V D^\mu V)$ proves the formula.

The eaten scalar in Result 6 illustrates the constrained directions precisely. Put

$$G = \frac{l}{2g_5^2}, \qquad q = c_L^2 + c_R^2, \qquad d = c_L - c_R.$$

Its KK action is

$$\frac{G}{4}(da)^2 + \frac{G}{2}(db + \kappa a)^2.$$

If both the gauge kinetic coefficient and the vector mass match,

$$4\eta q = G, \qquad 3\lambda^2 d^2 = \kappa^2 q,$$

then the spectral scalar coefficient automatically satisfies

$$Z_b^{\mathrm{spec}} = \frac{12\eta\lambda^2 d^2}{\kappa^2} = G.$$

Thus no separate tuning is needed for this eaten scalar once those two equalities hold.

A scalar direction transverse to the gauge orbit is not tested by that mass equation. In the commuting-bulk mixed model, A_5 supplies only the unitary link of Result 3, so there is no radial Higgs kinetic term to match. If boundary Yukawa matrices are promoted to independent scalar fields, their continuum dynamics, including any boundary kinetic form, must be specified separately. □

Consequence for T4b and hypothesis H

Gauge-covariant projection repairs the covariance defect and supplies an exact light-bundle connection, but it does not establish equality with the full finite-triple fluctuation space or with the interacting KK theory. At fixed length, vector-mass equality is possible precisely when the appropriate trace-and-overlap condition holds; covariance supplies no selection principle for it or for the remaining kinetic normalizations. Hypothesis H therefore requires additional geometric relations, and its collapse regime cannot reproduce a Neumann–Dirichlet vector of mass $\pi/(2l)$ from a leading finite spectral action whose retained fermion masses all remain bounded.

Part 15

Collapse in the spectral propinquity: a gap criterion and limits of commutative spaces

Lemmas R2 and R3 of the Farsi-Latrémolière assessment, written out and extended. Proved by Claude Opus 5.5 without Astra or Fable, from the definitions in the LaTeX source of arXiv:2404.00240v3, Section 2. The sketches came from the reading subagent; the lower bound in the sketch of R2 omitted the cap sqrt(2)/2 of the metric and is corrected here. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): Theorem 15.2 item 2 had a gap, closed by the referee; Remark 15.4 omitted the cap; Proposition 15.9 needed amendments; the T0 section first contained two invalid arguments (on unbroken symmetry and on Morita equivalence). All corrections are incorporated below. Proposition 15.10 and Counterexample 15.11 are the referee's.

Scope. Everything here concerns one metric, the spectral propinquity, applied to the whole algebra of the parent triple. The section "What this says about T0" states what follows for the founding problem and what does not; lemma 16 treats truncations, where other answers hold.

Conventions

These are the definitions of Farsi-Latrémolière, Section 2.

- A *quantum compact metric space* (A, L) is a unital C*-algebra A with a seminorm L on a dense subspace of sa(A) such that ker L = R 1; the metric mk_L(phi, psi) = sup{abs(phi(a) - psi(a)) : L(a) <= 1} metrizes the weak* topology of the state space S(A); L satisfies the Leibniz inequality for the Jordan product a o b = (ab + ba)/2 and the Lie product {a, b} = (ab - ba)/(2i); and {L <= 1} is closed.
- A *quantum isometry* is a *-epimorphism pi with L_B(b) = inf{L_A(a) : pi(a) = b}. Its dual map is an isometry of state spaces.
- A *tunnel* is a quantum compact metric space (E, L_E) with quantum isometries pi_1, pi_2 onto the two spaces. Its *extent* chi is the larger of the Hausdorff distances, in mk_{L_E}, from S(E) to pi_1^*S(A_1) and to pi_2^*S(A_2). The *propinquity* Lambda is the infimum of extents.
- A *metric spectral triple* T = (A, H, D) is a spectral triple for which L_T(a) = norm([D, a]) is an L-seminorm. In particular ker L_T = R 1. We write DN_T(xi) = norm(xi) + norm(D xi) and diam T for the diameter of S(A) in mk_{L_T}.

- A *metrical tunnel* between T_1 and T_2 consists of a C*-correspondence P from E to a unital C*-algebra B, with L_E and L_B as above and a norm DN_P on a dense subspace of P such that: the unit ball of DN_P is compact in P; DN_P dominates the module norm; DN_P(e Theta) <= (norm(e) + L_E(e)) DN_P(Theta) (*modular Leibniz*); L_B applied to the real and to the imaginary part of <Theta, Theta'>*B is at most 2 DN_P(Theta) DN_P(Theta') (*inner Leibniz*; this is the condition of Latrémolière, arXiv:1811.10843, in the definition of a metrical C*-correspondence, whereas the display in Farsi-Latrémolière applies L to the non-self-adjoint inner product and names L_A). It carries maps (Pi_j, pi_j, theta_j) onto (H_j, A_j, C), compatible with the actions and inner products, with DN*{T_j}(xi) = inf{DN_P(Theta) : Pi_j Theta = xi}. Its extent is the larger of the extents of (E, pi_1, pi_2) and (B, theta_1, theta_2).
- The *pseudo-states* of T are the functionals xi -> <omega, xi> with DN_T(omega) <= 1. The *separation* of two operator families (a_t), (b_t) is the Hausdorff distance between {(phi a_t Pi_1)_t} and {(psi b_t Pi_2)_t}, phi and psi ranging over pseudo-states, in the metric sup_t sup{abs(F_t(Theta) - G_t(Theta)) : DN_P(Theta) <= 1}. The *dispersion* is the larger of extent and separation.
- The *spectral propinquity* Lambda_spec(T_1, T_2) is the infimum of sqrt(2)/2 and of those eps > 0 for which some metrical tunnel has dispersion < eps for the families exp(i t D_1), exp(i t D_2), 0 <= t <= 1/eps.

Three elementary consequences.

(a) If tau is a tunnel of extent chi, then abs(diam_1 - diam_2) <= 2 chi. Take phi, phi' realizing diam_1; their pullbacks lie within chi of pullbacks of states of A_2, and the dual maps are isometries.

(b) Lambda_spec(T_1, T_2) >= min(sqrt(2)/2, Lambda(A_1, A_2)), because the dispersion dominates the extent of the top tunnel. So convergence in Lambda_spec implies convergence in Lambda.

(c) Combining: Lambda_spec(T_1, T_2) >= min(sqrt(2)/2, abs(diam T_1 - diam T_2)/2).

The cap is dimensionful: Lambda_spec has the units of a length and is cut off at sqrt(2)/2 in the chosen unit.

(d) If a = a* in (A, L), and z is the midpoint of the convex hull of Sp(a), then norm(a - z) <= (diam/2) L(a). The endpoints of the hull are values of states on a.

Comparison along a tunnel

Lemma 15.1. Let tau = (E, L_E, pi_1, pi_2) be a tunnel of extent chi, and let e = e* with L_E(e) finite. For {j, k} = {1, 2},

```
norm(pi_j e) <= norm(pi_k e) + chi L_E(e),     norm(e) <= norm(pi_j e) + chi L_E(e).
```

For a real polynomial p = sum p_m x^m, the element p(e) has finite L_E, with L_E(p(e)) <= L_E(e) sum_{m >= 1} abs(p_m) m norm(e)^{m-1}.

Proof. The norm of a self-adjoint element is the supremum of abs(state) on it. A state phi of A_j pulls back to S(E), and lies within chi of the pullback of a state psi of A_k, so abs(phi(pi_j e)) <= abs(psi(pi_k e)) + chi L_E(e). The second inequality is the same argument starting from a state of E. For powers, e^m = e o e^{m-1}, and induction with the Jordan-Leibniz inequality gives L_E(e^m) <= m norm(e)^{m-1} L_E(e); L_E(1) = 0. ■

Limits of commutative spaces (Lemma R3)

Theorem 15.2. Let (A_n, L_n) -> (A, L) in the propinquity.

1. If every A_n is commutative, then A is commutative.
2. If every A_n = C(X_n) with X_n having at most k connected components, then the spectrum of A has at most k connected components. In particular, a finite-dimensional limit is C^j with j <= k, and for k = 1 it is C.

Both statements hold for limits in Lambda_spec, by (b).

Proof. Let tau_n be tunnels of extent chi_n -> 0.

(1) Let a_1, a_2 in sa(A) have finite L. Choose lifts d_i in E_n with pi_A(d_i) = a_i and L_{E_n}(d_i) <= L(a_i) + 1. Lemma 15.1 bounds norm(d_i) uniformly. The Lie product e = {d_1, d_2} satisfies pi_n(e) = 0 and pi_A(e) = {a_1, a_2}, and L_{E_n}(e) <= norm(d_1) L(d_2) + norm(d_2) L(d_1) is uniformly bounded. So norm({a_1, a_2}) <= chi_n L_{E_n}(e) -> 0. Elements of finite L are dense in sa(A), so A is commutative.

(2) By (1), A = C(X). Suppose X has at least k + 1 components. Since components and quasicomponents agree in a compact Hausdorff space, X splits into nonempty clopen sets X_0, ..., X_k. Let a in sa(A) have finite L and norm(a - sum_j j 1_{X_j}) < 1/8, so that Sp(a) meets each [j - 1/8, j + 1/8] and lies in their union. Take lifts d_n as above, and b_n = pi_n(d_n), a real function on X_n. All spectra lie in a fixed compact interval I for large n, by Lemma 15.1.

Let g : I -> [0, 1] be continuous, 0 on the union of the [j - 1/8, j + 1/8] and 1 outside the union of the open intervals (j - 3/8, j + 3/8), and let p be a polynomial with abs(p - g) < 1/8 on I. Then norm(p(a)) <= 1/8, so Lemma 15.1 gives norm(p(b_n)) <= 1/4 for large n, because L_{E_n}(p(d_n)) is uniformly bounded. Hence the range of b_n lies in the union of the disjoint intervals (j - 3/8, j + 3/8), and each component of X_n maps into one of them. So b_n reaches at most k of them, and along a subsequence it misses one fixed j_*. Let h : I -> [0, 1] be continuous, 1 on [j_* - 1/8, j_* + 1/8] and 0 outside radius 3/8 around j_*, and let q be a polynomial with abs(q - h) < 1/8 on I. Then norm(q(a)) >= 7/8 and norm(q(b_n)) <= 1/8, contradicting Lemma 15.1 for large n. ■

The theorem is one-directional: noncommutative spaces may converge to commutative ones, as matrix algebras do to the sphere (Rieffel; Latrémolière) and spectral truncations of tori do to the tori in the Gromov-Hausdorff sense on state spaces (Leimbach-van Suijlekom). We did not find Theorem 15.2 stated in the sources we searched (arXiv:1302.4058, 1311.0104, 2404.00240, 2603.19128); its proof is elementary and it may be folklore.

The gap criterion (Lemma R2)

Theorem 15.3. Let T = (A, H, D) be a metric spectral triple with delta = diam T. Let Sigma be a finite subset of R, P = 1_Sigma(D) of rank N >= 1 and Q = 1 - P, and let g > 0 be such that abs(lambda) >= g for every lambda in Sp(D) outside Sigma. Let D_F be a Hermitian N x N matrix, W : PH -> C^N a unitary, extended by 0 on QH, and Delta = norm(W D W* - D_F). For T_F = (C, C^N, D_F),

```
min(sqrt(2)/2, delta/2) <= Lambda_spec(T, T_F) <= 1/g^2 + max(delta/2, Delta) + Delta + sqrt(Delta).
```

For Delta = 0 the upper bound reads delta/2 + 1/g^2. No anticommutation hypothesis enters.

Proof. The lower bound is (c), since diam T_F = 0. For the upper bound fix rho > 0 with rho >= max(delta/2, Delta), and put s = rho + 1/(2 g^2), c = 1/(1 + Delta), D_B = W D W*.

Top tunnel. On E = A (+) C let L_E(a, z) = max{L_T(a), norm(a - z)/rho}. Then ker L_E = R (1, 1). The Leibniz inequality for the Jordan product follows from a o b - z w = (a - z) o b + z (b - w), and for the Lie product from {a, b} = {a - z, b}. The set {L_E <= 1, mu(a) = 0}, for a fixed state mu of A, is contained in a product of compact sets, since abs(z) <= norm(a) + rho and norm(a) <= delta there, so L_E is an L-seminorm by Rieffel's criterion, with the closed ball following from lower semicontinuity. The projection onto A is a quantum isometry by (d), as rho >= delta/2; the projection onto C is one because L_E(z 1, z) = 0. A state of E has the form (a, z) -> t mu(a) + (1 - t) z. On L_E(a, z) <= 1 it differs from (a, z) -> mu(a) by (1 - t) abs(z - mu(a)) <= norm(a - z) <= rho, and from (a, z) -> z by t abs(mu(a) - z) <= rho. So the extent is at most rho.

Base tunnel. On B = C (+) C let L_B(x, y) = abs(x - y)/s. The two projections are quantum isometries onto (C, 0), and the extent is at most s.

Module. Let P_mod = H (+) C^N, a right Hilbert B-module with <(xi, eta), (xi', eta')> = (<xi, xi'>, <eta, eta'>) and left action (a, z)(xi, eta) = (a xi, z eta). On Dom D (+) C^N set

DN_P(xi, eta) = max{DN_T(xi), DN_F(eta), norm(W xi - eta)/rho}.

The unit ball is closed in a product of compact sets, because D has compact resolvent. The norm is dominated.

Modular Leibniz. DN_T(a xi) <= (norm(a) + L_T(a)) DN_T(xi) and DN_F(z eta) = abs(z) DN_F(eta). Further

norm(W a xi - z eta) <= norm(W (a - z) xi) + abs(z) norm(W xi - eta) <= norm(a - z) norm(xi) + abs(z) r

and dividing by rho gives at most (L_E(a, z) + norm(a, z)) DN_P(xi, eta). No commutation of P with A is needed, because the limit algebra acts by scalars.

Inner Leibniz. On Q H the spectrum has abs(lambda) >= g, so norm(Q xi) <= norm(D xi)/g <= DN_T(xi)/g. Then <xi, xi'> = <W xi, W xi'> + <Q xi, Q xi'> and

abs(<W xi, W xi'> - <eta, eta'>) <= norm(W xi - eta) norm(W xi') + norm(eta) norm(W xi' - eta') <= 2 rh

so abs(<xi, xi'> - <eta, eta'>) <= (2 rho + 1/g^2) DN_P DN_P' = 2 s DN_P DN_P'. This complex estimate implies the condition on real and imaginary parts.

Quotient conditions. Given xi, take eta = c W xi. Since P reduces D and D_F W = W D - (D_B - D_F) W on Dom D, we get DN_F(c W xi) <= c (norm(W xi) + norm(W D xi) + Delta norm(W xi)) <= DN_T(xi), and norm(W xi - c W xi) <= Delta norm(xi) <= rho DN_T(xi). So DN_P(xi, c W xi) = DN_T(xi). Given eta, take xi = c W* eta. Then DN_T(xi) = c (norm(eta) + norm(D_B eta)) <= DN_F(eta) and norm(W xi - eta) <= Delta norm(eta), so DN_P(c W* eta, eta) = DN_F(eta).

Separation. Let DN_T(omega) <= 1 and put omega' = c W omega, so that DN_F(omega') <= 1. For DN_P(xi, eta) <= 1 and real t, since P commutes with exp(i t D),

<omega, e^{itD} xi> - <omega', e^{itD_F} eta> = <Q omega, e^{itD} Q xi> + <W omega, (e^{itD_B} - e^{itD_

and the four terms are bounded by 1/g^2, abs(t) Delta, rho and Delta. Conversely, let DN_F(omega') <= 1 and put omega = c W* omega', so that DN_T(omega) <= 1 and Q omega = 0; the same decomposition without the first term gives at most abs(t) Delta + rho + Delta.

Conclusion. Put K = 1/g^2 + rho + Delta. The extent is at most s <= K and the separation on [0, 1/eps] is at most K + Delta/eps. If eps > K + sqrt(Delta), then Delta/eps < sqrt(Delta) (or Delta = 0), so the dispersion is below eps. Hence Lambda_spec <= K + sqrt(Delta), and we let rho decrease to max(delta/2, Delta). ■

Remark 15.4 (rescaling). For T_eps = (A, H, D/eps) with 0 in Sp(D), Sigma = {0} and D_F = 0, the theorem gives min(sqrt(2)/2, eps delta_1/2) <= Lambda_spec <= eps delta_1/2 + $\text{eps}^{2/g_1}2$, where delta_1 = diam T_1 and g_1 is the gap of D. This is Corollary 4.1 of Farsi-Latrémolière with a sharp leading rate for small eps. Counterexample R1 of the assessment violates the gap hypothesis, since the eigenvalue x stays bounded outside Sigma.

Corollary 15.5 (the tuned interval, Theorem 9.9). Let T_l = (C([0, l]), L^2([0, l]; C^2), D_{-mu(l)}) with condition (c) and mu(l) as in Theorem 9.9, c = mu(l) l, and let P project onto +/- m_f. For l small,

l/2 <= Lambda_spec(T_l, (C, C^2, m_f sigma_1)) <= l/2 + l^2/(pi^2 + c^2).

For the Connes-Lott triple CL(m_f) = (C + C, C^2, m_f sigma_1) of diameter 1/m_f,

Lambda_spec(T_l, CL(m_f)) >= min(sqrt(2)/2, (1/m_f - l)/2).

Proof. Multiplication by a Lipschitz function preserves condition (c), and the mass term commutes with functions, so L(f) = norm(f')_infinity and diam T_l = l. Theorem 9.6 gives W with W D W* = m_f sigma_1, so Delta = 0, and Theorem 9.4 gives g >= sqrt(mu^2 + (pi/l)^2). The second bound is (c). ■

So the whole triple converges to a point carrying a two-dimensional massive Dirac block, while the compression at each l is the Connes-Lott triple (Theorem 9.6). The two statements concern different objects, and both hold; see "What this says about T0".

Corollary 15.6 (towerless Yukawa block, Theorem 12.7). Fix the c's and Y_hat of Theorem 12.7, with T_{X,l} = (C([0, l]), L^2 (x) (C^{N_Q} (+) C^{N_U}), D_X). Then T_{X,l} is a metric spectral triple of diameter l, and

```
Lambda_spec(T_{X,l}, (C, C^{N_Q + N_U}, D_F)) = O(l),     D_F = [[0, M_F], [M_F^dagger, 0]],   M_F = Phi
```

Proof. Multiplication by f preserves the conditions at l because f takes one value there, and norm([D_X, f]) = norm(f')_infinity. Theorem 12.7 gives g >= min(L_Q, L_U)/l. Mapping eigenvectors of +/- lambda_k to the singular vectors of M_F gives W with Delta = max_k abs(lambda_k - sigma_k(M_F)) = O(l^2). Theorem 15.3 applies, with sqrt(Delta) = O(l). ■

This answers the remark in Theorem 12.7 that a limit triple needs a choice of algebra: in this metric the algebra of the limit is C.

The converse

Theorem 15.7. Let T_n = (A_n, H_n, D_n) be metric spectral triples with Lambda_spec(T_n, T_F) -> 0, T_F = (C, C^N, D_F). Then diam T_n -> 0; for every R > norm(D_F), D_n has exactly N eigenvalues in [-R, R] for large n, so that the gap g_n to the remaining spectrum tends to infinity (with g_n infinite if no spectrum remains); and there are unitaries W_n from the light space onto C^N with norm(W_n D_n W_n* - D_F) -> 0. With Theorem 15.3,

```
Lambda_spec(T_n, T_F) -> 0   if and only if   diam T_n -> 0, g_n -> infinity, Delta_n -> 0.
```

Proof. The diameter statement is (c). Let tau_n have dispersion < eps_n on [0, 1/eps_n], with eps_n -> 0.

Inner estimate. The characters theta_1, theta_2 are states of the base algebra B_n, and S(C) is one point, so mk_{L_B}(theta_1, theta_2) < eps_n. Applying this to the real and imaginary parts of <Theta, Theta'> and using inner Leibniz,

```
abs(<Pi_1 Theta, Pi_1 Theta'> - <Pi_2 Theta, Pi_2 Theta'>) <= 4 eps_n DN_P(Theta) DN_P(Theta').
```

At most N eigenvalues. Suppose D_n has N + 1 orthonormal eigenvectors xi_i with eigenvalues in [-R, R], so DN(xi_i) <= 1 + R. Lift them to Theta_i with DN_P(Theta_i) <= 2(1 + R), and put eta_i = Pi_2 Theta_i in C^N. The Gram matrix G of the eta_i has rank at most N, so norm(1 - G) >= 1, while the inner estimate gives norm(1 - G) <= 16 (N + 1)(1 + R)^2 eps_n. This fails for large n.

Concentration. Let e be a unit eigenvector of D_F with eigenvalue lambda, C = 1 + norm(D_F), and lift e to Theta with DN_P(Theta) <= 2C and xi = Pi_1 Theta. Apply the separation to the pseudo-state omega = xi/(2C): there is omega' with DN_F(omega') <= 1 and, for t in [0, 1/eps_n], abs(<omega, e^{itD_n} xi> - e^{it lambda} <omega', e>) <= 2C eps_n. Comparing with t = 0 and multiplying by 2C,

```
abs(<xi, e^{itD_n} xi> - e^{it lambda} norm(xi)^2) <= 8 C^2 eps_n.
```

Let nu be the spectral measure of D_n at xi. Taking real parts and averaging over t in [0, 1/eps_n] gives the integral of 1 - sinc((x - lambda)/eps_n) d nu(x) at most 8 C^2 eps_n. Since 1 - sinc(u) >= 1/2 for abs(u) >= 2, nu{abs(x - lambda) >= 2 eps_n} <= 16 C^2 eps_n.

Multiplicities. Apply this to an orthonormal eigenbasis e_i of D_F, with lifts xi_i. The inner estimate makes the Gram matrix of the xi_i tend to the identity, and concentration keeps it so after projecting each xi_i to the spectral window of radius 2 eps_n around its eigenvalue. So each window around an eigenvalue of multiplicity m contains at least m eigenvalues of D_n, for large n. The total is at most N, by the previous step, so equality holds. Mapping eigenvectors to the e_i gives W_n with norm(W_n D_n W_n* - D_F) <= 2 eps_n. ■

Corollary 15.8 (finite limits of connected commutative triples). Let T_n be metric spectral triples with A_n = C(X_n), X_n connected, converging in Lambda_spec to a triple with finite-dimensional Hilbert space. Then the limit is (C, C^N, D_F), and the convergence is characterized by Theorem 15.7. Every Hermitian D_F arises: on the circle of length l, the operator -i d/dx + diag(lambda_i) on $L^{2(S}$1_l; C^N)

has the block diag(lambda_i) on constants and gap at least 2 pi/l - norm(D_F). This construction concerns ungraded triples; a grading or real structure imposes its own compatibility conditions.

Proof. A metric triple is faithful: a self-adjoint a acting as 0 has [D, a] = 0, so a = t 1 by ker L = R 1, and then t = 0. So the limit algebra is finite-dimensional, and Theorem 15.2 gives C. ■

Corollary 9.10 fits the converse: at fixed mu every eigenvalue diverges, the spectra have no bounded sequences, and no limit exists. Theorem 9.8 lies outside the metric, because the diameter diverges.

Grading and real structure

The metric ignores gamma and J. They pass to the block exactly.

Proposition 15.9. In Theorem 15.3 let Sigma = -Sigma, and let D anticommute with a grading gamma and satisfy J D = eps' D J for an antiunitary J with J^2 = eps and J gamma = eps'' gamma J. Then P commutes with gamma and J, and gamma_F = W gamma W*, J_F = W J W* satisfy the same relations exactly with D_B = W D W*, and with D_F up to defects bounded by 2 Delta. The tunnel of Theorem 15.3 controls them: for DN_T(omega) <= 1 and DN_P(xi, eta) <= 1,

```
abs(<omega, gamma xi> - <W omega, gamma_F eta>) <= 1/g^2 + rho,     abs(<omega, J xi> - <W omega, J_F e
```

For Delta > 0, W omega need not be a pseudo-state of T_F; the normalized c W omega is one, and adds at most Delta to these bounds.

Proof. Spectral projections of D for a symmetric set commute with gamma and J, because gamma and J map the lambda-eigenspace to the eigenspaces of -lambda and eps' lambda. Decompose along P and Q as in Theorem 15.3, using additivity of J_F. ■

So gamma added to the families converges along the same tunnels, and the KO-signs of the parent reach the block unchanged when Delta = 0. For the interval, P Gamma P = sigma_3 (Theorem 9.6). Composition with the antilinear J gives antilinear functionals, which the definition of the separation does not cover; a statement of convergence for J needs an extension of the pseudo-state formalism, which we have not developed.

Unbroken symmetry and the metric class

Proposition 15.10 (referee). Let (A, H, D, J) be a finite-dimensional real triple with order zero, order one and J D = eps' D J, and let u be a unitary in A. If D commutes with u J u J^-1, then [D, u] = 0. If moreover the triple is metric, u is scalar, and the inner gauge transformation is trivial.

Proof. Put v = J u J^-1. From [D, u v] = 0, we get [D, u] v = - u [D, v], so a := u* [D, u] = - [D, v] v*. Now [D, v] = eps' J [D, u] J^-1 commutes with A by order one, and v* commutes with A by order zero. So a commutes with u, and [D, u^n] = n a u^n by induction. Then u^n D u^-n = D - n a. D is bounded, so a = 0. For a metric triple, the self-adjoint parts of u lie in ker L = R 1. ■

Counterexample 15.11 (referee; order one is needed). Let H = M_2(C) with the Hilbert-Schmidt product, A = M_2(C) acting by left multiplication, J x = x*, and D x = x - (1/2) tr(x) I. Applying [D, L_a] to I gives a - (1/2) tr(a) I, so [D, L_a] = 0 forces a scalar, and the triple is metric. Yet every conjugation x -> u x u* = L_u J L_u J^-1 commutes with D. This triple fails order one.

What this says about T0

Corollary 15.8 describes one notion of convergence: the state space of the whole algebra C(K_l) must converge in the Monge-Kantorovich metric. Under it, a collapsing connected parent yields a point, and the finite data survive as (C, C^N, D_F, gamma_F, J_F). Three limits on its reach:

1. **The truncation is a different object.** Along Theorem 9.9, the compression at each l is exactly Connes-Lott, with distance 1/m_f, while the parent has diameter l. The compression is not a quantum isometry, so this metric cannot see it. This is the towerless phenomenon itself: the two-sheet distance is unrelated to the size of the internal space.

2. **The class excludes a kind of target.** A metric spectral triple has ker L_T = R 1, so no nonscalar element of A commutes with D. The finite triple of the Standard Model lies outside the class: D_F commutes with the nonscalar subalgebra C_F = {(lambda, lambda, 0)}, which keeps the photon massless, and colour is unbroken (Chamseddine-Connes-Marcolli 2007, section 2.6). Its almost-commutative product lies outside as well, through the constant copy of such an element. A_F is moreover a real algebra, so applying the complex framework also needs a complexification. Among finite real first-order triples, the metric class has no nontrivial unbroken inner gauge transformation (Proposition 15.10); without order one this fails (Counterexample 15.11).
3. **The theorem concerns the scalar algebra.** Theorem 15.2 restricts limits of the whole scalar algebra C(K). Morita equivalence does not remove that restriction: C(K) and C(K, M_n) are Morita equivalent but not isomorphic, and their limits differ, and a bundle E can carry matrix-valued connections while C(K) still acts by scalars. Replacing the represented algebra by End(E) is a change of data, and a possible enlargement of T0. In that enlargement, Farsi-Latrémolière Theorem 4.6 makes an even metric finite (B, H_F, D_F) with noncommutative B the limit of B (x) C(K_l) with K_l collapsing, with nonzero multiplicity space ker D_K; there B is put in by hand.

Theorem 15.2 is a real constraint within its scope: this metric, on scalar function algebras. T0 also allows truncations, where it does not apply; lemma 16 treats these.

Part 16

Noncommutative finite real triples from commutative data: truncating the coarse correspondence

Proved by Claude Opus 5.5 without Astra or Fable, in response to the challenge that the T0 note overread lemma 15 and the Farsi-Latrémolière limits. Signs and the level-two algebra were checked with numpy on random data (section "Numerical check"). The term "coarse correspondence" is that of Connes, *Noncommutative Geometry* (1994), chapter V, for the bimodule of Hilbert-Schmidt operators, and the form D = Delta + J Delta J^-1 is in Krajewski (arXiv:hep-th/9701081) and Paschke-Sitarz (arXiv:q-alg/9612029); all checked against the sources. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): the examples and Corollary 16.4 are correct; amendments to Proposition 16.1, Theorem 16.2, Corollary 16.3 and the caveats are incorporated below; Remark 16.5 and Example 16.6 come from the referee's assessment of what the lemma achieves. The two-sheet section was refereed separately (Terra 6.1, 3 Nov 2026; report), with one error in Corollary 16.8 and one in Example 16.9, corrected below.

Outcome. Compressing functions on a connected commutative space to a fixed spectral window produces a noncommutative finite algebra with no limit taken. With the standard reality J a J^-1 = a*, the inherited structure fails order zero exactly when that algebra is noncommutative (Proposition 16.1). With the parent Hilbert space doubled to H_0 (x) conj(H_0), order zero and order one hold for every such algebra (Theorem 16.2, Corollary 16.3); these two conditions are formal consequences of the tensor square, available for any finite even representation (Remark 16.5). The KO-6 signs and the count dim H_F = 2 k^2 are reached by a product with a Clifford point, which is an input and fails finite orientability (Corollary 16.4), and more faithfully by a parent with two sheets, which gives the form of the Chamseddine-Connes classification (Theorem 16.7, Corollary 16.8, Example 16.9). This selects no particular algebra and gives no dynamics.

Setting

Let (A_0, H_0, D_0, gamma_0) be an even spectral triple, A_0 commuting with gamma_0, with D_0 of compact resolvent. Let Sigma = -Sigma be a finite subset of R and P_0 = 1_Sigma(D_0), of rank k, so that P_0 commutes with gamma_0. Put V = P_0 H_0, graded by gamma_V, with d = D_0 restricted to V, and

```
Q(a) = P_0 a P_0 on V,      O_F = Q(A_0),      A_F = C*(O_F) in B(V).
```

O_F is an operator system, in general not an algebra, and A_F consists of even operators. Compression is standard: for functions it is the Toeplitz compression, and its use in truncating spectral triples is due

to D'Andrea-Lizzi-Martinetti and Connes-van Suijlekom. The latter note that the C*-algebra generated by PAP is in many examples a noninformative full matrix algebra. Since Q(1) = 1, A_F is the bicommutant of O_F, so its structure is set by the operators commuting with all compressed functions, such as the grading in example (b) and j in Example 16.9. That is where the information lies, and the fact that A_F is noncommutative is modest.

Examples of A_F.

(a) *Chiral interval, lowest level.* For condition (c) with M = 0 and Sigma = {+/- pi/(2l)}, O_F = C + C = A_F (lemma 05).

(b) *Chiral interval, next level.* For Sigma = {+/- pi/(2l), +/- 3 pi/(2l)}, V has basis (c_1, 0), (c_2, 0), (0, s_1), (0, s_2), with c_r = (2/l)^{1/2} cos((r - 1/2) pi x/l) and s_r = (2/l)^{1/2} sin((r - 1/2) pi x/l). With C_n = (1/l) int f cos(n pi x/l),

Q(f) = (Q_L, Q_R), Q_L = [[C_0 + C_1, C_1 + C_2], [C_1 + C_2, C_0 + C_3]], Q_R = [[C_0 - C_1, C_1 - C_2], [C_1 - C_2, C_0 - C_3]].

The values C_0, C_1, C_2, C_3 are independent and determine Q(f), so O_F has dimension 4. Each of Q_L and Q_R alone ranges over all symmetric 2 x 2 matrices, which generate M_2(C), so A_F projects onto each summand. For f_1 = 1/2 + cos(pi x/l) + cos(2 pi x/l) + cos(3 pi x/l) and f_2 = 1/2 - cos(pi x/l) + cos(2 pi x/l) - cos(3 pi x/l),

```
Q(f_1) = ([[1, 1], [1, 1]], 0),     Q(f_2) = (0, [[1, -1], [-1, 1]]).
```

M_2(C) is simple, so multiplying Q(f_1) and Q(f_2) by elements of A_F gives M_2(C) + 0 and 0 + M_2(C). Hence A_F = M_2(C) + M_2(C), of dimension 8. The finite algebra of the interval itself is noncommutative at the second level.

(c) *Twistor space.* For the level-one Toeplitz compression on CP^3, Q(x_a) = Gamma_a/5 (Proposition 10.1), and these generate A_F = M_4(C) on V = C^4. Here V is the kernel of the Dolbeault-Dirac operator twisted by O(1), so d = 0 on V.

The standard correspondence

Proposition 16.1 (order zero fails with inherited reality). Let J_0 be a real structure on H_0 (antiunitary, J_0^2 = +/-1, with the grading sign), with J_0 a J_0^-1 = a* for all a in A_0 and J_0 D_0 = eps' D_0 J_0, and let J_F = J_0 restricted to V. Then J_F Q(a)* J_F^-1 = Q(a), and the data (A_F, V, d, J_F) satisfy order zero if and only if A_F is commutative. The same holds for a product with a fixed finite factor that itself satisfies order zero, with the compatible product real structure.

Proof. J_0 preserves V because Sigma = -Sigma. Then J_0 P_0 a* P_0 J_0^-1 = P_0 a P_0. Order zero would make Q(a) commute with Q(b)° = Q(b) for all a, b, so O_F and hence A_F would be commutative; the converse is immediate, since J_F A_F J_F^-1 = A_F. For the product, take the unit in the finite factor: [1 (x) Q(a), 1 (x) Q(b)] = 1 (x) [Q(a), Q(b)]. ■

Together with Amendment 2 of lemma 07 (first order fails for C + C with inherited J_2), this is the correct form of the obstruction. It concerns the standard correspondence L^2(A_0) with the pointwise real structure.

The coarse correspondence

Let conj(H_0) be the conjugate Hilbert space, with canonical antiunitary xi -> xi^c, and with gamma^c, D^c the conjugate operators, gamma^c xi^c = (gamma_0 xi)^c and D^c xi^c = (D_0 xi)^c. On H = H_0 (x) conj(H_0), which is the coarse correspondence of A_0, equivalently the Hilbert-Schmidt operators on H_0, define for homogeneous xi, eta

```
pi(a) = a (x) 1,
DD = D_0 (x) 1 + gamma_0 (x) D^c,
Gamma = gamma_0 (x) gamma^c,
```

J (xi (x) eta^c) = (-1)^{|xi| |eta|} eta (x) xi^c.

Theorem 16.2. (A_0, H, DD, J, Gamma) is a real even spectral triple with signs J^2 = 1, J DD = DD J, J Gamma = Gamma J (KO-dimension 0). For even a, b with bounded commutators,

a° := J pi(a)* J^-1 = 1 (x) (a*)^c, [pi(a), b°] = 0, [[DD, pi(a)], b°] = 0,

and norm([DD, pi(a)]) = norm([D_0, a]). So if the original triple is metric, the new triple is metric with the same Connes distance on A_0. If the Weyl exponent p = lim log N_0(R)/log R of D_0 exists, that of DD is 2 p, since N_0(R/sqrt(2))^2 <= N_DD(R) <= N_0(R)^2.

Proof. J is antilinear, and J^2 = 1 because the sign is applied twice with the degrees exchanged. Gamma has eigenvalue (-1)^{|xi| + |eta|}, symmetric under the exchange. For the Dirac operator,

J DD (xi (x) eta^c) = (-1)^{(|xi| + 1) |eta|} eta (x) (D_0 xi)^c + (-1)^{|xi|} (-1)^{|xi| (|eta| + 1)}
= (-1)^{|xi| |eta|} [D_0 eta (x) xi^c + (-1)^{|eta|} eta (x) (D_0 xi)^c] = DD J (xi

For even a, a* preserves degree, so J pi(a)* J^-1 (eta (x) xi^c) = eta (x) (a* xi)^c, which gives the formula for a° and order zero. Since a is even, [DD, pi(a)] = [D_0, a] (x) 1, which commutes with 1 (x) (b*)^c. Finally DD^2 = D_0^2 (x) 1 + 1 (x) ($D^{c)}$2, because gamma_0 anticommutes with D_0. An eigenbasis of D_0 decomposes DD into finite-dimensional invariant blocks, so DD is essentially self-adjoint on the algebraic tensor product, with compact resolvent, and the spectrum is +/- (lambda_i^2 + lambda_$j^{2)}${1/2}. ■

Geometrically, when H_0 = L^2(S_X) on an even-dimensional spin manifold X, a charge conjugation commuting with D_X identifies conj(H_0) with H_0, H is L^2(S_{X x X}) and DD is the ordinary product Dirac operator. (The grading may acquire an overall sign under this identification.) So the data are a product-manifold Dirac operator on X x X with a restricted observable algebra C(X) (x) 1, and a J covering the flip of the two factors, which differs from the pointwise spin real structure. They count as commutative Kaluza-Klein data only with that restriction and that J: they are neither the ordinary spin triple of X x X with its full function algebra nor a finite-rank twist over X. The restriction is essential: on all of C(X x X), the compressions generate A_F (x) conj(A_F), and order zero generally fails again.

Corollary 16.3 (truncation). Let P = P_0 (x) conj(P_0). It commutes with DD, Gamma and J. It is a joint cutoff in the two factors, and need not be a spectral projection of DD: for absolute eigenvalues 1, 5, 7, sqrt(73) of D_0 with P_0 selecting the first three, the selected pair (5, 7) and the excluded pair (1, sqrt(73)) both give DD^2 = 74. The compression of pi(A_0) is O_F (x) 1; passing to the generated algebra gives the finite real even triple

T_F = (A_F, V (x) conj(V), D_F, J_F, Gamma_F), D_F = d (x) 1 + gamma_V (x) d^c = Delta + J_F Delta

with A_F acting on the left. It has KO-dimension 0, and satisfies order zero and order one for every A_F, noncommutative or not, with dim H_F = k^2. The family is exact at each truncation; no limit is taken.

Proof. The proof of Theorem 16.2 applies verbatim on V with arbitrary even a, b in B(V), since it uses only the grading and the tensor structure. For J_F Delta J_F^-1, apply J to (d (x) 1) J (xi (x) eta^c): the signs combine to (-1)^{|xi|}, giving gamma_V (x) d^c. ■

So H_F has the bimodule form of Connes' finite geometries, with A_F on the left and its opposite on the right, and D_F has the first-order form of Krajewski and Paschke-Sitarz.

Example (a) gives (C + C, C^2 (x) C^2, m (sigma_1 (x) 1 + sigma_3 (x) sigma_1)). There [[D_F, e], e°] = 0 for e = diag(1, 0), which repairs the first-order failure of Counterexample 08.4 at the price of doubling. Example (b) gives (M_2(C) + M_2(C), C^16, D_F).

Corollary 16.4 (KO-6 by a Clifford point). The product of T_F with (C, C^2, 0, sigma_1 K, sigma_3) is

(A_F, V (x) conj(V) (x) C^2, D_F (x) 1, J_F (x) sigma_1 K, Gamma_F (x) sigma_3),

with signs (+, +, -), order zero, order one and dim H = 2 k^2. It fails the usual zero-dimensional orientability axiom: every zero-cycle built from the left and opposite actions acts as T (x) 1_2, while the grading is Gamma_F (x) sigma_3.

Proof. (sigma_1 K)^2 = 1 and sigma_1 K sigma_3 = -sigma_3 sigma_1 K, while the D term is untouched. a° is a° of T_F tensored with 1. ■

Remark 16.5 (the order conditions are formal). For every finite even representation (A, V, d, gamma), the map V -> V (x) conj(V), d -> d (x) 1 + gamma (x) d^c gives a real bimodule with order zero and order one, using no prior finite real structure. So these two conditions follow from choosing the tensor square, and say nothing specific about A_F. The construction does not recover an arbitrary prescribed finite (H, D, J). What is specific to T0 is that A_F is generated by compressed functions, that the tensor square is itself the truncation of a commutative parent in the restricted sense above, and families such as the following.

Example 16.6 (a tower-decoupling family, from the referee). Let D_0 be the Dolbeault-Dirac operator on CP^3 twisted by O(1), with V = ker D_0 = C^4, and let DD_l = DD/l on the doubled parent. Its kernel is V (x) conj(V), and its gap grows like 1/l. Here P is a spectral projection. Compression followed by algebra generation gives the constant finite data A_F = M_4(C), dim H_F = 16, D_F = 0, and the Clifford point gives dim H = 32 with the KO-6 signs. Order one is automatic, since D_F = 0. The whole triple converges to (C, C^16, 0) by Theorem 15.3, consistently with lemma 15: the two statements concern the whole algebra and the truncation.

Two sheets: the form of the classification

Chamseddine and Connes (*Why the Standard Model*, arXiv:0706.3688, sections 2-3) classify irreducible pairs (A, J) with order zero and a separating vector: the centre of the complexified algebra is C, with A_C = M_k(C) on itself and J x = x*, or C + C, with A_C = M_k(C) + M_k(C) on itself and J(x, y) = (y*, x*). With centre C, J gamma = -gamma J is impossible. The pair of Corollary 16.4 is therefore reducible: the projection 1 (x) (1 + sigma_1)/2 commutes with A_F and J. The irreducible form comes from a parent with two sheets. At the finite level the bimodule (W (x) conj(V)) (+) (V (x) conj(W)) with its adjoint-flip J is already stated in Remark 2.9 of that paper; what we add is its realization from parents and compressed functions.

Theorem 16.7 (two-sheet coarse parent). Let (A_X, H_X, D_X, gamma_X) and (A_Y, H_Y, D_Y, gamma_Y) be even spectral triples. On H = (H_X (x) conj(H_Y)) (+) (H_Y (x) conj(H_X)), for even a in A_X and b in A_Y, put

```
pi(a, b) = (a (x) 1) (+) (b (x) 1),
DD = (D_X (x) 1 + gamma_X (x) D_Y^c) (+) (D_Y (x) 1 + gamma_Y (x) D_X^c),
Gamma = (gamma_X (x) gamma_Y^c) (+) (-gamma_Y (x) gamma_X^c),
J (xi (x) eta^c) = (-1)^{|xi| |eta|} eta (x) xi^c,
```

where J maps each summand to the other. Then J^2 = 1, J DD = DD J and J Gamma = -Gamma J, the KO-6 signs. Moreover

```
(a, b)° = (1 (x) (b\*)^c) (+) (1 (x) (a\*)^c),
```

so the first summand is an (A_X, A_Y)-bimodule and the second an (A_Y, A_X)-bimodule, and order zero and order one hold.

Proof. The computation of Theorem 16.2 applies with D_Y in place of the second factor: on xi (x) eta^c in the first summand, J DD and DD J both give (-1)^{|xi| |eta|} [D_Y eta (x) xi^c + (-1)^{|eta|} eta (x) (D_X xi)^c]. The sign in Gamma on the second summand gives J Gamma = -Gamma J. For a°, apply J, then pi(a, b)*, then J, as in Theorem 16.2: an element of the first summand goes to the second, where b* acts on the left, and returns with b* on the right factor. Since a and b are even, [DD, pi(a, b)] = ([D_X, a] (x) 1) (+) ([D_Y, b] (x) 1), which commutes with the opposite actions. ■

Corollary 16.8 (truncation; the form of the classification). For symmetric windows with V_X, V_Y of dimensions k_X, k_Y, and even subalgebras A_{F,X} in B(V_X), A_{F,Y} in B(V_Y), the compression

```
T_F = (A_{F,X} + A_{F,Y}, (V_X (x) conj(V_Y)) (+) (V_Y (x) conj(V_X)), D_F, J_F, Gamma_F)
```

has the KO-6 signs, order zero and order one, and dim H_F = 2 k_X k_Y. If gamma_{V_X} lies in A_{F,X} and gamma_{V_Y} in A_{F,Y}, the real orientability relation holds: with c = (gamma_{V_X},

gamma_{V_Y}) and z = (1, -1), Gamma_F = pi(z c) c°, a zero-cycle with coefficients in the algebra and its opposite. This is weaker than Gamma_F in pi(A_F). If V_X and V_Y are ungraded with k_X = k_Y = k and the generated algebras are full, then (A_F, H_F, J_F) is exactly the irreducible pair with centre C + C: identify both summands with M_k(C), so that J_F(x, y) = (y*, x*); an invertible matrix in each summand is a separating vector, and a projection commuting with both full matrix actions is scalar on each summand, with J equating the scalars. In all cases D_F commutes with the two sheet projections, so the sheets are not connected. It need not commute with the whole centre: for V_X = C^2, gamma_X = sigma_3, d_X = sigma_1, A_{F,X} = C + C acting diagonally and V_Y = C, the central element (diag(1, 0), 0) does not commute with D_F.

Proof. Theorem 16.7 applies on the truncation with arbitrary even operators. For orientability, pi(c) c° = (gamma_{V_X} (x) gamma_{V_Y}^c) (+) (gamma_{V_Y} (x) gamma_{V_X}^c), and z flips the sign of the second summand. The last statements are direct. ■

Example 16.9 (twistor level one, and M_2(H) + M_4(C)). Let D_0 be the Dolbeault-Dirac operator on CP^3 twisted by O(1), with kernel $H^{0(CP}$3, O(1)) = C^4, and let j be the antiunitary with j^2 = -1 commuting with the Gamma_a of Proposition 10.1, which makes C^4 a quaternionic space H^2. Use this space twice, as W (with the functions of S^4 pulled back along the twistor projection) and as V (with the functions of CP^3).

1. A real function f on S^4 pulls back to a function invariant under w -> j w. So j Q(f) j^-1 = Q(f), and Q(f) is a Hermitian element of End_H(W). Since Q(x_a) = Gamma_a/5, the real algebra generated is End_H(W) = M_2(H), of real dimension 16.
2. For real functions on CP^3, Q(w* B w / w* w) = (B + tr(B) 1)/5 for Hermitian B, so Q is onto the Hermitian matrices, and the real algebra generated is End_C(V) = M_4(C), of real dimension 32.

Hence Corollary 16.8, with the parent gradings, which are trivial on the kernels, gives the real algebra A = M_2(H) + M_4(C) on H_F = Hom(V, W) (+) Hom(W, V) = E (-) E* of dimension 32, with J(x, y) = (y*, x*). These are the algebra, bimodule and real structure of Chamseddine and Connes. The generation is of real algebras from real functions: complex generation on the S^4 side would give M_4(C).

The nontrivial grading of W in that construction is not obtained from this parent. An even parent has [P gamma P, P a P] = 0 for every compressed function, while Gamma_5 = 5 Q(x_5) anticommutes with Q(x_1). So the grading has to be added after truncation, as abstract finite data, fixing an axis. The full algebra is then a graded prealgebra rather than the algebra of an even triple, with even part H + H + M_4(C), and Gamma_F(xi, eta) = (Gamma_5 xi, -eta Gamma_5), which is Proposition 3.2 of that paper; the real orientability relation holds on the even part. Similarly d = mu n^a Gamma_a with n orthogonal to the axis is the compression of a bounded real multiplication operator, but adding that operator to the parent Dirac operator keeps neither the kernel as a spectral projection nor the parent grading as an odd perturbation, so d too is abstract data. It commutes with the two sheet projections and with the centre R + C of the full prealgebra, and not with the centre of the even algebra.

This bears on the step left open in canonical section 4, where M_2(H) in M_4(C) is a nested pair on one four-space, with "the independent E, F module roles" to be derived. The example closes that abstract finite-module problem after adding a second sheet: the nested actions on one four-space become independent summands acting on two Hom spaces. It does not turn four global sections into rank-four fibres over S^4, derive the two sheets from the fibration, or produce the Dirac operator connecting the two pieces that the selection of A_F = C + H + M_3(C) requires. The fibre-rank problem remains open.

The inputs are the second sheet, the restriction of one algebra to base functions, real algebra generation, the four-dimensional window, the grading axis and d. The content beyond the formal bimodule is the concrete compression calculation in items 1 and 2, already in the corpus, and its explicit match with the classified data.

Caveats

These results should be read as narrowly as the note they correct.

1. **Metric.** T_F is usually not a metric spectral triple. For A_F = B(V_+) + B(V_-) with V_+ of dimension at least 2 and d odd, the commutant of d in A_F contains the pairs (x, B^-1 x B) with x commuting with B B*, when B is the invertible block of d. In example (b), (diag(1, 0), diag(1, 0)) has L = 0, so d leaves a residual U(1) x U(1) of U(2) x U(2) unbroken. The Standard Model finite triple has the same feature, through M_3(C).
2. **Spectrum.** The eigenvalues of D_F are +/- (lambda_i^2 + lambda_j^2)^{1/2}. In example (a) they are +/- sqrt(2) m with Connes distance 1/m, so the doubled spectrum differs from the masses in d, and the mass relations of Theorem 9.6 and lemma 11 would have to be redone.
3. **Parent.** The doubled parent is a choice. Its algebra sees one factor of X x X, its J covers the flip, which is outside the spin real structure of X x X, and generating A_F from O_F is an additional operation. We have no dynamical reason for it. For an ordinary compact spin X of dimension n, its Weyl exponent is 2 n with metric dimension n, which bears on the counting of 9, 10 and 11 and has not been analyzed.
4. **KO-6.** The Clifford point is an input, as F_0 is in Theorem D, and it fails finite orientability. In Theorem 16.7 the sign comes from the block structure of two sheets, which is itself an input. The documents checked here establish neither a construction of two sheets from one connected parent nor a no-go theorem. The grading-reversing structure of lemma 08 is a candidate we have not tested.
5. **Count.** For k = 4, dim H = 32, the value in the classification of Chamseddine-Connes, and k = 4 occurs in examples (b) and (c). For k = 4 in examples (b) and (c) alone this is a count and nothing more: the classification's algebra is M_2(H) + M_4(C), acting differently, and those examples do not produce it. Example 16.9 does, as abstract data with the listed inputs.

Consequence for T0 and T0b

T0 asks for finite real triples from commutative Kaluza-Klein data under specified notions of convergence. For one specified truncation procedure (compression, then generation, on the doubled parent with restricted algebra and flip J), this lemma gives a noncommutative finite algebra with an explicit bimodule, and a family with the tower decoupled (Example 16.6). It does not establish the target under every meaning of ordinary Kaluza-Klein data. For T0b, it gives examples with the requested signs and first-order relations; with two sheets it reaches the form of the classification, and in Example 16.9 the real algebra M_2(H) + M_4(C), bimodule and real structure of Chamseddine and Connes, with the grading added as abstract data. It leaves open an operator connecting the two pieces and hence the selection of A_F, a parent that yields the grading and nonzero mass data, and the beyond-first-order and twisted frameworks. See the background note.

Numerical check

Python (numpy), run on 3 Nov 2026 in a temporary directory, not kept. On random odd d and even a, b for dim V_+ = dim V_- = 2 with J = M K: J^2 = 1, J DD = DD J, J Gamma = Gamma J, DD Gamma = -Gamma DD, order zero and order one all hold to machine precision, with norm([a, b]) = 7.67; the KO-6 signs of Corollary 16.4 hold. For Theorem 16.7 and Example 16.9, with k = 4, W graded by Gamma_5 and V ungraded: J^2 = 1, J Gamma = -Gamma J, J D = D J, D Gamma = -Gamma D, D = Delta + J Delta J^-1, order zero and order one for random a, b, and Gamma_F = pi(z c) c° all hold to machine precision; j^2 = -1 commutes with the Gamma_a, and x_a(j w) = x_a(w); the real algebras generated by Gamma_a, by i Gamma_a and by all Hermitian matrices have real dimensions 16, 32 and 32; and sum_a x_a Gamma_a = 2 (P_w + P_{jw}) - 1. For example (b), by quadrature with 23 test functions: dim O_F = 4 and the generated algebra has dimension 8; the two displayed values Q(f_1) and Q(f_2) were reproduced by quadrature.

Part 18

T5b: vector masses in the spectral action at fixed cutoff

Prompt: prompts/18 (STRATEGY T5b). Attempted by gpt-6-astra (codex 0.164, xhigh, 3 Nov 2026), working alone. Checked by Claude Opus 5.5 with Python (numpy, scipy), by computations independent of

the kernel derived here: a direct heat trace with constant a reproduces equation (2) for five choices of (m_1, m_2), to about 0.03%; a direct heat trace on Fourier modes with a_2(x_1) = a cos(q x_1), including the F_{12} and [T, V] vertices, reproduces the kernel (37) at three values of (m, q), to 0.1-0.3%, the size of the discretization error. The closed forms (16)-(19), the bound (28), the limits (27), the neutral matrix (30) recomputed from Standard Model charges, and the identities behind (31) also check. The largest value of the Taylor mass is M_W^2 = 0.275 Lambda^2, at m_t^2 = 2.02 Lambda^2. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): all ten numbered results are correct, with small amendments incorporated below. The referee added Propositions 11 to 14 and Remark 15, which widen the scope in both directions; the ratio in Proposition 14 was added by Claude.

Scope. Within the three-generation, zero-Majorana, top-only family at fixed cutoff, these results refute an infinitely heavy W, Z, t limit for the stated quadratic mass definitions. They do not rule out other finite data (Proposition 14), a different action or constraint, a joint limit with Lambda -> infinity, or a quantum effective theory. They do not show unbroken SU(2) x U(1) on the light space (Remark 15).

For the heat cutoff, on $\mathbb{R}^4$, we obtain the exact quadratic vector form without expanding in m/Λ. In the three-generation Standard Model finite representation with only the top Dirac mass nonzero, both the mass defined by the q^0/q^2 Taylor ratio and the unique positive timelike pole mass satisfy $M_W \sim \Lambda^2/m_t$ as $m_t/\Lambda \to \infty$ at fixed Λ. The broken vectors consequently do not decouple by becoming heavy, although the heavy fermion contribution disappears from the gauge coefficients. These statements concern the classical bosonic quadratic action about a prescribed constant background; moreover, the positive Laplace class considered below does not itself select a finite nonzero stationary background when radial variations of that background are allowed.

1. Theorem (exact quadratic kernel).

Let

$$D_A = -i\gamma^\mu(\partial_\mu + iA_\mu) + \gamma_5 V, \qquad V = V_0 = V_0^\dagger,$$

on $\mathbb{R}^4$, with Hermitian Euclidean gamma matrices, $\{\gamma^\mu, \gamma^\nu\} = 2\delta^{\mu\nu}$, $\gamma_5^2 = 1$, and four spin components. Let the finite trace be the literal trace on H_F, including every stated multiplicity, without a subsequent projection; this agrees with the bosonic trace of Chamseddine-Connes-Marcolli, which counts particles, antiparticles and four spin components. Take A smooth and compactly supported.

We use the vacuum-subtracted action, or equivalently its quadratic variation, since the constant-background trace on $\mathbb{R}^4$ has infinite volume. Write

$$A_\mu(q) = a_\mu(q)T, \qquad T = T^\dagger, \qquad q^\mu a_\mu(q) = 0, \qquad \int_q = \int \frac{d^4q}{(2\pi)^4}.$$

Diagonalize V_0:

$$V_0 e_i = m_i e_i, \qquad m_i \in \mathbb{R}.$$

The signed eigenvalues are needed; their squares alone do not determine the answer.

For $S_t = \operatorname{Tr} e^{-tD_A^2}$, define

$$b(u) = u(1-u), \quad d_{ij} = m_i - m_j,$$

$$I_t(a, b; q^2) = \int_0^1 e^{-t[(1-u)a + ub + u(1-u)q^2]}\, du.$$

Then

$$S_t^{(2)} = \int_q |a_T(q)|^2\, C_{t,T}(q^2),$$

where

$$\boxed{C_{t,T}(q^2) = \frac{1}{8\pi^2} \sum_{ij} |T_{ij}|^2 \left[\left(\frac{2}{t} + d_{ij}^2 + q^2 \right) I_t(m_i^2, m_j^2; q^2) - \frac{e^{-tm_i^2} + e^{-tm_j^2}}{t} \right].} \tag{1}$$

For several generators, replace $|T_{ij}|^2$ by $\operatorname{Re}((T_a)_{ij}(T_b)_{ji})$.

In particular, writing $C(q^2) = C_0 + q^2 C_1 + O(q^4)$, we obtain

$$\boxed{C_{0,t} = \frac{1}{8\pi^2} \sum_{ij} |T_{ij}|^2 \left[\left(\frac{2}{t} + d_{ij}^2 \right) I_t(m_i^2, m_j^2; 0) - \frac{e^{-tm_i^2} + e^{-tm_j^2}}{t} \right]}, \tag{2}$$

and

$$\boxed{C_{1,t} = \frac{1}{8\pi^2} \sum_{ij} |T_{ij}|^2 \int_0^1 e^{-t[(1-u)m_i^2 + um_j^2]} \left[1 - 2b(u) - t d_{ij}^2 b(u) \right] du.} \tag{3}$$

These formulas also apply, after integration over $t = s/\Lambda^2$, to

$$f(z) = \int_{(0,\infty)} e^{-sz}\, d\nu(s), \quad 0 < f(0) = \nu((0,\infty)) < \infty, \quad \int (s^{-2} + s^2)\, d\nu(s) < \infty, \tag{4}$$

where ν is positive. These hypotheses suffice for the Euclidean action density and the Taylor coefficients used below. They do not, by themselves, guarantee analytic continuation to every negative q^2.

Proof. With $\nabla_\mu = \partial_\mu + iA_\mu$,

$$D_A^2 = -\nabla^2 + V_0^2 - \frac{i}{2}\gamma^{\mu\nu} F_{\mu\nu} + \gamma^\mu \gamma_5 [A_\mu, V_0].$$

Let $P_0 = -\partial^2 + V_0^2$ and write $D_A^2 = P_0 + P_1 + P_2$, according to degree in A. Duhamel expansion and cyclicity give

$$S_t^{(2)} = -t \operatorname{Tr}(e^{-tP_0} P_2) + \frac{t^2}{2} \int_0^1 \operatorname{Tr}\left(e^{-(1-u)tP_0} P_1 e^{-utP_0} P_1 \right) du. \tag{5}$$

The linear vertex from j, p to $i, p+q$ is

$$T_{ij} \left[(2p+q) \cdot a + \gamma^{\mu\nu} q_\mu a_\nu + (m_j - m_i) \gamma^\mu \gamma_5 a_\mu \right].$$

Completing the momentum square produces

$$(4\pi t)^{-2} e^{-t[(1-u)m_i^2 + um_j^2 + b(u)q^2]}.$$

For transverse a, the averaged spin trace of the vertex product is

$$4 \left(\frac{2}{t} + d_{ij}^2 + q^2 \right) |a|^2.$$

The P_2 term contributes

$$-\frac{1}{4\pi^2 t} \sum_{ij} |T_{ij}|^2 e^{-tm_i^2} |a|^2.$$

Symmetrizing in i, j proves (1). Differentiation proves (2)–(3).

The divided-difference structure is explicit:

$$I_t(a, b; 0) = \frac{e^{-ta} - e^{-tb}}{t(b-a)}, \qquad I_t(a, a; 0) = e^{-ta}, \tag{6}$$

and

$$\int_0^1 u(1-u) e^{-t[(1-u)a + ub]}\, du = \frac{1}{t^2} \partial_a \partial_b I_t(a, b; 0). \tag{7}$$

Thus (2) uses a first divided difference, while (3) also uses its mixed derivative. The full momentum dependence follows from the convergent series

$$I_t(a, b; q^2) = \sum_{n=0}^{\infty} \frac{(-q^2/t)^n}{n!} \partial_a^n \partial_b^n I_t(a, b; 0). \tag{8}$$

Repeated-node values are obtained continuously.

For completeness, the mass coefficient for (4) can be written entirely in dimensionless variables. Put

$$r_i = m_i/\Lambda, \quad x_i = r_i^2, \quad z(u) = (1-u)x_i + ux_j, \quad \Phi_1(z) = \int s^{-1}e^{-sz}\,d\nu(s).$$

The bracket replacing that in (2) is

$$\Lambda^2\left[2\int_0^1 \Phi_1(z(u))\,du + (r_i - r_j)^2\int_0^1 f(z(u))\,du - \Phi_1(x_i) - \Phi_1(x_j)\right]. \tag{9}$$

The corresponding bracket in (3) is

$$\int_0^1 \left[(1-2b)f(z) + (r_i - r_j)^2 b f'(z)\right]du. \tag{10}$$

Each unweighted integral is a divided difference of an antiderivative. The moment assumptions justify the integrations and differentiations. □

On a finite flat torus, (5) remains exact, but the Gaussian momentum integral must be replaced by the spin-structure-dependent lattice sum. Explicitly, for a transverse polarization a, its two-vertex contribution per unit volume is

$$\frac{2t^2}{\mathrm{Vol}}\sum_{ij,p}|T_{ij}|^2\int_0^1 e^{-t[(1-u)(p^2+m_i^2)+u((p+q)^2+m_j^2)]}\left(|(2p+q)\cdot a|^2 + (d_{ij}^2+q^2)|a|^2\right)du,$$

and the quadratic-vertex contribution is

$$-\frac{4t}{\mathrm{Vol}}\sum_{ij,p}|T_{ij}|^2 e^{-t(p^2+m_i^2)}|a|^2.$$

Consequently, the continuum formulas below are statements on $\mathbb{R}^4$, or in the infinite-volume limit. A finite torus can have additional holonomy dependence, including for constant commuting connections.

2. Lemma (gauge invariance and the Goldstone direction).

Let scalar fluctuations include the gauge orbit of V_0. Then the nonzero vector coefficient C_0 in (2) is compatible with gauge invariance.

For one generator, parameterize the orbit by

$$V(x) = e^{i\pi(x)T}V_0e^{-i\pi(x)T}.$$

With our conventions,

$$a_\mu \longmapsto a_\mu - \partial_\mu\alpha, \qquad \pi \longmapsto \pi + \alpha.$$

The invariant combination is

$$B_\mu = a_\mu + \partial_\mu\pi. \tag{11}$$

Proof. Gauge covariance gives

$$D_{A^U,V^U} = UD_{A,V}U^{-1}.$$

Applying $U = e^{-i\pi T}$ transforms V to V_0 and a to B. Hence

$$S[a, e^{i\pi T}V_0e^{-i\pi T}] = S[B, V_0]. \tag{12}$$

This establishes the Goldstone completion directly, without a derivative expansion.

For comparison, the longitudinal vector kernel is obtained from (1) by replacing the q^2 in its prefactor by $(1-2u)^2q^2$ inside the integral. If $[T, V_0] = 0$, its integral vanishes by

$$\int_0^1 [2 + tq^2(1-2u)^2]e^{-tq^2u(1-u)}\,du = 2.$$

For a broken generator the longitudinal kernel need not vanish; it belongs to the invariant combination (11).

If V_0 is not stationary, a Cartesian scalar Hessian alone need not annihilate the linear gauge-orbit tangent. The quadratic term in $e^{i\pi T}V_0e^{-i\pi T}$, contracted with the scalar tadpole, must also be retained. Equation (12) includes that term. Transverse vectors do not mix quadratically with scalar fluctuations on this translation-invariant background. □

3. Corollary (the asymptotic normalization).

Let all finite eigenvalues be small relative to Λ, let f satisfy (4), and let Q below be positive definite on the chosen generator space (otherwise pass to the quotient by its null space). Define

$$Q_{ab} = \mathrm{Re}\,\mathrm{Tr}(T_aT_b), \qquad K_{ab} = \mathrm{Re}\,\mathrm{Tr}([T_a, V_0]^\dagger[T_b, V_0]).$$

At leading order,

$$C_0 = \frac{f(0)}{8\pi^2}K, \qquad C_1 = \frac{f(0)}{12\pi^2}Q,$$

and therefore

$$\boxed{\mathcal{M}^2_{\mathrm{Taylor}} = \frac{3}{2}Q^{-1/2}KQ^{-1/2}.} \tag{13}$$

This reproduces Proposition 13.5.

For the full three-generation Standard Model representation,

$$Q_{11} = 12, \qquad K_{11} = \sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2),$$

so

$$\boxed{8M_W^2 = \sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2).} \tag{14}$$

Proof. In (2), the orbital expression

$$\frac{2}{t}I_t(a, b; 0) - \frac{e^{-ta} + e^{-tb}}{t}$$

starts at order $t(a-b)^2$. The leading term is consequently d_{ij}^2. In (3), the leading integral is $\int_0^1(1-2u(1-u))du = 2/3$, proving (13).

For $T_1 = \sigma_1/2$ on each left doublet, one generation of particles contributes 2 to Q_{11}. Charge conjugates and three generations give 12. A doublet with Dirac masses m_u, m_d contributes $\frac{1}{2}(m_u^2 + m_d^2)$ to the particle commutator trace, before color multiplicity. This proves (14).

Equivalently, the gauge coefficient is

$$\frac{f(0)g_2^2}{2\pi^2} = \frac{1}{4}.$$

For comparison with CCM's $\mathrm{Tr}\,F(D/\Lambda)$, take $F(v) = f(v^2)$; then $F(0) = f(0)$. The normalization and mass constraint are precisely equations (4.10) and (4.16), with the latter stated in Lemma 4.6 of the checked preprint. CCM describes the relation as "valid at unification scale." Chamseddine–Connes–Marcolli, pp. 37–39. □

The leading bound $M^2 \le 6\|V_0\|^2$ in Lemma 14 follows from (13) and $K \le 4\|V_0\|^2Q$. Formula (13), rather than an exact finite-cutoff identity, is its applicable approximation.

4. Proposition (the exact charged-vector Taylor mass).

Let the finite data be the full three-generation Standard Model representation, including charge conjugates. Let the Majorana matrix and every Dirac mass except one up-type color triplet be zero. Write

$$m = \mu\Lambda, \qquad x = \mu^2, \qquad r = e^{-x}.$$

Let $f(z) = e^{-z}$. Define

$$H(y) = \int_0^1 e^{-yu(1-u)}\, du,$$
$$g(y) = (2+y)H(y) - 2,$$
$$h_x(y) = (2+x+y)\int_0^1 e^{-xu-yu(1-u)}\, du - 1 - e^{-x}. \tag{15}$$

Then the full charged-vector kernel is

$$\boxed{C_W(q^2) = \frac{\Lambda^2}{4\pi^2}\left[\frac{9}{2}g(y) + \frac{3}{2}h_x(y)\right], \qquad y = q^2/\Lambda^2.} \tag{16}$$

Define

$$A(x) = h_x(0) = \frac{2[1-(1+x)e^{-x}]}{x},$$
$$w(x) = 3\int_0^1 u^2 e^{-xu}\, du = \frac{3[2-(x^2+2x+2)e^{-x}]}{x^3}. \tag{17}$$

Continuous values are understood at $x = 0$. The particle-trace kinetic coefficient is

$$K_W(x) = \frac{9}{2} + \frac{3}{2}w(x) > 0.$$

The Taylor mass, defined by $M^2_{W,\mathrm{Taylor}} = C_W(0)/C'_W(0)$, is

$$\boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{9A(x)}{4K_W(x)} = \frac{3[1-(1+x)e^{-x}]}{x[3+w(x)]}.} \tag{18}$$

Its limiting behavior is

$$\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{3}{8}x - \frac{23}{128}x^2 + O(x^3), \qquad \frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} \sim \frac{1}{x} \quad (x \to \infty). \tag{19}$$

Consequently,

$$\frac{m}{M_{W,\mathrm{Taylor}}} \longrightarrow \sqrt{\frac{8}{3}} \quad (\mu \to 0), \qquad \frac{m}{M_{W,\mathrm{Taylor}}} \sim \mu^2 \quad (\mu \to \infty). \tag{20}$$

Proof. On the particle space, $Q_{11} = 6$. The selected quark doublet contributes $3/2$, while the remaining doublets contribute $9/2$. Diagonalizing the massive left-right block produces eigenvalues $+m, -m$. The charged generator connects these to the massless down states; every such pair has $d_{ij}^2 = m^2$. Formula (1) therefore gives h_x for the selected doublet and g for the remaining doublets. Charge conjugation supplies the factor two in (16).

Integration by parts gives

$$h_x(y) = 2\int_0^1 u\,[x + 2y(1-u)]e^{-xu-yu(1-u)}\, du, \tag{21}$$

and

$$\boxed{h'_x(y) = 2\int_0^1 u^2[1 - y(1-u)]e^{-xu-yu(1-u)}\,du.} \tag{22}$$

The boundary term proving (22) is $\partial_u[u(1-u)e^{-xu-yu(1-u)}]$. Thus

$$h'_x(0) = \frac{2}{3}w(x), \qquad g'(0) = \frac{2}{3}.$$

This proves (18). Expanding the elementary expressions at zero and taking their large-x limits proves (19)–(20). □

The same Taylor result extends to every f in (4). Put

$$\ell = f(0), \qquad w_f(x) = \int w(sx)\,d\nu(s), \qquad A_f(x) = \int \frac{A(sx)}{s}\,d\nu(s).$$

Then

$$\boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{9A_f(x)}{4\left(\frac{9}{2}\ell + \frac{3}{2}w_f(x)\right)}.} \tag{23}$$

All denominators are positive, and dominated convergence gives

$$M^2_{W,\mathrm{Taylor}} \sim \frac{3}{8}m^2 \quad (x \to 0), \qquad \boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} \sim \frac{\int s^{-2}\,d\nu(s)}{\ell x} \quad (x \to \infty).} \tag{24}$$

Indeed, $A(z)/z \to 1$, $w(z) \to 1$ at zero, while $zA(z) \to 2$, $w(z) \to 0$ at infinity.

If the other Dirac masses are merely small and nonzero, the answer depends on those additional parameters and is given by (1). In particular, the vanishing limit in (19) assumes they are zero: masses of surviving light doublets can leave a residual vector mass.

5. Theorem (the charged-vector pole).

Under the heat-cutoff hypotheses of Proposition 4, define a timelike pole by analytic continuation of the quadratic inverse propagator:

$$C_W(-M^2_{W,\mathrm{pole}}) = 0. \tag{25}$$

For every $x > 0$, this equation has exactly one solution with $M^2_{W,\mathrm{pole}} > 0$. It is simple and has positive residue in the convention that the Euclidean propagator near the pole is a positive multiple of $(q^2 + M^2_{W,\mathrm{pole}})^{-1}$.

Let $z(x) = M^2_{W,\mathrm{pole}}/\Lambda^2$. Its exact specification is the unique positive root

$$\boxed{\frac{9}{2}g(-z) + \frac{3}{2}h_x(-z) = 0.} \tag{26}$$

Furthermore,

$$z(x) \sim \frac{3}{8}x \quad (x \to 0), \qquad z(x) \sim \frac{1}{x} \quad (x \to \infty), \tag{27}$$

and both mass definitions obey

$$\boxed{0 < M^2_{W,\mathrm{Taylor}},\ M^2_{W,\mathrm{pole}} \le \frac{\Lambda^2}{e}.} \tag{28}$$

Proof. Let

$$F_x(y) = \frac{9}{2}g(y) + \frac{3}{2}h_x(y).$$

Equation (22) implies $h_x'(y) > 0$ for $y \le 0$, and also $g'(y) > 0$, since $g = h_0$. Hence F_x is strictly increasing on that half-axis. We have

$$F_x(0) = \frac{3}{2}A(x) > 0.$$

As $y \to -\infty$, the prefactors in (15) become negative and both integrals grow exponentially, so $F_x(y) \to -\infty$. This proves existence, uniqueness, simplicity, and the residue sign.

For $y \le 0$, equation (22) gives $g'(y) \ge 2/3$, and therefore

$$F_x'(y) \ge 3.$$

The mean-value theorem yields

$$z(x) \le \frac{1}{2}A(x).$$

The Taylor denominator gives the same bound for $M^2_{W,\mathrm{Taylor}}/\Lambda^2$. Finally,

$$A(x) = 2\int_0^1 xu\, e^{-xu}\, du \le \frac{2}{e},$$

proving (28).

As $x \to 0$, the bound implies $z(x) \to 0$, and $F_x'(y) \to 4$ near the root. As $x \to \infty$, $z(x) \to 0$, $h_x'(y) \to 0$ near the root, and $F_x'(y) \to 3$. Together with $F_x(0) \sim 3x/2$ and $F_x(0) \sim 3/x$, respectively, these prove (27). □

The pole mass governs the linearized dispersion relation of this analytically continued quadratic action. The Taylor mass governs its two-derivative truncation; they need not coincide near the cutoff. We assert no closed-form inversion of (26), no classification of complex poles, and no Lorentzian unitarity theorem.

The pole proof also extends to positive measures supported in a compact subset of $(0, \infty)$, using

$$F_{f,x}(y) = \int \frac{1}{s}\left[\frac{9}{2}g(sy) + \frac{3}{2}h_{sx}(sy)\right] d\nu(s).$$

It gives the analogues of (24) for the pole mass. For the larger class (4), only the Euclidean and Taylor conclusions have been established here; its moment assumptions alone do not justify that continuation.

6. Proposition (unbroken coefficients, neutral vectors, and weak angles).

Retain the heat-cutoff Standard Model family of Proposition 4. Normalize color generators by $\mathrm{tr}_{\mathbf{3}}\, t_a t_b = \frac{1}{2}\delta_{ab}$, and use $Q = T_3 + Y$. Define the full-trace coefficients by

$$S^{(2)}_{\mathrm{unbroken}} = \int \left(\kappa_{\mathrm{em}} F^2_{\mathrm{em}} + \kappa_s \sum_a G_a^2\right).$$

Then

$$\boxed{\kappa_{\mathrm{em}} = \frac{1}{24\pi^2}\frac{16}{3}(5+r), \qquad \kappa_s = \frac{1}{24\pi^2}2(5+r), \qquad \frac{\kappa_{\mathrm{em}}}{\kappa_s} = \frac{8}{3}.} \tag{29}$$

Both coefficients approach 5/6 of their all-massless values.

Let K denote the particle-trace neutral q^2 coefficient in coordinates (A_3, A_Y). Then

$$\boxed{\begin{aligned} K_{33} &= \frac{21}{4} + \left(\frac{3}{4} - \frac{3x}{8}\right) r, \\ K_{YY} &= \frac{103}{12} + \left(\frac{17}{12} - \frac{3x}{8}\right) r, \\ K_{3Y} &= -\frac{1}{4} + \left(\frac{1}{4} + \frac{3x}{8}\right) r. \end{aligned}} \tag{30}$$

The matrix K, the charged coefficient K_W, and the unbroken coefficients are positive for every $x \geq 0$.

Define the neutral-current angle from the current of the massive direction after diagonalizing the two-derivative neutral kinetic form:

$$J_Z \;\propto\; T_3 - \sin^2\theta_{\mathrm{NC}}\, Q.$$

Then

$$\boxed{\cos^2\theta_{\mathrm{NC}} = \frac{K_{YY} + K_{3Y}}{K_{33} + 2K_{3Y} + K_{YY}} = \frac{5}{8}.} \tag{31}$$

A different continuation, namely

$$\cos^2\theta_{\mathrm{match}} := \frac{K_{YY}}{K_W + K_{YY}},$$

satisfies

$$\boxed{\cos^2\theta_{\mathrm{match}}(0) = \frac{5}{8}, \qquad \cos^2\theta_{\mathrm{match}}(\infty) = \frac{103}{157}.} \tag{32}$$

Thus these definitions cease to describe one common electroweak angle.

Proof. Photon and color generators commute with V_0. Their coefficients therefore follow from (3) with $d_{ij} = 0$ on each invariant mass block. The charge traces give (29).

A Dirac species of mass m, with neutral charges q_L^a, q_R^a, contributes to K

$$e^{-x}\left[q_L^a q_L^b + q_R^a q_R^b - \frac{x}{2}(q_L^a - q_R^a)(q_L^b - q_R^b)\right]. \tag{33}$$

Applying (33) to the three top colors, and retaining all other states, proves (30).

Write

$$v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad B_\infty = \begin{pmatrix} 21/4 & -1/4 \\ -1/4 & 103/12 \end{pmatrix}, \quad U = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 17/12 \end{pmatrix}.$$

Then

$$K = B_\infty + rU - \frac{3xr}{8} vv^T.$$

We have $B_\infty \geq 5I$, $U \geq 0$, and $xr \leq 1/e$. Hence

$$K \geq \left(5 - \frac{3}{4e}\right) I > 0.$$

Positivity of K_W follows from (17).

The neutral mass direction is v, while the photon direction is $(1, 1)$. The kinetic-orthogonal massive field is proportional to $K^{-1}v$, giving the first expression in (31). Directly,

$$K_{33} + K_{3Y} = 5 + r, \qquad K_{YY} + K_{3Y} = \frac{5}{3}(5 + r),$$

which proves (31). Taking limits proves (32). □

The neutral mass also tends to zero. More precisely,

$$\boxed{\frac{M_{Z,\mathrm{Taylor}}^2}{\Lambda^2} = \frac{9}{4} xr\, v^T K^{-1} v \sim \frac{2}{3} xe^{-x} \quad (x \to \infty).} \tag{34}$$

For $x > 0$ there is likewise one positive massive neutral pole, besides the photon pole. To prove this, put $B = B_\infty + rU$. The exact neutral kernel, apart from the common factor $\Lambda^2/(4\pi^2)$, is

$$B\, g(y) + \frac{3}{2} xr\, H(y) vv^T. \tag{35}$$

After congruence by $B^{-1/2}$, its massive eigenvalue is

$$g(y) + \beta H(y), \qquad \beta = \frac{3}{2} xr\, v^T B^{-1} v > 0.$$

Its timelike root obeys

$$z - 2 + \frac{2}{H(-z)} = \beta. \tag{36}$$

The left side increases from zero to infinity, because

$$\frac{d}{dz}\left(z - 2 + \frac{2}{H(-z)}\right) = 1 - \frac{2\,\mathbb{E}_z[b]}{H(-z)} \geq \frac{1}{2},$$

where $\mathbb{E}_z$ uses density proportional to $e^{zb(u)}$. The pole is therefore unique and has positive residue. Since $v^T B_\infty^{-1} v = 8/27$,

$$\frac{M_{Z,\mathrm{pole}}^2}{\Lambda^2} \sim \frac{2}{3} x e^{-x}.$$

In particular, M_W^2/M_Z^2 does not continue to equal $5/8$.

For every f in (4), equations (29)–(33) extend by replacing 1 by $\ell = f(0)$, r by $f(x)$, and xr by $-xf'(x)$. Positivity follows by averaging the positive heat-cutoff matrices. The ratio $8/3$, the neutral-current value $5/8$, and the matching limit $103/157$ persist. At the endpoint both neutral masses vanish, so the distinguished massive neutral direction—and hence its angle—requires continuation from finite x.

7. Counterexample (positive cutoff does not ensure positive broken kinetic coefficients).

Let $H_F = \mathbb{C}^2$,

$$V_0 = m\sigma_3, \qquad T = \sigma_1, \qquad x = m^2/\Lambda^2,$$

with the same spacetime and trace conventions. For the heat cutoff,

$$C_T(q^2) = \frac{\mathrm{Tr}(T^2)\Lambda^2}{8\pi^2} e^{-x} \left[(2 + 4x + y)H(y) - 2\right]. \tag{37}$$

Its q^2 Taylor coefficient is proportional to

$$e^{-x}(1 - x).$$

It vanishes at $x = 1$ and becomes negative for $x > 1$. Thus the two-derivative truncation has a wrong-sign vector kinetic term in this example.

Nevertheless, the full quadratic form has a unique positive real timelike pole with positive residue, satisfying

$$\boxed{4x < \frac{M_{\mathrm{pole}}^2}{\Lambda^2} < 4x + 2.} \tag{38}$$

Consequently $M_{\mathrm{pole}} \sim 2m$ diverges at fixed cutoff.

Proof. The generator connects the eigenvalues $m, -m$, so $d_{ij}^2 = 4m^2$ while both squared eigenvalues equal m^2. Substitution in (1) proves (37). Its Taylor ratio is

$$M_{\mathrm{Taylor}}^2 = \frac{6m^2}{1 - x},$$

where this is a positive mass only for $x < 1$.

The pole equation is (36) with $\beta = 4x$. The monotonicity proof there gives uniqueness and the residue sign. Evaluating (37) at $y = -4x$ gives $2H(-4x) - 2 > 0$, whereas at $y = -4x - 2$ it gives -2; this proves (38).

For general f satisfying (4), the Taylor kinetic coefficient in this example is proportional to

$$f(x) + xf'(x) = \frac{d}{dx}[xf(x)]. \tag{39}$$

It must be negative somewhere: $xf(x) > 0$ for $x > 0$, but $xf(x) \to 0$ at infinity under (4). A nonnegative derivative everywhere would contradict these facts. □

This counterexample concerns arbitrary finite data, not the top-only Standard Model family. It also distinguishes a wrong-sign **Taylor truncation** from a negative-residue pole of the **full** form factor. We have not proved absence of complex-pole or other nonlocal problems in either model.

8. Counterexample (heavy fermions do not dynamically select the centralizer).

Let the data be those of Proposition 4, and let

$$\mathfrak{g}_{V_0} = \{X \in \mathfrak{g}_{\mathrm{SM}} : [X, V_0] = 0\}.$$

For nonzero top mass this is the represented unbroken gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{\mathrm{em}}$. The limit $x \to \infty$ at fixed Λ does **not** produce a theory containing only these vector directions by quadratic mass decoupling.

Proof. For fixed y, equation (21) implies

$$h_x(y) \longrightarrow 0.$$

Consequently

$$\boxed{C_W(q^2) \longrightarrow \frac{\Lambda^2}{4\pi^2}\frac{9}{2}g(q^2/\Lambda^2).} \tag{40}$$

For positive y,

$$g(y) = 4y \int_0^1 u(1-u)e^{-yu(1-u)}\, du > 0.$$

Thus a charged generator outside $\mathfrak{g}_{V_0}$ retains a finite nonzero quadratic kinetic form and a nonvanishing propagator at every fixed positive momentum. Its mass approaches zero. Equation (35) similarly leaves the positive matrix $B_\infty g(y)$, including the additional neutral direction. These facts contradict decoupling of the broken vectors through diverging masses or vanishing propagators. □

A narrower decoupling statement is true. If we **restrict the connection from the outset** to the centralizer, the finite mass blocks are invariant and the heat trace factorizes. On the heavy block its contribution carries e^{-sm^2/Λ^2}, which tends to zero. The restricted action therefore converges to the light-block spectral action. This proves decoupling of a finite fermion block under that restriction; it does not prove dynamical elimination of the other gauge fields.

The amendment to Lemma 03 Part C is therefore:

- Its transverse kinetic weights, including $w_L(x, 0) = w(x)$, and its two weak-angle formulas are correct under their stated conventions.
- They must be accompanied by the mass coefficient (2), including the quadratic-vertex subtraction.
- A centralizer-only endpoint does not follow from those weights. In the present exact quadratic family, the broken-vector masses vanish while their remaining kinetic coefficients are positive.

9. No-go theorem (finite radial vacuum for the positive Laplace class).

Let f satisfy (4), and suppose the allowed scalar configurations include

$$V = cV_*, \qquad c > 0, \qquad V_* \neq 0,$$

with $A = 0$. Let the action be the literal spectral action, without an added scalar potential or a constraint fixing c. Then no finite $c > 0$ is stationary along this radial direction.

Proof. On $\mathbb{R}^4$, the constant-background action density is

$$\mathcal{U}(c) = \frac{\Lambda^4}{4\pi^2}\int s^{-2}\,\mathrm{Tr}_{H_F}\, e^{-sc^2V_*^2/\Lambda^2}\, d\nu(s).$$

Differentiating gives

$$\boxed{\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2}\int s^{-1}\,\mathrm{Tr}_{H_F}\left(V_*^2 e^{-sc^2V_*^2/\Lambda^2}\right) d\nu(s) < 0.} \tag{41}$$

The inequality is strict because $V_* \neq 0$ and ν is nonzero and positive. The same argument applies on a finite flat torus before performing its positive momentum sum. □

Thus the masses above are conditional quadratic-background quantities. In the top-only family, the ordinary Higgs radial variation supplies the direction used in this theorem. A finite symmetry-breaking minimum of the polynomial asymptotic scalar potential cannot automatically be regarded as a minimum of this full positive-Laplace spectral action.

10. Corollary (comparison of heavy limits).

Let the background family and mass definitions be those above.

At fixed Λ, with $\mu \to \infty$:

quantity, heat cutoff	behavior
m_t	$\mu\Lambda \to \infty$
$M_{W,\mathrm{Taylor}},\ M_{W,\mathrm{pole}}$	$\sim \Lambda/\mu \to 0$
$M_{Z,\mathrm{Taylor}},\ M_{Z,\mathrm{pole}}$	$\sim \sqrt{2/3}\,\Lambda\mu e^{-\mu^2/2} \to 0$
m_t/M_W	$\sim \mu^2 \to \infty$

(42)

At fixed $0 < \mu < \infty$, with $\Lambda \to \infty$, all four vector masses in (42) instead grow linearly with Λ, as does m_t. A controlled asymptotic family can also satisfy

$$\Lambda \to \infty, \qquad \mu \to 0, \qquad \mu\Lambda \to \infty;$$

then $M_W, m_t \to \infty$ while

$$m_t/M_W \to \sqrt{8/3}.$$

Proof. The first assertions follow from (19), (27), and (34)–(36). At fixed positive μ, their dimensionless mass functions are fixed and positive, so multiplication by Λ proves the joint-limit assertions. The final assertion follows from the small-μ limits. □

Accordingly, the obstruction has three distinct scopes. The CCM identity cannot be extrapolated to $\mu \to \infty$ at fixed cutoff because it belongs to the asymptotic truncation. The exact top-only quadratic action obstructs a fixed-cutoff heavy-W limit for the heat cutoff under both mass definitions, and for class (4) under the Taylor definition. A family of actual nonzero stationary vacua of the unmodified positive-Laplace action encounters the separate obstruction (41). Counterexample 7 excludes extending the first of these exact mass conclusions to arbitrary finite representations.

11. Proposition (the radial derivative for general f; referee). On flat $\mathbb{R}^4$, for a sufficiently integrable real cutoff f, with λ_j^2 the eigenvalues of V_*^2,

$$\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2} \sum_j \lambda_j^2 \int_{c^2\lambda_j^2/\Lambda^2}^{\infty} f(z)\,dz. \tag{43}$$

For nonnegative f, a finite $c > 0$ is stationary exactly when every tail associated with a nonzero λ_j vanishes. So if f has nonzero support arbitrarily far out, there is no finite stationary point on this ray; if f has compact support ending at R, there is a stationary plateau once every nonzero $c^2\lambda_j^2/\Lambda^2 \geq R$. Sign-changing cutoffs can have cancellations. A nonconstant cutoff flat near zero lies outside (4), since a nonzero positive measure gives $f'(0) < 0$; (43) covers the nonnegative ones. A radial path with an additional fixed noncommuting mass matrix need not have the form cV_*. □

12. Proposition (the top-only Taylor mass for general f; referee). For nonnegative smooth compactly supported f with $f(0) > 0$, the formulas of Proposition 4 hold with

$$A_f(x) = \frac{2}{x}\int_0^x z f(z)\,dz, \qquad w_f(x) = \frac{3}{x^3}\int_0^x z^2 f(z)\,dz,$$

by approximating f with its derivatives by real combinations of e^{-nz}. Hence

$$\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} \sim \frac{\int_0^\infty z f(z)\,dz}{f(0)\,x}.$$

This includes cutoffs flat near zero. For a nonanalytic cutoff there need be no analytic continuation defining the pole. □

13. Proposition (several heavy doublets; referee). For the heat cutoff, let P be the total particle-trace weight of complete massless doublets, and let heavy doublet i, of weight $d_i > 0$, have one Dirac member of mass m_i and one massless member. Then

$$C_W(y) = \frac{\Lambda^2}{4\pi^2}\Big(Pg(y) + \sum_i d_i h_{x_i}(y)\Big), \qquad \frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{3}{2}\,\frac{\sum_i d_i A(x_i)}{P + \sum_i d_i w(x_i)}.$$

For finitely many doublets, if $P > 0$ and all $x_i \to \infty$, the Taylor mass tends to zero. If $P = 0$ and all $x_i = x$, then $M^2_{W,\mathrm{Taylor}} \sim m^4/(2\Lambda^2)$, the form factor has a unique positive pole with $M^2_{W,\mathrm{pole}} \sim m^2/2$, and the whole unnormalized kernel tends to zero at every fixed Euclidean momentum. The divergent Taylor and pole scales differ, and neither alone establishes a healthy limiting gauge theory. □

14. Proposition (the fully massive family; referee; a complete proof by Astra 6 is in the validation, item 11). Give both members of every Standard Model doublet the same Dirac mass m, with no Majorana masses, and let z_W, z_Z be the charged and massive neutral poles in units of Λ^2. The charged form factor is proportional to $e^{-x}[g(y) + 2xH(y)]$, and

$$2x < z_W < 2x + 2, \qquad \tfrac{16}{5}x < z_Z < \tfrac{16}{5}x + 2.$$

So both pole masses diverge with m at fixed cutoff, and

$$\frac{M^2_{W,\mathrm{pole}}}{M^2_{Z,\mathrm{pole}}} \longrightarrow \frac{5}{8} \qquad (x \to \infty),$$

as also for $x \to 0$, where $z_W \sim 3x$ and $z_Z \sim \frac{24}{5}x$. In between the ratio differs, reaching about 0.66 near $x = 3$ (Python, solving the pole equations). The kernel tends to zero at every fixed Euclidean momentum, and the charged Taylor kinetic coefficient is proportional to $e^{-x}(1 - x/2)$, negative for $x > 2$.

Proof. The species with equal mass have eigenvalues $\pm m$. The charged generator connects them with $d^2 = 0$ and $d^2 = 4m^2$ in equal weights, which gives the charged form. For the neutral generators the diagonal charge has $d = 0$ and the off-diagonal charge $(q_L - q_R)/2 = T_3 v/2$ has $d^2 = 4m^2$; on the particle trace the kernel is $e^{-x}[\mathrm{diag}(6, 10)g + 12xHvv^T]$, so $\beta = 12x\,v^T\,\mathrm{diag}(6, 10)^{-1}v = \frac{16}{5}x$, and for the charged direction $\beta = 2x$. Equation (36) gives the bounds, since $H(-z) > 1$. □

This is a counterexample to any general no-go for diverging pole locations at fixed cutoff. It has no light fermions, and the physical quality of that limit is open.

15. Remark (the light space does not carry unbroken $SU(2) \times U(1)$; referee). In the top-only limit the broken masses vanish, but this does not restore the electroweak group on the whole light space. The light space retains b_L after removing t_L. If P_{light} projects off the top mass block, then on b_L we have $P_{\mathrm{light}}T_1P_{\mathrm{light}} = P_{\mathrm{light}}T_2P_{\mathrm{light}} = 0$ and $P_{\mathrm{light}}T_3P_{\mathrm{light}} = -\frac{1}{2}$, which do not obey the $SU(2)$ relations. Correspondingly $K_W = 9/2$, $K_{33} = 21/4$, $K_{3Y} = -1/4$ in the limit, while a linearly realized unbroken action would have equal weak coefficients and no mixing. The remaining complete massless doublets carry the ordinary representation, and their contribution is gauge invariant. If both members of every removed doublet are heavy, the retained light space can be invariant under the full group; a full limiting theory still needs control of the remaining fields, normalizations, interactions and vacuum. "The heavy fermion leaves the loop" is shorthand for the suppression in this classical trace.

The light-space statement should be read together with the separate role of the top (lemma 21): among single charged species only an up-type quark can be removed while keeping cos^2 theta_NC = 5/8 and the

ratio 8/3, and removing it leaves the five other quarks and the leptons, on which SU(3) x U(1)_em acts with b_L paired with b_R.

16. Remark (the CCM relation as the small-mass form of a weighted sum rule; Claude). For the heat cutoff and doublets with one massive member, Proposition 13 reads

$$8\kappa M_{W,\mathrm{Taylor}}^2 = \sum_f N_f m_f^2\, \rho(m_f^2/\Lambda^2), \qquad \rho(x) = \frac{A(x)}{x} = \frac{2[1-(1+x)e^{-x}]}{x^2},$$

with $N_f = 2d_f$ and $\kappa = (P + \sum_i d_i w(x_i))/6$. As $x \to 0$, $\rho, \kappa \to 1$, which is the CCM relation (14). The relation is therefore the leading term of an expansion, and its range is $m \ll \Lambda$. For the top alone, $\kappa = (3+w)/4$; m/M_W departs from $\sqrt{8/3}$ by 5% at $m \approx 0.45\Lambda$, and M_W peaks at about 0.52Λ at $m \approx 1.42\Lambda$ (Python). The maximum is unique, at $x = 2.024$ with $M_W/\Lambda = 0.5242$ (referee, by a sign argument and Python). A massive diagonal contribution to the trace is exponentially suppressed above the cutoff, while the mixed loop with a massless partner has the algebraic $\rho \sim 2/x^2$ at large x; a vacuum with no other mass leaves the W mass with no source. Whether the classical action is meaningful for $m > \Lambda$ is a separate question, on which these formulas do not decide. In the top-only heat-cutoff family, the ratio differs from $\sqrt{8/3}$ for every $x > 0$, since $3 + w - 4\rho$ vanishes at zero and is increasing, so the limit with M_W and m_t diverging together and $m_t/M_W \to \sqrt{8/3}$ needs $\Lambda \to \infty$ faster than m_t (Corollary 10). This necessity is for that family: with one complete massless lepton doublet and one member of each other doublet of common mass m, the weighted sum equals the unweighted CCM sum at some $x_* > 0$ (referee).

As algebraic checks, we used **Python 3 with SymPy** to verify the integration-by-parts identities, the displayed small-x Taylor series, and $v^T B_\infty^{-1} v = 8/27$. The arguments above provide the proofs; these computations introduce no fitted constants.

Consequence for T5b. At fixed cutoff, the specified top-only Standard Model spectral action does not realize an infinitely heavy W, Z, t family: the top grows while both vector masses tend to zero, for the heat cutoff and the stated quadratic mass definitions. A joint limit with $\Lambda \to \infty$ permits all three background mass scales to diverge, but neither centralizer-only decoupling nor a stationary Higgs vacuum follows from that scaling. These conclusions concern the classical bosonic action and contain no quantum corrections or claims about the physical Standard Model at those scales. Other finite data behave differently: with every doublet heavy and no light fermions, the pole masses diverge at fixed cutoff with M_W^{2/M_Z}2 -> 5/8 (Proposition 14). With a massless complete doublet retained, the Taylor mass tends to zero (Proposition 13). So in the families treated here, a heavy W and Z at fixed cutoff is found only without a massless complete doublet. Light nonzero masses are not treated: they leave a residual vector mass that depends on them.

Part 19

Inner fluctuations of the two-point data without order one

Step 1 of the next steps in the background note, for STRATEGY T0b. Proved by Claude Opus 5.5 without Astra or Fable; every identity checked with numpy on random data (section "Numerical check"). Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): Theorem 19.1 and the comparison with arXiv:1304.7583 are correct; Corollary 19.2 item 3 and the Wilson phase after the table were wrong as first written, and these and the amendments are incorporated below.

Outcome. The truncated two-point data satisfy order zero and fail order one, for each natural real structure. So the inner fluctuations of Chamseddine-Connes-van Suijlekom (arXiv:1304.7583), with their quadratic term, are the applicable ones. The result factorizes. With the inherited pointwise real structure, the fluctuated operator is m |Phi|^2 sigma_1: a real gauge-neutral radial field, with trivial gauge group. With the reflection real structure, which has the KO-6 signs, the link is m Phi^2: all of C is reached, and the gauge group U(1) acts with charge 2. Along a gauge orbit the quadratic term keeps the modulus of the link fixed. This parallels the fixed singular value of the covariant projection of lemma 14, which the fixed compression of lemma 13 lacks. So the first-order failure of Counterexample 08.4 points to the framework of arXiv:1304.7583 for these truncated data.

Setting

Let A = C + C act diagonally on H = C^2, with D = m sigma_1, m > 0, and gamma = sigma_3. For a real structure J = C K put x-hat = J x J^-1. We use two choices.

- **J_0 = K** (or sigma_3 K, the other sign of lemma 08), inherited from pointwise reality on the interval. Then x-hat = conj(x) on A, and J_0 commutes with gamma.
- **J_r = sigma_1 K**, inherited from the reflection real structure sigma_1 R_l K of Counterexample 08.4. Then diag(x_1, x_2)-hat = diag(conj(x_2), conj(x_1)), J_r^2 = 1, J_r D = D J_r and J_r gamma = -gamma J_r: the KO-6 signs.

Both satisfy order zero, since A is commutative and x-hat lies in A. Both fail order one: for e = diag(1, 0), [[D, e], e-hat] = m sigma_1 for J_0 and -m sigma_1 for J_r.

For a universal one-form A = sum_j a_j delta(b_j), the fluctuated operator of arXiv:1304.7583 is

```
D' = D + A_(1) + A~_(1) + A_(2),
A_(1) = sum a_j [D, b_j],   A~_(1) = sum a_j-hat [D, b_j-hat] = eps J A_(1) J^-1,   A_(2) = sum a_j-hat
```

with J D J^-1 = eps D. Gauge transformations are U = u J u J^-1 for unitary u in A, and U D(A) U* = D(gamma_u(A)), where gamma_u(A) = u delta(u*) + u A u*.

Put

```
phi_1 = sum a_{j,1} (b_{j,2} - b_{j,1}),     phi_2 = sum a_{j,2} (b_{j,1} - b_{j,2}),
```

so that A_(1) = m [[0, phi_1], [phi_2, 0]], which is self-adjoint exactly when phi_2 = conj(phi_1). The pair (phi_1, phi_2) takes every value in C^2, as a = diag(x, y) and b = diag(0, 1) show.

The fluctuated operator

Theorem 19.1. For every universal one-form,

```
J_0:   D' = m [[0, |1 + phi_1|^2], [|1 + phi_2|^2, 0]],
J_r:   D' = m [[0, (1 + phi_1)(1 + conj(phi_2))], [(1 + phi_2)(1 + conj(phi_1)), 0]].
```

For a self-adjoint form, with Phi = 1 + phi_1,

```
J_0:   D' = m |Phi|^2 sigma_1,          J_r:   D' = m [[0, Phi^2], [conj(Phi)^2, 0]].
```

Proof. For diagonal b, [X, b] for off-diagonal X = [[0, x], [y, 0]] is [[0, x (b_2 - b_1)], [y (b_1 - b_2), 0]].

For J_0: a-hat and b-hat are the conjugates. Then A~*(1) = m [[0, conj(phi_1)], [conj(phi_2), 0]], and the first row of A*(2) is m phi_1 sum conj(a_{j,1}) (conj(b_{j,2}) - conj(b_{j,1})) = m phi_1 conj(phi_1); the second row is m phi_2 conj(phi_2). So the (1,2) entry of D' is m (1 + phi_1 + conj(phi_1) + |phi_1|^2) = m |1 + phi_1|^2, and likewise (2,1).

For J_r: a-hat = diag(conj(a_2), conj(a_1)) and b-hat = diag(conj(b_2), conj(b_1)). Then A~*(1) = sigma_1 conj(A*(1)) sigma_1 = m [[0, conj(phi_2)], [conj(phi_1), 0]], and the first row of A_(2) is m phi_1 sum conj(a_{j,2}) (conj(b_{j,1}) - conj(b_{j,2})) = m phi_1 conj(phi_2), and the second is m phi_2 conj(phi_1). This gives the stated factorization. ■

Corollary 19.2 (gauge action and image). Let u = diag(e^{i alpha}, e^{i beta}) and psi = alpha - beta. Under gamma_u, Phi -> e^{i psi} Phi.

1. For J_0: U = 1. The image of the self-adjoint forms is {t sigma_1 : t >= 0}, so the link is real, and t is a gauge-neutral field. The value t = 0 is reached at Phi = 0.
2. For J_r: U = diag(e^{i psi}, e^{-i psi}), and the link m Phi^2 goes to e^{2 i psi} m Phi^2. So U(1) acts with charge 2 in the variable of the link, and the image is all of C, with Phi -> Phi^2 two-to-one away from 0. The image statement is pointwise: a globally prescribed smooth link need not have a global smooth square root.

3. The gauge orbit of the zero form has Phi = e^{i psi}. It gives D' = D for J_0, whose operator orbit is the single point D, and link m e^{2 i psi} for J_r. Without A_(2), the link is m (2 cos psi - 1) for J_0, of modulus m (1 - psi^2 + O(psi^4)), although U = 1; and m (2 e^{i psi} - 1) for J_r, of modulus m (5 - 4 cos psi)^{1/2} = m (1 + psi^2 + O(psi^4)), off the orbit of D.

Proof. gamma_u(0) has phi_1 = u_1 conj(u_2) - 1 = e^{i psi} - 1. The rest is direct. ■

Item 3 shows, in the smallest example, why the quadratic term is needed: it restores gauge covariance.

Comparison with the parent

Proposition 19.3. Consider the chiral interval at M = 0, with m = pi/(2l), and the links of lemmas 13 and 14.

1. *With no real structure*, the finite fluctuation is m (1 + phi_1), all of C, while the fixed compression of a gauge field gives only the line m + i B (Proposition 13.2).
2. *With pointwise reality*, the parent has no scalar fluctuation at all: the algebra C(I) is commutative with J f J^-1 = conj(f), it satisfies order one, and A + J A J^-1 = 0 for a scalar one-form. The truncated data do have a fluctuation, the neutral radial field t. So the finite fluctuation has a gauge-neutral radial parameter, while the specified pointwise-real parent has no surviving self-adjoint scalar inner fluctuation. This does not establish a dynamical scalar field or its action, and it says nothing about parent fields in other models. Compression is not multiplicative, so there is no canonical identification of finite potentials with parent potentials.
3. *With the reflection real structure*, the parent satisfies order one, since J_r f J_r^-1 = conj(f) composed with the reflection commutes with [D, g]. One finds J_r D_{A_5} J_r^-1 = D_{A_5 o R}, with R the reflection, so that the symmetrized field A_5 + A_5 o R survives, and the fixed compression gives m + i B^s with B^s = 2 B, because the weight sin(pi x/l) is even about l/2. The modulus along a pure-gauge field behaves as follows.

object	modulus of the link along a gauge orbit
covariant projection (lemma 14, Theorem 3)	constant: lambda W_l^* has fixed singular value
finite data with A_(2) (Corollary 19.2)	constant: m
fixed compression (Proposition 13.2), with B the integral of the symmetrized coefficient for J_r	$(m^2 + B^{2)}\{1/2\}$, growing at second order along paths where B has a linear term
finite data with J_r, linear terms only	m (5 - 4 cos psi)^{1/2}, growing at second order

Proof. Item 1 is lemma 13. For item 2, a represented self-adjoint parent one-form is E h(x) with h = i a and a real, and K (i E a) K^-1 = -i E a, so the real fluctuation cancels. An individual commutator need not change sign, since K [D, x] K^-1 = [D, x] = E: self-adjointness is the step that matters. For item 3, K changes the sign of i A_5, the reflection changes the sign of the derivative and composes A_5 with R, and sigma_1 E sigma_1 = -E for E = [[0, 1], [-1, 0]]. The table collects Theorem 3 of lemma 14, Corollary 19.2 and Proposition 13.2. ■

The table states a parallel and no identification. The variables differ: for constant unsymmetrized A_5 = a, the transported link is m e^{i a l} = m + i (pi a/2) + O(a^2), with Wilson phase a l, against m + i (2 a/pi) in the compression. For the reflection-real fluctuation from the same a, these become m e^{2 i a l} and m + i (4 a/pi). The phase of a link depends on the frame, while its modulus does not. In the parent, the covariance defect is repaired by moving the projection, and in the finite data by the quadratic term.

Reading

Spectral compression can break order one, and it can also break order zero: on the circle with J = K, compression to the zero mode preserves both, while compression to the modes -1, 0, 1 sends e^{ix} to a truncated shift S, with [S, S*] != 0, so that order zero fails. Where a represented algebra, a preserved real structure and order zero hold, as here, the framework of arXiv:1304.7583 applies to the truncated data as

they stand. So the first-order failure of Counterexample 08.4 has an escape, and this calculation is a first use of it. Two points were open. Whether the parallel in the table extends to an exact statement for a fluctuated parent and its truncation: this holds on gauge orbits, by Corollary 22.6 of lemma 22, and remains open for the nonunitary part. And whether the neutral radial parameter t is related to the singlet of the grand-symmetry model (Devastato-Lizzi-Martinetti, where sigma comes from a Majorana-operator fluctuation in a larger algebra). This is an analogy only: in arXiv:1304.7583 the quadratic term does not generate new scalar fields, and here too it depends on the same two parameters. An identification would need a map of algebras, representations, fields and gauge actions.

Numerical check

Python (numpy), run on 3 Nov 2026 in a temporary directory, not kept. On random complex a_j, b_j, with m = 1.3: the three real structures have the stated signs; the formulas of Theorem 19.1 hold to about 10^-15, without assuming self-adjointness; D(gamma_u(0)) = U D U* holds to 10^-16 for both real structures, and fails by 4.5 for the sample when A_(2) is dropped; U D(A) U* = D(gamma_u(A)) holds for a general form; the values of [[D, e], e-hat] are as stated.

Part 20

Sheet-connecting operators and boundary gluing

By Astra 6 (gpt-6-astra, effort xhigh), 3 Nov 2026; prompt in prompts/20_connecting_operator.md. Reproduced as delivered, with links made relative and one link to a temporary copy of a source written as plain text. The sections "Numerical check" and "Readings" at the end are by Claude Opus 5.5. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): Results 1 to 4 and 6 to 11 correct, with one amendment to Result 7 given in the note after its proof; the readings are corrected below.

A rank-one operator can connect the two finite summands with the required real signs and grading, and the Chamseddine–Connes maximal algebra follows from its intertwining algebra. A boundary coupling supplies such an operator in an explicit interval model. However, agreement of scalar functions at a wall generally does not survive compression as the finite order-one restriction: an exact one-flavour counterexample exhibits this failure. The interval construction realizes the classified rank-one block when its matrix algebra, grading and coupled channels are supplied as inputs; it does not derive those inputs from Example 16.9. The three requested obstruction checks hold in the scopes specified below.

1. Theorem — Finite connecting blocks and their admissible algebras

Hypotheses and conventions. Let

$$E = \mathrm{Hom}_{\mathbb{C}}(V, W), \qquad E^* = \mathrm{Hom}_{\mathbb{C}}(W, V),$$

with Hilbert–Schmidt inner products, conjugate-linear in the first argument. Here E^* denotes the second Hom space; operator adjoints also carry a superscript $*$.

Let $c : E \to E^*$ be the antiunitary occurring in the real structure:

$$J(x, y) = (c^{-1}y, cx), \qquad \Gamma = \begin{pmatrix} g & 0 \\ 0 & h \end{pmatrix}, \qquad h = -cgc^{-1}.$$

In Example 16.9, after choosing the grading of W,

$$cx = x^*, \qquad gx = \gamma_W x, \qquad hy = -y\gamma_W.$$

For the more general graded flip of Corollary 16.8, c includes its stated grading sign.

Let $\mathcal{A}^{\mathrm{ev}}$ act diagonally through π_1, π_2. Write $\rho_i(b)$ for its opposite action on the two summands. Assume that the sheet-diagonal D_F already satisfies self-adjointness, oddness, $JD_F = D_F J$, and order one on $\mathcal{A}^{\mathrm{ev}}$, as in Corollary 16.8.

For a complex-linear map $T : E \to E^*$, put

$$S_T = \begin{pmatrix} 0 & T^* \\ T & 0 \end{pmatrix}, \qquad D = D_F + S_T, \qquad \tau(T) = cT^*c.$$

Then D has the required signs and satisfies order one on a unital involutive subalgebra $B \subseteq \mathcal{A}^{\mathrm{ev}}$ precisely when

$$\boxed{T = \tau(T), \qquad hT + Tg = 0,} \tag{1}$$

and

$$\boxed{(T\pi_1(a) - \pi_2(a)T)\rho_1(b) = \rho_2(b)(T\pi_1(a) - \pi_2(a)T) \quad (a, b \in B).} \tag{2}$$

Self-adjointness is automatic from the definition of S_T. In bases related by c, the first equation in (1) says that the matrix of T is complex symmetric.

In the ordinary adjoint-flip convention, for $a = (a_W, a_V)$,

$$\pi_1(a)x = a_W x, \quad \pi_2(a)y = a_V y, \quad \rho_1(a)x = xa_V, \quad \rho_2(a)y = ya_W.$$

Thus (2) is an entirely explicit equation on Hom spaces.

For an operator $R : E \to E^*$, define exactly the commutation algebra used in section 4 of Chamseddine–Connes:

$$\mathcal{A}(R) = \left\{ a \in \mathcal{A}^{\mathrm{ev}} : \begin{array}{l} \pi_2(a)R = R\pi_1(a), \\ \pi_2(a^*)R = R\pi_1(a^*) \end{array} \right\}. \tag{3}$$

For fixed T satisfying (1), its maximal admissible algebras are the inclusion-maximal members of

$$\boxed{\{\mathcal{A}(R) : \ hR + Rg = 0, \quad T = R + \tau(R)\}.} \tag{4}$$

Consequently, maximal dimension for fixed T is obtained by maximizing the dimension of the algebras in (4). The distinction between T and R is necessary.

Proof. Block multiplication gives (1). The lower off-diagonal block of the order-one equation is exactly (2); adjoints give the other block.

To prove (4), suppose (2) holds. Average over the compact unitary group of the finite-dimensional real C^*-algebra B:

$$R_0 = \int_{U(B)} \pi_2(u)T\pi_1(u)^* \, du.$$

Then R_0 intertwines the left actions. Equation (2) implies that

$$T - \pi_2(u)T\pi_1(u)^*$$

intertwines the opposite actions. Therefore $Q_0 = T - R_0$ intertwines the opposite actions.

The involution τ exchanges these two intertwining conditions. Since $\tau(T) = T$,

$$R = \frac{1}{2}(R_0 + \tau(Q_0))$$

intertwines the left actions and satisfies $T = R + \tau(R)$. Averaging preserves oddness because B is even; τ also preserves oddness. Hence $B \subseteq \mathcal{A}(R)$ for an R occurring in (4).

Conversely, on $\mathcal{A}(R)$, the term R intertwines the left actions and $\tau(R)$ intertwines the opposite actions. Their sum satisfies (2). This proves the characterization and the maximality statement.

For an explicit reason why $\mathcal{A}(T)$ alone is insufficient, take ungraded $W = V = \mathbb{C}^2$, $g = 1$, $h = -1$, and

$$B = \{(\mathrm{diag}(\alpha, \beta), \mathrm{diag}(\alpha, \beta))\}.$$

Let $e_{ij} \in E$, $f_{ij} = ce_{ij}$, and define

$$Te_{11} = f_{12}, \qquad Te_{12} = f_{11},$$

with all other images zero. This T is symmetric and odd. The first arrow preserves the right B-character; the second preserves the left character. Each therefore satisfies order one. But the first arrow fails to intertwine the left actions when $\alpha \neq \beta$. ■

2. Theorem — The classified algebra is realized by a rank-one block

Hypotheses. Let

$$W = W_+ \oplus W_-, \qquad \dim_{\mathbb{H}} W_+ = \dim_{\mathbb{H}} W_- = 1, \qquad V = \mathbb{C}^4,$$

and use the adjoint-flip convention of Result 1. Thus

$$\mathcal{A}^{\mathrm{ev}} = \mathbb{H}_+ \oplus \mathbb{H}_- \oplus M_4(\mathbb{C}).$$

Choose unit vectors $w \in W_+$, $v \in V$, and $\kappa \neq 0$. Put

$$e = wv^* \in E, \qquad f = vw^* = ce,$$

and

$$\boxed{T_\kappa x = \kappa\,(w^* x v)\, vw^* = \kappa\,|f\rangle\langle e|x.} \tag{5}$$

Then T_κ has complex rank one, satisfies (1), and has intertwining algebra

$$\boxed{\mathcal{A}(T_\kappa) = \{(\lambda \oplus q, \lambda \oplus m) : \lambda \in \mathbb{C},\ q \in \mathbb{H},\ m \in M_3(\mathbb{C})\}.} \tag{6}$$

Here the first λ acts on the quaternionic line W_+ as

$$\lambda \longmapsto \begin{pmatrix} \lambda & 0 \\ 0 & \overline{\lambda} \end{pmatrix}$$

in a basis starting with w, and the second acts on $\mathbb{C}v \subset V$.

The algebra (6) has real dimension 24. Up to an automorphism of $\mathcal{A}^{\mathrm{ev}}$, it is the unique maximal-dimensional involutive subalgebra admitting a nonzero sheet-connecting Dirac block with order one. The off-diagonal self-adjoint operator S_{T_κ} has complex rank two.

Proof. Since $ge = e$ and $hf = -f$, T_κ is odd. Its matrix has the single symmetric entry κ in c-paired bases, so $\tau(T_\kappa) = T_\kappa$.

The two intertwining equations in (3) require that $\mathbb{C}w$ and $\mathbb{C}v$ be reducing subspaces for the respective actions, with the same scalar action on these lines. A quaternion preserving $\mathbb{C}w$ belongs to the displayed copy of $\mathbb{C}$. On V, the orthogonal complement $v^\perp$ carries an arbitrary $M_3(\mathbb{C})$-action. The other quaternionic summand is unrestricted. This proves (6), and $2 + 4 + 18 = 24$.

We reproduce the dimension argument in Chamseddine–Connes, section 4, whose local source defines (3) at line 875. A nonzero connecting block implies, by Result 1, a nonzero intertwiner R for B. Compress R on either side by projections in the commutants of the full even algebra. Some nonzero compression remains, and its support and range can be placed in irreducible representation spaces

$$K : \mathbb{C}^2 \longrightarrow \mathbb{C}^4.$$

These compressions enlarge the intertwining algebra.

If K has rank two, its image is a reducing two-plane in $\mathbb{C}^4$. The matrix action on that plane is determined by the quaternion action. The remaining two-plane contributes at most $M_2(\mathbb{C})$. Including the unused quaternionic summand gives

$$\dim_{\mathbb{R}} B \leq 4 + 4 + 8 = 16.$$

If K has rank one, its support and range are complex lines. The intertwining algebra is (6), of dimension 24. Unitary changes of these lines, and exchange of the quaternionic summands, give the stated uniqueness.

Finally, T_κ and T_κ^* have orthogonal source and target summands, giving rank $S_{T_\kappa} = 2$. ■

The connecting condition here is relative to the sheet projections of the original prealgebra: $[S_{T_\kappa}, e_1] \neq 0$. On the selected algebra (6), this particular Majorana block commutes with the algebra. Additional sheet-diagonal Dirac terms are needed for the usual nontrivial finite one-forms and Yukawa data.

3. Lemma — Boundary preservation is an intertwining condition

Hypotheses. Use the domain of Lemma 12:

$$Q_R(0) = 0, \quad U_L(0) = 0, \quad Q_R(l) = XU_R(l), \quad U_L(l) = -X^*Q_L(l). \tag{7}$$

Let multiplication act through smooth matrix functions F_Q, F_U, with the same matrix on both chiral components of each field.

Multiplication preserves this domain precisely when

$$\boxed{F_Q(l)X = XF_U(l), \qquad F_U(l)X^* = X^*F_Q(l).} \tag{8}$$

These equations define a unital involutive algebra. For scalar functions and $X \neq 0$, they reduce to

$$f_Q(l) = f_U(l). \tag{9}$$

For $X = 0$, no equality between the boundary values is required.

Proof. At $x = l$, the traces $Q_L(l)$ and $U_R(l)$ can be chosen independently. Applying multiplication to the first coupled condition in (7) gives

$$F_Q(l)XU_R(l) = XF_U(l)U_R(l)$$

for every $U_R(l)$. This is the first equation in (8). The other condition gives the second equation. Conversely, these equations preserve both boundary conditions; the conditions at zero are automatically preserved.

Taking the adjoint of the second equation gives

$$F_Q(l)^*X = XF_U(l)^*.$$

Thus (8) has precisely the two-equation form (3). Closure under products and adjoints follows directly. ■

For matrix-valued functions, "agreement at the wall" therefore means agreement through X on the coupled boundary channels. It need not mean equality of the entire boundary matrices.

4. Counterexample — Scalar gluing does not imply finite order one

Hypotheses. Let $N_Q = N_U = 1$, both bulk masses vanish, and $X = t > 0$. Set

$$\theta = \arctan t \in (0, \pi/2), \qquad m = \theta/l.$$

Take the full domain-preserving scalar algebra

$$\mathcal{A}_{\text{glue}} = \{(f_Q, f_U) \in C^\infty([0,l])^2 : f_Q(l) = f_U(l)\}. \tag{10}$$

The bulk operator is

$$D(Q_L, Q_R; U_L, U_R) = (Q_R', -Q_L'; U_R', -U_L')$$

on (7). Use physical chirality

$$\Gamma(Q_L, Q_R; U_L, U_R) = (Q_L, -Q_R; U_L, -U_R)$$

and the antiunitary

$$\mathcal{J}(Q_L, Q_R; U_L, U_R) = (\overline{U_R}, -\overline{U_L}; -\overline{Q_R}, \overline{Q_L}). \tag{11}$$

This parent satisfies the signs $(+,+,-)$, order zero and order one. Its lowest symmetric spectral window gives

$$A_{\mathrm{fin}} = \mathbb{C} \oplus \mathbb{C}, \qquad D_{\mathrm{fin}} = m\sigma_1, \qquad J_{\mathrm{fin}} = \sigma_1 K, \qquad \Gamma_{\mathrm{fin}} = \sigma_3, \tag{12}$$

and these finite data fail order one.

Proof. The boundary form proves self-adjointness, as in Lemma 12.1. Equation (11) preserves (7), squares to one, commutes with D, and reverses Γ. The opposite algebra acts by the two scalar functions exchanged. Hence it commutes both with the multiplication algebra and with its differential commutators.

The lowest window is spanned by the orthonormal chiral vectors

$$\begin{aligned} u_+(x) &= l^{-1/2}(\cos mx, 0; -\sin mx, 0), \\ u_-(x) &= l^{-1/2}(0, \sin mx; 0, \cos mx). \end{aligned} \tag{13}$$

Direct differentiation gives $Du_+ = mu_-$, $Du_- = mu_+$. The boundary conditions hold because $\tan(ml) = t$. Solving the same elementary ODE gives the next absolute eigenvalue $(\pi - \theta)/l$, so any symmetric cutoff between m and this value selects exactly (13).

Compression of (10) is

$$Q(f_Q, f_U) = \begin{pmatrix} \dfrac{1}{l}\displaystyle\int_0^l (f_Q \cos^2 mx + f_U \sin^2 mx)\, dx & 0 \\ 0 & \dfrac{1}{l}\displaystyle\int_0^l (f_Q \sin^2 mx + f_U \cos^2 mx)\, dx \end{pmatrix}. \tag{14}$$

Let f_Q be a nonnegative bump supported sufficiently near zero, and let $f_U = 0$. The wall equality holds, while the difference of the diagonal entries is

$$\frac{1}{l}\int_0^l f_Q(x)\cos(2mx)\, dx > 0.$$

Together with the identity, this gives the full diagonal algebra $\mathbb{C}^2$.

On (13), (11) gives $J_{\mathrm{fin}} = \sigma_1 K$. For $p = \mathrm{diag}(1, 0)$,

$$p^\circ = 1 - p, \qquad [[D_{\mathrm{fin}}, p], p^\circ] = -D_{\mathrm{fin}} \neq 0.$$

Thus finite order one fails.

There is also a precise distinction between the original field sheets and the two finite chiral summands. The projection onto the Q-field compresses to

$$\begin{pmatrix} \frac{1}{2} + \frac{\sin 2\theta}{4\theta} & 0 \\ 0 & \frac{1}{2} - \frac{\sin 2\theta}{4\theta} \end{pmatrix}, \tag{15}$$

which is not a projection. The coupled spectral window does not preserve the original field decomposition. ■

This proves failure of statement 2(c) for the full scalar gluing algebra, even when the parent has all the required real signs and order conditions. It also specifies the qualification needed in 2(b): the finite summands require an identification, such as continuation from the uncoupled zero modes.

5. Theorem — An exact interval parent supplies the rank-one connector

Hypotheses. Use the Hom representation, c, g, h of Result 1. Set

$$k = cgc^{-1}, \qquad h = -k.$$

Choose a unit vector $e \in E$ with $ge = e$, let $f = ce$, and put

$$S = |f\rangle\langle e|, \qquad X = tS^* : E^* \to E, \qquad t > 0. \tag{16}$$

Take two massless interval fields, Q valued in E and U valued in E^*, with boundary conditions (7).

As candidate multipliers, take **constant** elements of $\mathcal{A}^{\mathrm{ev}}$, acting by π_1 on Q and π_2 on U. Their domain-preserving algebra is

$$B = \mathcal{A}(S). \tag{17}$$

Define

$$\hat{\Gamma}(Q_L, Q_R; U_L, U_R) = (gQ_L, -gQ_R; kU_L, -kU_R), \tag{18}$$

$$\widehat{J}(Q_L, Q_R; U_L, U_R) = (c^{-1}U_R, -c^{-1}U_L; -cQ_R, cQ_L). \tag{19}$$

Let $n = \dim_{\mathbb{C}} E$, $\theta = \arctan t$, $m = \theta/l$, and choose

$$m < \Lambda_0 < (\pi - \theta)/l.$$

Then the spectral projection

$$P = 1_{[-\Lambda_0, \Lambda_0]}(D_X)$$

has rank $2n$. Under an explicit identification $P\mathcal{H} \simeq E \oplus E^*$, its finite data are

$$\boxed{\left(B,\ E \oplus E^*,\ \begin{pmatrix} 0 & mS^* \\ mS & 0 \end{pmatrix},\ J,\ \begin{pmatrix} g & 0 \\ 0 & -k \end{pmatrix}\right).} \tag{20}$$

They satisfy self-adjointness, oddness, order zero, order one, and the signs $(+, +, -)$. The connecting block is $T = mS$, of complex rank one.

Proof. Result 3, with $X = tS^*$, gives (17). The rank-one condition makes $\mathbb{C}e$ and $\mathbb{C}f$ reducing subspaces for B, with the same action on the two lines.

The coupled channels have exactly the eigenvectors (13), with 1 replaced by e in the Q-components and by f in the U-components. Every orthogonal Q-channel has boundary type (a), and every orthogonal U-channel has type (b). They contribute $2n - 2$ constant chiral zero modes; their next absolute eigenvalue is π/l. The coupled channels contribute the two eigenvalues $\pm m$. This proves the window and dimension assertions.

The identification in (20) sends e and f to these two coupled chiral vectors. It sends $e^{\perp} \subset E$ to constant Q_L-profiles and $f^{\perp} \subset E^*$ to constant U_R-profiles. Since B acts identically on the coupled lines, this identification intertwines its representations exactly. Direct application of (18) and (19) gives the displayed finite grading and real structure.

On the parent, (19) squares to one and commutes with the real differential expression. The relations $kf = f$ and $ge = e$ ensure grading preservation of the boundary conditions. The identity

$$cX = X^*c^{-1}$$

ensures their preservation by $\widehat{J}$. These establish the real signs on the operator domain.

The opposite representation is the original right Hom action, so order zero holds. Every admissible constant multiplier commutes with D_X, including preservation of its domain. Order one therefore holds on the parent and on (20). ■

The wall coupling can also be converted explicitly into a bounded bulk term. On the coupled field channels, let

$$F = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad E_{\text{spin}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad U(x) = \exp(mxF).$$

Apply the same field rotation to both spin components. It maps the uncoupled boundary domain to the coupled domain and gives

$$U^* D_X U = D_0 + mF \otimes E_{\text{spin}}. \tag{21}$$

Compression of the added bounded term to the uncoupled zero modes is exactly $m(S+S^*)$. Thus the parent supplies the connector through an explicit change of domain and an explicit unitary transport.

Two specializations specify the extent of this construction.

- **A commutative example.** Let $E = E^* = \mathbb{C}$, $g = 1$, $c = K$, and start with constant pairs $\mathbb{C}^2$. Boundary preservation selects their common scalar:

  $$B = \{(\lambda, \lambda)\} \simeq \mathbb{C}.$$

  The finite triple is

  $$(\mathbb{C}, \mathbb{C}^2, m\sigma_1, \sigma_1 K, \sigma_3).$$

  This is the simplest successful instance of 2(a)–(c), with the candidate observable algebra restricted to constant functions on each sheet.

- **The classified block.** Take $E = \mathrm{Hom}(\mathbb{C}^4, \mathbb{H}^2)$, the chosen quaternionic grading, and $e = wv^*$ from Result 2. Equations (17) and (20) give

  $$B = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}), \qquad \dim_{\mathbb{C}} H_{\text{fin}} = 32,$$

  and the rank-one connecting block (5) with $\kappa = m$. This parent contains the Hom fibres and constant matrix algebra as inputs. It is not a construction from scalar functions on the twistor parent.

An exact tower-decoupling family is obtained by fixing $m > 0$ and setting

$$t(l) = \tan(ml), \qquad 0 < ml < \pi/2.$$

The finite mass remains exactly m, while the next absolute eigenvalue is $\pi/l - m \to \infty$.

The usual zero-dimensional orientability axiom fails in these examples: the linked vectors e, f have identical left and opposite characters, whereas their grading eigenvalues are opposite. Every represented zero-cycle acts identically on these two lines. The claims above concern real even spectral triples with the stated order conditions and signs.

6. Proposition — Point evaluation bounds rank by the boundary-channel dimension

Hypotheses. Let K_i be finite-dimensional windows of smooth sections, let

$$\mathrm{ev}_i : K_i \to F_i$$

be evaluation at the proposed wall, and suppose a finite coupling is defined by

$$T = \mathrm{ev}_2^* \; X \; \mathrm{ev}_1 \,. \tag{22}$$

Then

$$\operatorname{rank} T \le \min\{\operatorname{rank} X, \dim F_1, \dim F_2\}. \tag{23}$$

A single scalar evaluation channel gives rank at most one. A point with several fibre components can give larger rank.

Proof. Equation (23) follows by factoring (22) through the range of X.

For a counterexample to an unrestricted rank-one claim, take Lemma 12 with

$$N_Q = N_U = N, \qquad M_Q = M_U = 0, \qquad X = tI_N.$$

It is the orthogonal sum of N copies of Result 4. Its lowest chiral block is

$$T = \frac{\arctan t}{l} I_N,$$

of rank N, although the coupling occurs at the single point $x = l$. ■

For an exact boundary-domain perturbation, formula (22) is an additional hypothesis. The operators D_X and D_0 have different domains, so their difference is not automatically a point-evaluation perturbation. The exact interval mass $\arctan(t)/l$ also distinguishes the finite operator from its small-t approximation t/l. An extended wall carries a space of boundary sections and supplies no rank-one conclusion by itself.

7. Proposition — Point gluing does not select the algebra in the unchanged twistor window

Hypotheses. Let $Q_i(f) = P_i M_f P_i$, where P_i project onto finite-dimensional spaces of smooth sections on compact manifolds. Let Z_i be closed subsets with empty interior. Impose any scalar gluing condition which allows independent functions vanishing near Z_1 and Z_2, including equality at two marked points.

Then the image of this glued algebra under the **unchanged separate compressions** $Q_1 \oplus Q_2$ equals the image of the unrestricted pair algebra.

Proof. We first show

$$Q_i(C_c^\infty(M_i \setminus Z_i)) = Q_i(C^\infty(M_i)). \tag{24}$$

If the left image were a proper subspace of the right image, finite-dimensional linear algebra would give a linear functional separating them. Applied to $Q_i(f)$, that functional has the form

$$f \longmapsto \int_{M_i} f(x)\, r(x)\, dx$$

for a smooth density r, because matrix entries of $Q_i(f)$ are integrals against products of smooth sections.

Vanishing for every test function supported outside Z_i forces $r = 0$ there. Since that complement is dense and r is continuous, $r = 0$ everywhere, a contradiction. The argument also applies over the real numbers.

By (24), arbitrary compressed values on either sheet can be obtained using functions vanishing near its wall. Such pairs satisfy the gluing condition independently, proving the assertion. ■

In Example 16.9, both analytic section spaces come from $\mathbb{CP}^3$; one observable algebra is restricted to pullbacks from S^4. The corresponding matrix-entry densities on S^4 are smooth after integration along the twistor fibres, so the same proof applies.

Neither closed space comes with the boundary required by Lemma 12. One may choose a base point $p \in S^4$, a point $q \in \mathbb{CP}^3$, or a twistor fibre, and impose scalar agreement there. In the original level-one windows, the generated real algebra remains

$$M_2(\mathbb{H}) \oplus M_4(\mathbb{C}).$$

Thus this operation derives no further algebra reduction.

A marked base point does provide a quaternionic line through

$$\Pi_p = \frac{1}{2}(1 + p_a \Gamma_a).$$

Choosing a complex line inside that quaternionic line, a complex line in the other four-space, and the rank-one map between the corresponding Hom vectors gives the data of Result 2. The grading $2\Pi_p - 1$ and the rule selecting this connector are then additional finite data. The stabilizer calculation derives $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ from those choices.

To obtain an analytic parent for Example 16.9, further work must specify an interface or interval collar, its boundary trace spaces, self-adjoint coupling conditions, a compatible real structure and grading, and the spectral window of the coupled operator. Result 7 concerns the unchanged windows. A new coupled window, boundary degrees of freedom, or a different observable representation lies outside its hypotheses.

Referee amendment. For a general subset upstairs in the twistor space, add the hypothesis that its image in S^4 has empty interior, so that independent base test functions exist.

8. Proposition — Scope of the Hořava–Witten comparison

Hypotheses. Compare chiralities using one common ten-dimensional chirality convention, and retain the unbroken boundary gauge group $E_8 \times E_8$.

The same-chirality interval analogue is type **(a) at both ends**, or type **(b) at both ends** after reversing the chirality convention. To model two separately localized multiplets, take two fields of that same type, with opposite signs of their localization masses. Lemma 12 at $X = 0$ contains one type-(a) and one type-(b) field, so its displayed pair has opposite chiralities.

Proof and scope. Hořava–Witten impose $\Gamma_{11}\chi = \chi$ on the boundary gaugino; the boundary vector multiplets are intrinsic boundary fields. Their chirality projector is stated in section 2, immediately before the boundary Yang–Mills Lagrangian. Labels based on outward normals change sign between the endpoints; this does not change the comparison in a fixed ten-dimensional convention.

Type (a) in Lemma 09 retains the L-zero mode at both ends, while type (b) retains R. Different mass signs localize two such fields near opposite ends. Type (c) instead produces opposite-chirality wall profiles.

On a space containing only the two same-chirality modes, an operator odd for that chirality grading has zero matrix elements between them. Moreover,

$$\mathrm{Hom}_{E_8\times E_8}((248,1),(1,248)) = 0,$$

because the first factor acts trivially on the target and has no invariant vector in its adjoint representation. Thus these boundary multiplets do not supply a gauge-invariant bilinear connector of the interval mass-block type. ■

Bulk propagation can define effective boundary-to-boundary interactions. Identifying one with the finite T requires a specified field, representation and grading, potentially after compactification or symmetry breaking. The Hořava–Witten comparison alone supplies no such identification.

9. No-go theorem — Corollary 11.2 under its matching-scale hypothesis

Verdict and hypotheses. The corrected Corollary 11.2 is correct. Assume a fixed finite set of Dirac mass parameters, N_g generations, and the matching-scale relation

$$8M_W^2 = \sum_{g=1}^{N_g}(m_{\nu,g}^2 + m_{e,g}^2 + 3m_{u,g}^2 + 3m_{d,g}^2). \tag{25}$$

For $N_g \neq 3$, the coefficient 8 is an explicitly assumed normalization; it does not follow automatically by replicating the three-generation gauge trace.

Then $M_W \to \infty$ is incompatible with keeping every Dirac mass bounded. More precisely,

$$\frac{m_{\max}}{2\sqrt{2}} \leq M_W \leq \sqrt{N_g}\, m_{\max}. \tag{26}$$

Proof. The weighted sum in (25) lies between $m_{\max}^2$ and $8N_g m_{\max}^2$, proving (26). Thus a fixed nonzero gauge coupling and $v \to \infty$, with all $y_f v$ fixed, contradict (25).

The conclusion concerns $m_{\max}$. One fixed species need only be unbounded along a subsequence. For example, with $M_W = t$, put

$$\alpha(t) = \frac{\pi}{4}(1 + \sin t), \quad m_e = 2\sqrt{2}\, t \cos\alpha(t), \quad m_\nu = 2\sqrt{2}\, t \sin\alpha(t).$$

Their squared sum is $8t^2$, while each mass vanishes on an unbounded sequence. ■

Scope and escapes. Equation (25) belongs to the stated normalization of the truncated asymptotic spectral action at matching. It concerns Dirac Yukawa masses, including the neutrino Dirac parameters, rather than physical seesaw eigenmasses. Changing the action or normalization, controlling the full finite-cutoff action, varying the particle content, or allowing Dirac masses to grow changes the problem. Extrapolation to a heavy-mass limit at fixed cutoff requires a separate calculation; the theorem directs attention to that calculation.

10. Theorem — Proposition 16.1 holds for inherited pointwise reality

Verdict and hypotheses. The corrected Proposition 16.1 is correct. Let

$$J_0 a J_0^{-1} = a^*, \qquad [P, J_0] = 0, \qquad Q(a) = PaP|_{P\mathcal{H}},$$

and let $J_F = J_0|_{P\mathcal{H}}$. Let A_F be the involutive algebra generated by the compressed functions. Then order zero holds for this inherited structure exactly when A_F is commutative.

Proof.

$$J_F Q(b)^* J_F^{-1} = P J_0 b^* J_0^{-1} P = Q(b).$$

Order zero therefore requires

$$[Q(a), Q(b)] = 0$$

for every pair of generators. Their generated algebra is then commutative.

Conversely, the opposite operation reverses products and fixes each compressed generator. On a commutative generated algebra its image lies in the same algebra, proving order zero. ■

For a product with a fixed finite factor, the unit of that factor gives the same necessity. Sufficiency also requires that the finite factor satisfy order zero and that the product real structure be compatible. The commutativity conclusion concerns the compressed function algebra; the separate finite factor may be non-commutative.

Scope and escapes. The theorem concerns compression followed by algebra generation with this inherited pointwise reality and a J_0-invariant window. The coarse correspondence supplies a separate opposite action and escapes the hypothesis, as demonstrated positively by Theorem 16.2. Other representations or real structures require their own order-zero calculation.

11. Proposition — The stated inherited grading cannot equal Γ_5

Verdict and hypotheses. The grading obstruction in Example 16.9 is correct under the following simultaneous requirements:

1. Every parent observable under consideration is even: $[\gamma, a] = 0$.
2. The finite grading is inherited as $g = P\gamma P|_{P\mathcal{H}}$.
3. This compression is an actual grading: $g^2 = 1$.
4. The retained observables include $Q(x_a) = \Gamma_a/5$.

Under these hypotheses, $g = \Gamma_5$ is impossible.

Proof. Even without initially assuming $[P, \gamma] = 0$, the grading identity implies

$$0 = P - (P\gamma P)^2 = P\gamma(1-P)\gamma P = \big((1-P)\gamma P\big)^*\big((1-P)\gamma P\big).$$

Hence $(1-P)\gamma P = 0$, and self-adjointness gives $[P,\gamma] = 0$. Therefore

$$[g, Q(a)] = P[\gamma, a]P = 0$$

for every retained even observable.

But

$$[\Gamma_5, Q(x_1)] = \frac{1}{5}[\Gamma_5, \Gamma_1] = \frac{2}{5}\Gamma_5\Gamma_1 \neq 0.$$

Thus the proposed grading and all the stated coordinate compressions cannot simultaneously be inherited in this manner. ■

Scope and escapes. The conclusion does not exclude Γ_5 as additional finite data or as a grading of a prealgebra. Possible changes include compressing a graded prealgebra with odd observables, restricting to the algebra commuting with the proposed grading, or constructing an additional grading. Changing the window alone does not remove the contradiction while retaining all four hypotheses. In the specified Dolbeault kernel, the inherited grading is already scalar.

The three checks above agree with the corrected repository statements. The recorded author is Claude Opus 5.5, and the existing independent Terra checks appear in the Lemma 11 report, the Proposition 16.1 report, and the Example 16.9 report. The present response adds a sequential Codex check; no subagents or additional model consultations were used.

Consequence for T0b and T11

For T0b, an exact interval parent supplies a rank-one connector with KO-6 signs, and a parent with chosen Hom fibres and constant matrix multipliers realizes the classified connecting block and its algebra. Deriving that algebra, grading and connector from the commutative twistor parent remains open: scalar wall agreement in the original window does not perform the selection. For T11, same-chirality boundary multiplets require the type-(a) or type-(b) comparison, and the unbroken Hořava–Witten boundary representations supply no identified finite mass connector.

Numerical check

By Claude Opus 5.5, in Python (numpy), 3 Nov 2026, in a temporary directory, not kept. On W = V = C^4 with random elements:

- T_kappa x = kappa K x K with K = v w*, the rank-one map (5): J D = D J, J Gamma = -Gamma J, D Gamma = -Gamma D; complex rank 1; real dimension of A(T_kappa) inside the 40-dimensional H + H + M_4(C) equal to 24, by linear algebra.
- K of rank two on W_+: dimension 16, as in the proof of Result 2, and x -> K x K has rank 4.
- On C + H + M_3(C): order zero, and order one with defect about 10^-15, and [D, a] = 0 with defect about 10^-15. For a random element off this algebra the order-one defect is about 40.
- The example at the end of Result 1: order-one defect 0, and a nonzero defect of left intertwining.
- Result 4: the values in (15), with a quadrature. By hand: the eigenvectors (13), their boundary conditions, the action of (11), and the value of the commutator.
- Equation (21): by hand, including the compression to m (S + S*).
- Pi_p is a projection of rank 2 and 2 Pi_p - 1 = p_a Gamma_a; the bound (26) on a random sample.

Against the source: (3) is equation (comrel) of arXiv:0706.3688, read in the LaTeX source. The proof there also takes an intertwiner distinct from the Dirac block, in line with the distinction between T and R in Result 1. The other statements are checked by the referee.

Readings

By Claude Opus 5.5, as proposals, corrected after the referee report.

1. The connector is the Majorana block. In the Standard Model dictionary (Chamseddine-Connes-Marcolli 2007, section 2, quoted from memory), W_+ holds (nu_R, e_R) and V = C^4 holds the lepton line

and the three colours, so that with w along nu_R and v along the lepton line, the vector e = w v* is nu_R, and the off-diagonal block of D_F acts on nu_R alone. After this identification, and an overall grading convention if needed, T = m S in Result 5 has the form of the one-generation Majorana block (equation (2.34) there, as the referee checked), with m = arctan(t)/l. In this window m < pi/(2l), so the first level of the tower bounds it, and for t of order one m is of the order of that level. As a pattern, this agrees with the placement of the Majorana mass near the cutoff in that model. Three identical copies give three equal masses, and a general three-generation Majorana matrix needs additional flavour data.

2. The connector as a marked pair of points and a coupling. Since T_kappa x = kappa K x K with K = v w*, the data of the connector are a complex line w in W = C^4, a complex line v in V = C^4, that is, a point of CP^3 on each sheet, and the coupling kappa. In Example 16.9, the point w lies over a point p of S^4 in the twistor fibration, and Pi_p = (1 + p_a Gamma_a)/2 projects onto its quaternionic line. Since Q(x_a) = Gamma_a/5, the grading is

```
2 Pi_p - 1 = p_a Gamma_a = 5 Q(p . x),
```

the compression of the height function at p. So one marked point fixes the quaternionic line and the complex line w, and a second one fixes v. Sp(2) acts transitively on the unit vectors of H^2, and U(4) on those of C^4, so at fixed nonzero |kappa| all line choices are conjugate, in agreement with the uniqueness in Result 2. The modulus |kappa| remains, rephasing w and v changes K x K by a phase, and other finite data can introduce relative moduli. The inputs are then: mark a point on each sheet, with a coupling and a prescription for evaluation and normalization. This has the form (22) with one channel per sheet.

The height function 5 Q(p . x) fails to commute with the other coordinate compressions, so it can serve as additional finite grading data only after restricting to its even algebra. This evades the inherited-grading hypothesis of Result 11, and it does not derive the grading from the even parent. Result 7 shows that scalar agreement at the points does not do this in the unchanged windows, so a coupled window with the data listed after Result 7 is needed. Whether a parent with an interface at those points yields this connector is open.

3. The wall coupling as a Wilson line. Equation (21) says that the wall coupling X = tan(m l) is unitarily equivalent to the uncoupled boundary conditions with a constant matrix connection m F, of relative endpoint rotation angle theta = m l, and the mass is that angle over the length, k = theta/l. On an interval, open-path transport depends on endpoint frames, and the physical quantity is the transport together with the boundary conditions. The comparison with gauge-Higgs unification (Y. Hosotani, Phys. Lett. B 126 (1983) 309; Hosotani, Noda and Takenaga, arXiv:hep-ph/0403106) is an analogy at this stage: the calculation supplies a fixed matrix connection, with no Higgs fluctuation or effective potential. It bears on the question of mass as internal momentum in the steer for the next paper, and compares with the link of lemma 19 and the covariant projection of lemma 14.

4. Next calculation. The connector commutes with B, so it gives no finite one-forms and no Higgs field. The next test is one interval parent carrying both the Dirac Yukawa block of lemma 12, odd for g and diagonal in the sheets, and this connector, for instance at opposite walls: whether order one survives on B, and what the joint form of (21) is.

Part 21

The separate role of the top: removal identities and counting

Prompted by the observation that the top has a separate role, which matters more than the loss of an electroweak partner for b_L. Proposition 21.1 and the counts below are exact calculations by Claude Opus 5.5, in Python with exact fractions. The readings in the last section are proposals, and nothing there is derived. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): one error, in Proposition 21.2, and several amendments, incorporated below.

Outcome. The spectral action detects the top: in the exact quadratic action of lemma 18, among single charged species only an up-type quark can be removed while keeping both the neutral-current value cos^2

theta_NC = 5/8 and the ratio 8/3 of the photon and gluon coefficients. Removing it leaves 84 of the 96 states of the finite Hilbert space. Those 84 are the seed of an SO(30), inside SO(32), whose adjoint contains the charged scalar counts of three generations, and 84 is also the number of on-shell components of the three-form of eleven-dimensional supergravity.

Removal identities

For a Dirac species with charges q_L, q_R in the coordinates (T_3, Y), equation (33) of lemma 18 gives a contribution to the neutral kinetic matrix K with weight e^{-x}. The off-diagonal charge is (q_L - q_R)/2 = (T_3/2)(1, -1), which is orthogonal to the photon direction (1, 1). So the combinations K_33 + K_3Y and K_YY + K_3Y depend on x only through the factor e^{-x} of each massive species, and cos^2 theta_NC = (K_YY + K_3Y)/(K_33 + 2 K_3Y + K_YY) is determined by the sums of T_3 Q and Y Q.

Proposition 21.1. Per species, with colour multiplicity, on the particle trace for the first two columns and with charge conjugates and both chiralities for the last two:

species	sum T_3 Q	sum Y Q	Q^2 coefficient	colour coefficient
up-type quark	1	5/3	16/3	2
down-type quark	1/2	1/6	4/3	2
charged lepton	1/2	3/2	4	0
neutrino	0	0	0	0
three generations	6	10	32	12

Hence the ratios of the whole, 5/3 and 8/3, are those of an up-type quark alone. As one species is removed, that is, as its mass tends to infinity at fixed cutoff in the setting of lemma 18:

removed	cos^2 theta_NC	photon/gluon coefficient
up-type quark	5/8	8/3
down-type quark	59/92	46/15
charged lepton	17/28	7/3

So among single charged species only an up-type quark preserves both values. For an up-type quark they hold at every x, as in Proposition 6 of lemma 18. Within one generation, the down-type quark and charged lepton together contribute exactly what the up-type quark contributes: (1, 5/3; 16/3, 2).

Proof. Arithmetic with the Standard Model charges, Y = Q - T_3. The last statement is direct addition. ■

The scope is that of lemma 18: classical bosonic trace, with these definitions of the coefficients and of the angle, three generations, and the other species with the mass assignments of that calculation. Other definitions of the weak angle do not stay at 5/8 (Proposition 6 there). At the endpoint where the neutral mass vanishes, the massive neutral direction is no longer distinguished, so its angle is read as the limit from finite mass, as lemma 18 states.

The state count in the action

Proposition 21.2. With three generations and nu_R, dim H_F = 96 (Chamseddine-Connes-Marcolli). With only an up-type Dirac mass m and no Majorana mass, the constant-background density of lemma 18 has finite trace Tr exp(-V_0^{2/Lambda}2) = 84 + 12 e^{-x} for the heat cutoff. So the coefficient of the vacuum density moves from 96 to 84 along the radial direction, with no finite nonzero stationary point on this free radial ray (No-go theorem 9 and Proposition 11 of lemma 18). This concerns the literal flat-space action with no added potential or constraint. Compact support can give a plateau; sign-changing cutoffs, additional terms, constraints and paths with an additional fixed noncommuting mass lie outside it. At m = 0 the derivative is zero.

Proof. The top block has 3 colours, 2 chiralities and charge conjugates, so 12 states, each with V_0^2 = m^2. ■

This also gives the top-only family a place in the radial problem, as the endpoint of a descending direction. A radial direction giving other species masses descends further, so this is the endpoint for that direction only.

SO(30)

Let the five quarks other than the top, in three colours, with their conjugates, form the vector 30 = (5, 3) + (5-bar, 3-bar) of SU(5) x SU(3) inside SO(30). The adjoint, of dimension 435, is

```
Lambda^2(30) = (24, 8) + (24, 1) + (1, 8) + (1, 1) + (10, 6) + (15, 3-bar) + conjugates of the last two
```

With Q = 2/3 for u, c and Q = -1/3 for d, s, b, that is, Q = diag(3/5, 3/5, -2/5, -2/5, -2/5) + (1/15) I_5, which includes the U(1) commuting with SU(5), the charges are:

representation	charge: number
(15, 3-bar)	-2/3: 6; +1/3: 6; +4/3: 3
(10, 6)	-2/3: 3; +1/3: 6; +4/3: 1
(24, 1) + (1, 1)	+1: 6; -1: 6; 0: 13

Three generations of scalar partners in a supersymmetric Standard Model with nu_R need, for antisquarks in the 3-bar, 6 of charge -2/3 and 6 of charge +1/3, and for sleptons 6 of charge +1, 6 of charge -1 and 12 neutral real states. So (15, 3-bar) and (24, 1) + (1, 1) contain exactly the charged counts. What is left over in them is three triplets of charge +4/3 and one neutral state. The counting of 15 diquarks and 25 mesons, with the charge +4/3 states, and the remark that the top is unable to form them, go back to an early preprint (hep-ph/0512065), which finds neither SO(30) nor a role for SO(32). What SO(30) adds is one group whose adjoint contains both families. It also sits in SO(32), with 496 -> (435, 1) + (1, 1) + (30, 2) under SO(30) x SO(2), which hints at a role for SO(32); we have not explored this. These are representation and charge counts: identifying the entries with physical scalar partners needs a field construction, spin and chirality assignments, interactions and dynamics.

The two families of 84

Proposition 21.3. Count four states per Dirac fermion and colour, and take Dirac neutrinos. Then

```
96 = 84 + 12:   five quarks and all leptons (60 + 12 + 12), and the top (12);
96 = 84 + 12:   all electrically charged fermions (72 + 12), and the neutrinos (12).
```

The two families of 84 overlap in 72 states, and each is the other with the top and the neutrinos exchanged. Without nu_R, the first family has 78 states and the second stays at 84. In eleven dimensions, the three-form C_3 has C(9, 3) = 84 on-shell components, and its dual C_6 has C(9, 6) = 84. The 84 in H_F is a complex finite-Hilbert-space dimension, and the three-form count is a real bosonic polarization count, so an identification needs a map of representations and degrees of freedom.

Readings, as proposals

These are proposals to test, with no derivation behind them.

1. **Seiberg duality.** For SU(3) with five flavours, N_f = 5 lies in the conformal window 3 N_c/2 < N_f < 3 N_c, and the magnetic dual is SU(N_f - N_c) = SU(2) with five flavours and a 5 x 5 gauge-singlet meson chiral superfield coupled by W = M q q-tilde (N. Seiberg, Nucl. Phys. B 435 (1995) 129, arXiv:hep-th/9411149, checked by the referee). The removal of the top is what gives N_f = 5, and SU(2) is the magnetic group there. To become a statement this needs an N = 1 supersymmetric theory with the required massless matter and dynamics, for example an emergent one, with an operator and

anomaly map. Five-flavour ordinary QCD alone does not supply the duality, and the magnetic SU(2) needs a separate identification with any electroweak group.

2. **Branes.** The first family of 84 could relate to C_3 or to an M2 or M5 brane, and the second family to the dual object. The exchange of the top and the neutrinos would then be the content of the duality on the Standard Model side. The counts match, and equality of the number is insufficient: a representation and dynamics must relate the finite fermionic states to the bosonic form or brane degrees of freedom. The pairing of item 1 is one candidate. The first family also needs Dirac neutrinos for its 84, and the second does not.
3. **Relation to the odd labels.** The removal of the top leaves the five quarks and the leptons, on which SU(3) x U(1)_em acts with b_L paired with b_R. So the loss of the electroweak partner of b_L has no role at the label 9, where SU(2) is absent. A statement here needs the required involution on this finite sector, including its action on D, J, the algebra and its opposite; the vectorlike charges alone do not establish it, and the neutral states still need treatment.

Numerical check

Python with exact fractions, run on 3 Nov 2026 in a temporary directory, not kept: all tables above, and the dimension $435 = 224 + 1 + 105 + 105$ with $105 = 60 + 45$ and $224 = 192 + 24 + 8$.

Part 22

The finite Dirac operator as a boundary condition on an interval

For STRATEGY T0, T0b, T10 and T13, following lemma 12 and lemma 20. Proved by Claude Opus 5.5, with the identities checked in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): two errors and one gap as first written, and amendments; all are incorporated below.

Outcome. Let (H_F, D_F, J_F, Gamma_F) be finite data of signs (+, +, -), and let l ||D_F|| < pi/2. Then D_F, with J_F and Gamma_F, is exactly the lowest spectral window of the pure derivative on an interval of length l, with no bulk mass and no Yukawa term, on the domain fixed by the boundary coupling tan(l D_F). So all the eigenvalues of D_F, Yukawa and Majorana blocks alike, are quantized internal momenta k with tan(k l) equal to a singular value of the coupling. The input is the finite triple: the content is that a consistent parent exists, with these properties:

- Preservation of the domain forces the observables at the wall into C_F, the commutant of D_F in A_F, while A_F acts freely in the bulk.
- The parent satisfies order one for any finite data with order zero.
- The compression is the exact average Q_l(a) = (2l)^-1 int_{-l}^{l} e^{ixD_F} a e^{-ixD_F} dx, so it tends to the finite representation as l -> 0, with defects of order l^2.
- The scalar inner fluctuations of the parent are pure gauge under its allowed endpoint gauge group. On the moving window, they act by the gauge orbit of D_F through the transport along the interval. This reproduces the quadratic term of Chamseddine-Connes-van Suijlekom on the orbit, and it leaves the singular values fixed, so the radial mode of the Higgs field is absent from this parent.

With the family X(l) = tan(l M) and l -> 0, the couplings of the light fermions vanish as m l at fixed mass, the tower decouples, and the broken gauge directions go to pi/(2l). The parent algebra is matrix-valued, so T0 for a commutative parent remains open.

Setting

Let H_F = H_+ (+) H_- be finite-dimensional, Gamma_F = diag(1, -1), let c : H_+ -> H_- be antiunitary, and put J_F(a, b) = (c^-1 b, c a), so that J_F^2 = 1 and J_F Gamma_F = -Gamma_F J_F. Let D_F = [[0, M*], [M, 0]] with M : H_+ -> H_-. Then J_F D_F = D_F J_F is equivalent to M = c M* c. By the Takagi factorization, in bases related by c, we may write M = V |M| with a unitary V satisfying V = c V* c.

Let l > 0 with l ||D_F|| < pi/2, and define X : H_- -> H_+ by

```
tan(l D_F) = [[0, X], [X\*, 0]],   that is,   X = tan(l |M|) V\*.
```

Since J_F commutes with real functions of D_F, c X = X* c^-1. Conversely, every X with c X = X* c^-1 arises from exactly one such D_F, namely D_F = l^-1 arctan of the block operator. In the limit, X = l M* + O(l^3).

The parent. On H_l = L^2([0, l]; C^2) (x) (H_+ (+) H_-), write psi = (Q_L, Q_R; U_L, U_R), with Q valued in H_+ and U in H_-. Let (D psi)_L = psi_R' and (D psi)_R = -psi_L' on both fields, the operator of lemma 12 with no bulk mass, on the domain

```
Q_R(0) = 0,   U_L(0) = 0,   Q_R(l) = X U_R(l),   U_L(l) = -X\* Q_L(l).
```

Call it D_X. Put Gamma^ psi = (Q_L, -Q_R; U_L, -U_R) and J^ psi = (c^-1 U_R, -c^-1 U_L; -c Q_R, c Q_L). Let (E psi)_L = psi_R and (E psi)_R = -psi_L, so that D_X = E d/dx on its domain.

In the Standard Model form of the classification, with E = Hom(V, W) and W = W_+ (+) W_-, one takes H_+ = E_+ (+) E_-* and H_- = E_- (+) E_+*, with c the adjoint map. At X = 0, the Q-fields have boundary type (a) and the U-fields type (b), in the terms of lemma 09; for X != 0 the modes continue from those zero modes, which generally lift. A single matrix X encodes the Yukawa blocks and the Majorana block together. Its entries need not keep the original block pattern, since the tangent brings in higher powers of D_F: for M = [[a, b], [b, 0]] real symmetric, X_22 = (l^3/3) a b^2 + O(l^5).

The window

Theorem 22.1. D_X is self-adjoint with compact resolvent, and J^^2 = 1, J^ D_X = D_X J^, J^ Gamma^ = -Gamma^ J^, Gamma^ D_X = -D_X Gamma^. Let R(x) = exp(x Gamma_F D_F), and define W : H_F -> H_l so that for phi = (a, b), the L-components (Q_L, U_L) of W phi at x are l^{-1/2} R(x)(a, 0), and the R-components (Q_R, U_R) are l^{-1/2} R(x)(0, b). Then W is an isometry and

```
D_X W = W D_F,    J^ W = W J_F,    Gamma^ W = W Gamma_F.
```

With sigma_k the singular values of M and theta_k = l sigma_k, the signed spectrum of D_X is (n pi +- theta_k)/l, n in Z, with multiplicities added when values coincide; a zero singular value gives a zero mode of multiplicity two. So the spectrum of |D_X| consists of the sigma_k and of (n pi +- theta_k)/l for n >= 1, and ran W is the spectral subspace of |D_X| <= ||D_F||, and the next level is at least (pi - l ||D_F||)/l.

Proof. Self-adjointness is Lemma 12.1. Since Gamma_F D_F is anti-Hermitian, R(x) is unitary, and with C(x) = cos(x |M|), C'(x) = cos(x |M*|), S(x) = V sin(x |M|),

```
R(x)(a, 0) = (C(x) a, -S(x) a),    R(x)(0, b) = (S(x)\* b, C'(x) b).
```

Isometry follows from the unitarity of R(x) and the orthogonality of the L- and R-components. The conditions at 0 hold since S(0) = 0. At l, using V* cos(l |M*|) = cos(l |M|) V*, we get X C'(l) = S(l)* and X* C(l) = S(l), which are the conditions at the wall. For the derivative, the L-part of D W phi is d/dx R(x)(0, b) = R(x) Gamma_F D_F (0, b) = R(x)(M* b, 0), and the R-part is -d/dx R(x)(a, 0) = R(x)(0, M a); together these are W D_F phi. The relations for Gamma^ are direct. For J^, use c |M| c^-1 = |M*| and V = c V* c, which give c^-1 C' = C c^-1 and c S(x)* = S(x) c^-1. The signs of the parent follow in the same way, using c X = X* c^-1 for the domain. For the spectrum, constant unitaries on Q and on U commute with E d/dx and bring X to diag(tan theta_k), which reduces to Example 12.5 in each channel. ■

Corollary 22.2 (conjugated form). D_X = R (D_0 + E Gamma_F D_F) R*, with R acting pointwise on the pair (Q, U) and D_0 the uncoupled operator, X = 0. So after this unitary change of domain, D_F is a constant matrix connection, odd for the grading, with relative endpoint rotation R(l). This is equation (21) of lemma 20, for general D_F.

Proof. D(R phi) = R(D phi + E R* R' phi), and R* R' = Gamma_F D_F. For phi in the uncoupled domain, the computation at the wall in the proof above shows that R phi satisfies the coupled conditions. ■

So in this family a fermion mass is a momentum: D_X has no mass term, and the light eigenvalue k = theta/l is fixed by tan(k l) = tan(theta), a singular value of X. In the finite description, the condition relating k to l is lost and the value appears as free data in D_F.

The parent as a real spectral triple

Let (A_F, H_F, D_F, J_F, Gamma_F) be finite data as above, with an even representation pi = pi_+ (+) pi_- of a unital finite-dimensional real or complex *-algebra A_F, faithful, unital and satisfying order zero, with pi°(b) = J_F pi(b)* J_F^-1. Order one is not assumed. Let

```
C_F = {a in A_F : [pi(a), D_F] = 0},    A_l = {f in C^infinity([0, l], A_F) : f(l) in C_F},
```

with f acting pointwise by pi_+(f(x)) on Q and pi_-(f(x)) on U.

Theorem 22.3.

1. A function f in C^infinity([0, l], A_F) preserves Dom D_X if and only if f(l) lies in C_F.
2. (A_l, H_l, D_X, J^, Gamma^) is a real even spectral triple, with [D_X, pi(f)] = E pi(f') and J^ pi(g)* J^^-1 = pi°(g) pointwise.
3. It satisfies order zero and order one, with signs (+, +, -).

So A_F acts freely in the bulk, and the boundary condition restricts the observables at the wall to the commutant of D_F. In the Standard Model data with nonzero Yukawa blocks, this is the algebra {(lambda, lambda, m)} of the photon and gluons; with vanishing Yukawa blocks the commutant is larger. Order one holds in this parent whatever the finite data, for its pointwise algebra: the finite condition appears only on the truncation, and arbitrary compressions need not preserve either order condition.

Proof. Item 1: at l, the traces Q_L(l) and U_R(l) are free, and applying f to the conditions gives pi_+(f(l)) X = X pi_-(f(l)) and pi_-(f(l)) X* = X* pi_+(f(l)), as in Result 3 of lemma 20. That is [pi(f(l)), tan(l D_F)] = 0, equivalent to [pi(f(l)), D_F] = 0 because l ||D_F|| < pi/2 makes D_F a function of tan(l D_F). At 0 the conditions are preserved since pi is even. Item 2: the commutator is the Leibniz rule. Conjugation by J^ acts on (Q, U) as J_F does, and pi(g) acts identically on the two spin components, which gives the second formula. This preserves the domain since J_F commutes with D_F. Item 3: pi(f(x)) and pi°(g(x)) commute at each x by finite order zero, and E commutes with both; so [E pi(f'), pi°(g)] = E [pi(f'), pi°(g)] = 0. ■

This is consistent with Counterexample 4 of lemma 20, whose finite data A_F = C + C, D_F = m sigma_1 fail order one while the parent satisfies it.

The compression

Let Q_l(T) = W* T W for a bounded operator T on H_l. For T(x) a bounded function with values in the even operators on H_F, acting pointwise and identically on both spin components, we write T for the multiplication operator.

Theorem 22.4.

```
Q_l(T) = l^-1 int_0^l cos(x ad_{D_F})(T(x)) dx,      Q_l(E T) = -l^-1 int_0^l sin(x ad_{D_F})(T(x)) dx.
```

For constant a in A_F,

```
Q_l(a) = sinc(l ad_{D_F})(pi(a)) = (2l)^-1 int_{-l}^{l} e^{ix D_F} pi(a) e^{-ix D_F} dx,
```

so in an eigenbasis of D_F with eigenvalues mu_i, the matrix elements are pi(a)_{ij} sin(l (mu_i - mu_j))/(l (mu_i - mu_j)). Moreover:

1. Q_l(A_l) = Q_l(C^infinity([0, l], A_F)), so the image of the parent algebra contains Q_l(a) for every a in A_F.
2. ||Q_l(a) - pi(a)|| <= (l^2/6) |||[D_F, [D_F, pi(a)]]|| <= (2/3) l^2 ||D_F||^2 ||a||.
3. Q_l([D_X, f]) = [D_F, Q_l(f)] for f in A_l.

4. With a' = Q_l(a) and b' = J_F Q_l(b)* J_F^-1, |||[a', b']|| <= (8/3) l^2 ||D_F||^2 ||a|| ||b||, and |||[[D_F, a'], b'] - [[D_F, pi(a)], pi°(b)]|| <= (16/3) l^2 ||D_F||^3 ||a|| ||b||.

So as l -> 0 with D_F fixed, D_F, J_F and Gamma_F stay exact, and the compressed observables converge to pi(A_F) with rate l^2. The finite order conditions hold in the limit exactly when they hold for the finite data.

Proof. From the definition of W, the expectation of T splits into the L- and R-parts, so Q_l(T) is l^-1 int of the Gamma_F-even part of R(x)* T(x) R(x). Put K = Gamma_F D_F. For even Z, ad_K Z = Gamma_F [D_F, Z], which is odd, and ad_K (Gamma_F [D_F, Z]) = -[D_F, [D_F, Z]]. So the even part of e^{-xK} T e^{xK} is cos(x ad_{D_F})(T). Likewise, E exchanges the L- and R-parts with a sign, so Q_l(E T) is l^-1 int of Gamma_F times the odd part of R* T R, and that odd part is -Gamma_F sin(x ad_{D_F})(T). The formula for constant a follows from cos(x ad_D)(Z) = (e^{ixD} Z e^{-ixD} + e^{-ixD} Z e^{ixD})/2.

Item 1: the map f -> Lambda(Q_l(f)), for a linear functional Lambda, is integration against a smooth density. If it vanishes on A_l, which contains all f of compact support in [0, l), the density vanishes on [0, l), hence everywhere by continuity. Item 2: by Taylor's formula with integral remainder, ||cos(x ad_D)(Z) - Z|| <= (x^2/2) ||ad_D^2 Z||, since cos(x ad_D) has norm at most 1 by the average formula; averaging over x gives l^2/6. Item 3: P commutes with D_X. It also follows from the two formulas by integration by parts, where the boundary term at l vanishes because f(l) is in C_F. Item 4: J_F commutes with D_F, so b' is the same average of pi°(b). Expand both commutators with the bound of item 2 and |||[D_F, Z]|| <= 2 ||D_F|| ||Z||. ■

The convergence is a statement about the maps Q_l, on the operator systems, in the setting of Connes and van Suijlekom (arXiv:2004.14115, sections 1 and 3). The generated algebra is not continuous at l = 0. An analytic example, from the referee: let M = [[0, 0, 1], [0, 0, 1], [1, 1, 0]], D = [[0, M], [M, 0]], with A_F = C^3 acting diagonally and equally on the two blocks. With P_1, P_2 the first two channel projections, [Q_l(P_1), Q_l(P_2)]_{12} = -l^2/3 + O(l^4), so the compressions of commuting projections generate a noncommutative algebra for small nonzero l. On the one-generation Standard Model data, a numerical closure with tolerance 10^-7 gives complex dimension 15 for the algebra generated by pi(B), with B = C + H + M_3(C), and 30 for Q_l(B) at l = 0.3; these are numerical values with no certified rank.

Scalar fluctuations

Theorem 22.5. Let h be smooth with values in the anti-Hermitian elements of A_F. The self-adjoint one-forms of the parent are exactly the E pi(h). Let u' = -h u, u(0) = 1, and let w = u(l), a unitary of A_F. Then

```
D_X + E pi(h) = pi(u) D_{X_w} pi(u)\*,      X_w = pi_+(w)\* X pi_-(w),
```

and the window of D_{X_w} carries pi(w)* D_F pi(w). With the real structure, for A = E pi(h) we have J^ A J^^-1 = E J_F pi(h) J_F^-1, and D_X + A + J^ A J^^-1 has the window U_w* D_F U_w, with U_w = pi(w) J_F pi(w) J_F^-1, and this commutes with J_F. Moreover, with u~(x) = u(x) w*, which lies in A_l since u~(l) = 1, and U~ = pi(u~) J_F pi(u~) J_F^-1,

```
D_X + A + J^ A J^^-1 = U~ D_X U~\*.
```

So every scalar connection in this model is gauge equivalent to the original parent under the allowed endpoint gauge group (referee).

Proof. A one-form is sum f_1 [D_X, f_2] = E pi(sum f_1 f_2'), and E pi(h) is self-adjoint exactly when h is anti-Hermitian, since E is anti-Hermitian and commutes with pi(h) and pi is faithful. Every smooth such h arises: with b(x) = int_l^x h, we have b(l) = 0, so b is in A_l, and [D_X, b] = E pi(h). Next, pi(u)* D pi(u) = D + E pi(u* u'), and pi(u)* E pi(h) pi(u) = E pi(u* h u), so the sum is D with no connection term. At 0 the conditions are unchanged. At l, the substitution psi = pi(u) phi gives the conditions with X_w. Then tan(l pi(w)* D_F pi(w)) has the block X_w. For general w, this operator need not commute with J_F, so we use the spectrum and the isometry of Theorem 22.1 without its real-structure part. For the real fluctuation, J^ commutes with E, u(x) is replaced by pi(u) J_F pi(u) J_F^-1, using finite order zero,

and U_w commutes with J_F, so the real window follows. The last formula follows by differentiating, with the endpoint condition. ■

Corollary 22.6 (the quadratic term as transport). By Chamseddine-Connes-van Suijlekom (arXiv:1304.7583, section III), for unitary v in A_F, with U = v J v J^-1, one has U D U* = D + A_(1) + A~*(1)* + *A*(2), for the universal form v delta(v*). Taking v = w*, on the gauge orbit the fluctuation of the parent, which satisfies order one and has no quadratic term, gives on the moving window exactly the fluctuated operator of that framework (equations (10)-(11) of section III.2 there), quadratic term included, for finite data that fail order one. This answers the first open point of lemma 19 on these gauge orbits and with the moving window. For the two-point data with J = sigma_1 K and w = diag(e^{i alpha}, e^{i beta}), it gives the upper link m e^{-2 i (alpha - beta)}, which is m e^{2 i psi} of Corollary 19.2 with psi defined from v = w*. Theorem 22.5 and the covariant projection of lemma 14 share a transport principle. The fixed compression, as in lemma 13, is a different object: by Theorem 22.4, it gives Q_l(E pi(h)) = -[D_F, l^-1 int_0^l x pi(h) dx] + O(l^3 ||D_F||^3 ||h||) for h fixed, so that for constant h it is -(l/2)[D_F, h], while the first derivative of the transported operator is -l [D_F, h]. The difference is the contribution of the moving spectral projection.

No-go theorem 22.7 (the radial mode). *Hypotheses:* the parent above, on one interval, with fixed X, no bulk masses, and scalar inner fluctuations from the even algebra A_l, with the moving spectral window. *Conclusion:* the fluctuations leave all singular values of the finite Dirac operator fixed. In the usual Standard Model representation with nonzero Yukawa data, the normalized shifted Higgs quaternion runs through the unit quaternions, a three-dimensional gauge orbit, with no radial mode. Degenerate data can have a smaller orbit. A fixed compression can show changes in singular values at second order, so the window has to be named.

Proof. Theorem 22.5: D_F and pi(w)* D_F pi(w) are unitarily equivalent. The same holds with U_w. ■

Known escapes.

These lie outside the hypotheses, and each needs its own construction.

- The coupling X as a field on the wall, with its own kinetic term and potential.
- Connections odd for the grading, as in Corollary 22.2 and lemma 23: the fluctuation K -> K + delta K changes the singular values.
- Bulk masses that do not commute with the gauge action.
- Two or more internal dimensions, where a connection has curvature, although curvature alone does not prove a radial mode.
- An additional composite or other degree of freedom; a quantum effective action on the same configuration space does not by itself add a radial coordinate.
- A fluctuation of the finite data alone, as with the radial parameter of lemma 19, which is not the moving window.

Direction. On the interval, the even connection is pure gauge up to its boundary effect, so the finite orbit is all it can reach. The physical Higgs modulus must then be sought at the wall, in odd directions, or in more dimensions. At the 10 -> 9 endpoint this agrees with T10, where the Higgs is frozen.

The gauge sector and the 10 -> 9 family

Proposition 22.8. Consider the product of M^4 with this parent, with algebra of functions f(y, x) valued in A_F and f(y, l) in C_F. Then the gauge one-forms sum f_1 d_mu f_2 take values in C_F at x = l. Assume, in addition, a flat product, a positive five-dimensional Yang-Mills kinetic form, the trivial background, unrestricted endpoint variation in the allowed directions, and no additional boundary kinetic or mass terms. Then the directions in the Lie algebra of C_F have masses n pi/l, n >= 0, and the orthogonal directions have masses (n - 1/2) pi/l, n >= 1. The physical gauge algebra comes from the skew-adjoint elements through the real gauge representation, with its kernel and any unimodularity condition.

Proof. The first statement is direct evaluation at x = l. The rest is the interval spectrum in the gauge A_5 = 0, with Neumann conditions at 0, and with Neumann or Dirichlet conditions at l. ■

In the Standard Model data with nonzero Yukawas, M_W = M_Z = pi/(2l). The Lie algebra of C_F has dimension 10 before unimodularity; with the unimodularity condition of Chamseddine-Connes-Marcolli (arXiv:hep-th/0610241, section 2.5), the massless modes are the photon and the gluons. These are the boundary conditions of Higgsless models (Csaki, Grojean, Murayama, Pilo and Terning, Phys. Rev. D 69 (2004) 055006), and the fermion formula tan(m L) = M_D L is equation (5.6) of Csaki, Grojean, Hubisz, Shirman and Terning (Phys. Rev. D 70 (2004) 015012); LITERATURE.md, section C. Boundary spectral geometry for the Standard Model has been studied by Chamseddine and Connes (arXiv:1008.3980, section 5). What is added here, as far as our searches went, is the combination: the real structure, the finite triple as an exact window, the wall algebra C_F, the compression formula and the fluctuations. A wider claim of novelty would need a fuller comparison.

Corollary 22.9 (the 10 -> 9 family). Fix a block M_0 for the light fermions and a block Theta/l on an orthogonal channel, compatible with J_F and the grading, with 0 < Theta < pi/2, for the top and if wanted for the Majorana block. Let D_F(l) have M(l) = M_0 (+) Theta/l, and let l -> 0. Then:

1. X(l) has the light block tan(l |M_0|) V_0* = l M_0* + O(l^3) and a heavy block with singular values tan(Theta): the light couplings vanish as m l at fixed mass m.
2. The heavy channel has mass Theta/l, the broken gauge directions have mass pi/(2l), and the other levels diverge.
3. The massless gauge sector is C_F. Let P_L be the projection reducing D_F(l) to the fixed light block D_L. For constant a, P_L Q_l(a) P_L = sinc(l ad_{D_L})(P_L pi(a) P_L), with ||P_L Q_l(a) P_L - P_L pi(a) P_L|| <= (2/3) l^2 ||D_L||^2 ||pi(a)||, also when pi(a) mixes light and heavy channels. The limiting light compression need not be a representation, since products can pass through the heavy subspace. The full estimate of Theorem 22.4 does not apply, because l ||D_F(l)|| tends to Theta. For example, with D_F(l) = sigma_1 (x) diag(m, Theta/l) and pi(a) = 1 (x) sigma_1, one has Q_l(a) = sinc(Theta - l m) pi(a), which tends to sinc(Theta) pi(a).
4. With l = pi/(2 M_W), we get X_q ~ (pi/2) m_q/M_W, which has the form of y_q = sqrt(2) m_q/v -> 0 at fixed mass, as the coupling vanishes while the mass stays. Identifying X_q with the four-dimensional Yukawa coupling is a hypothesis.
5. The ratio is m_t/M_W = 2 Theta/pi < 1, and M_W/M_Z = 1.

So the 10 -> 9 proposal is realized at this level as a limit l -> 0, with W, Z and top at the tower scale and the light fermions keeping explicit masses. Item 5 does not reach m_t/M_W = sqrt(8/3), which belongs to the interior. This is a statement in the stated family: lowest level, no bulk mass, strict wall algebra. Bulk masses for the top, and a changed gauge wall condition, are directions beyond it.

A numerical comparison for the interior (Python). A separate model, with a finite wall scalar and Neumann conditions at 0, lies outside Proposition 22.8 and No-go theorem 22.7. In it, the mass of a broken direction satisfies t tan t = s, with t = M l and s = g^2 v^2 l^2/4 (for the Z, s/cos^2 theta_0), which has the form of M_k tan(M_k pi R) = -V in Csaki et al. (arXiv:hep-ph/0305237, section on mixed boundary conditions, read in the LaTeX source), with the sign convention of V there. If g'^2/g^2 = 3/5 is taken at all scales, M_W^2/M_Z^2 goes from 5/8 as s -> 0 to 1 as s -> infinity. It equals the measured 0.777 at M_W l = 1.03, with 1/l = 78 GeV and a next W level at 280 GeV; whether collider searches allow this would need the couplings, branching fractions and an applicable limit, which we have not analyzed. The top needs M_W l < 0.96 for m_t/M_W = sqrt(8/3), or M_W l < 0.73 for the measured ratio, in the lowest level with no bulk mass. So in the flat tree-level toy without bulk masses or running, the top and gauge requirements are incompatible. It does show the interior interpolating between the spectral-action value 5/8 and the endpoint value 1. The directions are bulk masses, warping and running.

What is input and what is derived

The finite data (A_F, H_F, D_F, J_F, Gamma_F) are all inputs, so the existence of the window is close to a reparametrization: X = tan(l D_F). The derived statements are the wall algebra C_F for that input, the order conditions of the parent, the compression formula with its rate, the form of the fluctuations, and the gauge spectrum under the stated assumption. The window identifies D_F, J_F and Gamma_F exactly, while the represented algebra is obtained through compression and is not an exact representation of A_F at

l > 0. In particular, this does not select A_F. T0 asks for a commutative parent: a natural next step is a two-stage parent, compressing functions on S^4 and CP^3 at level one to get A_F-valued functions on the interval, and then taking this window, where the difficulty is the same interface problem as after Result 7 of lemma 20.

Numerical check

By Claude Opus 5.5, in Python (numpy and scipy), 3 Nov 2026, in a temporary directory, not kept.

- Random complex symmetric M on C^5, l ||M|| = 1.3: D u_+(a) = u_-(M a) and D u_-(b) = u_+(M* b) by finite differences, to about 10^-9; boundary conditions and J^ to about 10^-15; norm of W to 10^-13.
- Q_l(a) = sinc(l ad)(a), by quadrature, to 2 x 10^-8. The formula for Q_l(E T), with a nonconstant T, to 10^-14.
- One-generation Standard Model data with Yukawas and the rank-one Majorana block, with signs, order zero and order one exact on B = C + H + M_3(C), and commutant of D in B of real dimension 20. At l = 0.1 and 0.03, ||Q_l(a) - a + (l^2/6)[D, [D, a]]|| is 10^-5 and 4 x 10^-8, and the order-one defect of Q_l is 1.5 x 10^-4 and 1.2 x 10^-5, which scales as l^2.
- The dimensions of the generated algebras quoted above, 15 and 30, from a numerical closure with tolerance 10^-7, not retained and not certified; at smaller l the numerical rank depends on the tolerance.
- The roots of t tan t = s, and the values quoted in the comparison for the interior.
- On the two-point data with J = sigma_1 K: D + A_(1) + A~*(1)* + *A*(2) = U D U* to 4 x 10^-16, and the link of U_w* D U_w equal to m e^{-2 i psi}.
- The bounds of Theorem 22.4, items 2 and 4, on 30 random samples of the one-generation data: largest ratios to the bound 0.33, 10^-4 and 10^-3.

Part 23

Two gradings on an interval: masses as principal angles, the Higgs modulus, and exact W and Z masses

For STRATEGY T0b, T5b, T10 and T13, following lemma 22. Proved by Claude Opus 5.5, with the identities checked in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): four errors and one gap as first written, and amendments, all incorporated below. The simpler form of the mass formulas, and the proof of item 3 of Theorem 23.6, are by the referee.

Outcome. A self-adjoint boundary condition for the derivative on an interval is a grading at each wall. The whole finite Dirac operator is then the relative position of two gradings: its singular values are the principal angles between them, divided by l, and

```
D_F = (1/2l) Gamma_0 log(Gamma_1 Gamma_0).
```

With a bulk prealgebra, this gives four results.

1. Aligned walls compress a prealgebra stable under the grading to its even part, with the grading inherited, which is the step that Example 16.9 had to add by hand.
2. Odd scalar fluctuations move the second grading, so they change the singular values and supply the radial mode that No-go theorem 22.7 excludes for even fluctuations. For M_2(H) the field space is S^4, and the Higgs modulus is half the geodesic distance between the points chosen by the two walls.
3. With a wall connector on one channel and a five-dimensional Yang-Mills action with common normalization, the masses are exact: sin^2(M_W l) = sin^2(2 theta)/2 and sin^2(M_Z l) = (4/5) sin^2(2 theta), for universal fermions of mass theta/l. At small theta this gives M_W^2 = 2 m^2 and M_Z^2 = (16/5) m^2, the constants in the lower bounds of the fully massive family of lemma 18, with $M_W^{2/M_Z}2$ tending to 5/8 at both ends.
4. The heat-cutoff spectral action on the interval is a Jacobi theta function of theta, decreasing on [0, pi/2]. This family lies outside the hypotheses of the radial no-go of lemma 18: the radial direction is

compact, with a stationary point at m = pi/(2l), the Connes-Lott value, where M_W = M_Z = 0.

Everything finite is still an input, and the parent algebra is matrix-valued. For a commutative parent, lemma 24 finds the window exactly, with spectral walls, while direct compression of scalar functions gives only R + R on the S^4 factor. The formulas for gauge masses depend on the assumed action. The relation to measured values fails in the flat toy, as stated at the end.

Two gradings

Let H be a finite-dimensional Hilbert space, and let Gamma_0, Gamma_1 be self-adjoint involutions on H. On L^2([0, l]; C^2) (x) H, with psi = (psi_L, psi_R), let D = E d/dx, where (E psi)_L = psi_R and (E psi)_R = -psi_L, on the domain

```
Gamma_0 psi_L(0) = psi_L(0),   Gamma_0 psi_R(0) = -psi_R(0),   Gamma_1 psi_L(l) = psi_L(l),   Gamma_1 ps
```

Call it D(Gamma_0, Gamma_1), and let Gamma^ act as +1 on psi_L and -1 on psi_R.

Theorem 23.1.

1. D(Gamma_0, Gamma_1) is self-adjoint with compact resolvent, and it anticommutes with Gamma^.
2. Suppose Gamma_0 and Gamma_1 have eigenspaces of equal dimensions, and that -1 is not an eigenvalue of Gamma_1 Gamma_0. Let theta_k in [0, pi/2) be the principal angles between the +1 eigenspaces. Then K = (1/2l) log(Gamma_1 Gamma_0), with the principal branch, is anti-Hermitian, Gamma_0 K Gamma_0 = -K, and D_F = Gamma_0 K is self-adjoint, odd for Gamma_0, with singular values theta_k/l on the +1 eigenspace. Moreover Gamma_1 = e^{lK} Gamma_0 e^{-lK}.
3. With H_+, H_- the eigenspaces of Gamma_0, and X the H_- -> H_+ block of tan(l D_F), as in lemma 22, D(Gamma_0, Gamma_1) = D_X, and the spectral part of Theorem 22.1 gives the spectrum and the window, with no real structure needed. Conversely, every D_F odd for Gamma_0 with l ||D_F|| < pi/2 arises from exactly one such Gamma_1.
4. Let an antiunitary J_1 satisfy J_1^2 = -1, J_1 Gamma_i J_1^-1 = -Gamma_i for i = 0, 1. Then (J^ psi)_L = J_1 psi_R, (J^ psi)_R = -J_1 psi_L defines a real structure of signs (+, +, -) for D(Gamma_0, Gamma_1). With J_F = Gamma_0 J_1 up to sign, J_F^2 = 1, J_F Gamma_0 = -Gamma_0 J_F and J_F D_F = D_F J_F, which are the signs of lemma 22.

So J_F itself does not anticommute with Gamma_1 in general: J_F Gamma_1 J_F^-1 = -Gamma_0 Gamma_1 Gamma_0, the reflection of Gamma_1 through Gamma_0.

Proof. Item 1: the boundary form is [<psi_R, phi_L> - <psi_L, phi_R>] evaluated between 0 and l, which vanishes on the domain because L-values and R-values lie in orthogonal eigenspaces at each wall, and these are half of the boundary data.

Item 2: U = Gamma_1 Gamma_0 is unitary and Gamma_0 U Gamma_0 = Gamma_0 Gamma_1 = U^-1, so K is anti-Hermitian and anticommutes with Gamma_0. The eigenvalues of U are e^{+-2 i theta_k}, by the standard two-projection decomposition, and K then has eigenvalues +-i theta_k/l. Since e^{lK} Gamma_0 e^{-lK} = e^{2lK} Gamma_0 = U Gamma_0 = Gamma_1, the last formula follows.

Item 3: in the notation of Theorem 22.1, R(l) = e^{lK} carries H_+ to the graph {(C a, -S a)} = graph(-X*), and H_- to graph(X). These are the +1 and -1 eigenspaces of Gamma_1, which is the condition at the wall. The converse uses the same formula, with uniqueness from injectivity of the exponential on the stated range.

Item 4: J^ commutes with E by direct computation, and J^^2 = -J_1^2 = 1. The domain is preserved since J_1 exchanges the eigenspaces of each Gamma_i. For the last statement, J_1 U J_1^-1 = U, so as J_1 is antilinear, J_1 K J_1^-1 = K in the sense of the principal branch, and J_F = Gamma_0 J_1 then satisfies J_F K J_F^-1 = -K and commutes with D_F = Gamma_0 K. ■

When theta_k = pi/2 the channel has type (c) of lemma 09, with no zero mode and lowest level pi/(2l), the Connes-Lott mass of Theorem A in the paper. There the principal logarithm and the isolated window cease to be unique.

The bulk prealgebra

Let P be a *-subalgebra of End(H), which need not commute with Gamma_0. Let

```
A(Gamma_0, Gamma_1) = {f in C^infinity([0, 1], P) : [f(0), Gamma_0] = 0, [f(1), Gamma_1] = 0},
```

acting pointwise and identically on both spin components.

Theorem 23.2.

1. A function f in C^infinity([0, 1], P) preserves the domain exactly when it lies in A(Gamma_0, Gamma_1). The commutators are [D, f] = E f', and Gamma^ commutes with every f, including those with values odd for Gamma_0.
2. Under the hypotheses of Theorem 23.1, item 2, with W the isometry of Theorem 22.1, Q(T) = 1^-1 int_0^1 E_0(R(x)* T(x) R(x)) dx, where E_0(Z) = (Z + Gamma_0 Z Gamma_0)/2 is the conditional expectation onto the even part.
3. *Aligned walls.* If Gamma_1 = Gamma_0, then D_F = 0, the window consists of constant L-values in H_+ and constant R-values in H_-, and Gamma^ restricts on it to Gamma_0. For an ambient constant multiplier a in P, Q(a) = E_0(a), and the image of the boundary algebra A(Gamma_0, Gamma_0) is E_0(P), by compactly supported profiles. If Gamma_0 P Gamma_0 = P, then E_0(P) is the even part of P, so the compressions of the prealgebra are exactly its even part, with the grading inherited from the even parent operator Gamma^. Without that hypothesis this can fail: for H = C^2, Gamma_0 = sigma_z and P generated by sigma_x + sigma_z, E_0(P) = span{1, sigma_z}, while the even part of P is C.
4. *Order conditions.* Let J_1 be as in Theorem 23.1 and let P° = {J_1 b* J_1^-1 : b in P} commute with P. Then the parent has order zero and order one for A(Gamma_0, Gamma_1).

Proof. Item 1: as in Theorem 22.3, using that the conditions at a wall say that L- and R-values lie in the eigenspaces of Gamma_i. Item 2: as in the proof of Theorem 22.4. Item 3 is the case R = 1. Item 4: [D, f] = E f', and pointwise commutation gives both conditions. ■

Item 3 evades the inherited-grading obstruction of Result 11 of lemma 20, because hypothesis 4 there fails: for odd a, here Q(a) = 0, while there the coordinates are retained as Q(x_a) = Gamma_a/5. So at the level of a matrix-valued parent this addresses the question left in step 3 of the framework note: the prealgebra is the bulk algebra, the even algebra is what the window sees, and the grading is a boundary condition. The commutative-parent and retained-coordinate questions remain open, and Theorem 4 of lemma 24 shows a real difference: scalar functions compressed directly give R + R on the S^4 factor, and H + H needs the matrix algebra generated at the intermediate stage.

Remark 23.3 (the twist). On H = E (+) E*, with J(x, y) = (y*, x*), the operator J_1 = Gamma_0 J is the one that anticommutes with both gradings. For an odd anti-Hermitian h in P, J_1 h J_1^-1 is y -> +y h on E*, where J alone gives y -> -y h. So the opposite representation b° = J_1 b* J_1^-1 incorporates on the W-factor the grading automorphism sigma(b) = gamma_W b gamma_W. In this parent the twist comes from the requirement that J_1 preserve both boundary conditions. Twisted spectral triples for the grand symmetry (Devastato, Lizzi and Martinetti, JHEP 01 (2014) 042, arXiv:1304.0415; Devastato and Martinetti, Math. Phys. Anal. Geom. 20 (2017) 2, arXiv:1411.1320, Eqs. (1.8)-(1.11)) use a different automorphism: their rho exchanges left and right spinorial indices and algebra summands, while conjugation by gamma_W is inner on M_2(H) and fixes its centre. So this is a structural analogy between involutive automorphisms, and an identification needs a comparison of the representations, opposites and first-order conditions.

Odd fluctuations and the radial mode

Theorem 23.4. Let h be smooth with values in the anti-Hermitian elements of P, and let u' = -h u, u(0) = 1, w = u(1). Then

```
D(Gamma_0, Gamma_1) + E h = u D(Gamma_0, w\* Gamma_1 w) u\*.
```

So scalar fluctuations move the second grading along its orbit under the unitary group of P. If h has an odd part, the principal angles between Gamma_0 and w* Gamma_1 w change in general.

Proof. u* D u = D + E u* u', and u* E h u = E u* h u, so the connection cancels. Since u(0) = 1, the wall at 0 is unchanged. At l, psi = u phi gives the stated grading for phi. ■

This lies outside No-go theorem 22.7, whose hypotheses include the even algebra: an even connection gives a w commuting with Gamma_0, and the angles are kept, while an odd connection changes them. For a real fluctuation, add the conjugate one-form: D + E h + J^(E h)J^^-1, with generator of the form (L_F, R_F) in Example 23.5. The invariant is a double coset. With H_i = Stab(Gamma_i) in the connected unitary group G of P, a gauge transformation removing h has the form u(x) c, and it preserves both endpoints exactly when c is in H_0 and w c in H_1, that is, when w is in H_1 H_0. So the scalar moduli of the parent are the double cosets H_1 G / H_0, the relative position of the two gradings. This is with the moving window, as in lemma 22.

Example 23.5 (M_2(H) and S^4). Let W = H^2 = C^4, with the Clifford generators Gamma_a of lemma 16, and for a unit vector p in R^5 put Gamma_p = p_a Gamma_a. Each grading of W with one-dimensional quaternionic +1 and -1 eigenspaces is such a Gamma_p, so these gradings form S^4 = Sp(2)/(Sp(1) x Sp(1)), the quaternionic projective line. For two points, Gamma_q Gamma_p = p . q + (p wedge q), with eigenvalues e^{+-i d}, where d is the geodesic distance; so all the principal angles equal theta = d/2.

Take H = Hom(V, W) (+) Hom(W, V), J(x, y) = (y*, x*), Gamma_0(x, y) = (gamma x, -y gamma) with gamma = Gamma_{p_0}, and K(x, y) = (theta/l)(F x, y F), with F an odd anti-Hermitian element of M_2(H), F^2 = -1. Then:

- D_F = Gamma_0 K has all singular values equal to theta/l: a universal Dirac mass, for all colours.
- The second grading is Gamma_1(x, y) = (Gamma_p x, -y Gamma_{p'}), where p is at distance 2 theta from p_0 and p' is its mirror image through p_0. J_1 = Gamma_0 J satisfies Theorem 23.1, item 4, while J alone does not.
- The coordinates (theta, q) on S^4, with F determined by the unit quaternion q, are the polar coordinates of a Higgs quaternion: theta is the radial coordinate, bounded by pi/2, and q gives three angular directions, undefined at the poles. At theta = pi/2 the finite logarithm and the lowest window are degenerate. Quotienting by the stabilizer leaves the radial interval. A canonically normalized field needs the scalar kinetic term, which we have not computed.

With bulk masses, by Theorem 12.6, the light masses are about (tan(theta)/l) Phi_Q Phi_U when small compared with the tower thresholds, with a universal matrix before flavour data; general finite-angle eigenvalues need the secular equation. This field space is the coset of the minimal composite Higgs model (Agashe, Contino and Pomarol, Nucl. Phys. B 719 (2005) 165, sections 2-3, with their h/f_pi corresponding to 2 theta). Their construction uses a warped interval and additional fermion and boundary data, so the shared coset does not identify the actions or potentials. The appearance here of the S^4 of the twistor window is an observation, and no identification between the two roles of S^4 is made.

The weak bosons

Take P = M_2(H) (+) M_4(C), acting on E = Hom(V, W) on the left and on the right. Assume the following.

- A five-dimensional Yang-Mills field valued in sp(2) (+) su(4), with the trace form Tr_W + Tr_V, so with one coupling (the normalization of the spectral action, g_3 = g_2 = sqrt(5/3) g_Y).
- At x = 0: Neumann for the stabilizer of p_0, Dirichlet for its complement. At x = l: Neumann for the stabilizer of p_l and of a connector, that is, the a with [a_W, Gamma_p] = 0, a_W w = lambda w and a_V v = lambda v, as in Results 2 and 5 of lemma 20, for a line w in the +1 eigenspace of Gamma_p and a line v in V; Dirichlet for the complement.
- The connector is prescribed gauge-breaking wall data here. If it is instead a term in the fermion boundary operator, as for a unit vector e in the +1 eigenspace of Gamma_1, f = J_1 e, K_c = |f><e| - |e><f| and Gamma_{1,c} = e^{eta K_c} Gamma_1 e^{-eta K_c}, then J_1 still anticommutes with Gamma_{1,c}, the parent order conditions are kept, and the affected fermion tower changes, so the universal fermion formula must be recalculated for that channel.

Theorem 23.6. With theta the angle above, the modes that are massless for all theta are exactly nine, the photon and the eight gluons, with three additional electroweak zero modes at theta = 0 and theta = pi/2. The lowest charged and neutral massive modes satisfy

```
sin^2(M_W l) = sin^2(2 theta)/2,
sin^2(M_Z l) = (4/5) sin^2(2 theta),
```

equivalently tan^2(M_W l) = sin^2(2 theta)/(1 + cos^2(2 theta)) and tan^2(M_Z l) = 4 sin^2(2 theta)/(1 + 4 cos^2(2 theta)).

The photon has $1/e^2 = (8/3)/g^2$. Moreover:

1. As theta -> 0, with m = theta/l: M_W^2 -> 2 m^2 and M_Z^2 -> (16/5) m^2, so $M_W^{2/M_Z}2$ -> 5/8.
2. As theta -> pi/2: M_W, M_Z -> 0, with $M_W^{2/M_Z}2$ -> 5/8, while m -> pi/(2l).
3. At theta = pi/4: M_W l = pi/4, tan(M_Z l) = 2, and $M_W^{2/M_Z}2 = ((\pi/4)/\arctan 2)^2 = 0.5032\ldots$, the global minimum. So $M_W^{2/M_Z}2$ stays in [0.5032, 0.625), with 5/8 as a limit at the ends, where the ratio itself is undefined.
4. m/M_W goes from 1/sqrt(2) at theta -> 0 to infinity at theta -> pi/2. It equals sqrt(8/3) at theta = 1.0589.

Proof. In the gauge A_5 = 0 the field is free, with cos(k x) on the Neumann directions at 0 and sin(k x) on the Dirichlet ones, and the condition at l holds in the frame rotated by Ad exp(theta F). The generators split according to the frequency of Ad F. Write L, R for su(2) on W_- and W_+, S = (L + R)/sqrt(2), A = (L - R)/sqrt(2), and O for the odd generators; Ad F has frequency 0 on S, and frequency 2 on the plane (A, O). So S = s cos kx, A = alpha cos kx, O = beta sin kx.

At l, in the rotated frame, A' = cos(2 theta) A + sin(2 theta) O and O' = -sin(2 theta) A + cos(2 theta) O, with O' = 0 at l, which gives beta = alpha tan(2 theta) cot(k l), and so A'(l) = alpha cos(k l)/cos(2 theta).

Charged directions. The connector forces R^+- to vanish at l, with L^+- Neumann. So S(l) = A'(l), and d/dx (S + A') = 0 at l. Substituting gives sin^2(kl) (1 + cos^2(2 theta)) = sin^2(2 theta) cos^2(kl).

Neutral directions. Add B = b cos kx for the generator T_15 = diag(1, 1, 1, -3)/(2 sqrt 6) of su(4). With Y = sqrt(3/5) R + sqrt(2/5) B and Z' = -sqrt(2/5) R + sqrt(3/5) B, the connector forces Z' to vanish at l, with L and Y Neumann. The Neumann conditions give b = -sqrt(3) s, and Z' = 0 gives s = alpha/(4 cos(2 theta)). Substituting in the condition for L gives sin^2(kl) (1/4 + cos^2(2 theta)) = sin^2(2 theta) cos^2(kl).

Massless modes and limits. At k = 0, a zero mode is a constant in the intersection of the two stabilizers. This consists of a_W = diag(q, q) on W_+ and W_-, identified by F, with q preserving the line of w, together with a_V = lambda (+) m, and with the trace condition this has dimension 1 + 8. In it, L = R and B/L = sqrt(2/3), which gives $1/e^2 = (1 + 1 + 2/3)/g^2$. The limits follow by expanding. A second derivation, from the referee, uses the principal angles between the Neumann subspaces at the two walls: for each charged direction the squared sine of the angle is sin^2(2 theta)/2, and in the neutral sector the single nonzero value is (4/5) sin^2(2 theta); it also covers cos(2 theta) = 0, where the divisions above need a limit. For item 3, with a = sin(2 theta), the ratio is [arcsin(a/sqrt 2)/arcsin(2a/sqrt 5)]^2, and this decreases with a on (0, 1], since x/(sqrt(1 - x^2) arcsin x) increases. ■

The constants in item 1 are the values $||[D_F, a]||^{2/||a||}2$ in the Hilbert-Schmidt norm, for universal mass m, for a = L^+- and for the direction orthogonal to the photon and Z', which has A-component of squared length 4/5. These are the coefficients of the lower bounds for the poles of the fully massive family in Proposition 14 of lemma 18 (2 x < z_W, (16/5) x < z_Z); the small-x pole coefficients there are 3 and 24/5. The limit 5/8 is common to both, with different behaviour: the finite-action poles grow in the heavy limit of lemma 18, while the interval masses here tend to zero at the antipodal point. The spectral-action Taylor mass for universal m is M_W^2 = 3 m^2, which exceeds the value here by the factor 3/2 of Proposition F in the paper: there the mass matrix carries the relative heat coefficients of the action, and here we have assumed a five-dimensional Yang-Mills action, which gives the commutator norm alone. A direct matching claim needs the full parent spectral action, with its tower and boundary conditions.

Item 2 gives a geometric form of the behaviour that lemma 18 found: heavy fermions with light vectors. At antipodal points Gamma_p = -Gamma_{p_0}, so both walls have the same stabilizer in sp(2) again, and the fermion channels have type (c) with mass pi/(2l).

The spectral action on the interval

Proposition 23.7. For the heat cutoff and universal theta, with s = l Lambda, the trace over a channel is

```
Tr_channel exp(-D^2/Lambda^2) = 2 sum_{n in Z} exp(-(n pi + theta)^2/s^2).
```

This is proportional to the Jacobi function theta_3(theta, exp(-s^2)). It is strictly decreasing on [0, pi/2], with a maximum at theta = 0 and a minimum at theta = pi/2, where it is stationary, and it is symmetric under theta -> pi - theta. In the towerless regime s -> 0 at fixed x = theta$^{2/s}$2, it tends to 2 e^{-x}, the form of the vacuum density in lemma 18 and Proposition 21.2.

Proof. The eigenvalues in a channel are +-(n pi + theta)/l, by Theorem 22.1, and the sum is over Z after combining the signs. By Poisson summation the sum equals (s/sqrt(pi)) sum_{k in Z} exp(-k^2 s^2) e^{2 i k theta}, and the Jacobi triple product writes this as a constant times the product over n >= 1 of (1 + 2 q^{2n-1} cos(2 theta) + q^{4n-2}), with q = exp(-s^2). Each factor is positive and strictly decreasing in theta on [0, pi/2]. ■

In the form of the Jacobi function, the trace is (2 s/sqrt(pi)) theta_3(theta, exp(-s^2)). This family lies outside the hypotheses of No-go theorem 9 of lemma 18, which concerns finite internal data and a free ray V = c V_*: here the operator family is different, the radial direction is a compact orbit of relative gradings, and the tower, which the finite description omits, supplies a second branch with a stationary point at theta = pi/2. In the present family the literal action has its minimum at antipodal gradings, m = pi/(2l), with M_W = M_Z = 0 by Theorem 23.6, where the lowest eigenspace is degenerate. This is a vacuum with massless vectors, so it is not the physical point. The sign of the term agrees with the fermion loop in the Hosotani mechanism, where gauge-boson loops enter with the opposite sign. So an interior minimum could arise from that competition, which the classical spectral action does not contain. We have not computed it.

The one-loop potential in the same family

From the exact spectra, the products over the towers are, relative to theta = 0 and in Euclidean momentum x = p l,

```
fermion channel:  1 + sin^2(theta)/sinh^2(x),
charged vectors:  1 + sin^2(2 theta)/(2 sinh^2(x)),
neutral vector:   1 + (4/5) sin^2(2 theta)/sinh^2(x).
```

So in the gauge-boson and fermion-loop form of the Hosotani mechanism, assuming the usual one-loop formula with three polarizations per vector, and with n_f = 2 x (number of physical Dirac species),

```
8 pi^2 l^4 V(theta) = int_0^infinity x^3 dx [ 3 log(1 + s^2/(2 sinh^2 x)) + (3/2) log(1 + (4/5) s^2/sinh
```

with s = sin(2 theta). A Dirac top with three colours has n_f = 6, which needs a separately specified fermion sector, since the universal connection gives masses to all: one generation has n_f = 16 and three have n_f = 48. With I = int x$^{3/\sinh}$2 x dx = (3/2) zeta(3) and I' = int x$^{3/\cosh}$2 x dx = (9/8) zeta(3), near theta = 0 the bracket is (10.8 - n_f) I theta^2 + o(theta^2), and with epsilon = pi/2 - theta, near theta = pi/2 it is V(pi/2) + (10.8 I + n_f I') epsilon^2 + o(epsilon^2), where we have written V for the left side.

Proposition 23.8 (referee). In the massless flat family with this content, for every n_f > 0, theta = pi/2 is the unique global minimum on [0, pi/2], since the gauge logarithms are nonnegative and vanish there, and the negative fermion logarithm is minimized at sin^2(theta) = 1. For n_f >= 10.8, V is strictly decreasing on the interior. Indeed, with t = sin^2(theta) and J(z) = int_0^infinity x^3 dx/(sinh^2 x + z), the derivative with respect to t is, up to a positive factor,

```
(1 - 2t) [6 J(2t(1 - t)) + (24/5) J((16/5) t (1 - t))] - n_f J(t).
```

For t <= 1/2, both arguments in the bracket are at least t, and J decreases; for t > 1/2 both contributions have the required sign.

For 0 < n_f < 10.8, on grids of theta (Python), for n_f in {1, 2, 3, 4, 6, 8, 10}, V has one interior maximum, at about 0.33 for n_f = 8, and no interior minimum. Theorem 9 of lemma 24, by Astra 6, proves this for all n_f in that range: exactly one interior stationary point, a strict maximum, using a log-concavity property of the integral.

These are negative results in an exact scope: one interval, no bulk masses, fermions of frequency 1 under F, this gauge content, one loop, this condition at the connector wall. As a formal leading approximation, not uniform at x = 0, V has the form B sin^2(2 theta) - A sin^2(theta) with A, B > 0, whose interior stationary point cos(2 theta) = A/(4B) is a maximum. *Bulk masses.* With bulk masses, the fermion product becomes cos^2(theta) + sin^2(theta) G(x), with G = (w_Q coth w_Q - c_Q)(w_U coth w_U - c_U)/x^2 and w = (c^2 + $x^{2)}${1/2} in the secular form of lemma 12. For a = c_Q + c_U != 0, G = 1 - a/x + O(x^-2), so the potential has angle-dependent ultraviolet divergences, and a finite value depends on a regulator and subtraction: as the referee showed, an earlier scan of general pairs here did not establish a potential, and it is withdrawn. The exceptional case c_U = -c_Q = c has a common bulk mass on the two mixed components and G = 1 + $w^{2/(x}$2 sinh^2 w), which decays exponentially. Keeping the flat gauge products as a separate tree-level assumption, a scan for c in {0, 1/2, 1, 3/2, 2, 3, 5}, n_f in {1, 3, 6, 8, 10, 16, 48}, on a grid of 721 values of theta, finds no interior minimum (Python). General bulk masses need a gauge-invariant completion, a regulator and wall counterterms.

So an interior minimum needs another change: a negative coefficient of sin^2(2 theta), as from fermions of frequency 2 under F, which the bimodule E = Hom(V, W) does not contain; general bulk masses or other gauge form factors, as with warping; a wall scalar with its own potential; or tree-level terms at the walls. With h/f_pi = 2 theta, the leading flat form has the angular form alpha cos(h/f_pi) - beta sin^2(h/f_pi) of Agashe, Contino and Pomarol (section 2.1, Eqs. (19)-(23)) with spinorial SO(5) fermions, with alpha = A/2 and beta = -B, up to a constant. Their form factors can give the sign and competition needed for an interior minimum, so the flat results here do not exclude that construction. This points away from a pure flat Wilson line, toward those changed form factors or a wall scalar with its own potential, for the physical interior.

Comparison with measured values in the flat toy

These are numerical comparisons (Python), with no running and no bulk masses, in lowest levels, using the Particle Data Group 2026 values quoted by the referee, m_t = 172.60 GeV as a direct-measurement mass, M_W = 80.3625 GeV and M_Z = 91.1879 GeV, so m_t/M_W = 2.1478 and M_W^{2/M_Z}2 = 0.7767. The ratio m_t/M_W reaches sqrt(8/3) at theta = 1.0589 and the measured value at theta = 1.1626, where M_W l = 0.541, so 1/l is about 148.5 GeV. There, M_W^{2/M_Z}2 = 0.581. The wall-scalar model of lemma 22 covers [0.625, 1], and this family covers [0.5032, 0.625); in the wall-scalar model, fitting the measured vector ratio gives M_W l = 1.028, where the lowest no-bulk-mass fermion channel has m/M_W < 1.528. So neither flat family fits both ratios on its own, and the flat one-loop potential above does not select an interior point. Whether a tower at 1/l of this size is allowed would need the couplings and an applicable experimental limit, which we have not analyzed. The directions are bulk masses for the light fermions, warping, running of the couplings, and families combining a Wilson line with a wall scalar. These negative statements hold in the flat tree-level toy only.

Open questions

1. The interior vacuum: the effective potential with a wall scalar, or with warping, or with general bulk masses, with a gauge-invariant completion, a regulator and wall counterterms. The flat cases above have none on the grids.
2. The relation between the two roles of S^4: the base of the twistor window, and the field space of the Higgs quaternion.
3. A commutative parent: functions on S^4 x I and CP^3 x I, with the gradings as local boundary conditions.

4. The factor 3/2 between the five-dimensional normalization and the spectral action: whether the spectral action on the parent itself, with its tower, reproduces the formula of Theorem 23.6.
5. The comparison with twisted spectral triples, Remark 23.3.
6. The contribution of a real connector to the fermion tower, the degenerate antipodal window, and the scalar kinetic normalization needed for a canonically normalized Higgs field.

Numerical check

By Claude Opus 5.5, in Python (numpy and scipy), 3 Nov 2026, in a temporary directory, not kept.

- Random complex symmetric M on C^4 with l ||M|| = 1.25: Gamma_1 is an involution, Gamma_1 = exp(2 l K) Gamma_0; D_F = (1/2l) Gamma_0 log(Gamma_1 Gamma_0) to 3 x 10^-15; the principal angles over l equal the singular values of M to five digits.
- Sp(2) x SU(4), of dimension 25, with the Neumann and Dirichlet subspaces built from the stated stabilizers: the two-point boundary problem was solved as a linear system on a grid in k l. For theta = 0.2, pi/4, 1.06 and 1.3, the two lowest nonzero roots are (0.279, 0.356), (0.785, 1.107), (0.647, 0.868), (0.373, 0.479), equal to the formulas of Theorem 23.6 to three digits, with other roots at pi/2 and above. The intersection of the two stabilizers has dimension 9 at each theta tested.
- Small theta: M_W^{2/m}2 = 2.000 and M_Z^{2/m}2 = 3.200.
- The universal connection on E (+) E*: J D = D J, Gamma_0 D = -D Gamma_0, all |eigenvalues| equal to theta/l, Gamma_1 = (left Gamma_p, right mirror), J_1^2 = -1, J_1 Gamma_1 = -Gamma_1 J_1, while J alone fails; J_1 h J_1^-1 is right multiplication by +h on E*, for odd h.
- The theta function: monotone on a grid for s = 0.3, 1, 2, with the towerless value at s = 0.3 equal to e^{-x} to five digits; Poisson summation and the triple product to 10^-16.
- The ratio M_W^{2/M_Z}2 on a grid: range [0.5032, 0.6250], symmetric under theta -> pi/2 - theta to 10^-15, minimum at pi/4. At theta = 1.1625, M_W l = 0.541, and at theta = 1.0589, M_W l = 0.648.
- The one-loop integrals by adaptive quadrature (scipy), and by Gauss-Legendre quadrature on 400 nodes for the scan with c_U = -c_Q. The referee recomputed the derivative with scipy, found one sign change for each n_f in {1, 2, 3, 4, 6, 8}, with the stationary angle about 0.3382 at n_f = 8, and checked I = (3/2) zeta(3).

Part 24

Commutative cylinders, spectral walls, and the two-stage compression

By Astra 6 (gpt-6-astra, effort xhigh), 3 Nov 2026; prompt in prompts/24_commutative_parent.md. Reproduced as delivered, with links made relative. The section "Numerical check" at the end is by Claude Opus 5.5. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): no error in the displayed equations, and one gap in the general statement on compression maps. The record above the section "Referee amendments" stays as delivered, and those amendments apply to it.

A commutative cylinder admits an exact realization of the **operator window** of lemmas 22 and 23 when the two finite gradings are prescribed through nonlocal spectral boundary conditions. The higher K-modes then decouple as the radius of K tends to zero. Two qualifications remain essential: these boundary conditions restrict scalar functions at the walls, and compression of scalar functions through both stages differs from generating a matrix algebra between the stages. In particular, aligned walls give $\mathbb{R} \oplus \mathbb{R}$ on the quaternionic factor under direct scalar compression, whereas the intervening matrix-algebra construction gives $\mathbb{H} \oplus \mathbb{H}$. Local boundary gradings preserve all scalar functions, but their reduction to the unchanged four-section window has a precise restriction proved below. The radial no-go of lemma 22 survives rechecking in its stated scope. For the massless one-loop potential of lemma 23, an analytic argument strengthens the numerical statement to all $n_f > 0$.

1. Theorem — An exact spectral boundary construction.

Hypotheses and conventions. Let

$$K = \mathbb{CP}^3, \qquad \mathcal{H}_K = L^2\big(K, \Lambda^{0,*}T^*K \otimes \mathcal{O}(1)\big)\,,$$

and let

$$A = \sqrt{2}(\bar\partial + \bar\partial^*), \qquad \gamma_K = (-1)^{\deg}.$$

Use the Fubini–Study metric at a fixed reference scale. Write

$$V = \ker A \simeq \mathbb{C}^4, \qquad \Pi = P_K, \qquad \Delta_K = \inf(\operatorname{spec}|A| \setminus \{0\}) > 0.$$

Here $\gamma_K|_V = 1$. The radius parameter $r > 0$ replaces A by A/r, after the usual unitary identification of the L^2-spaces.

These are the analytic data of Example 16.9. Its S^4 option uses functions pulled back to this same twistor space. An actual operator on S^4 with the required kernel and compression map would need to be specified separately.

Let

$$E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

On $L^2(I_\ell; \mathbb{C}^2 \otimes \mathcal{H}_K)$, set

$$\mathcal{D}_r = E\partial_x + Z\,A/r, \qquad \hat\gamma = Z\gamma_K.$$

All unspecified tensor factors are identities.

Choose self-adjoint involutions Γ_0, Γ_1 on V, each with two-dimensional positive and negative eigenspaces, and assume

$$-1 \notin \operatorname{spec}(\Gamma_1\Gamma_0).$$

Extend Γ_i by zero on $V^\perp$, writing $C_i = \Pi\Gamma_i\Pi$. Define $\operatorname{sgn} A = 0$ on V, and impose

$$\chi_0\psi(0) = \psi(0), \qquad \chi_\ell\psi(\ell) = \psi(\ell),$$

where

$$\chi_0 = ZC_0 - S\operatorname{sgn} A, \qquad \chi_\ell = ZC_1 + S\operatorname{sgn} A. \tag{1}$$

Then:

1. The H^1 realization of $\mathcal{D}_r$ is self-adjoint with compact resolvent and anticommutes with $\hat\gamma$.
2. $\Pi \otimes 1$ reduces the operator, including its domain.
3. On this reducing subspace, the operator is exactly $D(\Gamma_0, \Gamma_1)$ of lemma 23.
4. Every eigenvalue on the complementary subspace satisfies

$$|\lambda| \geq \Delta_K/r. \tag{2}$$

Proof. Put $B = SA/r$. Since $ES = Z$,

$$\mathcal{D}_r = E(\partial_x + B).$$

The operators χ_i are self-adjoint involutions and anticommute with E. Their positive eigenspaces are therefore maximal isotropic for the boundary form

$$[\langle\psi, E\phi\rangle]_0^\ell.$$

On $V^\perp$, condition (1) selects the negative spectral subspace of B at 0 and its positive spectral subspace at ℓ. These are APS conditions with the inward-normal sign reversed between the two walls. On $\ker B = \mathbb{C}^2 \otimes V$, condition (1) supplies a Lagrangian subspace:

$$\Gamma_i\psi_L = \psi_L, \qquad \Gamma_i\psi_R = -\psi_R.$$

For q in the complementary domain, integration by parts gives

$$\|\mathcal{D}_r q\|^2 = \|q'\|^2 + \|Aq/r\|^2 + [\langle q, Bq\rangle]_0^\ell \geq \|q'\|^2 + \frac{\Delta_K^2}{r^2}\|q\|^2. \tag{3}$$

The boundary term is nonnegative by the spectral choices.

An eigenbasis of A decomposes the complementary problem into finite-dimensional interval problems with maximal isotropic endpoint conditions. Each is self-adjoint. Estimate (3) identifies the domain of their self-adjoint direct sum with the stated H^1 domain and gives compactness of its resolvent. The finite-dimensional tangential kernel contributes an ordinary self-adjoint interval problem. This proves self-adjointness, compactness, and (2).

Every boundary projector commutes with Π, proving reduction. Finally, $\{\gamma_K, A\} = 0$ implies $\{\hat{\gamma}, \mathcal{D}_r\} = 0$, and $\hat{\gamma}$ preserves both boundary subspaces. ■

Define

$$K_F = \frac{1}{2\ell} \log(\Gamma_1 \Gamma_0), \qquad D_F = \Gamma_0 K_F, \qquad R(x) = e^{x K_F},$$

with the principal logarithm. Let $P_\pm = (1 \pm \Gamma_0)/2$. The finite-window embedding is

$$W_\ell v(x) = \ell^{-1/2} \begin{pmatrix} R(x) P_+ v \\ R(x) P_- v \end{pmatrix}. \tag{4}$$

Direct differentiation gives

$$\mathcal{D}_r W_\ell = W_\ell D_F, \qquad \hat{\gamma} W_\ell = W_\ell \Gamma_0.$$

Thus the prescribed finite operator and grading are exact, provided $\Delta_K / r > \|D_F\|$. The next eigenvalue in absolute value is at least

$$\min \left\{ \frac{\Delta_K}{r}, \frac{\pi - \ell \|D_F\|}{\ell} \right\}. \tag{5}$$

---

2. Proposition — Local walls and spectral walls have different scalar domain algebras.

Hypotheses. Use the product differential expression of Result 1.

For the local option, let $G_i(y)$ be smooth self-adjoint bundle involutions and impose

$$G_i \psi_L = \psi_L, \qquad G_i \psi_R = -\psi_R \quad \text{at } x = i.$$

Assume

$$[G_i, \gamma_K] = 0, \qquad [G_i, \sigma_A(y, \xi)] = 0 \quad (\xi \in T_y^* K), \tag{6}$$

where σ_A is the tangential principal symbol.

Then these local conditions give a self-adjoint elliptic realization with compact resolvent, and **every** smooth scalar function on $K \times I_\ell$ preserves its domain.

For the spectral option (1), the maximal smooth scalar $*$-algebra preserving the domain is

$$\mathcal{A}_{\mathrm{APS}} = \{ f \in C^\infty(K \times I_\ell) : f(\,\cdot\,, 0), f(\,\cdot\,, \ell) \text{ are constant on } K \}. \tag{7}$$

The same conclusion holds for the subalgebra pulled back from $S^4 \times I_\ell$.

Proof. For the local option, the boundary involution is $Z G_i$. It anticommutes with the normal matrix E. Condition (6) makes it anticommute with the principal symbol of the adapted tangential operator SA/r. Hence its positive eigenspace is an elliptic Lagrangian boundary condition. The corresponding local ellipticity and adjoint-domain criteria are described in Bär–Ballmann, Corollary 4.18 and Example 4.20. Primary reference

Scalar multiplication commutes pointwise with G_i, so it preserves the endpoint conditions and the H^1 domain. Moreover,

$$[\mathcal{D}_r, f] = E\, \partial_x f + Z[A, f]/r$$

is bounded for each $r > 0$.

For spectral walls, preservation by a scalar $*$-algebra is equivalent to

$$[M_{f_i}, \chi_i] = 0.$$

Independence of the spin matrices Z, S gives

$$[M_{f_i}, C_i] = 0, \qquad [M_{f_i}, \operatorname{sgn} A] = 0.$$

Since $C_i^2 = \Pi$, the first equation implies

$$[M_{f_i}, \Pi] = 0. \tag{8}$$

Consequently multiplication by f_i preserves $H^0(K, \mathcal{O}(1))$. If s is such a section, then

$$0 = \bar\partial(f_i s) = (\bar\partial f_i)s.$$

The sections of $\mathcal{O}(1)$ have no common zero, so $\bar\partial f_i = 0$ everywhere. A holomorphic function on compact connected $\mathbb{CP}^3$ is constant. Constants plainly preserve (1), proving (7). ■

The hypotheses in (6) are a sufficient local ellipticity condition. An arbitrary bundle grading supplies a pointwise isotropic condition, but self-adjointness on an H^1 domain and compact resolvent still require an ellipticity check. Likewise, prescribing a grading only on V leaves the boundary problem incomplete until a condition on $V^\perp$ is supplied.

---

3. Proposition — Compression of a local grading and the unchanged level-one window.

Hypotheses. Let G be a self-adjoint bundle involution acting by multiplication, and let

$$g = \Pi G \Pi|_V, \qquad Q_K(f) = \Pi M_f \Pi|_V.$$

Keep the level-one space and scalar representation of Example 16.9 unchanged.

Then

$$1_V - g^2 = \big((1-\Pi)G\Pi\big)^*\big((1-\Pi)G\Pi\big). \tag{9}$$

In particular, g is a grading exactly when $[G, \Pi] = 0$. In that event,

$$[g, Q_K(f)] = 0 \quad \text{for every scalar } f. \tag{10}$$

For either the full $\mathbb{CP}^3$ scalar algebra or the S^4 pullback algebra, this forces

$$g = 1_V \quad \text{or} \quad g = -1_V. \tag{11}$$

Proof. Expand $\Pi G^2 \Pi = \Pi$ using $1 = \Pi + (1-\Pi)$; this gives (9). Vanishing of its positive right-hand side implies $(1-\Pi)G\Pi = 0$, and self-adjointness gives the other off-diagonal block.

Since scalar multiplication commutes with G, reduction then gives (10). The compressed $\mathbb{CP}^3$ functions generate $M_4(\mathbb{C})$. The compressed real S^4 functions generate $M_2(\mathbb{H})$, whose complex-linear commutant on $\mathbb{C}^4$ is also $\mathbb{C}1$. Thus a self-adjoint involution satisfying (10) is ± 1. ■

For **aligned local walls** with $[G, A] = 0$, the finite zero-mode window is explicitly

$$v \longmapsto \ell^{-1/2} \begin{pmatrix} P_+(g)v \\ P_-(g)v \end{pmatrix}.$$

The induced grading is g, and scalar compression is

$$P_+(g)Q_K(f)P_+(g) + P_-(g)Q_K(f)P_-(g) = Q_K(f).$$

By (11), the unchanged four-section window has trivial grading. Its generated algebra remains $M_2(\mathbb{H})$ or $M_4(\mathbb{C})$, according to the scalar algebra chosen. The globally constant choices $G = \pm 1$ give concrete self-adjoint examples with this window and a separated higher spectrum.

There is also an explicit local field illustrating failure of reduction. Add a trivial auxiliary $\mathbb{C}^4$ bundle and put

$$G(y) = \sum_{a=1}^{5} y_a \Gamma_a^{\mathrm{aux}},$$

where $y \in S^4$ is the twistor projection and the Γ_a satisfy $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$. This is a local involution satisfying (6), but now the tangential kernel has dimension 16, and

$$g = \frac{1}{5} \sum_{a=1}^{5} \Gamma_a^{\mathrm{section}} \otimes \Gamma_a^{\mathrm{aux}}.$$

Using $\mathrm{tr}(\Gamma_a \Gamma_b) = 4\delta_{ab}$,

$$\mathrm{tr}(g^2) = \frac{16}{5} \neq 16.$$

Thus g is not a grading. Equation (9) even gives

$$\|(1 - \Pi) G \Pi\| \geq \frac{2}{\sqrt{5}},$$

independently of r. This is a bound on boundary-trace mixing; it does not identify the actual low spectral window of the coupled local problem.

A local $G(y)$ specifies a **field of gradings**. It does not select a marked point p. In the tautological quaternionic line bundle,

$$\Pi_y = \frac{1 + y_a \Gamma_a}{2},$$

the natural field $2\Pi_y - 1$ acts as $+1$ on the fibre $\operatorname{ran} \Pi_y$. By contrast, the finite grading

$$\Gamma_p = p_a \Gamma_a = 5 Q_K(p \cdot y)$$

uses a separately chosen point p. Its action on global section coefficients is nonlocal.

The restriction proved here concerns the unchanged level-one space, a local multiplication grading, and its compression as an actual involution. A different coupled window, additional fibres, or spectral boundary conditions remain available directions.

---

4. Theorem — The composed compression map and the aligned-wall algebra.

Hypotheses. Use the exact spectral construction of Result 1. For a scalar multiplier $f(y, x)$, let

$$T_f(x) = Q_K(f(\,\cdot\,, x)), \qquad \mathcal{E}_0(T) = \frac{1}{2}(T + \Gamma_0 T \Gamma_0).$$

Use real-valued functions and real algebra generation when discussing $M_2(\mathbb{H})$.

Then the compression directly from the scalar cylinder to the finite window is

$$\boxed{\Phi_\ell(f) = \frac{1}{\ell} \int_0^\ell \mathcal{E}_0\big(R(x)^* T_f(x) R(x)\big)\, dx.} \tag{12}$$

It is independent of r.

For aligned walls $\Gamma_1 = \Gamma_0 = \Gamma_p$, direct scalar compression on the S^4 factor generates

$$\mathbb{R} \oplus \mathbb{R}. \tag{13}$$

If instead one first generates $M_2(\mathbb{H})$ and then applies the interval compression, the result is

$$\mathbb{H} \oplus \mathbb{H}. \tag{14}$$

Proof. Insert (4) on both sides of the multiplication operator. Orthogonality of the two interval spin components gives exactly (12).

For real S^4 functions, Example 16.9 implies

$$Q_K(C^\infty(S^4, \mathbb{R})) = \mathrm{span}_{\mathbb{R}}\{1, \Gamma_1, \dots, \Gamma_5\}. \tag{15}$$

Indeed all compressed functions are quaternionic Hermitian matrices; that space has real dimension six, and the displayed six matrices belong to the image and are independent.

For aligned walls $R = 1$. Clifford multiplication gives

$$\mathcal{E}_p(\Gamma_a) = p_a \Gamma_p.$$

Therefore the image of direct scalar compression is

$$\mathrm{span}_{\mathbb{R}}\{1, \Gamma_p\}.$$

This space is already the algebra $\mathbb{R} \oplus \mathbb{R}$, proving (13).

On the other hand,

$$\mathcal{E}_p(M_2(\mathbb{H})) = \{a \in M_2(\mathbb{H}) : [a, \Gamma_p] = 0\} = \mathbb{H} \oplus \mathbb{H},$$

proving (14). The noncommuting products needed to generate the quaternionic diagonal blocks have been inserted between the two compressions in this second procedure. ■

On the single $\mathbb{CP}^3$ factor, real scalar compression is onto the Hermitian 4×4 matrices. A chosen $2+2$ aligned grading therefore gives all Hermitian block-diagonal matrices and generates

$$M_2(\mathbb{C}) \oplus M_2(\mathbb{C}).$$

For the **chosen two-sheet coarse correspondence** of Example 16.9, with

$$H_0 = \mathrm{Hom}(V, W) \oplus \mathrm{Hom}(W, V), \qquad \Gamma_0(\xi, \eta) = (\Gamma_p \xi, -\eta \Gamma_p),$$

the same calculation gives

$$\boxed{\mathrm{alg}_{\mathbb{R}}(\text{direct aligned scalar compression}) = \mathbb{R} \oplus \mathbb{R} \oplus M_4(\mathbb{C}).} \tag{16}$$

The V-side matrices act on the left of η and commute with its right grading, so that factor survives unchanged.

Endpoint restrictions do not change these finite-dimensional images. If a linear functional vanishes on compressions of functions supported in $0 < x < \ell$, its smooth matrix-entry density vanishes in the interior and hence everywhere. Finite-dimensional separation then gives equality with the unrestricted image, exactly as in Result 7 of lemma 20.

At the intermediate stage, however, spectral walls still give scalar endpoint values. Even after algebra generation, those values remain scalar. Enlarging them to the full stabilizers of Γ_0, Γ_1, as in lemma 23, is another change of the parent algebra. Equality of finite compression images does not identify the parent gauge groups or one-forms.

---

5. Theorem — Radius collapse, interval collapse, and controlled mixing.

Hypotheses. First use the exact construction of Result 1. Identify the varying Hilbert spaces by metric rescaling. For resolvent formulas, fix units in which the nonreal spectral parameter is i.

Then, at fixed ℓ,

$$\left\|(\mathcal{D}_r - i)^{-1} - \iota_\Pi (D(\Gamma_0, \Gamma_1) - i)^{-1} \iota_\Pi^*\right\| \le \frac{r}{\Delta_K}. \tag{17}$$

Thus radius collapse gives the entire matrix interval operator in generalized norm-resolvent sense.

If D_F is fixed, $\ell \to 0$, and

$$\Gamma_1(\ell) = e^{\ell\Gamma_0 D_F}\Gamma_0 e^{-\ell\Gamma_0 D_F},$$

then $r \to 0$ and $\ell \to 0$ decouple both towers and leave the finite operator D_F. No relation between the two rates is required in this exactly reducing construction.

Proof. The resolvent is block diagonal with respect to Π. Estimate (2) bounds its complementary block by r/Δ_K, proving (17). For the finite-window limit, (5) bounds the complementary resolvent by

$$\max\left\{\frac{r}{\Delta_K}, \frac{\ell}{\pi - \ell\|D_F\|}\right\},$$

which tends to zero. ■

A useful mixing statement requires additional hypotheses. Suppose a transformed problem has a common domain and the form

$$\widetilde{\mathcal{D}}_r = \mathcal{D}_r + V_r, \qquad V_r = V_r^*, \qquad \|V_r\| \le b_r.$$

Write

$$v_r = \|(1 - \Pi)V_r\Pi\|.$$

For an eigenvector $\psi = p + q$, with $p = \Pi\psi$, $q = (1 - \Pi)\psi$, and $|\lambda| \le L$, assume

$$g_r = \Delta_K/r - b_r - L > 0.$$

Then

$$\boxed{\|q\| \le \frac{v_r}{g_r}\|p\|, \qquad \|\text{Schur correction at } \lambda\| \le \frac{v_r^2}{g_r}.} \tag{18}$$

Proof. The complementary eigenvalue equation gives

$$q = -(\mathcal{D}_{r,Q} + V_{QQ} - \lambda)^{-1} V_{QP} p.$$

The inverse has norm at most g_r^{-1}. Substituting this formula into the p-equation gives the second bound. ■

For bounded b_r, v_r, both errors are $O(r/\Delta_K)$. If a common-domain construction instead produces terms of order $1/\ell$, sufficient conditions include

$$\frac{r v_r}{\Delta_K} \to 0, \qquad \frac{r v_r^2}{\Delta_K} \to 0,$$

together with convergence of the Π-diagonal perturbation.

An arbitrary local $G_i(y)$ generally fails $[G_i, \Pi] = 0$. Although the differential expression commutes with Π, its domain then does not. Such a boundary change is not automatically a bounded perturbation satisfying (18). The explicit local field in Result 3 has nonvanishing boundary mixing at every radius. No convergence to the prescribed two-gradings operator is established here for that nonreducing local problem.

At the level of observables, radius collapse leaves (12) unchanged. For an x-independent $T = Q_K(f)$,

$$\|\Phi_\ell(f) - \mathcal{E}_0(T)\| \le \ell\|D_F\|\,\|T\|. \tag{19}$$

If T is even for Γ_0, the stronger formula of Theorem 22.4 applies:

$$\Phi_\ell(f) = \operatorname{sinc}(\ell \operatorname{ad}_{D_F})(T),$$

$$\|\Phi_\ell(f) - T\| \le \frac{\ell^2}{6}\|[D_F,[D_F,T]]\|. \tag{20}$$

Equation (19) follows by integrating $\|R^*TR - T\| \le 2x\|D_F\|\|T\|$; equation (20) follows by taking the even part of the conjugation series.

Consequently there is convergence of the **compression maps on their scalar operator systems**. The limit is $\mathcal{E}_0 Q_K$. It does not turn that image into every element of $\mathcal{E}_0(P)$; Result 4 computes the difference explicitly. Algebra generation remains a separate operation.

---

6. Proposition — The second sheet, reality, even algebra, and connector remain prescribed data.

Hypotheses. Consider the scalar cylinder construction above and, separately, its extension using the two-sheet coarse correspondence of Example 16.9.

The resulting distinctions are:

Structure	What follows under the stated choices
Second sheet and the two Hom spaces	They are inputs in the coarse correspondence. A single four-section cylinder window does not supply them.
$J_1^2 = -1$, $J_1\Gamma_i = -\Gamma_i J_1$	These are additional compatibility conditions on the prescribed gradings and antiunitary.
$\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C})$	This is the derived even part after choosing the quaternionic real form, the grading, and intermediate algebra generation. Direct aligned scalar compression gives (16).
Connector and $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$	The connector's lines and coupling are chosen. Their intertwining algebra is then derived.

Proof. The one-factor construction starts with $V = \mathbb{C}^4$, and (4) has domain V. The two-sheet construction instead starts with

$$(W \otimes \overline{V}) \oplus (V \otimes \overline{W}).$$

Geometrically, its analytic realization uses product factors and a restricted scalar algebra, as Theorem 16.7 specifies. It is additional correspondence data over the scalar spaces.

For reality, suppose an antiunitary $\mathcal{J}_1$ on the tangential Hilbert space satisfies

$$\mathcal{J}_1^2 = -1, \quad \mathcal{J}_1 A = -A\mathcal{J}_1, \quad [\mathcal{J}_1, \gamma_K] = 0,$$

and its restriction J_1 to V anticommutes with both Γ_i. Then

$$\widehat{J} = E\mathcal{J}_1$$

preserves (1) and obeys

$$\widehat{J}^2 = 1, \qquad [\widehat{J}, \mathcal{D}_r] = 0, \qquad \widehat{J}\hat{\gamma} = -\hat{\gamma}\widehat{J}.$$

On the window,

$$J_F = \Gamma_0 J_1.$$

These identities follow directly from the stated commutation relations. They specify sufficient conditions; the natural $\mathcal{O}(1)$ cylinder has not supplied such an extension or its order-zero property here.

In particular, the quaternionic antiunitary j used to recognize $M_2(\mathbb{H})$ commutes with the Γ_a. It does not itself satisfy the required anticommutation with Γ_p.

The even-algebra assertion is the centralizer calculation in Result 4 after intermediate generation.

Finally, choose

$$w \in W_+, \quad v \in V, \quad \|w\| = \|v\| = 1, \quad \kappa \neq 0,$$

and prescribe

$$T_\kappa \xi = \kappa (w^* \xi v)\, v w^*.$$

Its intertwining conditions require the lines $\mathbb{C}w, \mathbb{C}v$ to be reducing, with the same scalar action. A quaternion preserving $\mathbb{C}w$ lies in its chosen copy of $\mathbb{C}$; $v^\perp$ carries arbitrary $M_3(\mathbb{C})$, and W_- carries arbitrary $\mathbb{H}$. Hence

$$\mathcal{A}(T_\kappa) = \{(\lambda \oplus q, \lambda \oplus m)\} \simeq \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}),$$

as in Result 2 of lemma 20. ■

A marked $p \in S^4$ fixes a quaternionic line, but choosing w also requires a complex line within it. The second line v, the coupling, and an interface realizing the connector remain necessary. Scalar agreement at marked points has unchanged compression image by Result 7 of lemma 20, so it supplies none of this selection by itself.

---

7. No-go theorem — Recheck of the radial statement in lemma 22.

Hypotheses. Let the parent be exactly the one-interval model of lemma 22: fixed wall coupling X, zero bulk mass, the even algebra $\mathcal{A}_\ell$, scalar inner fluctuations Eh with $h^* = -h$, and the **moving spectral window**. For real fluctuations, include the conjugate one-form and assume the stated order-zero representation.

Then every fluctuated finite operator has the same singular values as D_F. This is an exact finite-amplitude spectral statement; it uses no action or loop expansion.

Proof. Solve

$$u' = -hu, \qquad u(0) = 1, \qquad w = u(\ell),$$

and set $v(x) = u(x)w^*$. Then

$$v(\ell) = 1, \qquad v(0) = w^*.$$

Since h is even, w preserves the boundary splitting at 0; the identity at ℓ preserves the fixed coupled wall. Thus v preserves the original domain, and

$$v D_X v^* = D_X + Eh.$$

For the real fluctuation, the analogous implementing unitary is

$$v J_F v J_F^{-1};$$

order zero separates the two factors in its derivative.

The full operators are unitarily equivalent, so their moving spectral windows are unitarily equivalent. In the finite identifications of lemma 22, the operators are $w^* D_F w$, or $U_w^* D_F U_w$ in the real case. Their singular values coincide. ■

The moving-window hypothesis matters. For example, with $D_F = m\sigma_1$ and $h = i\sigma_3$, fixed compression gives

$$Q_\ell(Eh) = -\frac{1 - \cos(2m\ell)}{2m\ell}\sigma_2.$$

Adding a multiple of this operator to the fixed $m\sigma_1$ changes its singular values at second order.

The verdict agrees with the author's amended theorem and Terra's independent endpoint-preserving proof. In the usual nondegenerate Standard Model representation, the normalized shifted Higgs quaternion lies on the unit-quaternion orbit; its norm is fixed.

The known escapes are a variable wall coupling, odd connection directions, suitable bulk masses, additional internal directions with nontrivial connection data, or additional elementary or composite fields. A quantum effective action on the same configuration space does not add a radial coordinate. The proposed $K \times I_\ell$ parent has tangential directions and tangential one-forms, so this theorem does not decide its full fluctuation space.

---

8. Lemma — A logarithmic concavity property of the one-loop integral.

Hypotheses. Let

$$\mathcal{J}(z) = \int_0^\infty \frac{x^3}{\sinh^2 x + z}\, dx, \qquad z > 0,$$

and define

$$e(z) = -\frac{z\mathcal{J}'(z)}{\mathcal{J}(z)}.$$

Then $0 < e(z) < 1$, and e is nondecreasing.

Proof. Positivity and

$$0 < \frac{z}{\sinh^2 x + z} < 1$$

show that $e(z)$, expressed as a weighted average of this fraction, lies strictly between zero and one.

Put $x = e^v$, $z = e^u$. Then

$$\mathcal{J}(e^u) = \int_{\mathbb{R}} \exp\Big(4v - \log(\sinh^2(e^v) + e^u)\Big)\, dv. \tag{21}$$

The function

$$b(v) = 2\log\sinh(e^v)$$

is convex, because, with $x = e^v$,

$$b''(v) = 2x(\coth x - x\ \operatorname{csch}^2 x) > 0.$$

The last inequality is equivalent to $\sinh x \cosh x > x$.

The function $(u, v) \mapsto \log(e^{b(v)} + e^u)$ is therefore convex. The integrand in (21) is jointly log-concave. Marginal integration preserves log-concavity by Prékopa's theorem, so $u \mapsto \log \mathcal{J}(e^u)$ is concave. We use precisely Theorem 6, p. 342, of Prékopa's original paper.

Finally,

$$\frac{d}{du}\log \mathcal{J}(e^u) = -e(e^u).$$

Concavity makes this derivative nonincreasing, proving the assertion. ■

---

9. Theorem — The massless one-loop potential has no interior local minimum.

Hypotheses. Let the action be exactly the one-loop determinant displayed in lemma 23: a flat interval, zero bulk masses, frequency-one fermions, the stated charged and neutral gauge towers and multiplicities, and the prescribed connector wall condition. Assume that no fermion connector modifies the displayed fermion tower, and add no wall potential or other angle-dependent term.

Let $n_f > 0$, set $t = \sin^2\theta$, and write

$$\mathcal{V}(t) = 8\pi^2\ell^4 V(\theta).$$

Then:

1. $t = 1$, equivalently $\theta = \pi/2$, is the unique global minimum.
2. If $n_f \geq 54/5$, the potential is strictly decreasing on the interior.
3. If $0 < n_f < 54/5$, there is exactly one interior stationary point, and it is a strict maximum.

Proof. Define

$$F(z) = \int_0^\infty x^3 \log\left(1 + \frac{z}{\sinh^2 x}\right) dx, \qquad F'(z) = \mathcal{J}(z).$$

The exact potential is

$$\mathcal{V}(t) = 3F(2t(1-t)) + \frac{3}{2}F\left(\frac{16}{5}t(1-t)\right) - n_f F(t). \tag{22}$$

Its gauge terms are nonnegative and vanish at $t = 1$. Since F is strictly increasing, the fermion term is uniquely minimized there. This proves item 1.

Differentiate:

$$\mathcal{V}'(t) = (1-2t)\left[6\mathcal{J}(2t(1-t)) + \frac{24}{5}\mathcal{J}\left(\frac{16}{5}t(1-t)\right)\right] - n_f \mathcal{J}(t). \tag{23}$$

For $t \geq 1/2$, this is strictly negative.

For $0 < t < 1/2$ and $c \geq 2$, put

$$R_c(t) = (1-2t)\frac{\mathcal{J}(ct(1-t))}{\mathcal{J}(t)}.$$

Let $a = ct(1-t)$. Then $a \geq t$, and Result 8 gives $e(a) \geq e(t)$. Consequently,

$$\begin{aligned}
\frac{d}{dt}\log R_c(t) &= -\frac{2}{1-2t} - \frac{1-2t}{t(1-t)}e(a) + \frac{e(t)}{t} \\
&\leq -\frac{2}{1-2t} + \frac{e(t)}{1-t} \\
&< -\frac{2}{1-2t} + \frac{1}{1-t} < 0.
\end{aligned}$$

Thus every R_c is strictly decreasing. Since $\mathcal{J}(0)$ is finite and positive,

$$R_c(0+) = 1, \qquad R_c(1/2-) = 0.$$

Therefore

$$H(t) = 6R_2(t) + \frac{24}{5}R_{16/5}(t)$$

decreases strictly from $54/5$ to zero. Equation (23) becomes

$$\mathcal{V}'(t) = \mathcal{J}(t)\,[H(t) - n_f] \quad (0 < t < 1/2).$$

For $n_f \geq 54/5$, its sign is negative. For $0 < n_f < 54/5$, it changes sign exactly once, from positive to negative. Since $t = \sin^2\theta$ is strictly increasing on the interior, the same stationary-point classification holds in θ. ■

This proves the massless statement for the entire parameter range, including the previously numerical range. No numerical computation is used in this proof.

The recorded author and Terra referee calculations agree on the sampled massless cases. The logarithmic-concavity proof above is an additional single-agent derivation; it has not received a fresh independent model review in this turn. The recorded agreement therefore should not be described as a multi-model review of this new proof.

The escapes remain changes to the stated determinant: frequency-two fermions, modified form factors through warping or other geometry, suitable bulk-mass sectors, wall potentials, and wall kinetic or interaction terms. The full effective action of the proposed $K \times I_\ell$ parent, including its K-tower, has not been computed here.

---

10. Proposition — Scope of the bulk-mass scans.

Hypotheses. Retain the flat gauge terms of (22) as an independent assumption. Replace the fermion factor by the expression in lemma 23:

$$1 + t(G(x) - 1), \qquad G(x) = \frac{(w_Q \coth w_Q - c_Q)(w_U \coth w_U - c_U)}{x^2}, \quad w_i = \sqrt{x^2 + c_i^2}.$$

If $c_U = -c_Q = c$, then the potential is finite and $t = 1$ is its unique global minimum for every $n_f > 0$. The reported absence of additional interior **local** minima at nonzero c remains a numerical statement here.

If $a = c_Q + c_U \neq 0$, the displayed unrenormalized integral has angle-dependent ultraviolet divergences.

Proof. For opposite masses,

$$G(x) = 1 + \frac{w^2}{x^2 \sinh^2 w}, \qquad w = \sqrt{x^2 + c^2}.$$

The added fraction is positive and decays exponentially at infinity. The negative fermion logarithm is strictly decreasing in t. The gauge logarithms are nonnegative and vanish at $t = 1$, proving the global-minimum assertion.

For general masses,

$$G(x) = 1 - \frac{a}{x} + \frac{a^2}{2x^2} + O(x^{-3}),$$

and hence

$$\log(1 + t(G - 1)) = -\frac{ta}{x} + \frac{t(1-t)a^2}{2x^2} + O(x^{-3}).$$

Multiplication by the integration measure $x^3 dx$ produces angle-dependent divergences. A finite potential therefore requires a specified completion, regulator, and counterterms. ■

The opposite-mass scan in lemma 23 samples finitely many masses, multiplicities, and angles. It cannot exclude unsampled local minima. General bulk masses and altered gauge form factors remain directions for constructing an interior vacuum; these conclusions leave that question open.

Consequence for T0 and T0b

For T0, a commutative scalar representation with prescribed spectral walls gives an exact interval operator window and controlled tower decoupling, while local walls preserve the full scalar algebra under the ellipticity hypotheses above. The derivation of a nontrivial local grading and the desired algebra from one coupled scalar window remains open, and intermediate algebra generation must be stated explicitly. For T0b, the second sheet, compatible reality, quaternionic grading choice, and connector remain inputs; their algebraic consequences can be derived once those choices are supplied. Neither the radial theorem nor the specified one-loop potential closes the commutative-parent problem.

Referee amendments

From Terra 6.1. These apply to the text above.

- **Result 2.** Write G_0 at x = 0 and G_l at x = l. Equation (6) supplies the principal-symbol condition of Bar-Ballmann, Corollary 4.18 and Theorem 4.11; their Example 4.20 assumes more. On the unaugmented Dolbeault bundle, the Clifford representation is irreducible, so a fibre endomorphism commuting with every sigma_A is scalar, and a smooth involution of this form is +1 or -1. Additional fibres and other local elliptic projectors lie outside this sufficient criterion.
- **Result 3.** The statement on marked points concerns the natural field displayed there. A local field can encode a chosen point: on the auxiliary bundle, G_p(y) = p_a Gamma_a^aux is a local constant field whose coefficients contain p.

- **Result 4.** Equation (16) has the two Hom spaces and the two scalar algebras of Example 16.9 as inputs. The endpoint argument proves equality of the finite-dimensional linear images, and does not identify the parent algebras, norms, gauge groups or one-forms.
- **Result 5, gap in the final assertion.** Convergence of compression maps needs a common domain and a collapse map. With aligned walls, D_F = 0 and f_l(y, x) = 6 (x/l)(1 - x/l), both endpoint values vanish while Phi_l(f_l) = 1. A correct form: for F(y, s) fixed on K x [0, 1] and f_l(y, x) = F(y, x/l), Phi_l(f_l) tends to E_0 Q_K(int_0^1 F(., s) ds). For x-independent ambient multipliers the limit is E_0 Q_K, by (19). The rate discussion after (18) also needs control of the denominator: a sufficient addition is r (b_r + L) <= (1 - epsilon) Delta_K. The case b_r = O(1/l) follows from the stronger estimate |D_{r,Q}| >= (Delta_K$^{2/r}$2 + pi^2/(4 l$^{2)}$){1/2}, from the one-ended Poincare inequality on the positive and negative B-spectral components.
- **Result 6.** At the finite level, if the gradings are Gamma_p, Gamma_q, a unit n orthogonal to p and q gives J_1 = Gamma_n j, with J_1^2 = -1 and the required anticommutation. Extending this choice to the parent remains a separate task.
- **Lemma 8 and Theorem 9.** An independent check of the key monotonicity, without log-concavity or Prekopa's theorem: with rho(s) = (arsinh sqrt(s))^3/(2 sqrt(s (1 + s))), J(z) = int_0^infinity rho(s)/(s + z) ds; with L(v) = log rho(e^v), L'(v) = 3 tanh(x)/(2x) - (1 + tanh^2 x)/2 for v = 2 log sinh x, which is strictly decreasing. For z = e^u, with the probability density in y proportional to e^y rho(e^{u+y})/(1 + e^y), e(e^u) is the mean of 1/(1 + e^y), and its derivative in u is Cov(1/(1 + e^y), L'(u + y)) > 0, as both are strictly decreasing. Theorem 9 is for theta in [0, pi/2]. At theta = pi/2 the determinant has its continuous value, while the principal-logarithm window is degenerate. Its hypotheses are the flat massless one-loop determinant, its gauge multiplicities and connector wall condition, frequency-one fermions, no fermion-connector modification, and no other angle-dependent term.
- **Result 10.** The divergence is stated for nonzero fermion multiplicity, with no cancellation by additional sectors. The opposite-mass scan in lemma 23 is a numerical result on its sampled parameters.

Numerical check

By Claude Opus 5.5, in Python (numpy, scipy), 3 Nov 2026, in a temporary directory, not kept: E_p(Gamma_a) = p_a Gamma_p to 10^-15, which gives equation (13); tr(g^2) = 16/5 for the auxiliary field of Result 3; on grids, e(z) lies in (0, 1) and is nondecreasing for z in [10^-3, 50]; H(t) is strictly decreasing on (0, 1/2), with H near 54/5 at small t and near 0 at t = 1/2; for n_f in {1, 3, 6, 8, 10}, H - n_f has one sign change, with the stationary angle 0.34 at n_f = 8, which agrees with the referee of lemma 23. Results 1, 2, 5 and 6 were read and not separately computed.

Part 25

The weak-boson sum rule and the effective dimension of the interval

For STRATEGY T5b, T13 and T1, following lemma 18, lemma 22 and lemma 23. Proved by Claude Opus 5.5, with checks in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): two errors and two gaps as first written, and amendments, all incorporated below.

Outcome. Unfolded, the interval parent is a circle of length 2l, and the parent operator is -i d/dx (x) 1 + 1 (x) D_F: the tower is D_F shifted by the momenta k_n = n pi/l. So the exact quadratic kernel of lemma 18 applies termwise over the tower. For small masses, at leading order, for the heat cutoff, the Taylor mass of a vector with constant profile in the transported frame is

```
M^2 = ((3 - d_eff)/2) ||[D_F, T]||^2 / ||T||^2,      d_eff(s) = d log S / d log s,      S(s) = sum_{n i
```

in Hilbert-Schmidt norms, where d_eff is the spectral dimension at the scale of the cutoff of the unfolded circle, equivalently of the paired Neumann and Dirichlet interval: 0 for l Lambda « 1 and 1 for l Lambda » 1. For the Standard Model representation, at the same order,

```
M_W^2 = ((3 - d_eff)/24) sum_f N_f m_f^2,       m_t/M_W = (8/(3 - d_eff))^{1/2} for the top alone,
```

with M_W$^{2/M_Z}$2 = 5/8 for all s. So the relation 8 M_W^2 = sum N_f m_f^2 of the asymptotic spectral action is the zero-dimensional value, with m_t/M_W = sqrt(8/3). A resolved interval gives 12 M_W^2 = sum N_f m_f^2, with m_t = 2 M_W, as five-dimensional covariance requires for this operator on the circle. The endpoint m_t = 2 M_W is known in flat gauge-Higgs models with suitable representations (references below); in the inspected records the referee did not find the coefficient (3 - d_eff)/24 or its crossover, and priority remains open pending a wider search. The factor 3/2 in Proposition F of the paper, the finite-action coefficient relative to the Yang-Mills commutator quotient, is the ratio of these two endpoints. The crossover is at l Lambda about 2.

Unfolding

Let D_X = E d/dx be the parent of lemma 22, and let D_0 + E Gamma_F D_F be its form after the transport of Corollary 22.2, where D_0 is the operator with X = 0.

Proposition 25.1. On the circle x in [-l, l] with periodic conditions, for psi in the domain of D_0, put

```
Psi_Q(x) = Q_L(|x|) + i sgn(x) Q_R(|x|),      Psi_U(x) = U_R(|x|) - i sgn(x) U_L(|x|).
```

After multiplying by 1/sqrt(2), this is a unitary from the parent Hilbert space onto L^2(circle) (x) (H_+ (+) H_-), which maps the H^1 domain of D_0 onto the periodic H^1 domain, and it carries D_0 + E Gamma_F D_F to

```
-i d/dx (x) 1 + 1 (x) D_F.
```

So the signed spectrum is k_n + mu_i, with k_n = n pi/l for n in Z and mu_i the eigenvalues of D_F. An even operator T = T_Q (+) T_U on H_F, constant in x in this transported frame and acting identically on both spin components, is carried to 1 (x) T.

Proof. Since Q_R(0) = Q_R(l) = 0 and U_L(0) = U_L(l) = 0, the functions Psi are continuous and periodic, and every smooth function on the circle arises from its even and odd parts. For x > 0, -i d/dx Psi_Q = Q_R' - i Q_L' = (D psi)_L + i (D psi)_R on Q, and -i d/dx Psi_U = -i U_R' - U_L' = (D psi)_R - i (D psi)_L on U, which are the transforms of D_0 psi; x < 0 is the same with the odd parts. For the connection term, with M* the block of D_F, the term E Gamma_F D_F sends Psi_Q to M* Psi_U and Psi_U to M Psi_Q. ■

In this form D_F is a constant connection on the circle, with holonomy exp(2 i l D_F) up to the convention for parallel transport, and the two walls are the fixed points of the reflection that carries D_F to -D_F. With U = exp(-i pi Gamma_0/4), the identity of lemma 23 reads Gamma_1 Gamma_0 = exp(2 l Gamma_0 D_F) = U exp(2 i l D_F) U*. This form of the interval as a circle with a reflection is the usual orbifold description. It also gives a one-line proof of the spectrum in Theorem 22.1.

The frame. By Corollary 22.2, D_X = R (D_0 + E Gamma_F D_F) R*, with R(x) = exp(x Gamma_F D_F). A generator T constant after transport is R(x) T R(x)* in the original frame of D_X, which generally varies with x and has an odd bulk component. Conversely, a constant generator in the original frame need not preserve the coupled wall, if it is broken. So the field used here requires an admissible enlarged gauge algebra, as in the prealgebra P of lemma 23, and it is generally not a field of the even algebra of lemma 22.

The kernel over the tower

Let A_mu(y) = a_mu(y) T on M^4 = R^4, with T even and constant along x, and consider D = -i gamma^mu (d_mu + i A_mu) + gamma_5 V with V = -i d/dx (x) 1 + 1 (x) D_F. Theorem 1 of lemma 18 has this form, with V_0 = V, whose eigenvalues are k_n + mu_i, and T_{n i, n' j} = delta_{n n'} T_{ij}.

Theorem 25.2. With d_ij = mu_i - mu_j, for the heat cutoff t = 1/Lambda^2, the kernel is

```
C_t(q^2) = (1/(8 pi^2)) sum_{n in Z} sum_{ij} |T_ij|^2 [ (2/t + d_ij^2 + q^2) I_t((k_n + mu_i)^2, (k_n +
```

with I_t as in lemma 18. The sum converges, and it holds without expansion in D_F. Expanding in D_F, with C_t = C_0 + q^2 C_1 + …, and with t = 1, so with k_n = n pi/(l Lambda) dimensionless,

C_0 = (1/(8 pi^2)) ||[D_F, T]||^2 sum_n e^{-k_n^2} (1 - (2/3) k_n^2) + O(||D_F||^4),
C_1 = (1/(8 pi^2)) (2/3) ||T||^2 sum_n e^{-k_n^2} + O(||D_F||^2).

So, with <k^2> = sum k_n^2 e^{-k_n2} / sum e^{-k_n2}, at fixed s and for fixed nonzero T,

M^2 = C_0/C_1 = (3/2 - <k^2>) ||[D_F, T]||^2/||T||^2 + O(||D_F||^4 / Lambda^2).

Proof. V and T split into finite-dimensional blocks indexed by n, so equation (1) of lemma 18 applies to symmetric finite sets of blocks, and the limit is justified by Gaussian bounds, uniform for D_F in a bounded neighborhood of zero. The same bounds justify the Taylor expansion under the sum. Terms odd in k cancel between n and -n: exactly, the bracket at q = 0 is d_ij^2 int_0^1 E(u) [1 - u(1-u)(2k + mu_i + mu_j)^2] du, with E(u) = exp(-(1-u)(k + mu_i)^2 - u (k + mu_j)^2), whose next term is d_ij^2 e^{-k2} ((2k^3 - 5k)/3)(mu_i + mu_j), and the bracket at order q^2 begins e^{-k2} (2/3 - (2k/3)(mu_i + mu_j)). For the leading term of the expansion, put a_i = (k + mu_i)^2 = k^2 + p mu_i + mu_i^2 with p = 2k. To second order in mu,

I(a_i, a_j) = e^{-k^2} [1 - (p/2)(mu_i + mu_j) - (mu_i^2 + mu_j^2)/2 + (p^2/6)(mu_i^2 + mu_j^2 + mu_i mu

and e^{-a_i} = e^{-k2} [1 - p mu_i - mu_i^2 + p^2 mu_i^2/2]. Therefore

2 I(a_i, a_j) - e^{-a_i} - e^{-a_j} = -(p^2/6) d_ij^2 e^{-k^2},

and adding d_ij^2 I gives d_ij^2 e^{-k2} (1 - p^2/6) = d_ij^2 e^{-k2} (1 - (2/3) k^2). Finally, sum_ij |T_ij|^2 d_ij^2 = ||[D_F, T]||^2. For C_1, at D_F = 0 the bracket in equation (3) of lemma 18 has d_ij = 0 and integrates to 2/3. ■

Differentiating directly, <k^2> = (1/2) d log S/d log s = (s/2) d log S/ds, with S(s) = sum_n exp(-n^2 pi$^{2/s}$2). This S is the heat trace of the Dirac operator of the unshifted circle of length 2l at t = 1/Lambda^2, equivalently the paired Neumann and Dirichlet trace on the interval, with S_N = (S + 1)/2 and S_D = (S - 1)/2 separately. So d_eff = d log S/d log s = -2 d log Tr e^{-t D^2}/d log t is its spectral dimension at that scale. For nonzero D_F, the parent trace has e^{-t (k_n + mu_i)^2}, and S is the unperturbed trace used in the leading expansion. Hence, at leading order,

M^2 = ((3 - d_eff)/2) ||[D_F, T]||^2/||T||^2.

The two limits. We first take the coefficient at small D_F/Lambda at fixed s, and then the limit in s. Explicitly, d_eff ~ (4 pi$^{2/s}$2) exp(-pi$^{2/s}$2) as s -> 0, and d_eff = 1 - 4 s^2 exp(-s^2) + O(s^2 exp(-2 s^2)) as s -> infinity.

1. For s « 1, only n = 0 contributes. Then d_eff = 0, the factor is 3/2, and this is the towerless regime of lemma 18 and Proposition F of the paper.
2. For s » 1, the sum is a Gaussian integral, with <k^2> = 1/2. Then d_eff = 1, and the factor is 1. This is forced by covariance on the unfolded circle: V is the fifth component of a five-dimensional Dirac operator with constant A_5 = D_F, and the local five-dimensional heat coefficient contains F_{MN} F^{MN}, with F_{5 mu} = i [D_F, A_mu] up to convention and with the same coefficient as F_{mu nu}^2. So the mass is the Hilbert-Schmidt commutator norm alone, as at small angle for the Yang-Mills parent of Theorem 23.6. On a general space with boundary, the bulk coefficient alone does not exclude boundary contributions; here the operator with these transported fields is unitarily equivalent to the circle problem, which is the reason they are absent.

In table form, at small masses:

s = l Lambda	0.3	1	1.5	2	2.5	3	5
(3 - d_eff)/2	1.5000	1.4990	1.3935	1.1414	1.0240	1.0022	1.0000

Corollary 25.3 (the sum rule). For the three-generation Standard Model representation, with the literal particle and antiparticle trace, with Dirac masses small compared with Lambda and with 1/l, and with T a

charged generator of SU(2)_L, at leading order,

```
M_W^2 = ((3 - d_eff)/24) sum_f N_f m_f^2,
```

which at d_eff = 0 is 8 M_W^2 = sum N_f m_f^2, and at d_eff = 1 is 12 M_W^2 = sum N_f m_f^2. For the top alone,

```
m_t/M_W = (8/(3 - d_eff))^{1/2},
```

from sqrt(8/3) = 1.633 at d_eff = 0 to 2 at d_eff = 1. The neutral direction scales with the same factor, so M_W$^{2/M_Z}$2 = 5/8 at all s to this order, and the massless directions keep zero mass coefficient, while their kinetic coefficients can change. These are leading-order statements: already in the towerless top-only family, M_W^2 = (3/8) m_t^2 - (23/128) m_t$^{4/\mathrm{Lambda}}$2 + … (equation (19) of lemma 18).

Proof. The trace gives ||T||^2 proportional to Q_11 = 12 and |||[D_F, T]||^2 proportional to K_11 = sum N_f m_f^2, which gives the coefficient. At d_eff = 0 this is the CCM relation (hep-th/0610241, Eq. (4.16)), as recovered in Proposition 4 and Remark 16 of lemma 18 at small x. The factor is common to all generators in Theorem 25.2. ■

In terms of Yukawa couplings, with m_t = y_t v/sqrt(2) and M_W = g v/2, this reads y_t^2 = 4 g^2/(3 - d_eff), so y_t = 2 g/sqrt(3) at d_eff = 0, the top-dominance value of CCM (Eq. (5.27) there), and y_t = sqrt(2) g at d_eff = 1.

Remark (positive Laplace cutoffs). For f(z) = int e^{-tz} dnu(t) as in equation (4) of lemma 18, with t measured in units of 1/Lambda^2, the same computation gives at leading order

```
M^2 = ((3 - d_nu)/2) ||[D_F, T]||^2/||T||^2,     S_nu(s) = int S(s/sqrt(t)) dnu(t),     d_nu = d log S_
```

where the average has weights dnu(t) e^{-t k_n^2}. Under the moment hypotheses there, dominated convergence gives the same endpoints, and the crossover and its corrections depend on nu. A general smooth cutoff, such as one flat near zero, need not belong to this class.

Scope

- This is the Taylor quotient of a prescribed constant-profile vector, at leading order in D_F, about a prescribed background and not a demonstrated vacuum. Taylor and pole masses differ in general. Calling it the leading term of a lowest-mode mass also needs an admissible gauge field, a positive kinetic form and an isolated lowest branch; on the circle, translation invariance separates the zero Fourier sector, within which the generator matrix must still be diagonalized. For finite angles in the Yang-Mills parent, the exact masses are in Theorem 23.6, which expands as M_W^2 = 2 m^2 - (4/3) m^4 l^2 + … and M_Z^2 = (16/5) m^2 - (64/75) m^4 l^2 + …, in agreement with d_eff = 1 at leading order. That theorem is for its own Yang-Mills boundary problem, and it does not establish the full finite-cutoff gauge spectrum of the spectral action.
- T must be even and constant in the transported frame, so it preserves both walls there, and it needs the enlarged gauge algebra noted above. Vectors that are Dirichlet at a strict wall, as in Proposition 22.8, are a different boundary problem.
- All fields have flat profiles on one interval, with no bulk masses. In particular, for the top-only statement, the kinetic term counts all twelve doublets of three generations on the same interval, with the same S. With bulk masses to localize light fields, the spectra and overlaps change, and the kinetic coefficient must be recomputed from all eigenstates and overlaps.
- The spectral action is taken as a function of the fermion operator at a matching scale, with no running.
- For a boundary with reflection, the walls contribute heat-kernel boundary terms in general. Here the unfolding removes them for this operator and these generators.
- Other priority: the endpoint m_t = 2 M_W appears at tree level in Cacciapaglia, Csaki and Park, Fully radiative electroweak symmetry breaking, arXiv:hep-ph/0510366, section 5, for a suitable rank-four representation, and a limit approaching 2 M_W in Adachi and Maru, arXiv:1804.06012, Eq. (3.23). These are representation-dependent mechanisms, and they do not give the spectral-action crossover. Universal propagation alone does not impose it in ordinary universal extra dimensions (Cordero-Cid et al., arXiv:1108.2926, section 2).

Reading

This connects three items of the project.

1. *The emergence of a dimension.* The heat dimension of the interval, which distinguishes the labels 10 and 11, enters the mass relation directly. As the cutoff resolves the interval, the coefficient moves from the finite-space value to the five-dimensional value.
2. *Proposition F and lemma 13.* The factor 3/2 is the finite-action coefficient relative to the Yang-Mills commutator quotient, and in this constant-profile calculation the two are the two ends of one function. Proposition F also contains a bounded-fermion against tower-scale obstruction and overlap conditions, which keep their separate hypotheses.
3. *The measured ratio.* With the Particle Data Group 2026 listings, m_t = 172.60 +- 0.27 GeV as a direct-measurement Monte Carlo mass, and M_W = 80.3625 +- 0.0077 GeV, the ratio is 2.148, which would formally correspond to d_eff = 1.27, outside the range. With a running mass of about 163 GeV (Yoon and Peskin, arXiv:1810.12352, section 3.3 and Eq. (23), for a pole mass of 173 GeV, with about 147 GeV at 2 TeV), it is 2.03, which corresponds to d_eff = 1.06. Dividing a running mass by an on-shell W mass is only a formal orientation and not a matching test at a common scale, and no running from the matching scale has been computed, so we draw no conclusion from this comparison. The known spectral-action value y_t = 2 g/sqrt(3) is the d_eff = 0 boundary condition. A tower near the weak scale, as in the separate unrefereed estimate of the steer note, item 9, would use the d_eff = 1 boundary condition y_t = sqrt(2) g, at a matching scale of that order, in a specified effective theory with thresholds and a renormalization prescription. In a one-loop orientation with universal bulk fields (the steer note, item 11, no referee), the running approaches an infrared quasi-fixed point near 211 to 232 GeV, nearly independently of the boundary value, so that family does not support the closeness of 2 M_W to the running top mass.

Numerical check

By Claude Opus 5.5, in Python (numpy), 3 Nov 2026, in a temporary directory, not kept. The kernel of lemma 18, with quadrature in u, was summed over the tower, with explicit 4 x 4 level blocks for the operator on the interval, independently of the unfolding.

- Random complex K_1 and Hermitian T_Q, T_U on C^3, with ||K|| = 10^-3 min(s, 1): the ratio of C_0/C_1 to the Hilbert-Schmidt quotient agrees with 3/2 - <k^2> to six digits, for s in {0.3, 1, 1.5, 2, 2.5, 3, 5, 10}.
- Three generations, top only, with T = sigma_x/2 on the U-fields: m_t/M_W agrees with (8/(3 - d_eff))^{1/2} to six digits on the same grid, from 1.63299 at s = 0.3 to 2.00000 at s = 5.
- The neutral direction, with weights for lepton and colour: $M_Z^{2/m}2$ goes from 4.801 to 3.200 for universal mass, and $M_W^{2/M_Z}2$ stays at 0.625 to four digits.
- Finite mass in the top-only family: $M^{2/m}2$ at s = 5 is 0.2500, 0.2498, 0.2477, 0.2409, 0.2241, 0.1478 for m/Lambda = 0.02, 0.1, 0.3, 0.6, 1, 2, and at s = 0.5 it is 0.3749, 0.3732, 0.3591, 0.3145, 0.2277, 0.0555, the latter matching lemma 18.

Part L (literature check and record of searches)

Literature check for claims A-I

Written by Claude Opus 5.5 on 3 November 2026, as recommendation 1 of evaluation2026-11-03.md. Method: web searches (standard mode, extended mode where results were thin), arXiv abstract pages and full texts, Crossref and INSPIRE metadata for every journal reference below. Where an equation or section number is given, it was read in the full text (arXiv version unless stated). Items that could not be checked are marked **unverified** and collected at the end.

Verdicts: **KNOWN** (same statement in print), **PARTIAL** (a close or weaker version in print; the difference is stated), **NOT FOUND** (no source located after the searches listed).

A. Connes-Lott as the minimal spectral truncation of a chiral interval

Claim (lemma 05, Part 3; lemma 09, Theorem 9.6). The minimal nonzero grading-preserving spectral truncation of the Dirac operator on [0, l] with R(0) = 0, L(l) = 0 is unitarily equivalent to the Connes-Lott two-point triple (C + C, D = [[0, m], [m, 0]]) with |m| = pi/(2l); the compressed operator system P C([0, l]) P equals C + C; the truncated end states are at distance 2l/pi. The deconstructed chain of Part 1 has Connes-Lott as its N = 2 member.

Verdict: PARTIAL. The interpretation of the two-point space as a discrete or Kaluza-Klein fibre, the truncation machinery, the deconstruction picture and the lowest mixed-boundary mass pi/(2l) are all in print. We found no source that identifies the Connes-Lott triple as a spectral truncation of an interval Dirac operator with chiral boundary conditions, computes the compression P C([0, l]) P for such an interval, or relates |m| to the interval length.

Interpretation of the two-point space (in print).

- A. Connes and J. Lott, *Particle models and noncommutative geometry*, Nucl. Phys. B Proc. Suppl. 18B (1991) 29-47, doi:10.1016/0920-5632(91)90120-4; author's copy at math.berkeley.edu/~lott/partmodels.pdf. Section I: "The corresponding geometric space can be thought of as a spacetime of Kaluza-Klein type, where the fiber is formed of a two-point set." Section II.B, item (9): for the two-point space, d(p, p') = L, where L^{-2} is the largest eigenvalue of M*M. Section III (p. 36): the character space of the product is two copies of Z, "Thus the picture is of two copies of Z separated by a distance L. (As one loses information in passing to the character space, this picture should not be taken too literally.)", with (d_Z(p, p')^2 + L^2)^{1/2} <= d(p, p') <= d_Z(p, p') + L.
- N. A. Viet and K. C. Wali, *Noncommutative geometry and a discretized version of Kaluza-Klein theory with a finite field content*, Int. J. Mod. Phys. A 11 (1996) 533-551, doi:10.1142/S0217751X96000249, arXiv:hep-th/9412220; and *A discretized version of Kaluza-Klein theory with torsion and massive fields*, Int. J. Mod. Phys. A 11 (1996) 2403-2418, doi:10.1142/S0217751X96001206, arXiv:hep-th/9508032. The second abstract: "We consider an internal space of two discrete points in the fifth dimension of the Kaluza-Klein theory." The "discrete fifth dimension" reading is therefore explicit in print since 1995.
- A. Dimakis and F. Müller-Hoissen, *Discrete differential calculus, graphs, topologies and gauge theory*, J. Math. Phys. 35 (1994) 6703-6735, doi:10.1063/1.530638, arXiv:hep-th/9404112. Abstract: the formalism "suggests that the [Connes-Lott two-point space] be regarded as an approximation of a manifold and thus opens a way to relate models with an 'internal' discrete space (à la Connes et al.) to models of dimensionally reduced gauge fields". Section III, Example 4: the universal differential algebra on two points has a Hasse diagram approximating the topology of S^2, linked to Manton's six-dimensional Yang-Mills reduction on S^2. Section VII: the Connes-Lott models should be understood "as approximations of higher-dimensional gauge theory models … The details have still to be worked out, however." This is the closest statement of the programme of lemma 05. The approximation there is topological (posets and Hasse diagrams) and the internal space is S^2; the passages we read contain no Dirac operator, boundary condition or mass-length relation.
- P. Martinetti and R. Wulkenhaar, *Discrete Kaluza-Klein from scalar fluctuations in noncommutative geometry*, J. Math. Phys. 43 (2002) 182-204, doi:10.1063/1.1418012, arXiv:hep-th/0104108. Section V, Theorem 4 and eq. (27): the distance between the same point on the two sheets is 1/||M|| (M the off-diagonal block of D_F), with a Pythagorean relation when [D_I, p] = 0. In the proof of Theorem 4 the authors embed the two sheets in a "virtual" five-manifold [0, 1] x M as a proof device and state that "the points between the sheets are not part of the geometry". Section VI: after scalar fluctuations the internal metric component becomes a function on M (the Higgs field); Section VII, Proposition 8, gives this component for the Standard Model. "Discrete Kaluza-Klein" refers to this Higgs-as-g_55 picture. Spectral truncation and boundary conditions do not occur.
- B. Iochum, T. Krajewski and P. Martinetti, *Distances in finite spaces from noncommutative geometry*, J. Geom. Phys. 37 (2001) 100-125, doi:10.1016/S0393-0440(00)00044-9: distances on finite spectral triples, including the two-point case.

- Matrix and discrete Kaluza-Klein precursors (bibliographic data verified; contents known from abstracts and secondary sources only): M. Dubois-Violette, R. Kerner and J. Madore, *Noncommutative differential geometry and new models of gauge theory*, J. Math. Phys. 31 (1990) 323 (Higgs fields as components of connections in matrix directions); J. Madore, *Kaluza-Klein aspects of noncommutative geometry* (1989 proceedings, posted as arXiv:1510.03915); J. Madore and J. Mourad, *Noncommutative Kaluza-Klein theory*, arXiv:hep-th/9601169; R. Coquereaux, G. Esposito-Farèse and G. Vaillant, *Higgs fields as Yang-Mills fields and discrete symmetries*, Nucl. Phys. B 353 (1991) 689 (**unverified** beyond citations); A. H. Chamseddine, J. Fröhlich, B. Schubnel and D. Wyler, *Dimensional reduction without continuous extra dimensions*, J. Math. Phys. 54 (2013) 012302, arXiv:1201.2661.

Spectral truncation (in print, for other spaces).

- A. Connes and W. D. van Suijlekom, *Spectral truncations in noncommutative geometry and operator systems*, Commun. Math. Phys. 383 (2021) 2021-2067, doi:10.1007/s00220-020-03825-x, arXiv:2004.14115. Section 1 (Introduction, pp. 2-3 of the arXiv version): "a natural spectral truncation is given simply by a spectral projection P onto eigenspaces of D … The natural truncation of the action of the -*algebra A to the Hilbert space PH is given by the space PAP. Since P does not commute with A, this is not an algebra anymore. Moreover in many interesting examples if one takes the C*-algebra generated by PAP one gets a non-informative full matrix algebra. However, not all is lost: the space PAP is a *-closed subspace in B(H), i.e. it is a so-called operator system". The same passage notes that symmetries of (A, D) induce symmetries of (PAP, PDP). The worked example is the circle (Toeplitz operator systems, Sections 4-5). Relation to lemma 05: the compression map Q of eq. (18) is non-multiplicative, as in this general picture, but for the chiral interval the minimal truncation has range exactly C + C, an algebra; this exceptional case is absent from the paper.
- F. D'Andrea, F. Lizzi and P. Martinetti, *Spectral geometry with a cut-off: topological and metric aspects*, J. Geom. Phys. 82 (2014) 18-45, doi:10.1016/j.geomphys.2014.03.014, arXiv:1305.2605. Cut-off by a projection on D and/or A; points are approximated by non-pure truncated states at finite distance; on the circle these are Fejér distributions; applications to the Moyal plane and the fuzzy sphere. The interval does not appear.
- Follow-ups on truncations (all circle, tori or groups): E.-M. Hekkelman, *Truncated geometry on the circle*, Lett. Math. Phys. 112 (2022), doi:10.1007/s11005-022-01514-5; L. Glaser and A. Stern, *Reconstructing manifolds from truncations of spectral triples*, J. Geom. Phys. 159 (2021) 103921; M. Leimbach and W. D. van Suijlekom, *Gromov-Hausdorff convergence of spectral truncations for tori*, Adv. Math. 439 (2024) 109496, arXiv:2302.07877.

Deconstruction and lattices.

- N. Arkani-Hamed, A. G. Cohen and H. Georgi, *(De)constructing dimensions*, Phys. Rev. Lett. 86 (2001) 4757-4761, doi:10.1103/PhysRevLett.86.4757, arXiv:hep-th/0104005.
- C. T. Hill, S. Pokorski and J. Wang, *Gauge invariant effective Lagrangian for Kaluza-Klein modes*, Phys. Rev. D 64 (2001) 105005, doi:10.1103/PhysRevD.64.105005, arXiv:hep-th/0104035. The abstract states that the chain of gauge groups linked by bifundamentals "matches the continuum compactified theory truncated in momentum space", and treats chiral fermions. This is the physics statement "deconstruction = momentum truncation of the KK tower".
- H.-C. Cheng, C. T. Hill, S. Pokorski and J. Wang, *The standard model in the latticized bulk*, Phys. Rev. D 64 (2001) 065007, arXiv:hep-th/0104179 (split-generation fermions on a transverse lattice).
- J. Dai and X.-C. Song, *Spontaneous symmetry broken condition in (de)constructing dimensions from noncommutative geometry*, ICFP 2001 proceedings, doi:10.1142/9789812777379_0044, arXiv:hep-ph/0105280. The only paper found that applies NCG (differential calculus on cyclic groups, in the Dimakis-Müller-Hoissen/Sitarz style) to the Arkani-Hamed-Cohen-Georgi model; it treats the bosonic vacuum, without spectral triples or fermion truncation.
- NCG distances on lattices: G. Bimonte, F. Lizzi and G. Sparano, *Distances on a lattice from noncommutative geometry*, Phys. Lett. B 341 (1994) 139-146, arXiv:hep-lat/9404007 (the anomalous lattice distances that lemma 05, Part 1 calls "the familiar lattice-distance anomaly"); E. Atzmon, *Distances on a one-dimensional lattice from noncommutative geometry*, Lett. Math. Phys. 37 (1996) 341-348, doi:10.1007/BF00343197.

- The lowest Dirac (or gauge) mass pi/(2l) for mixed boundary conditions on an interval is textbook interval physics; see C. Csáki et al., *Fermions on an interval* (reference in C), Section 3 and Appendix B.

What remains unmatched. The identification "minimal grading-preserving spectral truncation of the chiral interval = Connes-Lott with |m| = pi/(2l)", the computation P C([0, l]) P = C + C, the truncated end-state distance 2l/pi against the geodesic length l, and the chain family of Part 1 with an exact Connes-Lott member at N = 2 and Gromov-Hausdorff convergence to [0, l].

Searched: "two-point space" with "spectral truncation", "interval", "boundary conditions"; "discrete Kaluza-Klein"; "deconstruction" with "noncommutative geometry", "spectral triple"; "gauge-Higgs unification" with "Connes-Lott"; "spectral truncation" with "interval [0,1]" (extended mode); NCG plus orbifold S^1/Z_2.

B. The two limits of the massive chiral interval

Claim (lemma 09, Theorems 9.8-9.9, Corollary 9.10). As l -> infinity at fixed bulk mass, the light pair becomes two chiral wall modes and the rest of the spectrum obeys a one-dimensional Weyl law above the gap mu. As l -> 0 with mu(l) l ~ log(1/(m_f l)), a Connes-Lott pair of fixed mass m_f survives while the interval collapses; at fixed mu no light fermion survives.

Verdict: PARTIAL. The spectral physics is classical: wall zero modes, exponentially small mixing of the two boundary modes on a finite slab, and the resulting light mass ~ 2 mu e^{-mu l}. Collapse of spectral triples is an active topic in NCG, with a general result for collapse to a point. We found no source stating either limit as a theorem about families of finite or interval spectral triples, and no source for the tuned collapse mu(l) that keeps a Connes-Lott pair.

References.

- R. Jackiw and C. Rebbi, *Solitons with fermion number 1/2*, Phys. Rev. D 13 (1976) 3398-3409, doi:10.1103/PhysRevD.13.3398: the wall zero mode.
- D. B. Kaplan, *A method for simulating chiral fermions on the lattice*, Phys. Lett. B 288 (1992) 342-347, doi:10.1016/0370-2693(92)91112-M, arXiv:hep-lat/9206013: domain-wall fermions.
- Y. Shamir, *Chiral fermions from lattice boundaries*, Nucl. Phys. B 406 (1993) 90-106, doi:10.1016/0550-3213(93)90162-I, arXiv:hep-lat/9303005. Section 2: a Weyl mode bound to the boundary of a half-infinite fifth direction, profile b^s with |b| < 1 (eq. (14)). Section 3: with a finite fifth direction 0 <= s <= N, "a second chiral fermion appears on the new boundary at s = N … this fermion has the opposite chirality … Strictly speaking, there is a tiny mixing between the two chiral modes, which vanishes like (1 - M)^N." This is the lattice version of Lemma 9.5 and Theorem 9.8.
- C. G. Callan and J. A. Harvey (1985), see section H: the continuum domain-wall zero modes and anomaly inflow.
- C. Csáki, C. Grojean, J. Hubisz, Y. Shirman and J. Terning, *Fermions on an interval: quark and lepton masses without a Higgs*, Phys. Rev. D 70 (2004) 015012, doi:10.1103/PhysRevD.70.015012, arXiv:hep-ph/0310355. Section 2: boundary conditions from the variational principle; Section 4.1, eqs. (4.8)-(4.9): a bulk mass profile realizing Dirichlet-Neumann conditions, with a tan(m_n L) quantization; Appendix B: KK decompositions in flat space.
- C. Farsi and F. Latrémolière, *Collapse in noncommutative geometry and spectral continuity*, J. London Math. Soc. (2) 114 (2026) e70628, doi:10.1112/jlms.70628, arXiv:2404.00240. Collapse of the vertical part of spectral triples in the spectral propinquity, with spectral continuity; Section 4.1, Corollary 4.1: collapse of a spectral triple to a point gives a nontrivial limit acting on ker D (read in the published text). The published text (first published 3 July 2026, read on 3 Nov 2026 from the journal HTML) agrees with arXiv v3 in the statements and equations compared (copy-editing, and L_A corrected to L_B in Definition 2.13): Theorem 3.4 (Theorem 2.5 in v1) keeps the same six hypotheses, and the step at issue is printed as eq. (3.11), so Counterexample R1 of the assessment applies to the printed statement. Relation to lemma 09: Corollary 9.10 (fixed mu, condition (c), ker D = 0, no light

sector) agrees with this picture. Theorem 9.9 keeps a nonzero light pair because the Dirac operator itself changes with l (mu = mu(l)); that family lies outside the metric-rescaling collapse treated there. **Assessed in detail:** reviews/2026-11-03_farsi_latremoliere_assessment.md. Corollary 4.1 (a fixed triple scaled by 1/eps converges to (C, ker D, 0), and has no limit when ker D = 0) is sound. The printed hypotheses of Theorem 3.4 (Theorem 2.5 in v1) are insufficient; anticommutation of D_h and D_v, or a uniform relative bound, supplies a sufficient additional hypothesis; Counterexample R1 there, checked by hand by Claude and by Terra 6.1, satisfies the six listed hypotheses and contradicts Corollary 3.5. The anticommuting applications avoid this defect; Theorem 4.6 and Corollary 4.7 also need a nonzero vertical kernel for the displayed target to be a spectral triple. The published version keeps the statement without the hypothesis. Validated by Astra 6 (Astra validation, items 2 and 4). Our tuned collapse (Theorem 9.9) lies outside its hypotheses.

- B. Ammann and C. Bär, *The Dirac operator on nilmanifolds and collapsing circle bundles*, Ann. Global Anal. Geom. 16 (1998) 221-253, doi:10.1023/A:1006553302362, arXiv:math/9801091: in commutative geometry, collapse either sends all Dirac eigenvalues to infinity or keeps a convergent part, depending on the spin structure.

What remains unmatched. The statement of both limits for the interval spectral triple with condition (c) as theorems (Theorems 9.8-9.9), the formula c = log(2c/(m_f l)) + o(1) for the bulk mass needed to keep a fixed light mass, and the identification of the surviving pair with Connes-Lott at each l (Theorem 9.6).

Searched: domain-wall fermions on a finite slab (exponentially small mass); "collapse" with "spectral triple"; "decompactification" and "emergent dimension" with "almost-commutative"; Ammann-Bär collapse and its NCG follow-ups.

C. Towerless Yukawa textures

Claim (lemma 12). For bulk fields Q, U on [0, l] with bulk masses and a boundary Yukawa matrix X at x = l: an exact secular equation det(lambda^2 - A X B X^dagger) = 0 (Theorem 12.4); the one-flavour, massless spectrum (n pi +- arctan|X|)/l (Example 12.5); singular-value bounds (Theorem 12.6); in the limit l -> 0 at fixed c and Y_hat = X/l the low spectrum converges to the finite triple with Yukawa block M_F = Phi_Q Y_hat Phi_U (Theorem 12.7); graded singular values of F X G (Lemma 12.8).

Verdict: PARTIAL. The physics is in print, in more elaborate (warped) form: exact KK determinant equations with brane Yukawa matrices, the effective Yukawa F(c_Q) Y F(c_u), masses from minors and mixings F(c_Qi)/F(c_Qj). The one-flavour flat spectrum is Csáki et al. eq. (5.6) (KNOWN). We found no NCG paper that derives the finite Dirac operator D_F, or its Yukawa texture, from localization on an extra dimension.

References.

- C. Csáki et al., *Fermions on an interval* (full data in B). Section 5.1, eqs. (5.4)-(5.6): flat interval, no bulk mass, boundary conditions psi_eL(0) = 0, chi_eR(0) = 0, psi_eL(L) = -M_D L psi_eR(L), chi_eR(L) = M_D L chi_eL(L), spectrum tan(m_n L) = M_D L. With X = M_D L and signed 4D masses, the roots are (n pi +- arctan|X|)/L, which is Example 12.5. Section 2, eqs. (2.8)-(2.9): general boundary conditions with matrices; the spectra are computed one field at a time (Section 7.1: "a much richer spectrum of possibilities for such mixings which we will however not deal with here"). Section 6.2.2, eq. (6.35): secular equation for bulk mass plus brane Majorana mass (warped). Appendix C.2, eqs. (C.9)-(C.14): the boundary parameter equals M_D L, 2 tanh(M_D L/2) or M_D L/(1 + M_D^2 L^2/4) depending on how the brane mass is regularized, so the X of lemma 12 is a boundary-condition parameter whose relation to a brane Lagrangian mass is scheme dependent.
- S. Casagrande, F. Goertz, U. Haisch, M. Neubert and T. Pfoh, *Flavor physics in the Randall-Sundrum model I*, JHEP 10 (2008) 094, doi:10.1088/1126-6708/2008/10/094, arXiv:0807.4937. Section 3.2: IR boundary conditions with the Yukawa matrices (eqs. (55), (63)) and the exact determinant condition for the KK masses, eq. (65), for N generations; eq. (71): profile normalization F(c); eq. (87): Y_eff = F(c_Q) Y F(c_u); eq. (90): zero-mode approximation det(m_n^2 - (v^2/2) Y_eff Y_eff^dagger)

= 0; eqs. (94)-(96): masses as F(c_Q) F(c_u) times ratios of minors of Y; eq. (97): (U_q)_ij ~ F(c_Qi)/F(c_Qj). This is the warped analogue of Theorems 12.4, 12.7, Corollary 12.9 and Remark 12.10.

- Flat and warped localization: N. Arkani-Hamed and M. Schmaltz, *Hierarchies without symmetries from extra dimensions*, Phys. Rev. D 61 (2000) 033005, doi:10.1103/PhysRevD.61.033005; T. Gherghetta and A. Pomarol, *Bulk fields and supersymmetry in a slice of AdS*, Nucl. Phys. B 586 (2000) 141-162, doi:10.1016/S0550-3213(00)00392-8; Y. Grossman and M. Neubert, *Neutrino masses and mixings in non-factorizable geometry*, Phys. Lett. B 474 (2000) 361-371, doi:10.1016/S0370-2693(00)00054-X; S. J. Huber and Q. Shafi, *Fermion masses, mixings and proton decay in a Randall-Sundrum model*, Phys. Lett. B 498 (2001) 256-262, doi:10.1016/S0370-2693(00)01399-X; K. Agashe, G. Perez and A. Soni, *Flavor structure of warped extra dimension models*, Phys. Rev. D 71 (2005) 016002, doi:10.1103/PhysRevD.71.016002; D. E. Kaplan and T. M. P. Tait, *New tools for fermion masses from extra dimensions*, JHEP 11 (2001) 051, doi:10.1088/1126-6708/2001/11/051, arXiv:hep-ph/0110126 (flat compact extra dimension, zero modes localized by odd bulk masses, Higgs on a boundary; the flat-space setting closest to lemma 12).
- Graded matrices: C. D. Froggatt and H. B. Nielsen, *Hierarchy of quark masses, Cabibbo angles and CP violation*, Nucl. Phys. B 147 (1979) 277-298, doi:10.1016/0550-3213(79)90316-X; J. Demmel, M. Gu, S. Eisenstat, I. Slapničar, K. Veselić and Z. Drmač, *Computing the singular value decomposition with high relative accuracy*, Linear Algebra Appl. 299 (1999) 21-80 (singular values of D_1 X D_2 with well-conditioned X are determined to high relative accuracy; theorem numbers **unverified**). The leading-order "masses = products of profiles times ratios of minors" is Casagrande et al. eq. (96). We did not find the two-sided compound-matrix bound of Lemma 12.8 in this explicit form; its proof (Cauchy-Binet on the p-th compound) is routine.
- NCG side: Yukawa matrices enter D_F as free moduli. T. Krajewski (1998) and A. Connes (2006, Theorem 2.7) classify the admissible D_F; see section I. No derivation of these moduli from localization was found.

What remains unmatched. The flat-space exact secular equation for an arbitrary N_Q x N_U boundary matrix with bulk masses, stated with multiplicities; the two-sided bound of Theorem 12.6 (via concavity of alpha, beta in lambda^2); the towerless limit l -> 0 at fixed c and Y_hat as a convergence theorem with error sigma^3 l^2 (1/L_Q^2 + 1/L_U^2) onto a finite even spectral triple; Corollary 12.3 on vectorlike residual theories; and the reading "D_F is the towerless remnant of internal momentum".

Searched: "noncommutative geometry" or "spectral triple" with "split fermions", "fermion localization", "Randall-Sundrum", "Yukawa overlap" (standard and extended); Csáki et al. full text; Casagrande et al. full text; graded-matrix SVD literature.

D. KO-dimension of products: 9, 10 or 11

Claim (lemmas 06, 07, 08). A finite KO-6 real triple is T_n x P_k with (s, n, k) = (+1, 7, 7) or (-1, 5, 1) iff a grading-reversing involution U commutes with A, A°, D and satisfies UJ = sJU (Theorem 06.2); J alone does not fix s; an unpaired nu_L forces the KO-7 reduction (No-go 06, with [U, Q] = 0); in the internal-first product with the interval, the bare reflection has s = -1 (Theorem 08.5).

Verdict: PARTIAL. Products of real spectral triples of all parities, including the doubled odd x odd product on H_1 x H_2 x C^2, the freedom J versus J gamma in even dimension, and the total KO-dimension 10 = 2 mod 8 of the Standard Model are in print. The converse direction, a criterion for when a given even KO-6 triple factors as (odd) x (odd point), was NOT FOUND, nor were the nu_L statement, the reflection sign, or any discussion of 9 or 11 as alternatives to 10.

References.

- L. Dąbrowski and G. Dossena, *Product of real spectral triples*, Int. J. Geom. Methods Mod. Phys. 8 (2011) 1833-1848, doi:10.1142/S021988781100597X, arXiv:1011.4456. Introduction: in even dimensions "there are always two real structure operators J, that differ by multiplication by the grading operator.

None of them should be preferred as they are perfectly on the same footing." Section 4.1 (Tables 2-3) and 4.2 (Tables 4-5): even-even and even-odd products. Section 4.3, eq. (25) and Table 6: the odd-odd product H = (H_1 x H_2) x C^2, D = D_1 x 1 x sigma_1 + 1 x D_2 x sigma_2, chi = 1 x 1 x sigma_3, J = J_1 x J_2 x M K. Section 6: metric and KO-dimension need not agree mod 8. Lemma 06.1 already cites eq. (25) and Table 6 for its product convention. The paper treats the forward direction (building products); the question of when a given triple is a product is left open there.

- F. J. Vanhecke, *On the product of real spectral triples*, Lett. Math. Phys. 50 (1999) 157-162, doi:10.1023/A:1007690509512, arXiv:math-ph/9902029: the naive product J may violate the sign table; a corrected product real structure.
- F. J. Vanhecke, A. R. da Silva and C. Sigaud, *On the finite spectral triple of an almost-commutative geometry*, Braz. J. Phys. 42 (2012) 471-474, doi:10.1007/s13538-012-0101-2, arXiv:1011.4925. Section 2.1: with J = J_1 x J_2, a Minkowski factor of signature 2 or 6 forces an odd internal factor (sigma_2 in {1, 5} or {3, 7}) and "the chirality paradigm is lost"; Section 2.2: with J = J_1 x J_2 Omega_2 an even internal factor is allowed and sigma_2 = 6 gives total signature 0 (Barrett's case). This is the closest printed discussion of odd internal factors, framed as a consequence of the product convention.
- S. Farnsworth, *The graded product of real spectral triples*, J. Math. Phys. 58 (2017) 023507, doi:10.1063/1.4975410, arXiv:1605.07035: graded tensor product, commutative and associative, always a spectral triple.
- B. Ćaćić, *Real structures on almost-commutative spectral triples*, Lett. Math. Phys. 103 (2013) 793-816, doi:10.1007/s11005-013-0616-7, arXiv:1209.4832: real reconstruction theorem; compact oriented manifolds carry real (almost-)commutative triples of every KO-dimension.
- A. Connes, *Noncommutative geometry and the standard model with neutrino mixing*, JHEP 11 (2006) 081, doi:10.1088/1126-6708/2006/11/081, arXiv:hep-th/0608226, Section 1: "the KO-dimension of the product space M x F is in fact equal to 10 ~ 2 modulo 8. Of course the above 10 is very reminiscent of string theory".
- J. W. Barrett, *Lorentzian version of the noncommutative geometry of the standard model of particle physics*, J. Math. Phys. 48 (2007) 012303, doi:10.1063/1.2408400, arXiv:hep-th/0608221. Section 1: J^2 = 1 and J gamma = gamma J "hold only if the total signature is equal to zero"; Section 2: the internal space is "determined to be a real spectral triple of signature six".
- A. H. Chamseddine and A. Connes, *Why the Standard Model*, J. Geom. Phys. 58 (2008) 38-47 (section I): the finite geometry raises "the K-theoretic dimension from four to ten (modulo eight)".
- K. van den Dungen and W. D. van Suijlekom, *Electrodynamics from noncommutative geometry*, J. Noncommut. Geom. 7 (2013) 433-456, doi:10.4171/JNCG/122: the KO-6 two-point triple for electrodynamics, the natural reference for Counterexample 06.3.
- Background: the complex analogue of Theorem 06.2 (an even triple with a commuting, grading-reversing Clifford generator is an odd triple doubled) is the standard identification of odd Fredholm modules with Cl_1-graded ones in Kasparov theory; we did not locate a reference stating the real KO-6 version with the sign s of UJ = sJU.
- Terminology: "doubling" in lemma 06 (H = H_0 x C^2) differs from "fermion doubling" in the NCG literature (F. Lizzi, G. Mangano, G. Miele and G. Sparano, Phys. Rev. D 55 (1997) 6357, section I; resolved in Connes 2006 by the Pfaffian). A paper should use a different word.

What remains unmatched. Theorem 06.2 (criterion), Result 06.3 (both s for electrodynamics), the nu_L no-go under [U, Q] = 0, Theorem 08.5 (s = -1 for the bare reflection in the internal-first product), Counterexample 08.11 and Theorem 08.12.

Searched: "real spectral triple" with "factorization", "product of odd", "Clifford", "KO-dimension 6"; "9" and "11" with "noncommutative standard model"; full texts of Dąbrowski-Dossena, Vanhecke-da Silva-Sigaud, Connes 2006 and Barrett 2007.

E. Spectral action: the W mass relation, the top ratio, heavy-fermion weights

Claim (lemma 11; lemma 03). 8 M_W^2 = sum N_f m_f^2 at unification; m_t/M_W -> sqrt(8/3) when the top dominates; the bounds m_max/(2 sqrt 2) <= M_W <= sqrt(N_g) m_max; gauge kinetic terms

weighted by sum_j f(m_j$^{2/\mathrm{Lambda}}$2) tr_{H_j} F^2.

Verdict: KNOWN for the mass relation and the top ratio; Proposition 11.1 and Corollary 11.2 are one-line consequences of it. **NOT FOUND** for the exact heavy-fermion weighting in this form.

References.

- A. H. Chamseddine, A. Connes and M. Marcolli, *Gravity and the standard model with neutrino mixing*, Adv. Theor. Math. Phys. 11 (2007) 991-1089, doi:10.4310/ATMP.2007.v11.n6.a3, arXiv:hep-th/0610241. Eq. (1.5) (Introduction): sum over generations of m_e^2 + m_nu^2 + 3 m_d^2 + 3 m_u^2 = 8 M_W^2. Lemma 4.6, eq. (4.16), proved from the normalization (4.17) Tr(k* k ...) = 2 g^2. Section 5.4 ("The fermion-boson mass relation"): eq. (5.24) m_top = v y_t/sqrt 2 with v = 2M/g; eqs. (5.25)-(5.26): Y_2(S) = 4 g^2; eq. (5.27): neglecting all Yukawas but the top, y_t = (2/sqrt 3) g at unification; eq. (5.28): y_t ~ 0.597. With M_W = g v/2 this is m_t/M_W = sqrt(8/3), which follows at once from (5.24) and (5.27); CCM state the relation in the form y_t = (2/sqrt 3) g. Eq. (5.29): if the tau-neutrino Dirac Yukawa equals the top Yukawa, y_t ~ g, i.e. m_t/M_W = sqrt 2.
- A. H. Chamseddine and A. Connes, *Conceptual explanation for the algebra in the noncommutative approach to the standard model*, Phys. Rev. Lett. 99 (2007) 191601 (abstract: a mass relation linking the fermion masses to the W mass, used to predict the top mass).
- Heavy-fermion weights: CCM Section 3 expands Tr f(D_A/Lambda) with D_F entering polynomially through the coefficients a, b, c, d, e, and Section 5.3 places the Majorana mass at the scale Lambda (Lemma 5.2, eq. (5.20)). The resummed weighting sum_j f(m_j$^{2/\mathrm{Lambda}}$2) of lemma 03 Part A was not found. Checked: M. Eckstein and B. Iochum, *Spectral Action in Noncommutative Geometry*, Springer-Briefs in Mathematical Physics 27 (2018), doi:10.1007/978-3-319-94788-4, arXiv:1902.05306 (abstract and contents only); B. Iochum, C. Levy and D. Vassilevich, *Spectral action beyond the weak-field approximation*, Commun. Math. Phys. 316 (2012) 595-613, doi:10.1007/s00220-012-1587-8 (second order in the gauge field with full momentum dependence; abstract only). The physics counterpart is decoupling: T. Appelquist and J. Carazzone, *Infrared singularities and massive fields*, Phys. Rev. D 11 (1975) 2856-2861, doi:10.1103/PhysRevD.11.2856. The evaluation already classes lemma 03 as a routine computation; a paper should present it as such.

F. Weinberg angle from a squashed S^3 / lens space and from M^{pqr}

Claim (canonical_research.md §14.3). For the Berger metric with fibre scaled by xi^2, K_W = V R^2 (2 + xi^2)/3, K_Y = V R^2 xi^2, hence cos^2 theta_W = 3 xi^2/(4 xi^2 + 2); the lens order p cancels; 5/8 needs xi^2 = 5/2.

Verdict: KNOWN (same formula, same parameterization).

- A. J. Niemi and S. Slizovskiy, *On relations between electroweak hierarchy problem, fluctuating three branes and general covariance*, arXiv:1004.0212 (no journal version found on INSPIRE; earlier titles "Hiding Higgs with brane" and "Embedding Weinberg-Salam in Kaluza-Klein"). Section IV, eqs. (77)-(78): internal metric dtheta^2 + sin^2 theta dpsi_-^2 + xi^2 (dpsi_+ - cos theta dpsi_-)^2, isometry SU(2)_L x U(1)_R for xi != 1; Section V, eqs. (94)-(95): q^2 = 3/(xi^2 + 2), q'^2 = 1/xi^2; Section VI: sin^2 theta_W = (xi^2 + 2)/(4 xi^2 + 2), i.e. cos^2 theta_W = 3 xi^2/(4 xi^2 + 2). They fix xi by a custodial-symmetry condition and quote sin^2 theta_W = 0.296 (that step, in Section VIII, was not read). The ratios 1/q^2 : 1/q'^2 = (2 + xi^2)/3 : xi^2 coincide with K_W : K_Y of §14.3. The cancellation of the lens order p in the common volume factor is immediate.
- E. Witten, *Search for a realistic Kaluza-Klein theory*, Nucl. Phys. B 186 (1981) 412-428, doi:10.1016/0550-3213(81)90021-3: eleven-dimensional supergravity on the seven-dimensional M^{pqr} with SU(3) x SU(2) x U(1) isometry; gauge couplings determined by the internal metric; the chirality obstruction. Read on 3 Nov 2026 from the PDF supplied by the user, and checked by Astra 6 (Astra validation, item 1). Eq. (3): gauge fields as components of the metric along Killing vectors. Pp. 416-417: seven is the minimum internal dimension with this symmetry, so eleven in total. P. 418: M^{pqr} = G/H with H = SU(2) x U(1) x U(1) and the deleted generator p lambda_8 + q

T_3 + r Y; M^{001} = CP^2 x S^2 x S^1, M^{101} = S^5 x S^2, M^{011} = CP^2 x S^3. Pp. 421-423: "One would inevitably get vector-like rather than V - A weak interactions", for a reason that "depends only on the fact that the number of extra dimensions is odd": gamma_i -> -gamma_i fixes sigma_ij and flips Gamma_7, on an orientable manifold, with the gauge group acting by coordinate transformations. Escapes considered: non-orientable internal spaces (none suitable for this symmetry, p. 423) and torsion (left open, pp. 423, 426-427). Pp. 425-426: the three parameters of the invariant metric would determine the three gauge couplings. The paper computes no weak angle, so section F cites it for the method only.

- L. Castellani, R. D'Auria and P. Fré, *SU(3) x SU(2) x U(1) from D = 11 supergravity*, Nucl. Phys. B 239 (1984) 610-652 (INSPIRE): the Einstein metrics on M^{pqr}, with parameters depending on q/(3p). Content from abstract and secondary sources.

G. The protected zero mode on S^5 (requested by the review)

Claim (connes_kk_research_state.md, around the S^5 zero mode): for the nontrivial quaternionic line V -> S^5, dim ker D_{S^5, V} is odd (real mod-two index); for SU(3)-invariant data the invariant line is a zero mode; the text attributes the mechanism to Witten 1982, footnote 2.

Verdict: KNOWN, for the topological statement (a nontrivial SU(2) bundle on S^5 has odd mod-two index) and for the attribution. Footnote 2 of Witten 1982 (p. 327, read on 3 Nov 2026 from the PDF supplied by the user) reads: "Actually, in a special case one can easily find the zero mode. If one conformally compactifies S^4 x R to the five-sphere S^5, then on S^5 one can choose the instanton field to be invariant (up to a gauge transformation) under an SU(3) subgroup of the symmetry group O(6) of the five-sphere. The fermion zero mode is then the unique SU(3) invariant spinor field that can be defined." The wording of the corpus matches it.

- E. Witten, *An SU(2) anomaly*, Phys. Lett. B 117 (1982) 324-328, doi:10.1016/0370-2693(82)90728-6. As summarized by Wang, Wen and Witten (2019), Section 2: "The main result of [1] is that the fermion path integral for a fermion field on M in a given representation of SU(2) is phi-invariant if and only if the mod 2 index of the Dirac operator on Y – in that representation – vanishes. Here the mod 2 index is defined as the number of zero modes of the five-dimensional Dirac operator, reduced mod 2", where Y is the mapping torus of the gauge transformation phi (the mapping torus S^4 x S^1, with phi in the nontrivial class of pi_4(SU(2)) = Z_2). The original, read on 3 Nov 2026, gives the criterion for other isospins on p. 328: the fermion integration is odd exactly when tr T^2, with tr T^2 = 1/2 on the doublet, is a half-integer.
- I. García-Etxebarria and M. Montero, *Dai-Freed anomalies in particle physics*, JHEP 08 (2019) 003, doi:10.1007/JHEP08(2019)003, arXiv:1808.00009. Section 3.1, eq. (3.4): Omega_5^Spin(BSU(2)) = Z_2, detected by the mod-two index; immediately after: "A simple example with non-vanishing mod-2 index was constructed in [2] [Witten 1982]. While S^5 itself is trivial in Omega_5^Spin ..., there is a bundle over it such that the total space is no longer null-bordant in Omega_5^Spin(BSU(2)) = Z_2." Section 3.1.2: SU(3)/SU(2) = S^5, so in five dimensions every SU(3) bundle reduces to an SU(2) bundle.
- Y. Sato, *Majorana fermion zero modes and anomalies in string and M theories* (PhD thesis, Tokyo, 2024), arXiv:2404.18138, Appendix C: the five-dimensional configuration of Witten's argument (S^4 x R with pure-gauge ends) compactifies to S^5; G-bundles on S^5 are classified by pi_4(G), and for G = Sp(N) the mod-two index is an isomorphism onto KO^{-1}(pt) = Z_2.
- J. Wang, X.-G. Wen and E. Witten, *A new SU(2) anomaly*, J. Math. Phys. 60 (2019) 052301, doi:10.1063/1.5082852, arXiv:1810.00844: mod-two index on mapping tori, extended to non-spin manifolds. The corpus cites it for the parity of the index at isospin j = 2k + 1/2. That parity follows from the criterion of Witten 1982: tr T^2 = j(j + 1)(2j + 1)/3 equals (4k + 1)(2k + 1)(4k + 3)/6, a half-integer, at j = 2k + 1/2, and (4k + 3)(k + 1)(4k + 5)/3, an integer, at j = 2k + 3/2. We did not reread the 2019 paper for this statement.

Answer to the review's objection. Witten's original construction uses S^4 x R, with the specified

asymptotic gauge fields; footnote 2 discusses its conformal compactification to S^5. The mapping-torus formulation uses S^4 x S^1, with the gauge bundle glued by the gauge transformation. Both that mapping torus (odd index for one doublet) and the nontrivial SU(2) bundle on S^5 (García-Etxebarria-Montero, sentence after eq. (3.4)) are nonzero in Omega_5^Spin(BSU(2)) = Z_2, so they represent the same generator; the mod-two index is a bordism invariant and is odd on both. Sato's Appendix C describes the S^5 bundle as the compactification of the S^4 x R configuration of Witten's argument. García-Etxebarria-Montero state the S^5 version and attribute the example to Witten 1982. What remains specific to the corpus is the identification of V = rho^*Q with this generator via the Hopf map from CP^3 data and the SU(3)-invariant zero-mode check.

H. Anomaly inflow as the five-dimensional parent of chiral boundary modes (requested by the review)

Relevance: claim B (chiral walls of Theorem 9.8) and Corollary 9.3 (equivariant index from boundary types).

Verdict: KNOWN as background; lemma 09 contains no gauge-field inflow, so these references supply context.

- C. G. Callan and J. A. Harvey, *Anomalies and fermion zero modes on strings and domain walls*, Nucl. Phys. B 250 (1985) 427-436, doi:10.1016/0550-3213(85)90489-4. Chiral zero modes of the Dirac equation on domain walls in 2n+1 dimensions and on strings in 2n+2 dimensions; their anomalies are cancelled by inflow from the bulk through an induced Chern-Simons (topological mass) term. Content from the abstract fragment and secondary sources; full text not read.
- S. G. Naculich, *Axionic strings: covariant anomalies and bosonization of chiral zero modes*, Nucl. Phys. B 296 (1988) 837-867, doi:10.1016/0550-3213(88)90400-2: the consistent/covariant anomaly bookkeeping in inflow.
- N. Arkani-Hamed, A. G. Cohen and H. Georgi, *Anomalies on orbifolds*, Phys. Lett. B 516 (2001) 395-402, doi:10.1016/S0370-2693(01)00946-7: anomalies of bulk fermions on S^1/Z_2 localized at the fixed points (content **unverified** beyond title and abstract-level knowledge).
- Kaplan (1992) and Shamir (1993), section B, built lattice chiral fermions on this mechanism.

For the corpus: Theorem 9.2 and Corollary 9.3 are the index-theoretic shadow of Callan-Harvey (boundary type fixes the chiral zero mode); an inflow statement would require bulk gauge fields with a Chern-Simons term on the interval, which lemma 09 does not include.

I. The current Connes et al. framework (requested by the coordinator)

The corpus treats Connes-Lott 1990 as the obsolete layer. The following define the present framework. All bibliographic data verified (Crossref or arXiv) unless marked.

- **A. Connes, *Noncommutative geometry and the standard model with neutrino mixing*, JHEP 11 (2006) 081**, arXiv:hep-th/0608226. Reverses the grading on the antiparticle sector so that the finite space has KO-dimension 6, separating metric dimension (0) from KO-dimension. Theorem 2.7 classifies the Dirac operators D_F (Yukawa matrices, CKM and PMNS data, a Majorana mass); the fermion doubling problem is solved by the Pfaffian in the fermionic action. The total KO-dimension 10 = 2 mod 8 is noted in Section 1.
- **J. W. Barrett, *Lorentzian version ...*, J. Math. Phys. 48 (2007) 012303**, arXiv:hep-th/0608221. Independently arrives at internal signature 6 from the Lorentzian requirement that physical fermions be simultaneous eigenvectors of J and gamma (total signature 0 mod 8), which removes the quadrupling of fermions.
- **A. H. Chamseddine, A. Connes and M. Marcolli, *Gravity and the standard model with neutrino mixing*, Adv. Theor. Math. Phys. 11 (2007) 991-1089**, arXiv:hep-th/0610241.

Section 2: the finite geometry, with A_F = C + H + M_3(C), Lemma 2.5 (the irreducible representations of the left-right bimodule algebra have total dimension 32 per generation), Section 2.4 (the order-one condition selecting A_F), Sections 2.6-2.7 (classification and moduli of D_F, i.e. the Yukawa parameters) and Section 2.8 (KO-dimension and Poincaré duality); N = 96 in Section 3.7.1; Section 3: the spectral action on M x F and its heat-kernel coefficients; Section 4: the full Lagrangian; Section 5: predictions at unification (gauge coupling relation, Higgs quartic, the mass relation (4.16)).

- **A. H. Chamseddine and A. Connes, *Conceptual explanation for the algebra in the noncommutative approach to the standard model*, Phys. Rev. Lett. 99 (2007) 191601**, doi:10.1103/PhysRevLett.99.191601, arXiv:0706.3690. Version 1 of arXiv:0706.3690 carried the title *A dress for SM the beggar*; the published version is this PRL (the coordinator's "PNAS 2007" does not match any record we found). Classifies irreducible finite geometries with reality and chirality and no fermion doubling, requiring total K-theoretic dimension 10.
- **A. H. Chamseddine and A. Connes, *Why the Standard Model*, J. Geom. Phys. 58 (2008) 38-47**, doi:10.1016/j.geomphys.2007.09.011, arXiv:0706.3688. Section 2: the order-zero condition; Proposition 2.5 (centre C: dim H = k^2) and Proposition 2.8 (centre C + C: dim H = 2k^2, A_C = M_k(C) + M_k(C)); Lemma 3.1: the KO-6 relation J gamma = -gamma J excludes centre C; eq. (3.1): with a quaternionic, nontrivially graded W of quaternionic dimension 2, A = M_2(H) + M_4(C), k = 4, dim H = 2k^2 = 32 per generation; Theorem 4.1: the order-one condition with an off-diagonal D selects the unique maximal subalgebra A_F = C + H + M_3(C). For the project this is the statement that D_F must connect the two components of the centre: the "two sheets" of the finite space are forced by the classification.
- **A. H. Chamseddine, A. Connes and W. D. van Suijlekom, *Inner fluctuations in noncommutative geometry without the first order condition*, J. Geom. Phys. 73 (2013) 222-234**, doi:10.1016/j.geomphys.2013.06.006, arXiv:1304.7583. Without the first-order condition, inner fluctuations acquire a quadratic term and form a semigroup depending only on A. Relevant to Amendment 2 of lemma 07 (the compressed two-point data fail the first-order condition).
- **A. H. Chamseddine, A. Connes and W. D. van Suijlekom, *Beyond the spectral Standard Model: emergence of Pati-Salam unification*, JHEP 11 (2013) 132**, doi:10.1007/JHEP11(2013)132, arXiv:1304.8050. Dropping the first-order condition, the classification gives an SU(2)_R x SU(2)_L x SU(4) model; the Standard Model is recovered when the vacuum restores the first-order condition.
- **A. Devastato, F. Lizzi and P. Martinetti, *Grand symmetry, spectral action and the Higgs mass*, JHEP 01 (2014) 042**, doi:10.1007/JHEP01(2014)042, arXiv:1304.0415. A larger "grand" algebra mixing gauge and spin degrees of freedom produces the extra scalar sigma that brings the Higgs mass near 126 GeV; requiring an almost-commutative geometry breaks it to the Standard Model.
- **A. H. Chamseddine, A. Connes and V. Mukhanov, *Quanta of Geometry: Noncommutative Aspects*, arXiv:1409.2471** (journal data unverified; read from the LaTeX source, 3 Nov 2026). Clifford algebras C_kappa on n + 1 gammas with kappa = +-1; in dimension 4, C_+ = M_2(H) and C_- = M_4(C), "in perfect agreement with the algebraic constituents of the Standard Model"; two maps Y_+- : M -> S^4 of slashed form, with det e = Omega_+ + Omega_-. Bears on the twistor line of canonical section 4 and on current_framework.md.
- **J. Baptista, *Chiral interactions of fermions and massive gauge fields in Kaluza-Klein models*, JHEP 05 (2026) 008, arXiv:2506.09126** (found by the referee; introduction read from the LaTeX source). The no-go arguments against chiral fermions in Kaluza-Klein models (Witten 1983; Atiyah-Hirzebruch) assume that the gauge fields are linked to exact Killing fields; gauge fields linked to non-Killing fields are massive, can mix fermions of different masses, and can couple asymmetrically to the two chiralities, with examples on even-dimensional K = S^2 and T^2. Bears on the reading of no-go results in the paper and on T4.
- **Lemma 22: the finite Dirac operator as a boundary condition.** Known: the one-flavour formula (Csáki, Grojean, Hubisz, Shirman and Terning, section C here) and the gauge boundary conditions and the form M tan(M pi R) = -V (Csáki, Grojean, Murayama, Pilo and Terning, *Gauge theories on an interval: unitarity without a Higgs*, Phys. Rev. D 69 (2004) 055006, arXiv:hep-ph/0305237, read in the LaTeX source). Not found: the real structure with a finite triple as exact window, the wall algebra C_F, the average formula for the compression. **Searched** (arXiv API, 3 Nov 2026): "non-

commutative geometry" with Higgsless; "spectral triple" with "boundary conditions", Higgs, interval; "spectral triple" with Yukawa, "boundary conditions"; "almost-commutative" with interval, boundary; "spectral triple" with "Kaluza-Klein", "boundary conditions": no hits. "noncommutative geometry" with "gauge-Higgs": one hit, arXiv:2511.05909, a survey, abstract only. Absence from these queries does not establish absence. The referee adds operator-system compression in Connes-van Suijlekom (arXiv:2004.14115, sections 1, 3), and A. Chamseddine and A. Connes, *Noncommutative geometric spaces with boundary: spectral action*, J. Geom. Phys. 61 (2011) 317, arXiv:1008.3980 (read in the LaTeX source: sections 3 to 5). There the boundary is that of the manifold in a product with a finite space, with the condition Pi_- Psi = 0, Pi_- = (1 - chi)/2, for a local chirality projector chi, and the spectral action acquires the extrinsic-curvature terms. The internal boundary with finite gradings of lemmas 22 and 23 is a different setting.

- **Lemma 25: the sum rule and the effective dimension.** Known: m_t = 2 M_W at tree level in flat gauge-Higgs models with suitable representations (Cacciapaglia, Csáki and Park, arXiv:hep-ph/0510366, section 5; Adachi and Maru, arXiv:1804.06012, Eq. (3.23)). Not found in the records inspected by the referee: the coefficient (3 - d_eff)/24 and its theta-function crossover. **Searched** by the referee: web-index queries for spectral action, heat kernels, compactification and gauge-Higgs mass relations, including a 2026 query, and the reference chain from Adachi, Kurahashi and Maru (arXiv:1404.4281). A forward-citation search, an author sweep and a broader database search remain open. This does not establish priority.
- **Twisted spectral triples:** A. Connes and H. Moscovici, *Type III and spectral triples*, arXiv:math/0609703 (journal data **unverified**); G. Landi and P. Martinetti, *On twisting real spectral triples by algebra automorphisms*, Lett. Math. Phys. 106 (2016) 1499-1530, doi:10.1007/s11005-016-0880-4, arXiv:1601.00219; A. Devastato and P. Martinetti, *Twisted spectral triple for the Standard Model and spontaneous breaking of the grand symmetry*, Math. Phys. Anal. Geom. 20 (2017) 2, doi:10.1007/s11040-016-9228-7, arXiv:1411.1320. The twist bounds the unbounded vector terms of the grand-symmetry model and introduces a twisted first-order condition; the breaking to the Standard Model becomes dynamical.
- **Classification of finite triples:** T. Krajewski, *Classification of finite spectral triples*, J. Geom. Phys. 28 (1998) 1-30, doi:10.1016/S0393-0440(97)00068-5, arXiv:hep-th/9701081 (Krajewski diagrams: vertices are chiral gauge multiplets, links are Yukawa couplings); M. Paschke and A. Sitarz, *Discrete spectral triples and their symmetries*, J. Math. Phys. 39 (1998) 6191-6205, doi:10.1063/1.532623, arXiv:q-alg/9612029 (classification of 0-dimensional triples over complex and real algebras). For the project, a Krajewski link is exactly the Yukawa block that lemma 12 derives as Phi_Q Y_hat Phi_U.
- **Fermion doubling:** F. Lizzi, G. Mangano, G. Miele and G. Sparano, *Fermion Hilbert space and fermion doubling in the noncommutative geometry approach to gauge theories*, Phys. Rev. D 55 (1997) 6357-6366, doi:10.1103/PhysRevD.55.6357, arXiv:hep-th/9610035: unphysical degrees of freedom in H = L^2(S) x H_F, analogous to lattice doublers; the 2006-2007 KO-6 framework resolves them (Connes 2006, Barrett 2007).
- **Standard reference:** W. D. van Suijlekom, *Noncommutative Geometry and Particle Physics*, Mathematical Physics Studies, Springer (2015), doi:10.1007/978-94-017-9162-5 (a second edition exists; its data were **not verified**). Covers finite spectral triples and Krajewski diagrams, almost-commutative manifolds, the spectral action, and the electrodynamics and Standard Model examples.

Items not verified in this session

- Castellani-D'Auria-Fré 1984 (section F): equation-level content not read.
- Banks-Casher 1980: the original was not read (the publisher returned HTTP 403). The form of the relation in the chiral limit note is checked against L. Giusti and M. Lüscher, JHEP 03 (2009) 013, arXiv:0812.3638, Section 2 (LaTeX source), which counts the modes of one flavour and states the order of limits lambda -> 0, m -> 0, V -> infinity.
- Dashen 1971 and Witten 1983 (*Fermion quantum numbers in Kaluza-Klein theory*): not read.
- Coquereaux-Esposito-Farèse-Vaillant 1991, Dubois-Violette-Kerner-Madore 1990 (section A): bibliographic data from secondary citations, content from secondary sources.

- Callan-Harvey 1985 and Arkani-Hamed-Cohen-Georgi 2001 (section H): full texts not read.
- Demmel et al. 1999: theorem numbers for the D_1 X D_2 results.
- Niemi-Slizovskiy, Section VIII (custodial fixing of xi).
- Connes-Moscovici journal reference; second edition of van Suijlekom's book.

Summary table

Claim	Verdict	Key references
A. Connes-Lott as minimal truncation of the chiral interval,	m	= pi/(2l)
B. l -> infinity (walls, 1D Weyl law) and l -> 0 with mu(l)	PARTIAL: wall modes and exponentially small mixing known; NCG statement and tuned collapse not found	Jackiw-Rebbi 1976; Kaplan 1992; Shamir 1993 §3; Csáki et al. 2004; Farsi-Latrémolière 2026 Cor. 4.1
C. Towerless textures Phi_Q Y_hat Phi_U, secular equation, one-flavour spectrum	PARTIAL: one-flavour spectrum KNOWN (Csáki et al. eq. (5.6)); multi-flavour warped analogue KNOWN (Casagrande et al. eqs. (65), (87), (90), (94)-(97)); NCG derivation of D_F not found	Csáki et al. 2004 §5.1, App. C.2; Casagrande et al. 2008 §3; Kaplan-Tait 2001; Arkani-Hamed-Schmaltz 2000
D. KO-6 = (odd) x (Clifford point) criterion; s from U; nu_L; reflection sign	PARTIAL: products, J vs J gamma, odd x odd product and 10 = 2 mod 8 known; factorization criterion and the 9/11 results not found	Dąbrowski-Dossena 2011 §4.3, eq. (25), Table 6; Vanhecke 1999; Vanhecke-da Silva-Sigaud 2012 §2; Farnsworth 2017; Connes 2006 §1; Barrett 2007
E. 8 M_W^2 = sum N_f m_f^2; m_t/M_W = sqrt(8/3); heavy weights	KNOWN (mass relation, top ratio); NOT FOUND (exact $f(m^{2/\mathrm{Lambda}}2)$ weights)	CCM 2007 eq. (1.5), Lemma 4.6 eq. (4.16), §5.4 eqs. (5.24)-(5.29)
F. cos^2 theta_W = 3 xi^2/(4 xi^2 + 2)	KNOWN	Niemi-Slizovskiy 2010 eqs. (94)-(95), §VI; Witten 1981 (method)
G. Odd mod-two index on S^5	KNOWN (topology and footnote attribution)	García-Etxebarria-Montero 2019 §3.1; Witten 1982; Wang-Wen-Witten 2019
H. Anomaly inflow	KNOWN (background)	Callan-Harvey 1985; Naculich 1988; Arkani-Hamed-Cohen-Georgi 2001
I. Current framework	reference list	Connes 2006; Barrett 2007; CCM 2007; Chamseddine-Connes 2007, 2008; CCvS 2013 (two papers); Devastato-Lizzi-Martinetti 2014; Krajewski 1998; Paschke-Sitarz 1998; van Suijlekom 2015

Part R1 (assessment of Farsi-Latremoliere, with the counterexample to their Theorem 3.4)

Farsi-Latrémolière, "Collapse in noncommutative geometry and spectral continuity": referee assessment

Paper. C. Farsi and F. Latrémolière, arXiv:2404.00240: v1 30 Mar 2024, v2 13 Oct 2025, v3 15 Oct 2025. Published as J. London Math. Soc. (2) 114 (2026), no. 1, e70628, 54 pp. (journal reference on the arXiv record).

What was read. The LaTeX sources of v1 and v3, in full. The v2 and v3 sources are byte-identical. The journal page (doi:10.1112/jlms.70628) returned HTTP 403, so we have not seen the published text. Theorem numbers below are those of v3, with v1 numbers in brackets. Read by Claude Opus 5.5 on 3 Nov 2026 for lemma 09 (Theorems 9.8-9.10) and STRATEGY T0. **Update (3 Nov 2026):** the published text was read later that day, from the journal HTML supplied by the user; see Section 3.6.

Summary. - The convergence notion is Latrémolière's spectral propinquity on metric spectral triples (A, H, D). These carry no grading and no real structure. - One collapse is treated: D = D_h + D_v, with only D_v rescaled by 1/eps. The limit is (B, ker D_v, p D_h p), where B ⊆ A and p projects onto ker D_v. - The general Theorem 3.4 is false as stated: it needs a hypothesis that its proof uses but does not list (Counterexample R1). All of the paper's applications satisfy that hypothesis. - Corollary 4.1 (collapse to a point, limit (C, ker D, 0)) is sound. - Our Theorem 9.9 lies outside every theorem of the paper. The paper's method adapts to it (Lemma R2, ours). The limit is (C, C^2, m_f sigma_1), and the Connes-Lott two-point triple stays at distance at least about 1/(2 m_f). - For T0, limits of commutative triples in this metric have commutative algebras (Lemma R3, ours).

1. Setting and main results

1.1 Prerequisites (Section 2)

- **Spectral triple** (Definition 2.1). This is a unital C*-algebra A, a Hilbert space H that is a left A-module, and a self-adjoint D. The Lipschitz algebra A_D = {a : a Dom D ⊆ Dom D, [D, a] bounded} is dense, and (D + i)^{-1} is compact. No grading, real structure or summability enters anywhere in the paper.
- **Quantum compact metric space** (Definition 2.3). This is a pair (A, L), with L a seminorm on a dense subspace of sa(A), such that:
  - ker L = R 1;
  - the Monge-Kantorovich metric mk_L(phi, psi) = sup{|phi(a) - psi(a)| : L(a) <= 1} metrizes the weak* topology on the state space;
  - the Leibniz inequality holds for Jordan and Lie products;
  - the unit ball {L <= 1} is closed.
- **Metric spectral triple** (the definition after 2.4). A spectral triple is metric when (A, L_D) with L_D(a) = |||[D, a]||| is a quantum compact metric space, i.e. when Connes' distance metrizes the weak* topology.
- **Tunnels, extent, propinquity** (Definitions 2.5-2.10).
  - A quantum isometry is a *-epimorphism pi with L_B(b) = inf{L_A(a) : pi(a) = b}.
  - A tunnel is a quantum compact metric space (D, L_D) with quantum isometries onto both spaces.
  - Its extent is the larger of the two Hausdorff distances, in mk_{L_D}, between S(D) and the pulled-back state spaces.
  - The propinquity is the infimum of extents over tunnels. The paper records that "it induces the same topology as the Gromov-Hausdorff distance on the class of classical metric spaces".
- **Metrical C*-correspondences** (Definition 2.13). A metric spectral triple yields (H, DN, A, L_D, C, 0), where DN(xi) = ||xi|| + ||D xi|| is the graph norm. Metrical tunnels and their extents follow in Definitions 2.16-2.17. Separation and dispersion of operator families are measured through pseudo-states xi -> mu(<omega, xi>) with DN(omega) <= 1 (Definitions 2.20-2.21).

- **Spectral propinquity** (Definition 2.22, taken from Latrémolière, Adv. Math. 404 (2022), and Math. Ann. 389 (2024), Theorem 3.5). The distance Lambda_spec between two metric spectral triples is the infimum of sqrt(2)/2 and of those eps > 0 for which some metrical tunnel has dispersion < eps between the families (exp(i t D_1))*{0 <= t <= 1/eps} and (exp(i t D_2))*{0 <= t <= 1/eps}.
- **Properties listed after Definition 2.22.**
  - "(1) Lambda_spec = 0 if the two metric spectral triples are unitarily equivalent." The paper states only this, the trivial direction. Latrémolière (Adv. Math. 2022) proves both directions: the distance is null exactly on unitarily equivalent pairs, and it dominates the propinquity of the underlying quantum metric spaces.
  - "(2) if Lambda_spec(T_n, T_infinity) -> 0 then for all bounded continuous f, Lambda_op(f(D_n), f(D_infinity)) -> 0, and Sp(D_infinity) = {lim lambda_n : (lambda_n) convergent with lambda_n in Sp(D_n) for all n}."

1.2 The collapse setting and Theorem 3.4 [v1: Theorem 2.5] A unital C*-subalgebra B ⊆ A plays the base, and a positive map E : A -> B plays the bundle projection. The Dirac operator splits as D = D_h + D_v, and only the vertical part is scaled: D_eps = D_h + (1/eps) D_v. The limit object is (B, ker D_v, D_B) with D_B = p D_h p. Quoted, with notation transcribed:

> **Theorem 3.4.** Let (A, H, D) be a metric spectral triple, and let B ⊆ A be a unital C*-subalgebra of A, and such that D = D_h + D_v, where D_v is self-adjoint and such that 0 is isolated in Sp(D), together with the following assumptions. Setting D_eps := D_h + (1/eps) D_v for all eps in (0, 1), the triple (A, H, D_eps) is a spectral triple, such that: 1. for all eps in (0,1), |||[D_h, a]|| <= |||[D_eps, a]|||, 2. there exists M > 0 such that for all eps in (0,1), |||[D_v, a]|| <= M eps |||[D_eps, a]|||, 3. [D_v, b] = 0 for all b in B, 4. writing p for the projection onto ker D_v, we assume that [p, b] = 0 for all b in B and [D_h, p] = 0, 5. (B, ker D_v, D_B), where D_B := p D_h p, is a metric spectral triple, 6. there exists a positive linear map E : A -> B, whose restriction to B is the identity, and a constant k > 0 such that for all a in A: ||a - E(a)|| <= k |||[D_v, a]|||, and ||p [D_h, E(a)] p|| = |||[D_h, E(a)]||| <= |||[D_h, a]|||.
>
> Then (A, H, D_eps) is a metric spectral triple, and lim_{eps -> 0+} Lambda_spec((A, H, D_eps), (B, ker D_v, D_B)) = 0.

"0 is isolated in Sp(D)" is a typo for Sp(D_v). The proof uses a gap delta with Sp(D_v) ∩ (-delta, delta) = {0}. The typo is present in v1, in v3 and in the introduction.

Structure of the proof. - The quantum-metric tunnel is A ⊕ B with seminorm max{L_eps(a), L_B(b), ||a - b||/(k M eps)}, and its extent is at most k M eps. - The module tunnel is H ⊕ ker D_v with norm max{DN_eps(xi), DN_B(eta), ||p xi - eta||/K_eps}. Here K_eps = max{k M eps, 4 sqrt(2 eps/delta)}, and the 4 sqrt(2 eps/delta) comes from a Fourier estimate (Lemma 3.3). - Separation is controlled through exp(i t D_eps) p = exp(i t p D_h p) (Lemma 3.1), uniformly in t. - The resulting bound is Lambda_spec <= alpha_eps + K_eps = O(sqrt(eps/delta)).

1.3 Meaning of "spectral continuity" (Corollary 3.5 [v1: 2.7])

> **Corollary 3.5.** Under the same hypothesis as in Theorem 3.4, for any f in C_b(R), and for any sequence (eps_n) in (0, infinity) converging to 0, we have lim_n Lambda_op(f(D_{eps_n}), f(D_B)) = 0, and in particular, Sp(D_B) = {lim_n lambda_n : (lambda_n) convergent with lambda_n in Sp(D_{eps_n}) for all n}.

The corollary has two parts. - The bounded continuous functional calculus converges in Latrémolière's operational propinquity, a metric between operators on different metrical C*-correspondences. - The spectra converge as sets. The statement holds for every sequence eps_n -> 0, so applying it to subsequences gives Kuratowski convergence of Sp(D_{eps_n}) to Sp(D_B).

Multiplicities and eigenvectors are not addressed. The corollary is a direct citation of Latrémolière, Math. Ann. 2024.

1.4 Applications

result (v3 [v1])	data	limit	anticommutation of D_h and D_v
Corollary 4.1 [3.1]	D_h = 0, D_v = D, B = C	(C, ker D, 0)	trivial
Theorem 4.6 [4.5]	(A, H, D) even with grading gamma, (B', J, S) metric, one of them Abelian; D_eps = D x 1 + (1/eps) gamma x S	(A, H x ker S, D x 1)	yes (gamma anticommutes with D)
Corollary 4.7 [4.6]	odd version via two Clifford generators gamma_1, gamma_2 on E	(A, H x ker S x E, D x 1 x gamma_1)	yes
Theorem 5.5 [5.5]	Schwieger-Wagner triple on a free action of compact Lie G on A, B = A^G; D_v = sum_j d_j x gamma_j, D_h = (…) x gamma_0	(B, H_B x H_spin, D_B x gamma_0)	yes (Remark 5.1)
Corollaries 5.6, 5.7	B x_alpha Z^d with equicontinuous alpha; quantum 4-torus collapsing to a quantum 2-torus	as in 5.5	yes
Theorem 6.3 [6.3]	M -> N a principal U(1)-bundle (projectable spin structure, constant fibre length, dim N even); D_eps = (1/l_eps) D_v + D_h	(C(N), L^2(Sigma N), D_N)	yes (Ammann-Bär)

2. Section 4 and Corollary 4.1: the shape of the limit

> **Corollary 4.1** [v1: 3.1]. If (A, H, D) is a metric spectral triple with 0 in Sp(D), then lim_{eps -> 0} Lambda_spec((A, H, (1/eps) D), (C, ker D, 0)) = 0.

The proof takes D_h = 0, D_v = D, B = C and E = mu, a state, and uses ||a - mu(a)|| <= diam |||[D, a]|||. The paper follows it with this remark:

> If 0 is not in Sp(D), then there is no bounded sequence (lambda_n) with lambda_n in n Sp(D) […] So by Corollary 3.5, it is not possible for (A, H, n D), and even less (A, H, (1/eps) D), to converge to anything for the spectral propinquity.

The limit object. The limit is a point (algebra C, state space a single point) with Hilbert space ker D, which is finite-dimensional because D has compact resolvent, and with zero Dirac operator. Before the corollary the paper puts it this way: "(C, C^n, 0) […] arise naturally as limits […] of arbitrary metric spectral triples collapsed to a point. Moreover, in that case, n is the dimension of the space of harmonic spinors." This corollary does not depend on the defect in Section 3.1 below, since D_h = 0. A direct proof is short: - the quantum-metric extent is at most eps diam(A, D); - by the spectral theorem, ||xi - p xi|| <= ||D xi||/delta = eps ||(1/eps) D xi||/delta; - so Lambda_spec = O(eps).

The product case (Theorem 4.6). The limit is (A, H x ker S, D x 1), i.e. the base triple with multiplicity dim ker S and no internal Dirac operator. The internal factor survives only as the multiplicity space ker S.

Can a nontrivial finite part survive? - Finite algebra. No. The limit algebra is always B ⊆ A, acting on ker D_v, so the finite part never carries an algebra of its own. - **Finite Dirac operator.** Possibly yes. The limit operator is p D_h p, and it can have a nonzero finite piece when D_h contains a bounded term that commutes with A, preserves ker D_v and anticommutes with D_v. Example (ours, not in the paper): - Take A = C(S^1), H = $L^{2(S}$1) x C^2 x C^k, D_v = -i d/dtheta x sigma_1 x 1 and D_h = 1 x sigma_2 x Phi with Phi = Phi* a k x k matrix. - All six hypotheses hold with B = C and E(f) = (1/2pi) integral of f, and D_h anticommutes with D_v. - The limit is (C, C^2 x C^k, sigma_2 x Phi), a finite spectral triple over C with spectrum ±Sp(Phi). - The paper contains no example with p D_h p != 0 and B = C.

3. Skeptical reading

3.1 Theorem 3.4 is missing a hypothesis Counterexample R1. - Let H have orthonormal basis e_0, e_1, e_2, ..., and let A = B = C act by scalars. - Set D_v e_n = n e_n and D_h e_n = -n^2 e_n for n >= 0. Then D_eps e_n = (n/eps - n^2) e_n, which has compact resolvent for every eps. - D_v has spectral gap 1 at 0, and D = D_1 has Sp(D) ∩ (-2, 2) = {0}. The projection p onto ker D_v = C e_0 commutes with D_h. - Hypotheses (1)-(6) hold trivially, since commutators with scalars vanish and E = id. (C, C e_0, 0) is a metric spectral triple. - Theorem 3.4 and Corollary 3.5 would therefore give Sp(D_B) = {0}. - Fix x > 0 and let eps_n = 1/(n + x/n) -> 0. Then D_{eps_n} e_n = x e_n, so the constant sequence x lies in Sp(D_{eps_n}), and Corollary 3.5 would put x in Sp(D_B) = {0}.

Corollary 3.5 follows from Theorem 3.4 through Latrémolière's published continuity theorem, so the conclusion of Theorem 3.4 fails for this family.

The failing step is equation (main-eq-1) in the proof (v3 only):

```
||xi - p xi|| <= 4 sqrt(2 eps/delta) (||xi|| + ||(1/eps) D_v xi||) <= 4 sqrt(2 eps/delta) DN_eps(xi).
```

The second inequality needs ||(1/eps) D_v xi|| <= ||D_eps xi||, which is neither assumed nor proved. For xi = e_n at eps = eps_n, the left-hand side is 1 while the right-hand side is 4 sqrt(2 eps_n)(1 + x) -> 0.

Repair. Add the hypothesis that D_h and D_v anticommute on a common core. Then ||D_eps xi||^2 = ||D_h xi||^2 + eps^{-2} ||D_v xi||^2. A relative bound ||D_v xi|| <= C eps (||xi|| + ||D_eps xi||) also suffices. With anticommutation, the spectral theorem gives ||xi - p xi|| <= ||D_v xi||/delta <= (eps/delta) ||D_eps xi||; with the relative bound it gives ||xi - p xi|| <= (C eps/delta)(||xi|| + ||D_eps xi||). (The estimate for the relative bound is corrected on 3 Nov 2026, as Astra 6 noted.) Either one makes the Fourier Lemma 3.3 unnecessary and improves the rate from O(sqrt(eps)) to O(eps).

All applications in Section 1.4 satisfy anticommutation, so they are not affected by this defect. Theorem 3.4 itself, as printed in both arXiv versions and in the journal (Section 3.6), may not be cited without the extra hypothesis.

3.2 Other points in Section 3

- **Lemma 3.1** [2.1]. v3 added the hypothesis "D and D + S both have discrete spectra". The lemma is applied with D = D_h, and the paper itself says D_h has no compact resolvent in Section 5. The conclusion holds without discreteness: pH reduces both operators, and D_eps p = D_h p. So the proof is fixable and the statement is fine.
- **Inner Leibniz.** Definition 2.13 is ill-typed: it applies L to the non-self-adjoint inner product and names L_A where L_B is meant (the journal text corrects L_A to L_B and keeps the rest). Latrémolière's own definition (arXiv:1811.10843) bounds the real and imaginary parts. Noticed by the referee of lemma 15; the proofs are unaffected.
- **Time window.** The proof concludes with separation over 0 <= t <= M_eps where 1/M_eps is meant. The estimate is uniform in t, so this is harmless. "1/t-covariant reach" is leftover terminology from Latrémolière 2022.
- **Hypotheses that are strong or hard to verify in examples.**
  - (4) requires D_h to preserve ker D_v exactly. A mass or Yukawa term that mixes zero modes with excited modes violates it.
  - The exact kernel with a fixed gap excludes near-zero modes.

- The fixed D_h excludes horizontal operators that change with eps: the Ammann-Bär term V_eps, curvature terms, non-constant fibre length.
- (6) requires the horizontal derivative of base elements to be seen entirely on ker D_v.
- (2) is automatic in Clifford-graded constructions and otherwise needs an argument.

3.3 Sections 4-6: sketched or delegated steps, and asserted examples

- **Theorem 4.6.** The check of hypothesis (6) is garbled; it bounds |phi x psi(c - E(c))| by "mk_D(phi, phi) + eps mk_S(psi, mu)". The needed bound ||c - E(c)|| <= diam(B', S) |||[1 x S, c]|| follows from Lemma 4.4. The statement writes "gamma x S" for gamma ⊗ S.
- **Section 5.** The spectral triple, its compact resolvent and the two sup-bounded commutator conditions are taken from Schwieger-Wagner (Adv. Math. 396 (2022)). Its representation is "a technical matter to check" (Section 5.3).
  - Lemma 5.2 uses |||[D, alpha^g(a)]|| = |||[D, a]|| for D in {D_h, D_v, D_A}.
  - For D_v = sum_j d_{e_j} x gamma_j this needs u^g D_v u^{g-1} to be unitarily equivalent to D_v by a unitary commuting with A.
  - Such a unitary exists when the inner product on the Lie algebra of G is Ad-invariant, but the paper fixes only "some inner product" (a left-invariant metric).
  - All worked examples are tori, where the issue is absent. We have not checked Schwieger-Wagner's conventions to see whether they impose Ad-invariance.
- **Section 6, Theorem 6.3.**
  - (a) The theorem concerns D_eps = (1/l_eps) D_v + D_h. The Dirac operator of the shrunk metric is D_{M_eps} = D_eps + V_eps. The bounded term V_eps tends to 0 under the Ammann-Bär condition, but the passage to D_{M_eps} is not made (the abstract says "a bounded variation").
  - (b) The verification of the hypotheses is sketched and contains errors. Item (0) checks that 0 is isolated in Sp(D_eps) instead of Sp(D_v), and argues self-adjointness from "one of them being bounded", although both summands are unbounded. Item (1) cites the Fourier Lemma 3.3, which is irrelevant, and then Proposition 3.6, whose hypotheses (constant anticommuting unitaries commuting with the D_j) do not literally cover local Clifford frames. Item (2) "proves" [D_v, b] = 0 by displaying [D'_{h,k}, b] = 0, which is false for nonconstant b (it is Clifford multiplication by grad b). Item (3) cites an unrefereed web note (Orduz). Item (5) is delegated ("Verified in the same way as in the proof of Theorem 5.5"), and its displayed inequalities swap D_h and D_v.
  - The conclusions are correct classically, by pointwise Clifford anticommutation and fibre averaging, but the written proof does not establish them.
  - (c) Odd dim N is asserted "mutatis mutandi", and non-constant fibre length is left "to the interested reader".
- **Asserted examples.** The introduction of v1, of v3 and of the journal text lists "homogeneous spaces of compact Lie groups" (the abstract does not; an earlier version of this line said it did). The corresponding passage is commented out in both sources, so no such result is proved.

3.4 Dependencies The proofs rest on published work: - Latrémolière, Adv. Math. 404 (2022) and Math. Ann. 389 (2024); - Farsi-Latrémolière-Packer, Adv. Math. (Paper No. 109442); - Rieffel 1998, 1999, 2005; - Schwieger-Wagner, Adv. Math. 396 (2022); - Ammann 1999 and Ammann-Bär 1998.

No step relies on an unpublished or forthcoming result of the same authors. The preprints Schwieger-Wagner arXiv:2406.08145, Kaad arXiv:2410.03475 and Mesland-Rennie are cited for context only. One proof step (Theorem 6.3, item (3)) cites the web note by Orduz.

3.5 Changes from v1 to v3 v2 and v3 have identical sources. The arXiv comment says: restructured, introduction split, "Section 2 results were cleaned up and some statements corrected", references changed. From the diff:

- **Restructuring.** The background moved from the introduction into a new Section 2. Numbering shifts:
  - main theorem 2.5 -> 3.4;
  - spectral corollary 2.7 -> 3.5;
  - "Collapsing to a point", a separate section in v1 (Corollary 3.1), became subsection 4.1 (Corollary 4.1);
  - product theorem 4.5 -> 4.6;
  - comparison Lemma 2.4 -> Proposition 3.6.
  - Sections 5 and 6 keep their numbers.
- **Theorem 3.4.**
  - The statement is unchanged in substance: the same six hypotheses and the same Sp(D) typo.
  - v1's Remark 2.6 on domains ("Dom D = Dom D_h ∩ Dom D_v") was removed.
  - In the proof, v1's (main-eq-1), ||xi - p xi|| <= 4 sqrt(2 eps/delta), had no norm of xi on the right. v3 added (||xi|| + ||(1/eps) D_v xi||) <= DN_eps(xi), which makes the missing step of Section 3.1 explicit without justifying it.
  - v3 added the argument that p commutes with D_eps, so that DN_B(p xi) <= DN_eps(xi).
  - v3 corrected the circular "M_eps := alpha_eps + M_eps" to alpha_eps + K_eps.
- **Lemma 3.1.** The discrete-spectra hypothesis was added; v1's proof had silently used compact resolvent.
- **Corollary 3.5.** The stray notation D_{x,eps_n} was corrected to D_{eps_n}.
- **Lemma 4.4** [4.3]. v1's proof argued that psi([S, b_j]) = 0 for an extended state, which fails for non-self-adjoint b_j. v3 replaced it with an argument using the grading. The roles of phi and psi in the statement were corrected.
- **Section 5.** Conjugate representations were corrected (Phi_sigma : Gamma_A(sigma) x conj(V_sigma) -> A(conj sigma)), with a footnote reporting a typo in Schwieger-Wagner Eq. (16). Lemma 5.2 was extended to the L^1 bound for all three operators.
- **Section 6.** v1's Lemma 6.4 was folded into the proof with its errors unchanged. v1's Remark 6.5, which claimed the method covers Roos's weaker collapse condition, is commented out in v3. That claim has been withdrawn.
- The abstract was rewritten and references were added.

3.6 Published version The arXiv record carries the journal reference, and an announcement page lists the paper as accepted. v2/v3 (Oct 2025) is therefore probably the accepted text. We could not see the journal version, so whether the hypothesis missing from Theorem 3.4 was added in print is unknown.

Update (3 Nov 2026). The journal text (J. London Math. Soc. (2) 114 (2026) e70628, first published 3 July 2026) has now been read, and the hypothesis was not added. We compared its prose word by word with the v3 source (Python) and read the following statements directly: Theorem 3.4, the step (main-eq-1), the constant K_eps, Corollaries 3.5 and 4.1, and the remark after 4.1. In those statements and in the prose, the differences are copy-editing: macros written out, step headings in the proof of Theorem 3.4, and L_A corrected to L_B in Definition 2.13. Astra 6 compared the main statements of Sections 3 to 6 and found no further change (validation, item 4); other equations were not compared. The printed Theorem 3.4 has the same six hypotheses and the same typo "0 is isolated in Sp(D)". The step of Section 3.1 is printed as equation (3.11). So Counterexample R1 applies to the printed statement, and the conclusions above on v3 carry over to the statements compared. Anticommutation and the relative bound are sufficient repairs, as Terra 6.1 and Astra 6 both note; the anticommuting applications avoid this defect. Astra 6 also notes that Theorem 4.6 and Corollary 4.7 need a nonzero vertical kernel for the displayed target to be a spectral triple, in v3 and in print.

4. Relevance to lemma 09 and STRATEGY T0

4.1 Our interval as a metric spectral triple **The framework's prerequisites hold.** T_l = (C([0,l]), L^2([0,l]; C^2), D_M), with condition (c), is a metric spectral triple: - the boundary conditions sit in Dom D_M, and multiplication by C^1 functions preserves them; - the mass term M sigma_1 commutes with C([0,l]), so ||[D_M, f]|| = ||f'||_inf and Connes' distance is |x - y|; - compact resolvent is Lemma 9.1.

What the framework does not see. T_l is even with grading Gamma, but the spectral propinquity ignores Gamma. No chirality statement (Theorem 9.2, P_0 Gamma P_0 = sigma_3 in Theorem 9.6) is visible to it, and neither is a real structure or a KO-dimension.

Boundary identity. Write D_M = D_0 + M sigma_1 with D_0 the massless operator. For psi = (L, R) in Dom under condition (c):

```
||D_M psi||^2 = ||D_0 psi||^2 + M^2 ||psi||^2 + M (|R(1)|^2 + |L(0)|^2),
```

since 2 Re <D_0 psi, sigma_1 psi> = [|R|^2]_0^1 - [|L|^2]_0^1. After the dilation x = l y, D_M = (1/l) D_0^{[0,1]} + M sigma_1. This is the paper's form with eps = l, D_v = D_0 and D_h = M sigma_1. The anticommutator {D_0, sigma_1} vanishes in the interior, and its quadratic form is the boundary term above. So the hypothesis missing from Theorem 3.4 fails for our interval exactly by the term that creates the wall modes: - For M = -mu the term is negative, and Theorem 9.4.1 gives one eigenvalue below mu. - For M = +mu the identity gives ||D_M psi||^2 >= ((pi/(2l))^2 + mu^2) ||psi||^2. This is a one-line proof of the bound in Theorem 9.4.3.

4.2 Corollary 9.10 (fixed mu, l -> 0) In dilated form this is the paper's setting with D_v = D_0 under condition (c). Hypothesis 0 in Sp(D_v) fails, because ker D_0 = 0 (Theorem 9.2(c)). - D_l is a bounded perturbation, of norm mu, of (1/l) D_0, so every eigenvalue diverges. - The remark after Corollary 4.1, together with Corollary 3.5, then excludes any limit in the spectral propinquity (apart from a degenerate triple on the zero Hilbert space).

This agrees with Corollary 9.10. **The paper covers it** only in this indirect sense: its remark treats pure rescalings, and the bounded-perturbation step is ours.

4.3 Theorem 9.9 (mu(l) tuned, l -> 0) Here D_l = (1/l)(D_0 - c(l) sigma_1) with c(l) = mu(l) l -> infinity. In the paper's form D_h(l) = -mu(l) sigma_1 has norm mu(l) ~ log(1/l)/l -> infinity, and D_v = D_0 has no kernel. Three hypotheses of Theorem 3.4 fail: a fixed D_h, an exact kernel of D_v, and anticommutation. **The paper says nothing about this family**; this confirms the earlier note in LITERATURE.md. Its method does adapt:

Lemma R2 (ours, proof sketch; to be written out and refereed). - Statement. Let P_l project onto the light pair ±m_f of Theorem 9.9, and let W : P_l H -> C^2 be the coordinates in the basis (u_L, u_R) of Theorem 9.6. Then Lambda_spec(T_l, (C, C^2, m_f sigma_1)) <= 2l for small l. - **Quantum-metric tunnel.** Take C([0,l]) ⊕ C with seminorm max{||f'||_inf, ||f - z||/l}. Its extent is at most l, since Connes' metric of D_l is the Euclidean one. - **Module tunnel.** Take H ⊕ C^2 with norm max{DN_l(xi), DN(eta), ||W xi - eta||/l}. - **Key estimates.** - P_l is a spectral projection of D_l, and the next eigenvalue satisfies lambda_1(l) >= sqrt(mu^2 + (pi/l)^2) >= pi/l. Hence ||xi - P_l xi|| <= ||D_l xi||/lambda_1 <= l ||D_l xi||/pi, and exp(i t D_l) P_l = W* exp(i t m_f sigma_1) W for all t. - The modular Leibniz inequality needs no commutation of P_l with C([0,l]), because the limit algebra acts by scalars: ||W((f - z) xi)|| <= ||f - z|| ||xi||. - Inner Leibniz and separation follow as in the paper's proof, uniformly in t. - **Consequences.** - Theorem 9.9 becomes a spectral-propinquity limit with rate O(l): the light sector survives as a finite triple over C with D_F = m_f sigma_1. - The limit algebra is C. The two-point algebra C ⊕ C of Theorem 9.6 is the range of the compression f -> P_l f P_l, and the limit object does not contain it. - |diam_A - diam_B| <= 2 Lambda for any tunnel. The Connes-Lott triple (C ⊕ C, C^2, m_f sigma_1) has diameter 1/m_f, and (C, C^2, m_f sigma_1) has diameter 0. Hence Lambda_spec(T_l, Connes-Lott(m_f)) >= 1/(2 m_f) - 2l. - So in this metric the tuned interval converges to a point carrying a two-dimensional massive Dirac block, and it stays at a fixed positive distance from the Connes-Lott triple. - **Wording consequence for lemma 09.** The sentence "the light pair stays Connes-Lott with mass m_f, while the interval collapses to a point" in Theorem 9.9 is correct as a statement about the compression at each l. It should not be read as convergence to the Connes-Lott triple.

4.4 Theorem 9.8 (l -> infinity at fixed mu) Outside the paper's scope. The propinquity concerns compact quantum metric spaces. The diameter l diverges, and |diam_A - diam_B| <= 2 Lambda rules out

any limit. Decompactification would need a pointed or locally compact version of the metric, and we did not check whether one exists in Latrémolière's work.

4.5 As the convergence notion for T0 Correction (3 Nov 2026). The conclusions of this subsection on what T0 "needs" overread the metric; see the assessment of the T0 note. Lemmas R2 and R3 are written out in lemma 15, which also corrects the lower bound of R2 (the cap sqrt(2)/2 of the metric was omitted, and the bound improves to (1/m_f - l)/2). The original text follows as a record.

What it offers. - A genuine metric on metric spectral triples, with distance zero exactly at unitary equivalence. - Convergence of the bounded functional calculus, and Kuratowski convergence of spectra. - Corollary 4.1, Theorem 4.6 and Lemma R2 show it handles towerless limits whose finite part is a multiplicity space carrying a finite Dirac block over the base algebra. - Latrémolière, arXiv:2603.19128 (Mar 2026; abstract only read), proves spectral-propinquity continuity for almost-commutative manifolds M x F as the metric and D_F vary. It is the nearest existing treatment of M x F in this metric.

What it cannot do for T0.

Lemma R3 (ours, proof sketch). - **Statement.** If commutative quantum compact metric spaces (A_n, L_n) converge in the propinquity to (A, L), with Leibniz tunnels as in Definition 2.7, then A is commutative. - **Proof sketch.** - Let a_1, a_2 be self-adjoint with finite L. Take a tunnel (D, L_D, pi_A, pi_n) of extent chi_n -> 0, and lifts d_i with pi_A(d_i) = a_i and L_D(d_i) <= L(a_i) + 1. - For self-adjoint e in D, every state of A pulls back to within chi_n of a state of A_n. Hence ||pi_A(e)|| <= ||pi_n(e)|| + chi_n L_D(e), and similarly ||d_i|| <= ||a_i|| + chi_n L_D(d_i). - Apply this to the Lie product e = (d_1 d_2 - d_2 d_1)/2i. Then pi_n(e) = 0 because A_n is commutative, and L_D(e) <= L_D(d_1) ||d_2|| + ||d_1|| L_D(d_2) by the Leibniz rule. - So ||[a_1, a_2]|| <= C chi_n -> 0. Self-adjoint Lipschitz elements are dense, so A is commutative. - **Consequence.** The spectral propinquity dominates the propinquity, so every spectral-propinquity limit of commutative triples has a commutative algebra.

Implications for the T0 family (C^infinity(K_l), L^2(S x E_l), D_l, J_l, gamma_l): - (a) Its limits in this metric can never carry a noncommutative A_F. - (b) With K_l connected and D_l of Dirac type (Clifford symbol plus a zero-order endomorphism), Connes' metric on points is the Riemannian distance. A collapse of K_l therefore yields the base algebra alone, and the finite part appears as (C, H_F, D_F) tensored over the base. Several sheets (C^k) need disconnected K_l or a non-local D_l. - (c) Gamma and J are not part of the data. KO-dimension and chirality would need an extension of the metric, for instance by adding gamma and J to the families whose separation is measured. We found no such extension in this paper. - (d) Noncommutative limits do occur in Section 5, but only when the parent algebra is already noncommutative (the limit is its fixed-point algebra), as in Corollary 5.7.

So spectral propinquity can serve T0 for spectral and metric convergence of the Dirac data. Producing the finite algebra needs another mechanism: compressions or operator systems (Connes-van Suijlekom truncations, STRATEGY T6), or a noncommutative parent algebra.

Verdict

We may cite the paper for: - the framework of vertical collapse in the spectral propinquity (Sections 2-3, as a notion); - Corollary 4.1: a fixed metric spectral triple scaled by 1/eps converges to (C, ker D, 0), and to nothing when ker D = 0. This is sound and independent of the defect below; it supports Corollary 9.10 in the indirect sense of Section 4.2; - Theorems 4.6 and 5.5 (for tori) as statements in their anticommuting settings; - Theorem 6.3 as a statement about D_h + D_v/l_eps, with the caveats of Section 3.3.

For spectral continuity we should cite Latrémolière, Math. Ann. 389 (2024), directly.

We may not cite it for: - Theorem 3.4 in its printed generality, since it needs anticommutation of D_h and D_v or a relative bound (Counterexample R1); - anything about Theorem 9.9, which lies outside all its hypotheses; - l -> infinity (Theorem 9.8); - grading, chirality, real structure or KO-dimension; - Connes-Lott (or any noncommutative finite triple) as a limit; - homogeneous spaces of compact Lie groups; - the actual Dirac operator of the shrunk U(1)-bundle metric.

Open doubts. - We have not seen the published version, which may have repaired Theorem 3.4. *(Answered on 3 Nov 2026: the published version keeps it; see Section 3.6.)* - Lemma R2 (tuned interval -> (C, C^2, m_f sigma_1), at distance at least 1/(2 m_f) - 2l from Connes-Lott) and Lemma R3 (commutative limits are commutative) are our sketches. Both are short, and both bear directly on lemma 09 and on T0, so they should be written out and refereed before use. - The G-invariance step in Lemma 5.2 for nonabelian G is unverified.

Part R2 (referee report, lemmas 08, 13, 14)

Referee report on lemmas 08, 13 and 14 and on the matching parts of the paper

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_08_13_14.md. Reproduced as delivered, with links relativized. The corrections are incorporated in the lemmas and in the paper. Claude checked by hand or with numpy: the counterexample to the unqualified neutrino claim, the invisible profile cos(2 pi x/l), and the ratio 27/26.

We count the 29 numbered results in Lemmas 08, 13 and 14. The paper is assessed separately. Our checks below are analytic; we ran no numerical computation.

The spectral-action calculations use the ordinary trace over the stated finite Hilbert space and the leading four-dimensional a_4 coefficient. The relevant coefficient is in Chamseddine–Connes–Marcolli, arXiv v1, Theorem 6.1, equation (6.10). The KO-product comparison uses the conventions in Dąbrowski–Dossena, arXiv v1, equations (18) and (25).

Lemma 08

08.1 — Classification under pointwise reality **Correct.** Write $J_I = WK$. Pointwise reality makes W a multiplication operator. Comparing derivative and zeroth-order terms in $J_I D_I = tD_I J_I$ gives $WE = tEW$ and $W' = 0$. The endpoint conditions then force W to be diagonal, and $WE = tEW$ gives $\alpha = t\beta$. Thus the signs are $(+, t, +)$.

For $t = -1$, the designation 0_ is an extended convention; the standard Connes KO-0 signs have $t = +1$.

08.2 — Spectral compression **Correct.** The relation $J_I D_I = tD_I J_I$ implies commutation with D_I^2 and its spectral projections. Solving the boundary problem gives eigenvalues $\pm(r - \frac{1}{2})\pi/l$, $r \geq 1$. Direct differentiation of the displayed normalized profiles gives $D_I u_L = m u_R$ and $D_I u_R = m u_L$, with $m = \pi/(2l)$. The diagonal antiunitary compresses as claimed.

08.3 — Reflection sign **Correct.** Reflection reverses the derivative and exchanges the endpoint conditions. For the off-diagonal phase matrix, commutation with $E\partial_x$ requires $a = b$; involutivity then gives $a = b = \pm 1$. The relation $\alpha = t\beta$ gives $\sigma J_I = tJ_I\sigma$, and the lowest profiles are exchanged, so the compressed reflection is $\pm\sigma_1$.

08.4 — Two counterexamples **Correct.** The antiunitary $\sigma_1 R_l K$ preserves the domain, commutes with D_I, squares to 1, and anticommutes with Γ_I. Its opposite action is multiplication by the reflected function, so order zero and first order hold for the scalar interval algebra.

For the compressed full diagonal algebra, $e^o = e$ for $e = \mathrm{diag}(1, 0)$, and

$$[[m\sigma_1, e], e] = m\sigma_1 \neq 0.$$

The first-order failure is therefore exact.

08.5 — Internal-first KO-6 product **Correct.** For the stated grading-twisted tensor structures,

$$J_{a,b}^2 = (-1)^a, \qquad J_{a,b}(\Gamma_0 \otimes D_I) = -t(-1)^b(\Gamma_0 \otimes D_I)J_{a,b}.$$

The standard KO-6 signs therefore require $a = 0$ and $t(-1)^b = -1$. Since $\sigma\Gamma_I = -\Gamma_I\sigma$, the bare reflection has

$$(1 \otimes \sigma)J_F = -J_F(1 \otimes \sigma).$$

This conclusion uses the stated product convention and the specified class of real structures.

08.6 — Grading and lift of reflection **Correct.** The interval grading alone commutes with the displayed tensor real structures, so it cannot supply the standard KO-6 grading. For $U_1 = \Gamma_0 \otimes \sigma$,

$$[U_1, D_F] = [\Gamma_0, D_0] \otimes \sigma = 2\Gamma_0 D_0 \otimes \sigma,$$

which vanishes exactly when $D_0 = 0$. The displayed unitary W transports both D_F and U; consequently it preserves the sign against the transported J.

08.7 — Reconciliation with the odd Clifford point **Correct.** With the scalar interval algebra, the displayed map R is unitary onto the positive-grading space and gives

$$R^*UD_FR = D_0 + m\Gamma_0, \qquad R^*UJ_FR = J_0\Gamma_0.$$

The latter antiunitary squares to -1 and anticommutes with $d_5 = D_0 + m\Gamma_0$. Its opposite action is that of J_0, and $[d_5, a] = [D_0, a]$, so the order conditions descend.

In the resulting two-block basis, the matrices are those of the odd–odd product with $B = -i\sigma_2$. The zero Dirac operator on the point makes the P_1 label a KO-sign convention. The factorization does not extend to the full compressed two-sheet algebra with its inherited real structure: 08.4 shows the first-order failure there.

08.8 — Literal collapse **Correct.** Anticommutation gives

$$D_F^2 = D_0^2 \otimes 1 + 1 \otimes D_I^2 \geq \frac{\pi^2}{4l^2}.$$

Thus every fixed finite energy window is empty for sufficiently small l. This is a conclusion about the stated product, with no compensating bulk mass or modified cutoff prescription.

08.9 — Spacetime grading twists **Correct.** The two summands of $D_M \otimes 1 + \gamma_M \otimes d$ have signs $(-1)^a$ and s against the proposed real structure, so $(-1)^a = s$ gives the stated odd-product convention. Its square is $-s$, giving the claimed KO-1 and KO-3 signs. The displayed unitary V also transports the complete products because $\sigma_1 B_s = sB_s\sigma_1$. Neither calculation selects a finite involution.

08.10 — Euclidean and Lorentzian charge conjugation **Correct.** The stated Euclidean and Lorentzian operators have the displayed squares and chirality relations in the specified matrix basis; in particular $C_L = -i\gamma_E^0 C_E$. Charge conjugation preserves the real Dirac mass while changing the charge, whereas subsequent multiplication by γ_5 reverses the scalar mass sign.

The source identifications are supported by CCM, arXiv v1, §§3.1–3.2 and D'Andrea–Kurkov–Lizzi, arXiv v2, equations (3.13)–(3.15).

08.11 — Fixed real structure with both signs **Correct.** The stated D_F, J_F, Γ_F have the standard KO-6 signs and D_F commutes with the represented algebra. Both $U_- = 1 \otimes \sigma_1$ and $U_+ = \tau_3 \otimes \sigma_1$ commute with D_F, Q and the two algebra actions, reverse the grading, and square to 1. Direct multiplication gives the opposite signs against J_F. The block unitary W gives the displayed electrodynamics matrices.

08.12 — Particle-number pairing **Correct.** Under all the stated hypotheses, $[Q, U] = 0$ makes QU a unitary involution. The commutators of Q with D, Γ, the algebra and its opposite preserve the remaining requirements, while

$$(QU)J = sQJU = -sJ(QU).$$

The final bimodule obstruction also follows: a particle-preserving, grading-reversing U must identify the two particle bimodules.

Lemma 13

13.1 — Inner fluctuations and boundary conditions **Correct.** The commutator with a matrix-valued function has the five displayed Clifford components, and matrix-valued one-forms are generated locally by sums $a\,db$. Self-adjointness gives Hermitian A_μ, A_5. The fluctuation is bounded and self-adjoint, hence preserves self-adjointness on the original Dirac domain without having to map that domain into itself.

Applying the original boundary conditions to $D_A\psi$ yields the two covariant derivative conditions in the proof. They constrain ψ, not endpoint values of the potentials.

13.2 — Fixed-projection compression **Correct.** The diagonal spacetime matrix elements give the two weighted averages, and the vertical off-diagonal element is

$$m + i\int_0^l \frac{\sin(\pi x/l)}{l} A_5(x)\,dx.$$

The proposed cosine lift realizes arbitrary Hermitian A_L, A_R, and $A_5 = (\pi/2)B$ realizes arbitrary Hermitian B. Thus the exact link image is $\phi + \phi^* = 2mI_n$. An unrestricted finite two-sheet fluctuation can in particular set $\phi = 0$, so its image is larger.

13.3 — Failure of gauge covariance of fixed compression **Correct.** For $A_5 = a\cos(2\pi x/l)$, the integral of A_5 vanishes, hence $W(l) = 1$. But

$$B = \frac{a}{\pi}\int_0^\pi \sin t\cos(2t)\,dt = -\frac{2a}{3\pi}.$$

The resulting compressed eigenvalues are $\pm\sqrt{m^2 + 4a^2/(9\pi^2)}$, rather than $\pm m$. Thus even an endpoint-fixed gauge transformation changes the fixed compression. The transported-frame link and the Schur-complement term in the proof follow by direct substitution.

13.4 — Mixed-boundary vector mass **Correct.** Neumann at 0 gives $\cos(kx)$, and Dirichlet at l gives $kl = (r + \frac{1}{2})\pi$. The lowest mass is $\pi/(2l)$. These gauge boundary conditions are additional data; 13.1 does not derive them from the fermion domain.

13.5 — Spectral-action vector-mass formula **Correct.** In the stated convention, the Clifford traces in the proof give

$$\mathcal{L}_{a_4} = \eta\ \mathrm{Tr}\, F_{\mu\nu}F^{\mu\nu} + 3\eta\ \mathrm{Tr}\, D_\mu V D^\mu V, \qquad \eta = \frac{f_0}{24\pi^2}.$$

At V_0, the quadratic terms are $\eta Q_{ab} f^a f^b + 3\eta K_{ab} A^a A^b$. Setting $A_{\mathrm{can}} = 2\sqrt{\eta}\, Q^{1/2} A$ gives

$$\mathcal{M}^2 = \frac{3}{2} Q^{-1/2} K Q^{-1/2}.$$

The coefficient agrees with CCM, arXiv v1, equation (6.10). It is conditional on the stated trace, vacuum and a_4 approximation.

13.6 — Minimal two-sheet action **Correct.** For each generator, the stated trace matrices are

$$Q = \kappa I_2, \qquad K = 2\kappa m^2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

The mass-squared eigenvalues are therefore 0 and $6m^2$, independent of the common multiplicity. For the additional profile restriction, $c_L : c_R = 2 : 1$, and the one-field formula gives $3m^2/5$. That restricted field is not the massive eigenvector of the unrestricted two-sheet action.

13.7 — Conditional CCM comparison **Correct.** The full three-generation trace gives $Q_{11} = 12$, while one massive colour-triplet Dirac fermion gives $K_{11} = 3m^2$. Hence

$$M_W^2 = \frac{3}{2}\frac{3m^2}{12} = \frac{3}{8}m^2.$$

Keeping the massless charged states in Q is essential. This reproduces the conditional relation in CCM, arXiv v1, Lemma 4.6, equation (4.16). The normalized interval matrix element is already m; the separate A_5 overlap does not rescale that background mass.

13.8 — Electroweak doublet and neutral-vector ratio **Correct with amendment.** The doublet calculation gives the displayed W, Z, and ρ formulas. The full trace has $Q_2 = 12$, $Q_Y = 20$, so $g_Y^2/g_2^2 = 3/5$ and $M_W/M_Z = \sqrt{5/8}$.

The last paragraph must say that **the link of the independently gauged finite two-sheet completion** transforms as $g_L \phi g_R^*$. The fixed compression of the original continuum field does not in general obey that endpoint transformation law: 13.3 gives an endpoint-fixed gauge transformation that changes ϕ. The $1 \oplus 3$ decomposition under the diagonal $SU(2)$ remains correct.

Lemma 14

14.1 — Covariant projection and spectral data **Correct with amendment.** The operator identity

$$D_{A^g,\mathsf{M}^g,\mathcal{B}^g} = G_g D_{A,\mathsf{M},\mathcal{B}} G_g^\dagger$$

and the Riesz formula prove covariance of the gapped projection. Conjugation by W gives the stated axial-gauge bulk matrix and domain. In particular, a gauge-invariant bulk mass and the mixed conditions are unchanged, so their spectrum is independent of A_5.

The double-coset statement needs the hypothesis that endpoint transformations act **independently** through H_0 and H_l, with the corresponding extendability understood; it is a local pointwise classification unless global topology is specified. For example, periodic boundary conditions require $g(l) = g(0)$. Their holonomy is classified by conjugation, not by $G\backslash G/G$. For $G = U(1)$, a periodic constant A_5 has a nontrivial gauge-invariant Wilson phase despite both endpoint projections being $U(1)$. The lemma's subsequent discussion of identified endpoints gives the appropriate distinction.

14.2 — Induced connection and finite operator **Correct.** Acting on $U\chi$ gives

$$\nabla_\mu^{\mathcal{E}} \chi = \partial_\mu \chi + U^\dagger \partial_\mu U\, \chi + i U^\dagger A_\mu U\, \chi,$$

which is the displayed $\mathcal{A}_\mu$. Expansion of $P\nabla_\mu P \nabla_\nu P - (\mu \leftrightarrow \nu)$ gives the curvature formula, and $U^g = G_g U h^\dagger$ gives the stated frame transformations.

The sums involving ∇^S and the internal connection are tensor-product connection sums; in a local spin frame the derivative appears once. The Schur-complement correction is the standard result of eliminating the complementary block.

14.3 — Finite algebra and fluctuation image **Error.** The displayed formulas, the compression range, and the **local** fixed-singular-value statement are correct. The unqualified claim that the global continuum link image consists *exactly* of all links with $\Phi^\dagger\Phi = \lambda^2 I$ is too strong.

Take $M_4 = S^1 \times S^3$, $n = 1$, and the globally defined fixed-singular-value link

$$\Phi(\theta, z) = \lambda e^{i\theta}.$$

For any globally smooth real A_5 on the stated trivial bundle,

$$\Phi = \lambda W_l^\dagger = \lambda \exp\left(i \int_0^l A_5(y, x)\, dx \right)$$

has a globally defined real phase and zero winding around S^1. It cannot equal the displayed link. More generally, $W(y, x)$ is a homotopy from the identity to $W_l(y)$.

The corrected statement is that every fixed-singular-value link is obtained **locally in spacetime**; globally, the unitary link must admit the corresponding path from the identity in the chosen trivialization. The reducing-subalgebra criterion is correct: the compression identity in the proof forces both a and a^* to preserve the range. Since p_L is nonzero in the interior, this gives precisely $a(x) = W(x)bW(x)^\dagger$, and conversely such multipliers reduce the light space.

14.4 — Boundary Yukawa data **Correct with amendment.** The transformation law for X, its axial-gauge form, and the limiting texture follow from the boundary equations. In the towerless scaling, the normalized wall values are $l^{-1/2}\Phi_Q$ and $l^{-1/2}\Phi_U$, giving

$$Y_0 = \Phi_Q \widehat{X} \Phi_U.$$

Expanding the transfer functions at fixed bounded λ changes this matrix by $O(l^2)$; after grading-preserving changes of frame the light block converges to Y_0. Positive diagonal Φ_Q, Φ_U give the stated attainability.

Two hypotheses must be made explicit:

1. To obtain the **entire** displayed limiting block, retain the full zero-mode-descended cluster of dimension $N_Q + N_U$, not an arbitrary smaller gapped grading-invariant cluster.
2. The **nonzero** singular values of Y are the retained positive eigenvalues. Zero singular values and possible unmatched chiral zero modes must be included separately.

For the first point, with two massless flavors and $X = l\,\mathrm{diag}(1, 2)$, the cluster containing only the lowest $\pm$ pair is gapped but has a 1×1 limiting block, not the full 2×2 texture.

14.5 — Quadratic matching condition **Correct with amendment.** On a common retained profile space where both kinetic forms are positive, the two mass matrices are the stated generalized eigenvalue problems. If the KK mass matrix is $\kappa^2 I$, equality of all its masses is equivalent to

$$K = \frac{2}{3}\kappa^2 Q.$$

The equality of actions in the same coordinates gives the stronger conditions $G^{(5)} = 4\eta Q$, $H^{(5)} = 6\eta K$. Block multiplication gives the stated C_a, Q, K, and one-vector sum rule.

The comparison must explicitly restrict **both** quadratic forms to the same profile space on which $G^{(5)}$ and Q are positive. A KK profile can be invisible to compression. For the massless mixed fermion pair, $f(x) = \cos(2\pi x/l)$ satisfies Neumann conditions at both endpoints and has positive five-dimensional norm and derivative norm, but

$$\int w_L f = \int w_R f = 0.$$

Thus $Q = K = 0$ for that nonzero KK mode, and the finite action has no corresponding canonically normalized vector.

14.6 — Covariant lowest-mode reduction **Correct.** The displayed gauge transformation closes on the retained Abelian a_μ, b fields. The transported frame gives

$$D\Phi = i\Phi(c_L - c_R)\left(a + \frac{db}{\kappa}\right),$$

including the Berry terms. Canonical normalization then gives

$$M^2_{\text{spec}} = 3\lambda^2 \frac{(c_L - c_R)^2}{c_L^2 + c_R^2}.$$

At $M = 0$, $c_L = 8/(3\pi)$, $c_R = 4/(3\pi)$, and $\lambda = \kappa$, so the ratio is exactly $3/5$.

14.7 — Tuned bulk mass **Correct.** For $M \geq 0$, the lowest root satisfies $M = -k\cot(kl)$, with $t = kl \in [\pi/2, \pi)$, and $\lambda = k/\sin t$. At $t = 3\pi/4$, the displayed normalization integral and overlaps give

$$c_L = \frac{9}{3\pi + 2}, \qquad c_R = \frac{6}{3\pi + 2}, \qquad \frac{M^2_{\text{spec}}}{M^2_{\text{KK}}} = \frac{27}{26}.$$

The ratio is $3/5$ at $M = 0$ and varies continuously, so an M_* exists in the stated open interval. No uniqueness follows from this argument. At such a solution, $\kappa < \lambda < (3/\sqrt{2})\kappa$, so the retained fermion is at the tower scale.

14.8 — Bounded-fermion no-go **Correct.** For any Hermitian linear combination T of retained couplings,

$$\|[T, V_0]\|_{\text{HS}} \leq 2\|V_0\|\,\|T\|_{\text{HS}}.$$

Consequently $K \leq 4\|V_0\|^2 Q$ as quadratic forms, and every finite mass satisfies

$$M^2_{\text{spec}} \leq 6\|V_0\|^2.$$

The constant is sharp: take $V_0 = C\sigma_3$ and $T = \sigma_1$. Thus bounded V_0 cannot match the divergent Neumann–Dirichlet mass within the stated leading-action approximation.

14.9 — Gauge and scalar normalizations **Correct.** With fixed generator and profile normalizations, $\eta q_s F_s^2$ gives $g_{4,s}^{-2} = 4\eta q_s$, whereas integrating the five-dimensional kinetic term gives $J_s/g_{5,s}^2$. Rescaling a five-dimensional coupling changes both of its quadratic forms by the same factor and leaves its KK mass unchanged, so mass matching alone does not determine the relative coupling.

At vanishing ambient A_μ, $D_\mu V = H_i \partial_\mu \varphi^i$, giving $Z_{ij} = 6\eta\,\mathrm{ReTr}(H_i H_j)$. In the eaten-scalar example, imposing both displayed gauge-kinetic and mass equalities gives $Z_b = G$. No corresponding conclusion follows for an independent physical scalar direction.

Comparison with the paper

Section 3: Theorem A and Proposition A.1 Theorem A is **correct with amendment**. Its operator, spectrum, compression range and two-state distance are correct. It should identify the two states as states of the compressed operator system. Their pullbacks are the smeared measures $w_L dx, w_R dx$, not literal endpoint evaluations; for example, $\cos(3\pi x/l)$ compresses to zero but has endpoint values $1, -1$. The theorem also omits that the full diagonal compressed algebra with the inherited J fails first order, as 08.4 proves.

Proposition A.1 correctly states 08.8 for the specified massless product and fixed cutoff.

Section 6: Theorem D

- **D.1 — Correct, with its finite-triple scope made explicit.** The criterion agrees with Lemma 06.2. The point has zero Dirac operator; its KO label is part of the product convention.
- **D.2 — Error as a general statement.** The assertion about *a finite triple with an unpaired left-handed neutrino* is broader than the particular reduced representations treated in Lemma 06. For a counterexample, take

  $$H = \mathbb{C}^2_{\text{gen}} \otimes \mathbb{C}^2, \quad A = \mathbb{C}, \quad D = 0, \quad \Gamma = 1 \otimes \sigma_3, \quad J = (1 \otimes \sigma_1)K.$$

  Regard the two positive-grading states as two unpaired left neutrinos and the negative-grading states as their conjugates. There is no Majorana term. Both

  $$U_+ = 1 \otimes \sigma_1, \qquad U_- = \sigma_2 \otimes \sigma_1$$

  satisfy the factorization criterion, with signs $+1$ and -1, respectively. Conversely, an unpaired neutrino does not alone guarantee the $+1$ factorization: the full chiral algebra can obstruct both, as in Lemma 06.8. The paper must specify the reduced algebra, charged bimodules and neutral multiplicity for its KO-7-only claim.
- **D.3 — Correct with amendment.** The sign calculation is correct. Calling both interval choices "KO-dimension 0" needs the extended 0_+, 0_- convention: under the paper's stated standard convention only $t = +1$ has the standard KO-0 signs. The bare reflection is generally not linear over the full interval algebra.
- **D.4 — Correct with amendment.** The exact factorization is for the **scalar-algebra** interval truncation. With the full compressed two-sheet algebra, the inherited real data fail first order and the reflection does not commute with that algebra.
- **D.5 — Correct with amendment.** The fixed-product example is correct. The general pairing statement also needs Q to be a self-adjoint involution commuting with Γ, the algebra and its opposite, in addition to the stated conditions and particle preservation. The omitted algebra hypothesis matters: with $A = \mathbb{C} \oplus \mathbb{C}$ acting diagonally on a multiplicity factor, $D = 0$, $\Gamma = 1 \otimes \sigma_3$, $J = (1 \otimes \sigma_1)K$, $U = 1 \otimes \sigma_1$, and $Q = \sigma_2 \otimes 1$, the conditions printed in D.5 hold and U preserves Q, but QU does not commute with A.

The paragraph after Theorem D repeats the unqualified neutrino claim and the claim that adding ν_R with Dirac masses gives both signs; both need the reduced-representation hypotheses. Its statement that the Majorana case is open is also less precise than Lemma 06: that lemma gives a one-generation result and a singular-value multiplicity criterion for its specified neutral sector. A broader model can remain open.

Section 7: Propositions E and F and the following paragraph Proposition E states its CCM mass relation as an assumption and correctly limits its use to the matching-scale heat-kernel regime.

Proposition F is **correct with amendment** for its fixed-compression and gauge-covariance claims, but its mass sentence contains a dimensional error: $\mathcal{M}^2$ cannot equal $\sqrt{6}\,m$ or $\sqrt{3/8}\,m$. Those are masses; their squares are $6m^2$ and $3m^2/8$. It must also say that the minimal two-sheet model has a massless diagonal vector, and that the CCM value is for the W under the full three-generation trace, specified vacuum and one-mass assumption. The Neumann–Dirichlet vector boundary conditions are separate inputs.

The following paragraph reports the 3/5 example, tuned mass, and norm bound correctly, but its opening sentence is too general: the two masses can agree in the tuned example described immediately afterward. The 3/5 and M_* claims concern the specified one-fermion Abelian mixed-boundary reduction. The no-go requires **all fermion masses retained in the finite trace** to be uniformly bounded, and applies to the stated leading spectral action. Covariant projection and one mass equality do not establish equality of the full gauge or scalar actions.

The abstract and introduction repeat the overly broad claims that an unpaired neutrino selects KO-7 and that the gauge sector does not reproduce the tower. They need the same qualifications.

Amendments and errors

- **Lemma amendments:** restrict 13.8's $g_L \phi g_R^*$ law to the independently gauged finite completion; add independent endpoint action and extendability to 14.1's double coset; specify the full light cluster and zero modes in 14.4; compare both positive kinetic forms on the same profile space in 14.5.
- **Lemma error:** 14.3's global fixed-singular-value link image omits the homotopy restriction. Its local statement is valid.
- **Paper:** qualify Theorem A's state and real-structure interpretation; restrict Theorem D's neutrino, interval-algebra and particle-pairing claims; correct Proposition F's mass-versus-mass-squared sentence and state the model, trace, vacuum and boundary assumptions; limit the claimed gauge-sector disagreement and bounded-fermion no-go to the cases proved.

Overall: 29 numbered lemma statements checked — 24 correct, 4 correct with amendment, 0 gaps, and 1 error.

Part R3 (referee report, lemmas 15, 16)

Referee report on lemmas 15 and 16 and on the T0 note assessment

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_15_16.md. Reproduced as delivered, with links relativized. The corrections are incorporated in lemma 15, lemma 16 and the assessment. Claude checked the two external citations against the arXiv LaTeX sources: the inner Leibniz condition in arXiv:1811.10843 and the commuting subalgebra C_F in hep-th/0610241, section 2.6.

We use the definitions and notation of Lemma 15 and Lemma 16. All calculations below are analytic; we ran no numerical computation.

The statements involving spectral propinquity require one correction to the supplied Farsi–Latrémolière source. Its displayed inner Leibniz condition is ill-typed as written. We state the correction under Theorem 15.3 and use it throughout.

Part 1. Statements and proofs

Lemma 15.1 Correct.

For self-adjoint e, its norm is the supremum of $|\varphi(e)|$ over states. The definition of extent therefore gives both norm inequalities. The Jordan-Leibniz inequality gives

$$L_E(e^m) \leq m\|e\|^{m-1} L_E(e)$$

by induction, and the polynomial estimate follows by the triangle inequality.

Theorem 15.2 Gap, closed here.

The commutativity argument in item 1 is correct. Bounded-Lipschitz lifts of two self-adjoint elements have uniformly bounded norms. Their Lie product maps to zero in every commutative approximating algebra, so Lemma 15.1 forces the Lie product of the two limit elements to vanish.

In item 2, the stated conditions on g do **not** imply that b_n stays within 3/8 of an integer. They control neighborhoods of the half-integers, but permit values beyond the outermost integers.

We can repair the proof with the same constants. Choose $g : I \to [0,1]$ such that

$$g = 0 \quad \text{on} \ \bigcup_{j=0}^{k} [j - \tfrac{1}{8}, j + \tfrac{1}{8}], \qquad g = 1 \quad \text{outside} \ \bigcup_{j=0}^{k} (j - \tfrac{3}{8}, j + \tfrac{3}{8}).$$

A polynomial p with $\|p - g\|_I < 1/8$ has $\|p(a)\| < 1/8$. Lemma 15.1 then gives $\|p(b_n)\| < 1/4$ for large n, so the range of b_n lies in the union of the disjoint open 3/8-neighborhoods. Each connected component of

X_n maps into one neighborhood. At most k neighborhoods are reached, and a fixed one is missed along a subsequence. The polynomial q constructed in the note then gives the stated contradiction.

The clopen partition used at the start exists because components and quasicomponents agree in a compact Hausdorff space. Both conclusions follow.

Theorem 15.3 **Correct with amendment.** The estimates hold under the standard inner Leibniz definition. The supplied source's literal definition must be corrected before its tunnels can be checked as stated.

At source lines 602–604 (arXiv:2404.00240v3 LaTeX source), the inner product is required to lie in the domain of L_B, although that domain was defined on self-adjoint elements; the displayed inequality then names L_A. Literally, the first requirement fails for every nonzero complex Hilbert module: $\langle \omega, i\omega \rangle$ is a nonzero skew-adjoint element. The required condition is

$$\max\{L_B(\mathrm{Re}\langle \Theta, \Theta' \rangle), L_B(\mathrm{Im}\langle \Theta, \Theta' \rangle)\} \leq 2\,\mathrm{DN}_P(\Theta)\mathrm{DN}_P(\Theta').$$

This is the condition in Latrémolière's Definition 2.2.

With that correction, the tunnel has the required properties:

- The top seminorm is Leibniz for Jordan and Lie products, has scalar kernel and a closed unit ball. Its centered unit ball is compact. Choosing the midpoint of the spectrum of a proves the quotient condition onto A; scalar lifts prove it onto $\mathbb{C}$. Its extent is at most ρ.
- The base seminorm on $\mathbb{C} \oplus \mathbb{C}$ is an L-seminorm with extent at most s.
- The module D-norm dominates the module norm and has a compact unit ball. The displayed modular estimate is valid without assuming P commutes with A.
- The inner-product calculation proves the stronger complex estimate

  $$|\langle \xi, \xi' \rangle - \langle \eta, \eta' \rangle| \leq 2s\,\mathrm{DN}_P(\xi, \eta)\mathrm{DN}_P(\xi', \eta'),$$

  which implies the corrected real-and-imaginary-parts condition.
- The lifts $\eta = cW\xi$ and $\xi = cW^*\eta$ attain the two D-norm quotients. The coordinate maps also preserve the left and right actions and inner products.

The pseudo-state choices in both directions are valid and are independent of t. Consequently they bound the **Hausdorff distance between families**, as required, by

$$g^{-2} + \rho + \Delta + \Delta/\varepsilon.$$

The conclusion follows by taking ε larger than $g^{-2} + \rho + \Delta + \sqrt{\Delta}$ and then decreasing ρ. The lower bound correctly includes the cap:

$$\Lambda_{\mathrm{spec}} \geq \min\left\{ \frac{\sqrt{2}}{2}, \frac{\delta}{2} \right\}.$$

Remark 15.4 **Error.** The uncapped lower bound is asserted for every $\varepsilon > 0$, although spectral propinquity is at most $\sqrt{2}/2$.

The correct statement is

$$\min\left\{ \frac{\sqrt{2}}{2}, \frac{\varepsilon\delta_1}{2} \right\} \leq \Lambda_{\mathrm{spec}} \leq \frac{\varepsilon\delta_1}{2} + \frac{\varepsilon^2}{g_1^2}.$$

For example, take the periodic Dirac operator on a circle of length 1. It has zero modes and $\delta_1 = 1/2$. At $\varepsilon = 4$, the asserted uncapped lower bound is $1 > \sqrt{2}/2$. The claimed leading asymptotic remains correct for sufficiently small ε.

Corollary 15.5 **Correct.**

For the stated sufficiently small ℓ, including $\ell < \sqrt{2}$ in the chosen normalization, the cap does not alter the lower bound. The interval diameter is ℓ, the light block is exactly $m_f\sigma_1$, and the next eigenvalue satisfies

$$g^2 \geq \mu(\ell)^2 + \frac{\pi^2}{\ell^2} = \frac{c(\ell)^2 + \pi^2}{\ell^2}.$$

Thus the upper correction is exactly bounded by $\ell^2/(\pi^2 + c^2)$, as displayed. The Connes–Lott lower bound has the correct factor $1/2$ and cap.

Corollary 15.6 **Correct.**

The boundary conditions are preserved by scalar Lipschitz functions, and the commutator seminorm is $\|f'\|_\infty$. The zero modes and positive and negative light modes in Theorem 12.7 have the correct total dimension $N_Q + N_U$. Matching them to the singular vectors of M_F gives $\Delta = O(\ell^2)$, including zero singular values. The gap is $O(\ell^{-1})$, so Theorem 15.3 gives the asserted $O(\ell)$ bound.

Theorem 15.7 **Correct with amendment.** The proof uses the corrected real-and-imaginary-parts inner Leibniz condition stated under Theorem 15.3.

The two base characters have distance below ε_n, since either is a state of the tunnel base algebra and the other is the sole state in its quotient image. Applying the corrected inequality to the real and imaginary parts proves the displayed $4\varepsilon_n$ inner estimate. Indeed, rotating one vector by a phase gives the stronger constant $2\varepsilon_n$; the stated constant is sufficient.

The upper multiplicity argument is valid: the Gram matrix of $N + 1$ target vectors has rank at most N, while the entrywise estimate gives

$$\|I - G\| \leq 16(N + 1)(1 + R)^2\varepsilon_n.$$

For concentration, Hausdorff separation supplies **one** pseudo-state matching the entire time family. Comparing its estimates at t and 0 gives the displayed $8C^2\varepsilon_n$ bound. Averaging over $0 \leq t \leq 1/\varepsilon_n$ gives

$$\int \left(1 - \operatorname{sinc} \frac{x - \lambda}{\varepsilon_n}\right) d\nu(x) \leq 8C^2\varepsilon_n.$$

Since $|\operatorname{sinc} u| \leq 1/2$ for $|u| \geq 2$, the stated tail bound follows.

For each eigenvalue of D_F, projecting the lifted vectors to its $2\varepsilon_n$-window changes them by $o(1)$; their Gram matrix remains invertible. Distinct eigenvalues have disjoint windows eventually. The lower counts and the preceding upper count give the exact total multiplicity N, after which the eigenvectors can be matched with operator error at most $2\varepsilon_n$. The gap to the remaining spectrum tends to infinity. If no spectrum remains, the gap is interpreted as infinite.

Corollary 15.8 **Correct.**

Faithfulness and finite-dimensionality of the limiting Hilbert space make the limiting algebra finite-dimensional. Theorem 15.2 then gives $\mathbb{C}$. On a circle of length ℓ, the constant modes of

$$-i\,\frac{d}{dx} + \operatorname{diag}(\lambda_1, \ldots, \lambda_N)$$

carry the prescribed Hermitian matrix, while the other modes have absolute eigenvalue at least $2\pi/\ell - \|D_F\|$. The circle diameter tends to zero, so Theorem 15.7 applies. The construction concerns ungraded triples; additional real or even structures would impose their own compatibility conditions.

Proposition 15.9 **Correct with amendment.**

The symmetric spectral projection commutes with γ and J, and the displayed estimates hold for $D_B = WDW^*$. Two qualifications are needed for the subsequent claim about metric control.

First, if $\Delta > 0$, $W\omega$ need not be a pseudo-state for the graph norm of D_F. The normalized choice $cW\omega$, with $c = (1+\Delta)^{-1}$, is a pseudo-state and adds at most Δ to the stated bounds.

Second, the induced γ_F, J_F satisfy the Dirac relations exactly with D_B, not necessarily with the independently chosen D_F. The defects with D_F are bounded by 2Δ. The case $\Delta = 0$ is exact; when $\Delta \to 0$, the defects vanish.

Finally, composition with an antiunitary produces an antilinear functional. The source's operational propinquity is defined for linear functionals, so convergence of J requires the extension acknowledged in the note.

Example 16(a) **Correct.**

At the lowest massless interval level, the two chiral profiles give

$$Q(f) = \operatorname{diag}(C_0 + C_1, C_0 - C_1).$$

The coefficients C_0, C_1 can be chosen independently, so the range and generated algebra are both $\mathbb{C} \oplus \mathbb{C}$.

Example 16(b) **Correct.**

The product-to-sum identities give the two displayed matrices. Their four coefficients $C_0, \dots, C_3$ are independently recoverable, so the operator system has complex dimension 4.

For the two displayed functions, direct substitution gives

$$Q(f_1) = \left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, 0 \right), \qquad Q(f_2) = \left(0, \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \right).$$

Each coordinate projection of the generated algebra contains all symmetric 2×2 matrices, hence all of $M_2(\mathbb{C})$. The first displayed element generates a nonzero two-sided ideal supported on the first summand; simplicity of $M_2(\mathbb{C})$ makes that ideal the entire first summand. The second element does the same for the other summand. Therefore

$$A_F = M_2(\mathbb{C}) \oplus M_2(\mathbb{C}).$$

Example 16(c) **Correct.**

For a uniformly distributed rank-one projector P_w on $\mathbb{C}^4$,

$$\mathbb{E}(P_w \otimes P_w) = \frac{I + \mathrm{Flip}}{20}.$$

Together with $\operatorname{tr} \Gamma_a = 0$, this gives $Q(x_a) = \Gamma_a/5$. Four of the gamma matrices generate $M_4(\mathbb{C})$. For the twisted Dolbeault operator, the kernel is $H^0(\mathbb{CP}^3, \mathcal{O}(1))$, of dimension 4; the higher cohomology groups vanish. Thus $d = 0$ on the selected subspace.

Proposition 16.1 **Correct with amendment.**

The order-zero equivalence is correct. The hypotheses imply $J_F Q(a)^* J_F^{-1} = Q(a)$, so order zero forces all compressed generators to commute. Conversely, if their generated algebra is commutative, the opposite image is the same algebra and order zero holds.

For the product assertion, specify that the fixed finite factor itself satisfies order zero and that the product real structure is the compatible one. Also, the hypotheses on J_0 alone do not assert $J_0^2 = \pm 1$ or a grading sign. Those relations must be added if J_F is called a real structure in the full sense.

Theorem 16.2 **Correct with amendment.** The Weyl-exponent assertion requires the original exponent to exist.

The graded flip is well defined on the complex tensor product. On each homogeneous summand it is the ordinary conjugate flip times a sign; it is therefore antiunitary. Direct application to homogeneous tensors gives

$$J^2 = 1, \qquad J\mathcal{D} = \mathcal{D}J, \qquad J\Gamma = \Gamma J.$$

For even a, $a^\circ = 1 \otimes (a^*)^c$. Hence order zero and order one hold, and

$$[\mathcal{D}, \pi(a)] = [D_0, a] \otimes 1.$$

The signs $(+,+,+)$ are the usual KO-0 signs.

An eigenbasis of D_0 decomposes $\mathcal{D}$ into finite-dimensional invariant blocks. On each block,

$$\mathcal{D}^2 = D_0^2 \otimes 1 + 1 \otimes (D_0^c)^2.$$

This proves essential self-adjointness on the algebraic tensor core and compact resolvent of the closure.

Writing $N_0(R)$ and $N_{\mathcal{D}}(R)$ for absolute-eigenvalue counting functions with multiplicity,

$$N_0(R/\sqrt{2})^2 \le N_{\mathcal{D}}(R) \le N_0(R)^2.$$

Thus, if $\lim_{R\to\infty} \log N_0(R)/\log R = p$ exists, the new exponent is $2p$. Compact resolvent alone does not guarantee such a limit: eigenvalue multiplicities can be chosen to make the logarithmic ratio oscillate.

Corollary 16.3 **Correct with amendment.** Assume $k \ge 1$, and distinguish the product projection from a spectral projection of $\mathcal{D}$.

The identities in the corollary and both order conditions hold for every even subalgebra $A_F \subseteq B(V)$. In particular,

$$J_F(d \otimes 1)J_F^{-1} = \gamma_V \otimes d^c, \qquad [D_F, a \otimes 1] = [d, a] \otimes 1.$$

The original compression has range $Q(A_0) \otimes 1$; the corollary uses the **generated** algebra $A_F \otimes 1$.

The projection $P_0 \otimes \overline{P_0}$ reduces $\mathcal{D}$, but need not equal $1_S(\mathcal{D})$ for any set S. For a counterexample, take an even finite D_0 with positive absolute eigenvalues

$$1, \quad 5, \quad 7, \quad \sqrt{73},$$

and let P_0 select the first three absolute values. The selected pair $5, 7$ and the mixed pair $1, \sqrt{73}$ both give squared product eigenvalue 74. Both reducing subspaces contain the eigenvalues $\pm\sqrt{74}$, so the product projection cannot be a spectral projection of $\mathcal{D}$. It is a joint cutoff in the two factors. In the kernel example used below, it *is* a spectral projection.

Corollary 16.4 **Correct.**

The displayed operators have signs $(+,+,-)$, the usual KO-6 signs. Both algebra actions are trivial on the last factor, so order zero and order one persist, and the Hilbert-space dimension is $2k^2$.

The signs alone do not establish the usual zero-dimensional orientability axiom. Every represented zero-cycle built from the left and opposite algebra actions acts as $T \otimes 1_2$, whereas the grading is $\Gamma_F \otimes \sigma_3$. No such zero-cycle can equal that grading.

Caveat 1: commutant **Correct.**

For

$$d = \begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix},$$

with B invertible, the relation $\operatorname{diag}(x, y)d = d \operatorname{diag}(x, y)$ is equivalent to

$$y = B^{-1}xB, \qquad [x, BB^*] = 0.$$

Invertibility entails equal dimensions of V_+ and V_-. In example (b), $B = \operatorname{diag}(\pi/(2\ell), 3\pi/(2\ell))$; its squared singular values are distinct. The commuting pairs are diagonal with the same entries in both summands, giving the stated $U(1) \times U(1)$.

Caveat 2: spectrum **Correct.**

The square of the doubled Dirac operator is the sum of the two squared factors, with multiplicities counted in the tensor product. In example (a), $D_F^2 = 2m^2 I$, so its eigenvalues are $\pm\sqrt{2}\, m$. Its commutator with the left copy of $\mathbb{C}^2$ still has norm $m|a - b|$, giving distance $1/m$.

Caveat 3: parent **Correct with amendment.**

For an ordinary compact spin example of dimension n, the underlying product Dirac operator has Weyl exponent $2n$, while the restricted algebra has the metric of one n-dimensional factor. The statement about the product manifold requires the charge-conjugation identification and algebra restriction specified in Part 2(iii). The discrepancy also means that the restricted triple is generally not the ordinary finite-rank twisted spin triple over the single factor.

Caveat 4: KO-6 **Correct with amendment.**

The Clifford point is an input to Corollary 16.4. The claim that obtaining the sign from a geometric parent is "open" should be scoped to the work checked here: the cited documents do not establish such a construction or a no-go theorem.

Caveat 5: dimension count **Correct.**

For $k = 4$, the dimension after adding the Clifford point is 32. Neither example produces the stated classified algebra or its representation. Moreover, the C^*-algebras generated in Lemma 16 are complex algebras; a quaternionic real summand requires additional real-algebra data.

Part 2. Interpretation

(i) Unbroken symmetry and the Standard Model **Error, for the unrestricted claim.** A metric spectral triple excludes nonscalar elements of its represented algebra that commute with D. That does not exclude every symmetry implemented on its Hilbert space.

For a counterexample within the spectral-propinquity domain, let $H = M_2(\mathbb{C})$ with Hilbert–Schmidt inner product, let $A = M_2(\mathbb{C})$ act by left multiplication, and let

$$D(x) = x - \tfrac{1}{2} \operatorname{tr}(x) I.$$

If $[D, L_a] = 0$, applying the commutator to I gives a scalar. Finite dimensionality then makes this a metric spectral triple. Yet every conjugation $x \mapsto uxu^*$, $u \in SU(2)$, commutes with D. With $Jx = x^*$, these are the transformations $L_u J L_u J^{-1}$. This example fails order one, which spectral propinquity itself does not require.

A narrower statement does hold for a **finite real first-order** triple. If $uJuJ^{-1}$ commutes with D, order one and $JD = \varepsilon' DJ$ imply that $a = [D, u]u^*$ commutes with u. Thus

$$u^n D u^{-n} = D - na.$$

Since D is bounded, $a = 0$. If the triple is metric, u is scalar and the induced transformation is trivial.

The Standard Model finite triple is outside the metric class. In the usual construction, the color action commutes with the finite Dirac operator; independently, the nonscalar subalgebra C_F is explicitly required to commute with it in Chamseddine–Connes–Marcolli, §2.6. The constant copy of such an element also obstructs the almost-commutative product. The original algebra is a real algebra, so applying a framework formulated for complex C^*-algebras also requires specifying a complexification.

(ii) Morita equivalence Error, as an argument that removes the scalar-algebra restriction. The Morita equivalence $C(K) \sim C(K, M_n(\mathbb{C}))$ is correct, but Morita equivalence does not make the two algebras isomorphic or identify their spectral propinquity limits. Theorem 15.2 remains a restriction on limits of the whole **scalar** algebra $C(K)$.

A vector bundle E can carry matrix-valued connections while $C(K)$ still acts by scalar multiplication on sections. Replacing the represented algebra by $\mathrm{End}(E)$ is an additional change of data. It may be a useful enlargement of T0, but it is not a consequence of the presence of E.

The product-collapse example in Farsi–Latrémolière (arXiv:2404.00240v3 LaTeX source) concerns an already noncommutative finite algebra B in $B \otimes C(K_\ell)$. Its finite triple must be metric, and the limiting Hilbert space has the additional factor $\ker D_K$, which must be nonzero. A one-dimensional kernel recovers the stated finite multiplicity. The example does not produce a noncommutative whole-algebra limit from scalar parent algebras.

(iii) The geometry of the doubled parent Correct with amendment.

For a compact even-dimensional spin manifold, an appropriate charge-conjugation identification of $\overline{H_0}$ with H_0 identifies the displayed $\mathcal{D}$ with the product Dirac operator on $X \times X$. The grading may acquire an overall sign under that identification. The real structure in Lemma 16 covers the exchange of factors and is not the usual pointwise spin real structure.

The spectral triple used in the construction represents only $C(X) \otimes 1$. It is therefore a product-manifold Dirac operator equipped with a restricted observable algebra. If "commutative Kaluza–Klein data" includes that restriction and the flip real structure, the description is accurate. If it means the ordinary spin triple of the total manifold with its full function algebra, or a finite-rank spinor twist over X, it is too broad. With the full product algebra, the generated compression is $A_F \otimes \overline{A_F}$, and the flip generally fails order zero when A_F is noncommutative.

(iv) What Lemma 16 advances Correct with amendment.

For every finite even representation (A, V, d, γ), the construction

$$V \longmapsto V \otimes \overline{V}, \qquad d \longmapsto d \otimes 1 + \gamma \otimes d^c$$

gives a real bimodule satisfying order zero and order one. It uses no pre-existing finite real structure. Consequently, those two order conditions follow formally from the chosen tensor-square representation; the construction does not recover an arbitrary prescribed finite H, D, J, or their multiplicities.

The examples supply more than that formal observation. In particular, the $\mathbb{CP}^3$ kernel example gives a simple tower-decoupling family under the note's generated-compression rule. Put

$$\mathcal{D}_\ell = \ell^{-1}\mathcal{D}$$

on the doubled parent. Its kernel is $V \otimes \overline{V}$, its complementary gap grows like $1/\ell$, and compression followed by algebra generation gives the constant finite data

$$A_F = M_4(\mathbb{C}), \qquad \dim H_F = 16, \qquad D_F = 0.$$

After adding the stated Clifford point, the dimension is 32 and the KO-6 signs hold. Here order one is automatic because the finite Dirac operator vanishes. The whole parent triple has scalar algebra in its spectral-propinquity limit, consistently with Lemma 15.

Thus Lemma 16 advances T0 for a specified truncation procedure, including a noncommutative finite algebra and an explicit bimodule. It does not establish the target under every meaning of ordinary Kaluza–Klein data: the parent algebra is restricted, the real structure flips factors, and algebra generation is an additional operation. For T0b, it gives examples with the requested signs and first-order relations, but leaves the desired algebra and representation, orientability, nonzero mass data, classification, and the beyond-first-order and twisted conditions unestablished. The Clifford point also fails the finite orientability condition described above. The assessment is right to reject an obstruction inferred from one metric; its description of a "concrete origin" should retain these qualifications.

(v) Prior sources **Gap.** We cannot identify a source that states either the full Theorem 15.2, including its component bound, or the exact graded-flip construction of Theorem 16.2. We make no originality claim.

Two constituent results are identifiable in the supplied source: Farsi–Latrémolière state the kernel collapse to $(\mathbb{C}, \ker D, 0)$ at lines 1231–1235 (arXiv:2404.00240v3 LaTeX source), and use the product Dirac formula at lines 1262–1267 (arXiv:2404.00240v3 LaTeX source). Those citations do not establish prior publication of either full statement.

Amendments and errors

- Correct the supplied source's inner Leibniz condition to use L_B on real and imaginary parts; Theorems 15.3 and 15.7 then have the stated estimates.
- Repair the choice of g in Theorem 15.2 as above. The theorem follows.
- Restore the cap in Remark 15.4.
- In Proposition 15.9, normalize the comparison pseudo-state when $\Delta > 0$, and distinguish the exact relations for D_B from the $O(\Delta)$ relations for D_F.
- Add the stated real-structure and product hypotheses to Proposition 16.1; assume a Weyl exponent exists in Theorem 16.2; distinguish a joint cutoff from a spectral projection in Corollary 16.3.
- Qualify the parent geometry and the scope of the claimed open KO-sign problem. Record the orientability limitation of the Clifford-point construction.
- Restrict the unbroken-symmetry claim to the conditions actually proved, and do not use Morita equivalence to erase the distinction between scalar and matrix-valued parent algebras.

Overall: 26 statements and interpretive items checked: 11 correct, 10 correct with amendment, 2 gaps, and 3 errors.

Part R4 (referee report, lemma 18, two sheets, framework note)

Referee report on lemma 18, the two-sheet section of lemma 16, and the background note

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_18_16b.md. Reproduced as delivered, with links relativized. The corrections are incorporated in lemma 18, lemma 16, the note and the paper. Claude checked with Python the equations (R1) and (R3), the fully massive kernel and its poles, and read Remark 2.9 of arXiv:0706.3688.

We checked the numbered statements in lemma 18, the two-sheet section of lemma 16, and the claims and table in current_framework.md. We read the relevant passages of the cited papers, identified below. We did not run a computation.

The principal conclusions are:

- The fixed-cutoff quadratic calculation in lemma 18 is correct in its stated top-only family. Its conclusion extends to a broad class of nonnegative smooth cutoffs, including compactly supported cutoffs flat near zero.
- The limiting top-only quadratic action does **not** by itself have unbroken $SU(2) \times U(1)$ acting on the entire light space. It contains an incomplete top-bottom doublet.

- The two-sheet construction reproduces the classified bimodule and real structure as abstract finite data. Example 16.9 does not derive its nontrivial grading as an even spectral truncation of the parent it specifies.
- Corollary 16.8 incorrectly says that its Dirac operator commutes with the whole centre in every case. It always commutes with the two *sheet projections*.

1. Lemma 18, Theorem 1

Correct with amendment. State suitable regularity and decay assumptions on A, for example smooth compactly supported fields, and interpret the trace on $\mathbb{R}^4$ as the stated relative trace or quadratic variation. For several generators, contract the coefficient matrix with the corresponding polarizations.

The Duhamel expansion has the two terms displayed in (5). For transverse a, shifting the loop momentum gives the orbital contribution $2/t$. At the two opposite vertices, the spin traces of the curvature and commutator terms give q^2 and $(m_i - m_j)^2$, respectively; their mixed spin traces vanish. The A^2 vertex contributes

$$-\frac{1}{4\pi^2 t}\sum_{ij}|T_{ij}|^2 e^{-tm_i^2}|a|^2.$$

Symmetrizing this term gives the subtraction in (1), including its factor. Differentiation gives (2)–(3). The divided-difference identities (6)–(8) also follow as written.

The moment assumptions (4) are sufficient for the displayed Euclidean coefficients: $\int s^{-2}\,d\nu$ controls the action density, while the positive moments provide the required differentiability and domination. They do not supply arbitrary timelike analytic continuation.

The trace convention agrees with CCM's bosonic spectral action. Their calculation explicitly counts both particle and antiparticle sectors and four spin components; see CCM, Lemma 3.14 and §3.7.6. The fermionic chirality projection must not be inserted into that bosonic trace.

2. Lemma 18, Lemma 2

Correct. With the stated convention, $A_\mu \mapsto A_\mu - \partial_\mu\alpha$ and $\pi \mapsto \pi + \alpha$, so $A_\mu + \partial_\mu\pi$ is invariant. Conjugating by $e^{-i\pi T}$ proves (12), including the scalar contribution that is missed by using only the linear orbit tangent at a nonstationary background.

The stated longitudinal replacement follows from retaining the longitudinal part of the shifted orbital vertex. For a commuting generator, its integral vanishes by the displayed integration-by-parts identity. Translation invariance and rotation symmetry exclude quadratic mixing of a transverse vector with a scalar.

3. Lemma 18, Corollary 3

Correct with amendment. Formula (13) requires Q to be positive definite on the chosen generator space. Otherwise it applies after quotienting by its null space.

The orbital part of (2) starts at second order in the squared-mass difference, whereas the commutator term starts with d_{ij}^2. Also $\int_0^1 (1-2u(1-u))\,du = 2/3$. This gives the stated C_0, C_1, and factor $3/2$.

For the full Standard Model finite trace, one generation contributes 2 to the particle trace of T_1^2; three generations and charge conjugation give $Q_{11} = 12$. The commutator trace gives (14). This agrees with CCM, equations (4.10) and (4.16), with $F(v) = f(v^2)$. It is a leading expansion, not an identity at fixed m/Λ.

4. Lemma 18, Proposition 4

Correct. In one color of the massive block, let

$$|+\rangle = (t_L + t_R)/\sqrt{2}, \qquad |-\rangle = (t_L - t_R)/\sqrt{2}.$$

Each has squared charged-generator matrix element $1/8$ to b_L, and the reverse ordered entries have the same value. Their total is $1/2$ per color. Thus the massive color triplet contributes $3/2$ to the particle trace, and the other complete doublets contribute $9/2$. The charge-conjugate block supplies the factor two used in (16).

Integration by parts gives $h'_x(0) = 2w(x)/3$ and $g'(0) = 2/3$. Consequently (18) follows. Expanding A and w gives the small-x coefficient $-23/128$, and $A(x) \sim 2/x$, $w(x) \sim 6/x^3$ give the large-x limit. The positive-Laplace extension (23)–(24) follows by dominated convergence under (4).

5. Lemma 18, Theorem 5

Correct. Equation (22) makes h'_x and g' strictly positive for $y \le 0$. Since $F_x(0) > 0$ and $F_x(y) \to -\infty$ as $y \to -\infty$, there is exactly one positive timelike root, and its derivative has the sign giving the stated Euclidean residue.

For $y \le 0$, $g'(y) \ge 2/3$, hence $F'_x(y) \ge 3$. The mean-value theorem gives $z \le A(x)/2$; the Taylor ratio has the same bound. Since $xue^{-xu} \le e^{-1}$, $A(x) \le 2/e$, proving (28). The limiting derivatives are 4 and 3 at small and large x, respectively, yielding (27). The result says nothing about complex poles or Lorentzian unitarity.

6. Lemma 18, Proposition 6

Correct with amendment. The statement about one *positive massive* neutral pole requires $x > 0$. At $x = 0$ there is only the massless neutral system.

For one Dirac species, diagonalizing the left-right mass block gives diagonal neutral charge $(q_L + q_R)/2$ and off-diagonal charge $(q_L - q_R)/2$. In (3), the latter connects eigenvalues $+m$ and $-m$, giving exactly (33). Adding three top colors gives (30). The commuting photon and color traces give (29), including charge conjugates.

The angle in (31) is a sensible **neutral-current definition**: the kinetic-orthogonal massive direction is $K^{-1}(1,-1)^T$, and its current can be written $T_3 - \sin^2\theta_{\mathrm{NC}} Q$. Here

$$K_{33} + K_{3Y} = 5 + r, \qquad K_{YY} + K_{3Y} = \frac{5}{3}(5 + r),$$

so the value $5/8$ is constant. This is a charge-trace identity for this family; it does not make all finite-cutoff definitions of the weak angle coincide.

The neutral mass form is rank one in $v = (1,-1)^T$. Its generalized eigenvalue gives (34), using $v^T B_\infty^{-1} v = 8/27$. For the pole, the left side of (36) has derivative at least $1/2$, so the positive massive root is unique and has positive residue. The substitutions stated for (29)–(33) are valid for (4).

7. Lemma 18, Counterexample 7

Correct. The only nonzero entries of T connect m to $-m$, so (37) follows from (1). Its q^2 coefficient is proportional to $e^{-x}(1-x)$, while its Taylor ratio is $6m^2/(1-x)$.

The pole equation is (36) with $\beta = 4x$. Since $H(-z) > 1$ for $z > 0$, its solution lies strictly between $4x$ and $4x + 2$, and the derivative is positive.

For (39), positive Laplace mixtures give $f(x) + xf'(x)$. The assumptions imply $xf(x) \to 0$, while $xf(x) > 0$ for $x > 0$; therefore its derivative is negative somewhere. A negative Taylor kinetic coefficient does not itself establish a negative-residue pole of the full form factor.

8. Lemma 18, Counterexample 8

Correct. For fixed y, (21) gives $h_x(y) \to 0$, leaving (40), which is positive at fixed positive y. The neutral matrix likewise leaves nonzero kinetic terms. The broken vector directions therefore do not disappear by their masses diverging or their fixed-momentum propagators vanishing.

The narrower decoupling statement is also correct: after restricting the connection to the centralizer, the mass blocks are invariant and the heavy block carries its heat suppression. "Decoupling" here is a statement about the classical bosonic quadratic action, not a quantum effective-action theorem. The symmetry of the unrestricted limiting light space is addressed below.

9. Lemma 18, No-go theorem 9

Correct. Differentiation of the heat-trace density gives (41), which is strictly negative for $c > 0$, nonzero V_*, and the positive measure in (4). The theorem is fairly limited by the phrase "positive Laplace class."

The conclusion can nevertheless be sharpened on flat $\mathbb{R}^4$. For a sufficiently integrable real cutoff f, let λ_j^2 be the eigenvalues of V_*^2. Direct integration of the momentum variable gives

$$\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2} \sum_j \lambda_j^2 \int_{c^2\lambda_j^2/\Lambda^2}^{\infty} f(z)\, dz. \tag{R1}$$

For **nonnegative** f, a finite $c > 0$ is stationary exactly when every tail in (R1) associated with a nonzero λ_j vanishes. Thus:

- if f has nonzero support arbitrarily far out, there is no finite stationary point on this ray;
- if f has compact support ending at R, there is a stationary plateau once every nonzero $c^2\lambda_j^2/\Lambda^2 \geq R$.

This argument uses the continuum momentum integral; it is not the proof given for a finite torus. Sign-changing cutoffs can have cancellations in (R1), and a radial path with an additional fixed, noncommuting mass matrix need not have the form cV_*.

A nonconstant cutoff flat near zero cannot satisfy (4): a nonzero positive measure on $(0, \infty)$ gives $f'(0) = -\int s\, d\nu(s) < 0$. CCM use positive test functions and their asymptotic moments; profiles flat near zero are allowed and are outside the theorem's stated class. For nonnegative profiles, however, (R1) supplies the separate conclusion above. See CCM, Introduction and §6.2.

10. Lemma 18, Corollary 10

Correct. Equations (19), (27), and (34)–(36) give every entry of (42). At fixed positive μ, all dimensionless pole and Taylor masses are fixed and positive, so their dimensionful values grow with Λ. In the last joint limit, the small-μ expansion gives $M_W \sim \sqrt{3/8}\, m_t$; the neutral masses also diverge, with $M_Z^2 \sim (3/5)m_t^2$.

These are scaling statements for prescribed backgrounds. They do not establish stationary vacua.

11. Lemma 16, reducibility of Corollary 16.4's pair

Correct. The projection

$$P = 1 \otimes \frac{1 + \sigma_1}{2}$$

commutes with the represented algebra and with $J_F \otimes \sigma_1 K$, and is neither zero nor one. It proves reducibility of the **pair** (A, J). The projection need not commute with the grading; irreducibility of the pair is the issue asserted in the text.

12. Lemma 16, Theorem 16.7

Correct. The graded flip is an antiunitary involution. The two direct-summand Dirac operators square to the corresponding sums of squares, and the graded-flip signs give $J\mathcal{D} = \mathcal{D}J$. The minus sign on the second grading gives $J\Gamma = -\Gamma J$.

For even a, b, the opposite action is on the other tensor factor, as displayed. Both order conditions follow from the tensor-factor separation. The product operators have compact resolvent under the spectral-triple hypotheses.

When the two finite spaces are ungraded and have equal dimension, the identification $\xi \otimes \bar{\eta} \leftrightarrow (v \mapsto \xi\langle \eta, v\rangle)$ gives left composition and $J(x, y) = (y^*, x^*)$, exactly as in Chamseddine–Connes, Proposition 2.8 and Remark 2.9.

13. Lemma 16, Corollary 16.8

Error. The KO signs, dimension, order conditions, real orientability formula, and the full-matrix special case are correct. The assertion that D_F commutes with the **whole centre** in every case is false. It always commutes with the two projections distinguishing the sheets.

For a counterexample, take

$$V_X = \mathbb{C}^2, \quad \gamma_X = \sigma_3, \quad d_X = \sigma_1, \quad A_X = \mathbb{C} \oplus \mathbb{C}$$

acting diagonally, and take $V_Y = \mathbb{C}$, $d_Y = 0$. The element $(\mathrm{diag}(1, 0), 0)$ is central in $A_X \oplus A_Y$, but its commutator with D_F is nonzero.

The orientability formula is the usual **real** zero-dimensional relation $\Gamma = \sum_i \pi(a_i) J \pi(b_i) J^{-1}$, with opposite-algebra coefficients. It is not the stronger assertion $\Gamma \in \pi(A)$, which can fail. In the full-matrix, equal-dimension special case, an invertible matrix in each summand is a separating vector; the pair is irreducible because a projection commuting with both left and right full matrix actions is scalar on each summand, and J equates the two scalars.

14. Lemma 16, Example 16.9

Error. The two algebra-generation calculations and the abstract classified bimodule are correct. The claimed application of Corollary 16.8 with the subsequently chosen nontrivial grading is not justified by the specified even parent, and the compressed potential does not repair it.

For the generation calculations:

- The matrices $1, \Gamma_a, \Gamma_a\Gamma_b$ for $a < b$ are 16 real-linearly independent elements commuting with j. Since $\dim_{\mathbb{R}} \mathrm{End}_{\mathbb{H}}(W) = 16$, the real algebra generated by the five Γ_a is $M_2(\mathbb{H})$.
- For Hermitian B, the fourth moment gives $Q(w^*Bw/w^*w) = (B + \mathrm{tr}(B)1)/5$. This ranges over all Hermitian matrices. Their products generate $M_4(\mathbb{C})$ as a real algebra.

The use of two copies of the four-space is mathematically legitimate. It supplies independent direct summands and the 32-dimensional bimodule. For this particular grading of W and trivial grading of V, the graded flip is the ordinary adjoint flip, and

$$\Gamma_F(\xi, \eta) = (\Gamma_5\xi, -\eta\Gamma_5),$$

exactly the graded representation in Chamseddine–Connes, Proposition 3.2. The full algebra is a **graded prealgebra**; its even part is

$$\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C}).$$

The obstruction is that an even parent has $[P\gamma_X P, PaP] = 0$ for every compressed parent function. But the proposed grading $\Gamma_5 = 5Q(x_5)$ anticommutes with $Q(x_1) = \Gamma_1/5$. Thus the full compressed $M_2(\mathbb{H})$ and that grading cannot both arise from the stated even parent. They can be chosen **after** truncation as abstract graded-algebra data, and the even subalgebra then gives the real zero-cycle orientability relation.

Similarly, $d = \mu\Gamma_1$ is the compression of a bounded real multiplication operator, but adding that operator to the original parent Dirac operator does not retain the original kernel as a spectral projection: multiplication by the nonconstant real coordinate does not preserve the holomorphic kernel. It also does not preserve the original parent grading as an odd perturbation.

Finally, d commutes with the centre of the full prealgebra, namely $\mathbb{R} \oplus \mathbb{C}$. It does **not** commute with the whole centre after passing to the even algebra: for $e_+ = (1 + \Gamma_5)/2$, $[\Gamma_1, e_+] \neq 0$. It still commutes with the two sheet projections.

15. Framework note: real structure and KO-dimension

Correct with amendment. The KO-degree and antiparticle grading statements are right. The Euclidean Pfaffian construction uses the fermionic action on the appropriate chiral subspace. Barrett's Lorentzian treatment removes the quadrupling through compatible reality and chirality conditions; it should not be presented as the same Euclidean Pfaffian argument. See Barrett, §§1–3 and CCM, §4.4.

16. Framework note: classification of (A, J)

Correct with amendment. The classification is under order zero, a separating vector, irreducibility, and $J^2 = 1$. "Complexified algebra" here means the represented complex span used in the paper. The exclusion of the one-centre case also uses compatibility of the grading with the algebra—normalization in the graded-prealgebra formulation, or commutation in an even triple. Without compatibility, an arbitrary anticommuting grading is not excluded. See Chamseddine–Connes, §§2–3.

17. Framework note: real form and grading

Correct with amendment. The quaternionic grading and smallest choice $\dim_{\mathbb{H}} W = 2$ are hypotheses and choices in the source. The displayed full algebra is graded; $\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C})$ is its even part. On $\mathrm{Hom}(W, V)$, the grading acts by **right** composition, with the minus sign.

18. Framework note: order one

Correct. Under the hypotheses of Chamseddine–Connes, Theorem 4.1, the stated maximal even subalgebra is selected by the existence of a centre-connecting Dirac operator satisfying order one. The subsequent gauge group uses the represented unimodular lift; the statement is an algebraic selection, not a dynamical derivation of a vacuum.

19. Framework note: unbroken symmetry

Correct with amendment. CCM explicitly impose commutation with the nonscalar $\mathbb{C}_F = \{(\lambda, \lambda, 0)\}$ and identify its physical purpose with the massless photon; see CCM, equation (2.30) and Remark 2.19. The subalgebra is not unital in the full algebra, and the physical electromagnetic lift also uses the unimodularity condition. The nonscalar element in the kernel of $a \mapsto [D_F, a]$ explains the failure of the metric condition cited in the note; a treatment formulated only for complex algebras must specify its complexification.

20. Framework note: beyond first order

Correct with amendment. The quadratic fluctuation, gauge covariance, transitivity, and semigroup statements are supported by Chamseddine–Connes–van Suijlekom, §§I and III. The Pati–Salam construction is in the separate arXiv:1304.8050, §§III and VIII. Its discussion of a vacuum selecting the Standard Model subalgebra is model- and vacuum-dependent: the paper explicitly says that determining all vacua is difficult

and checks a restricted Hessian. It should not be stated as a general theorem that every vacuum restores order one.

21. Framework note: quanta of geometry

Correct. In the conventions of Chamseddine–Connes–Mukhanov, equations (2)–(5), the two Clifford algebras in dimension four are $M_2(\mathbb{H})$ and $M_4(\mathbb{C})$, and the two-sided relation gives the sum of the pulled-back volume densities. This is a quantization condition with two maps, not a truncation result.

22. Framework note: twisted triples and grand symmetry

Correct. The larger algebra mixing spin and internal degrees of freedom and the extra scalar are described in Devastato–Lizzi–Martinetti, §§3–5. A twisted commutator replaces the ordinary bounded-commutator requirement in the later Devastato–Martinetti construction, §3. The note properly says that these constructions have not been studied here.

23. Table: noncommutative finite algebra

Correct. The interval and projective-space examples give the stated generated algebras, with algebra generation an additional operation. The distinction between PAP as an operator system and its generated C^*-algebra is also emphasized in Connes–van Suijlekom, §1.

24. Table: inherited pointwise reality

Correct. Under Proposition 16.1's pointwise-reality and invariant-window hypotheses, order zero for the generated algebra holds exactly when that algebra is commutative. This is scoped to that inherited correspondence.

25. Table: order zero, order one, centre $\mathbb{C}$

Error. Corollary 16.3 gives the order conditions and dimension k^2, but it does not generally give centre $\mathbb{C}$. For its interval example $A_F = \mathbb{C} \oplus \mathbb{C}$, the centre is $\mathbb{C} \oplus \mathbb{C}$. The stated centre requires an additional condition, such as a simple generated algebra.

26. Table: KO-6 signs

Correct. Both constructions give the signs. The first pair is reducible and fails the displayed real orientability relation; the two-sheet input is explicit.

27. Table: centre $\mathbb{C} \oplus \mathbb{C}$, dimension $2k^2$

Correct with amendment. Corollary 16.8 reaches this classified pair when the two spaces have equal dimension and the represented complex spans are full matrix algebras. Arbitrary generated algebras in that corollary can have larger centres and need not give an irreducible pair.

28. Table: orientability

Correct. The displayed zero-cycle works under the stated condition that both finite gradings lie in the even represented algebras. It is real zero-dimensional orientability, in the sense specified in section 13 of this report.

29. Table: $M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$

Correct with amendment. The real generated algebras and the 32-dimensional abstract bimodule are reached. The nontrivial grading is added after the stated truncation, and the full algebra with that grading is a graded prealgebra. It is not an even spectral-triple algebra, nor is the grading derived from the specified

even parent. The construction requires a **real** generated algebra on the S^4 side; complex C^*-generation would also include $i1$.

30. Table: selection of A_F

Correct with amendment. An off-diagonal term is missing, but an arbitrary off-diagonal term alone is not enough. One must impose the required order-one condition and the maximal-subalgebra hypothesis of the classification.

31. Table: Yukawa block

Correct with amendment. Theorem 12.7 gives the texture in a towerless limit, up to grading-preserving unitary identifications. An algebra and its representation must also be specified before calling the result a finite spectral triple. It is not embedded in Example 16.9.

32. Table: $[D_F, \mathbb{C}_F] = 0$

Correct with amendment. Proposition 17.3 explains the consequence **when** a chosen subalgebra commutes with D_F; it does not make an arbitrary interval or Yukawa block commute with a Standard Model representation. The appropriate common action on the paired blocks is additional data.

33. Table: bosonic action

Correct with amendment. The leading-order results are correctly described, but "near-cutoff in progress" is stale: lemma 18 now gives the exact quadratic heat-cutoff calculation and the stated positive-Laplace extension. Its scope and limitations are those checked above.

34. Table: beyond first order

Correct. The inherited two-point data supply an order-zero, first-order-failing example to which the generalized-fluctuation question can be posed. The table correctly records that the calculation has not been done and that relaxing order one does not cure order-zero failure.

35. Table: quanta of geometry

Correct with amendment. The same five Clifford matrices occur, and the projector identity is valid. For the quaternionic rank-one projector $\Pi = P_w + P_{jw}$, expansion in the real basis $1, \Gamma_a$ gives $\Pi = (1 + \sum_a x_a \Gamma_a)/2$. This algebraic connection does not derive the two-sided quantization condition from truncation.

36. Table: twisted triples and grand symmetry

Correct. The table records no result and marks the subject untested.

Part 2: scope of the heavy-limit conclusion

Within the three-generation, zero-Majorana, top-only family of Proposition 4, at fixed cutoff and for the heat function, **both** the charged Taylor mass and its unique positive timelike pole tend to zero. The neutral Taylor and pole masses also tend to zero. The result refutes an infinitely heavy W, Z, t limit **for these background data and these quadratic mass definitions**. It also rules out a stationary nonzero radial vacuum in the positive-Laplace class. It does not rule out other finite data, a different action or constraint, a joint $\Lambda \to \infty$ limit, or a quantum effective theory.

The top-only Taylor conclusion is broader than (4). For a nonnegative smooth compactly supported f with $f(0) > 0$, define

$$M_1 = \int_0^\infty z f(z)\, dz, \qquad M_2 = \int_0^\infty z^2 f(z)\, dz.$$

The heat formulas extend by linearity and smooth functional calculus to

$$A_f(x) = \frac{2}{x}\int_0^x zf(z)\,dz, \qquad w_f(x) = \frac{3}{x^3}\int_0^x z^2 f(z)\,dz. \tag{R2}$$

One way to justify the extension is to approximate a smooth compactly supported $f(z)$, with the derivatives needed for the quadratic variation, by finite real linear combinations of e^{-nz}: write $f(z) = e^{-z}g(e^{-z})$ and approximate the smooth g and its derivatives by polynomials. The divided-difference formula for the variation permits passage to the limit. Thus

$$\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} \sim \frac{M_1}{f(0)x}.$$

This includes a nonnegative cutoff flat near zero. For a nonanalytic cutoff there need not be a prescribed analytic continuation defining the pole used in Theorem 5.

Several heavy doublets change the answer. Suppose, for the heat cutoff, that P is the total particle-trace weight of complete massless doublets, and heavy doublet i, of weight $d_i > 0$, has one Dirac member of mass m_i and one massless member. Then

$$C_W(y) = \frac{\Lambda^2}{4\pi^2}\left(Pg(y) + \sum_i d_i h_{x_i}(y)\right), \qquad \frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{3}{2}\,\frac{\sum_i d_i A(x_i)}{P + \sum_i d_i w(x_i)}. \tag{R3}$$

For any fixed finite number of doublets, if $P > 0$ and all $x_i \to \infty$, the Taylor mass tends to zero. If $P = 0$ and all $x_i = x$, (R3) instead gives

$$M^2_{W,\mathrm{Taylor}} \sim \frac{m^4}{2\Lambda^2}.$$

The full heat form factor also has a unique positive pole in this equal-mass case. To find its scale, put $z = cx$, $0 < c < 1$, in (21). Concentration near $u = 0$ gives

$$h_x(-cx) \sim \frac{2(1-2c)}{x(1-c)^2}.$$

Monotonicity then places the root at $z/x \to 1/2$, so $M^2_{W,\mathrm{pole}} \sim m^2/2$. At every fixed Euclidean momentum, however, the whole unnormalized quadratic kernel tends to zero. The divergent Taylor and pole scales differ, and neither alone establishes a healthy limiting gauge theory.

There is also a fully massive example. Give **both** members of every Standard Model doublet the same Dirac mass m, with no Majorana masses. The charged form factor is proportional to

$$e^{-x}\big[g(y) + 2xH(y)\big].$$

Its positive pole obeys $2x < z_W < 2x+2$. The neutral massive pole obeys $(16/5)x < z_Z < (16/5)x+2$. Both pole masses therefore diverge with m. Yet the fixed-momentum action tends to zero, and the charged Taylor kinetic coefficient is proportional to $e^{-x}(1-x/2)$, which is negative for $x > 2$. This is a counterexample to a general fixed-cutoff no-go for diverging pole locations, while leaving the physical quality of that limit open.

The phrase "the heavy fermion leaves the loop" is only a shorthand for the suppression in this **classical spectral trace**. The light complete doublets supply the surviving charged kinetic form, but the top-only light space also retains b_L after removing t_L. If P_{light} is the projection off the top mass block, then on b_L

$$P_{\text{light}} T_1 P_{\text{light}} = P_{\text{light}} T_2 P_{\text{light}} = 0, \qquad P_{\text{light}} T_3 P_{\text{light}} = -\tfrac{1}{2}.$$

These compressed operators do not obey the $SU(2)$ commutation relations. Correspondingly, the limiting particle-trace coefficients have $K_W = 9/2$, $K_{33} = 21/4$, and $K_{3Y} = -1/4$; a linearly realized unbroken electroweak gauge action would require equal weak coefficients and no $3Y$ mixing. Vanishing vector masses therefore do **not** prove unbroken $SU(2) \times U(1)$ on the entire light space.

The remaining **complete** massless doublets do carry the ordinary electroweak representation, and their separate contribution is gauge invariant. If finite data instead make both members of every removed doublet heavy, the retained light subspace can be invariant under the full group; a full limiting theory still requires control of the remaining fields, normalization, interactions, and vacuum.

Part 4: what the two-sheet construction establishes

Example 16.9 is substantive as a concrete compression calculation: real base functions generate a quaternionic matrix algebra, real total-space functions generate a complex matrix algebra, and two copies give the classified 32-dimensional bimodule. The calculation does not select those ingredients dynamically. The restriction to base functions on one sheet, the second independent sheet, real algebra generation, the four-dimensional window, and the grading axis are inputs.

At the finite algebraic level, the two-sheet bimodule is the standard construction from two representations:

$$(W \otimes \bar{V}) \oplus (V \otimes \bar{W}).$$

The graded flip supplies the real structure and the block sign supplies the KO-6 relation. The calculation's additional content is its realization from the particular compressed function systems and its explicit matching to the classified data. The abstract bimodule and its adjoint-flip real structure are already stated in Chamseddine–Connes, Remark 2.9; the report makes no broader novelty claim.

The note appropriately lists several choices and says that no ordinary Kaluza–Klein vacuum is derived. It overstates the reach where it treats the chosen nontrivial grading and nonzero compressed d as consequences of the stated even-parent truncation. It also labels the centre of Corollary 16.3 too generally. Lemma 16's later caveat that its generated algebras are complex and a quaternionic summand still needs additional real data is stale after Example 16.9's real-algebra construction.

For canonical section 4, the example closes the **abstract finite-module** problem after adding a second sheet: it turns the nested actions on one four-space into independent summands acting on two Hom spaces. It does not turn four global sections into rank-four fibres over S^4, derive the two sheets from the fibration, or produce the centre-connecting Dirac operator required for the subsequent algebra selection. The fibre-rank problem remains open.

Amendments and errors

- Add field regularity to Theorem 1, $Q > 0$ or a quotient by its null space to Corollary 3, and $x > 0$ to the massive neutral-pole assertion.
- Keep the neutral-current angle distinct from other finite-cutoff angle definitions.
- Replace Corollary 16.8's claim about the whole centre with commutation with the two sheet projections.
- In Example 16.9, distinguish the correct abstract graded prealgebra from an even spectral truncation; the proposed grading and nonzero compressed perturbation are not derived by the stated parent. The nonzero d also fails to commute with the whole centre of the even algebra.
- Correct the table's unrestricted centre-$\mathbb{C}$ claim for Corollary 16.3, and add the full-matrix and equal-dimension hypotheses to its classified two-sheet row.
- Update the table's near-cutoff entry and qualify the claims about the Yukawa representation, off-diagonal Dirac operator, real-algebra generation, and the vacuum discussion beyond first order.
- Distinguish the Euclidean Pfaffian and Lorentzian fermion reductions, and retain the grading-compatibility hypothesis in the classification.

- On flat $\mathbb{R}^4$, equation (R1) extends the radial analysis to nonnegative non-Laplace cutoffs; compact support permits a stationary plateau.
- The top-only limiting quadratic action does not establish an unbroken electroweak representation on the full light space.

Overall: 36 statements checked: 18 correct, 15 correct with amendment, 0 gaps, and 3 errors.

Part R5 (referee report, main draft and note)

Referee report on the main draft and the note

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_paper_note.md. Reproduced as delivered, with links relativized. The corrections are incorporated in the paper, the note and lemmas 17 and 18.

The main draft contains useful exact calculations, but it needs substantial revision before it is ready for a Physical Review journal. The most consequential issues are the identification of the formal KO labels with geometric endpoints, the claim that collapse with a fixed light mass requires a tuned bulk mass, and several statements whose scope changes between the theorems, Table I, the abstract, and the discussion.

The note preserves the corrected reduced-mass formula and the sign of j_-. Its main mathematical defect is that the stated condition on the condensate subtraction does not suffice for the claimed $O(m)$ remainder.

The numbered verdicts below count distinct statements, including separate occurrences in Table I and the abstract. We read the LaTeX sources as text and did not compile or edit any file. The underlying lemmas were consulted for the relevant statements and hypotheses; their proofs were not re-refereed in full.

Part 1. Main draft

Chirality at the odd labels The paragraph is at towerless_intervals.tex:184.

1. Classical objection — Correct with amendment. An irreducible spinor representation in odd spacetime dimension has no intrinsic Weyl decomposition: the Clifford volume element is central, so there is no nontrivial chirality operator that anticommutes with every Clifford generator. In the usual smooth, closed, product compactification of an odd-dimensional bulk, the standard fermion reduction therefore has a vectorlike four-dimensional content.

State the compactification and fermion assumptions. Odd-dimensional bulk theories can have chiral modes localized on defects or boundaries, and "no chiral fermions" should not be read as excluding all localized modes. The full text of the cited Witten 1981 article was inaccessible in this review; its precise attribution remains unchecked. The mathematical amendment above follows directly from the Clifford representation.

2. Full chiral algebra — Correct with amendment. For the *specified Standard Model representation* of $\mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$, neither proposed odd factorization exists. Let $e_{\mathbb{H}} = (0, 1_{\mathbb{H}}, 0)$. Its represented range lies in one grading sector and has no counterpart in the other. A grading-reversing U commuting with the algebra would have to identify those two ranges. This is the argument in lemma 06, Result 8.

The conclusion depends on the representation, not on the abstract algebra alone. Name the Standard Model bimodule in the sentence.

3. Label 9 and the absence of needed chiral fermions — Correct with amendment. The charged surviving sectors are vectorlike under the reduced $SU(3)\times U(1)_{\mathrm{em}}$ representation and each admits the $s=-1$ factorization. This establishes a sectorwise formal label 9.

It does not give that factorization for the strict *whole* light spectrum with one or three unpaired neutrinos. A neutral Weyl fermion has a real gauge representation but remains a chiral spacetime fermion; it may be expressed in Majorana form. State explicitly whether the neutral sector is removed, paired, or treated separately.

4. Neutral representation and parity — Correct. The neutral gauge representation is real in the reduced model. If the descended antiunitary j obeys $j^2 = -1$, its complex representation space has even dimension. Indeed, writing $j = UK$ gives

$$|\det U|^2 = \det(-1) = (-1)^{\dim_{\mathbb{C}} H_0}.$$

For a Majorana block the stronger condition is even multiplicity of **each** singular value, including zero, because j commutes with d^2. Odd total multiplicity is one obstruction.

5. Spectrum at fixed distance from a wall — Error. At fixed $\mu > 0$, the normalized left-wall profile tends to

$$\sqrt{2\mu}\, e^{-\mu y}.$$

It is nonzero at every fixed finite distance y from the wall. Thus a region at fixed distance from a wall still sees the chiral wall mode, even though its amplitude is exponentially small when μy is large.

The gapped bulk statement applies when the observation region recedes from **both** walls as $l \to \infty$. In that limit the translation-invariant bulk Dirac operator has spectral threshold μ, while the two half-line limits retain the chiral zero modes.

6. Assignment of the wall limit to label 11 — Gap. The wall-mode limit and the KO-7 factorization are established separately. The text does not construct an identification that assigns KO label 11 to the decompactifying interval. The interval product in Theorem D also has a stated KO-5 factorization. Present the label-to-endpoint connection as an interpretation, or give the missing compatibility and limiting construction.

7. Hořava–Witten comparison — Correct with amendment. Boundary chirality is a fair qualitative comparison. With a common orientation and common identification of the ten-dimensional tangent spaces, the two Hořava–Witten boundary gaugino sectors have the same ten-dimensional chirality. In the interval model here, the two limiting modes of one type-(c) fermion have opposite Γ-chirality. Boundary-induced orientations reverse between the two ends, so the chirality convention must be stated.

The constructions also have different content: Hořava–Witten has an E_8 vector multiplet on each boundary and anomaly cancellation involving the bulk, whereas the interval calculation follows two modes of a bulk Dirac operator. The comparison should identify that common feature and those limits. See Hořava and Witten, hep-th/9603142, §§1–2 and their preceding orbifold paper, §2.

8. Physical point at finite l in an even interior — Gap. The results do not relate l quantitatively to the measured Fermi constant or weak angle, nor do they identify the physical Standard Model point with a member of this interval family. State this as a proposed interpretation. The finite KO-6 structure does accommodate chirality; its identification with a measured finite-l point remains to be shown.

Labels by sector **9. Charged paired sectors — Correct.** In the reduced representation, each charged species and its conjugate form an invariant sector for the represented algebra, its opposite, D, and J. The displayed paired-sector construction admits both signs, including with flavour mixing within a species.

10. One or three unpaired neutrinos — Correct with amendment. With the stated zero Majorana block and reduced algebra, the strict light triple admits $s = +1$ and excludes $s = -1$. Add that the neutral $s = +1$ involution does **not** preserve particle number. If particle preservation is imposed on the strict unpaired spectrum, neither proposed reduction is available; see lemma 06.

11. Right-handed neutrinos with pure Dirac masses — Correct. Retaining the Dirac partners gives paired singular values in the neutral block, so both signs occur under the stated reduced representation.

12. Majorana masses — Correct with amendment. A Majorana term breaks the particle-number hypothesis used to pair U and QU. Generic nondegenerate neutral Majorana data obstruct $s = -1$, while $s = +1$ remains available under the scalar-character assumptions. State the exact criterion: $s = -1$ exists precisely when every neutral-block singular value has even multiplicity. A nonzero Majorana term alone does not exclude it in several generations.

Weighted sum rule and cutoff limit The claims are at towerless_intervals.tex:202 and towerless_intervals.tex:228; their source is lemma 18, Remark 16.

13. Weighted identity — Correct with amendment. The identity

$$8\kappa M_{W,\mathrm{Taylor}}^2 = \sum_f N_f m_f^2\, \rho(m_f^2/\Lambda^2), \qquad \rho(x) = \frac{2[1-(1+x)e^{-x}]}{x^2},$$

follows from the displayed kernel for the heat cutoff $f(z) = e^{-z}$. Its scope is the specified finite collection of doublets, each massive doublet having one massive Dirac member and one massless member, with the trace weights and mass-compatible doublet decomposition used in Proposition 13 of lemma 18. The mass is the **Taylor ratio** $C_W(0)/C_W'(0)$, not the exact pole mass.

Also define the trace on $\mathbb{R}^4$ through a relative heat trace or action density: the unsubtracted vacuum heat trace has infinite volume.

14. Definition of κ — Correct with amendment. In the three-generation normalization, write

$$\kappa = \frac{P + \sum_i d_i w(x_i)}{6}, \qquad N_i = 2d_i, \qquad w(x) = 3\int_0^1 u^2 e^{-xu}\, du,$$

where P is the particle-trace weight of complete massless doublets and $P + \sum_i d_i = 6$. Thus κ is the charged Taylor kinetic coefficient divided by its all-massless value. For the top alone, $P = 9/2$, $d_t = 3/2$, giving $(3+w)/4$.

"Normalization of the kinetic term" without these weights does not define κ for the displayed general sum.

15. Small-mass form — Correct with amendment. If **all** $x_i = m_i^2/\Lambda^2 \to 0$, then $\rho(x_i) \to 1$ and $\kappa \to 1$, giving the CCM relation to leading order. This proves the stated relation for the heat-cutoff Taylor calculation. The CCM asymptotic formula also has its own cutoff and trace conventions; the displayed ρ is not a general-cutoff form factor.

16. Five-percent departure — Correct. For the top alone,

$$\frac{M_W^2/m^2}{3/8} = \frac{\rho(x)}{\kappa(x)}.$$

The expansions

$$\rho(x) = 1 - \frac{2}{3}x + \frac{1}{4}x^2 + O(x^3), \qquad w(x) = 1 - \frac{3}{4}x + \frac{3}{10}x^2 + O(x^3)$$

give a departure of about 5% in m/M_W at $x = 0.45^2$.

17. Peak — Correct. The stated peak is supported by the formula. For an analytic uniqueness check, set

$$B = 1-(1+x)e^{-x}, \qquad D = x^3 + 2 - (x^2+2x+2)e^{-x}.$$

Apart from a positive factor, the derivative has the sign of

$$T(x) = 4 - x^3 + (x^5 + x^4 - 8x - 8)e^{-x} + (x^3 + 4x^2 + 8x + 4)e^{-2x}.$$

The power series of $e^{2x}T(x)$ has positive coefficients at degrees 5, 6 and 7, and negative coefficients at every degree at least 8. Dividing by x^7 therefore gives a strictly decreasing function on $x > 0$, with one zero. The Taylor mass has one positive maximum and falls thereafter.

One small **Python 3** computation solved this derivative equation and evaluated the stated mass formula. It gave $x = 2.02428$, $m/\Lambda = 1.42277$, and $M_W/\Lambda = 0.52422$. The computation tests the location; the sign argument gives uniqueness.

18. Heavy suppression — Correct with amendment. A massive diagonal contribution to the unperturbed heat trace is exponentially suppressed. In a mixed massive–massless contribution, the form factor instead has the algebraic behavior

$$\rho(x) \sim 2/x^2.$$

The corresponding mass-source contribution behaves as $m^2\rho(m^2/\Lambda^2) \sim 2\Lambda^4/m^2$. State which trace contribution is exponentially suppressed so that the sentence does not suggest exponential suppression of the mixed loop.

19. Necessity of Λ growing faster than m_t — Correct with amendment. It is an exact necessity for the **top-only heat-cutoff Taylor family** if the limit requires $m_t/M_W \to \sqrt{8/3}$. To see this, use

$$\rho(x) = 2\int_0^1 ue^{-xu}\,du, \qquad w(x) = 3\int_0^1 u^2e^{-xu}\,du.$$

The function $3 + w - 4\rho$ vanishes at zero and has derivative $\int_0^1 u^2(8-3u)e^{-xu}\,du > 0$. Hence the exact top-only ratio differs from its CCM value for every $x > 0$; it approaches that value only if $x \to 0$.

The necessity is broader than the calculation if stated without the top-only and cutoff hypotheses. For example, retain one complete massless lepton doublet and give one member of each other doublet the common mass m. Then $\kappa = (1+11w)/12$. Near zero, $\rho - \kappa = x/48 + O(x^2) > 0$, while at infinity it tends to $-1/12$. Continuity gives a positive x_* at which the full weighted identity equals the unweighted CCM sum exactly. Scaling $m = \sqrt{x_*}\Lambda$ preserves that equality with a nonzero m/Λ. This is a different mass family.

Generated algebra and the cited remark **20. Bicommutant — Correct with amendment.** For a unital complex representation on the finite-dimensional V,

$$C^*(Q(A_0)) = Q(A_0)''$$

by the finite-dimensional bicommutant theorem. State that the commutants are taken in $\mathrm{End}_{\mathbb{C}}(V)$, and retain the unital or nondegenerate-representation hypothesis. The quaternionic real algebra in item 4 needs a separate real-algebra statement.

21. Attribution to Connes and van Suijlekom — Correct. Their introduction says that in many interesting examples the C^*-algebra generated by PAP is a "non-informative full matrix algebra." The draft preserves both the scope and the substance of the observation. Their paper also emphasizes the information retained by the operator system itself. Connes and van Suijlekom, arXiv:2004.14115v2, Introduction.

22. Interval block structure — Correct with amendment. The grading commutes with every compressed function, so it forces the generated algebra to be block diagonal. The explicit interval compressions then show that it fills both blocks, giving $M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$.

The grading alone gives containment in that block algebra; it does not prove equality. The two matrices displayed in the main proof sketch lie in different blocks and commute, so those two examples alone do not generate both full blocks. Add the independent symmetric matrices within each block and the argument separating the two summands; lemma 16, example (b) supplies it.

23. Quaternionic form on S^4 — Correct with amendment. For **real functions**, the compressed operators commute with the antiunitary j, $j^2 = -1$, and generate the real algebra $M_2(\mathbb{H})$. The five compressed coordinate matrices provide the generators. The complex C^*-algebra generated by the same coordinates is $M_4(\mathbb{C})$.

Thus j identifies a quaternionic real form; it is an antiunitary symmetry and is not an element of the ordinary complex-linear commutant in item 1. State the real-versus-complex distinction at towerless_intervals.tex:296, not only earlier in item 4.

Reading the negative results The directions in towerless_intervals.tex:306 are reasonable research directions when kept within their stated scope.

24. Odd-dimensional chirality → boundaries and an even interior — Correct with amendment. Boundaries are one way to obtain localized chiral modes. The claimed even interior is an additional proposal; singularities, defects and other changes of hypotheses are also possible directions.

25. Bare interval limits → bulk mass — Correct with amendment. A negative bulk mass with the stated localization supplies a light pair. The boundary-Yukawa family supplies another route; see check 34.

26. Fixed-compression covariance failure → covariant projection — Correct. Projecting onto a gauge-transformed spectral subspace addresses the demonstrated covariance defect. Further matching of actions remains a separate question.

27. Bounded-fermion collapse → tower-scale fermion or changed action/order — Correct with amendment. This follows for the leading spectral action, specified trace, common profile setting, and tower matching considered in Proposition F. It is a useful direction beyond those assumptions, not a general theorem about every gauge action.

28. Inherited order-zero failure → doubled parent — Correct. The tensor-square parent supplies commuting left and opposite actions and thereby gives a constructive route. The two-sheet and grading inputs remain explicit.

29. Fixed-cutoff behavior → joint cutoff limit — Correct with amendment. The joint limit is a valid route in the top-only family. At fixed cutoff the lemma also contains fully massive families with diverging pole locations, so the direction must retain the mass-family and mass-definition hypotheses.

30. No finite radial vacuum → a potential or constraint — Correct. Adding a specified radial potential or imposing a constraint could alter the radial stationarity problem. Neither follows from the spectral action already analyzed.

31. Commutative whole limits → truncation — Correct with amendment. Compression followed by algebra generation is a constructive way to obtain noncommutative finite data. The commutative-limit statement concerns the specified spectral propinquity and whole scalar algebras; it does not rule out other notions of convergence or other parent algebras.

Table I The table begins at towerless_intervals.tex:46.

Check	Entry and verdict	Amendment
32	Two-point space — **Correct.**	The stated mass and minimal truncation are the content of Theorem A.
33	10 to 11 — **Correct with amendment.**	Say "in the type-(c) family, at fixed $\mu > 0$." The KO-label identification remains the gap in check 6.
34	10 to 9 — **Error.**	A tuned bulk mass is necessary for the stated one-field type-(c) family, not for every interval collapse. In Theorem C's one-flavour model take zero bulk masses and $X(l) = \tan(ml)$. The light eigenvalue is exactly $\arctan\|X\|/l = m$ for small l, and the higher levels diverge.
35	Yukawa block — **Correct with amendment.**	Replace "D_F is the towerless limit" by "the displayed Dirac Yukawa block of D_F is obtained in the boundary-Yukawa limit." The full finite algebra, Majorana block and real triple are not derived there.

Check	Entry and verdict	Amendment
36	9 and 11 — **Correct with amendment.**	Specify the reduced representation and the particle-number caveat in check 10. The full chiral algebra admits neither proposed factorization.
37	Heavy W — **Correct with amendment.**	The weighted formula is for the heat cutoff and Taylor mass with the stated doublets. The top/W limit needs top dominance in the CCM expansion.
38	W against the tower — **Correct with amendment.**	State the leading action, trace, gauge-boundary input and common-profile matching assumptions.
39	Chirality — **Gap.**	The table promotes the proposed assignment of the two odd labels to limits and of the physical point to an even interior into a result. Supply the identification or label it as an interpretation.
40	Finite algebra — **Correct with amendment.**	Distinguish the complex generated algebra from the real quaternionic form, and retain the two sheets, restricted algebra and added grading as inputs.

Abstract **41. Collapse and boundary Yukawa data — Error.** The abstract repeats the unrestricted "only along a tuned bulk mass" claim and makes the boundary-Yukawa result sound like the continuation of that same tuned family. The counterexample in check 34 applies. State the two different constructions and their scalings separately.

42. KO labels and weighted mass relation — Correct with amendment. State that the charged and neutral factorizations use the reduced representation, and identify the heat cutoff and Taylor mass for the weighted sum. The complete chiral Standard Model algebra has neither factorization.

43. Whole-triple limit and two-sheet parent — Correct with amendment. "Whole triples converge … only to points" needs restriction to the finite limits of the connected commutative metric triples considered in Proposition G. The two-sheet construction reproduces the specified algebraic data with the listed inputs; it does not derive a connected parent or the full finite Dirac operator.

Discussion, literature, and one additional main-text statement **44. Masses as momenta combined in quadrature — Error.** The discussion at towerless_intervals.tex:302 applies the real-momentum formula to the localized light mode. For that mode $k = i\kappa_0$, and

$$\lambda_0^2 = \mu^2 - \kappa_0^2.$$

The formula $\lambda_n^2 = \mu^2 + k_n^2$ with real k_n applies to the excited tower.

45. Localization alone determines the texture — Error. At towerless_intervals.tex:149, order-one **entries** of $\hat{Y}$ do not by themselves give the stated individual mass pattern. Let

$$F = G = \mathrm{diag}(\varepsilon, 1), \qquad \hat{Y} = \begin{pmatrix} 1 & 1 \\ 1 & 1+\eta \end{pmatrix}.$$

All entries are order one and the trailing minors are nonzero for $\eta > 0$, but $\det(F\hat{Y}G) = \varepsilon^2\eta$. The small singular value carries the additional arbitrarily small factor η. Require quantitative lower bounds on the relevant minors, as well as the stated profile hierarchy, or present the formula as a generic numerical expectation.

46. Index and open problems — Correct with amendment. The ordinary index is stable under the boundary coupling in the stated family. The equivariant statement is made for the stabilizer of X: symmetry breaking changes the group on which the representation-valued index is read. The discussion should distinguish those statements and retain its explicit qualifications on the parent, grading, dynamics and action.

47. Claims about what was not found — Gap. The statements about the chiral-interval compression, tuned collapse, localization-derived Yukawa block and factorization criterion may be accurate, but the draft does not say which databases, years, synonyms, or citation chains were searched. We cannot infer a comprehensive absence from the text.

The source check here was limited. Queries included `"noncommutative geometry" "interval" "spectral truncation" two point chiral 2026`, a localization/Yukawa query, and an odd/even real-triple factorization query. We opened the relevant sections of the sources cited in checks 7, 21, 48 and 49. Forward and backward citation chains and author recency searches were not completed, and the full texts of Witten 1981 and Banks–Casher 1980 were unavailable. Consequently this review does not establish priority or absence in the literature.

A relevant recent neighboring paper is Baptista, arXiv:2506.09126v2, §§1 and 5: it studies chiral couplings to massive gauge fields in smooth Kaluza–Klein models. Its examples use even-dimensional internal spaces; it does not refute the odd-dimensional Clifford statement. It is relevant to the broader discussion of routes to chiral interactions.

48. Positive attributions checked — Correct. Csáki et al., hep-ph/0310355v1, Eq. (5.6) give $\tan(m_n L) = M_D L$, yielding the one-flavour arctangent spectrum after identifying the boundary parameter. Chamseddine and Connes, arXiv:0706.3688v1, Remark 2.9, give the two Hom-space bimodule and flip real structure cited in the draft. Check 21 establishes the Connes–van Suijlekom attribution.

49. Farsi–Latrémolière theorem — Correct with amendment. The issue is visible in arXiv:2404.00240v3, Theorem 3.4 and Eq. (3.11). The proof's displayed estimate requires control of the vertical operator by the total rescaled operator. A scalar example shows the need for an additional condition: on e_n, take $D_v e_n = n e_n$, $D_h e_n = -n^2 e_n$, and $\epsilon_n = (n + 1/n)^{-1}$. Then $D_{\epsilon_n} e_n = e_n$ although the proposed limit on $\ker D_v$ has spectrum $\{0\}$.

Anticommutation or a suitable relative bound are **sufficient repairs**, rather than the only conceivable repairs. Identify the checked preprint version; the published text was not examined here.

50. Proposition G converse — Error. The statement at towerless_intervals.tex:263 uses the previously arbitrary unitary W_n in an "if and only if." Take the constant triple

$$T_n = T_F = (\mathbb{C}, \mathbb{C}^2, \operatorname{diag}(1, 2)),$$

with $P = 1$, and let every W_n swap the two basis vectors. The spectral propinquity is zero, while $\Delta_n = \|\operatorname{diag}(2, 1) - \operatorname{diag}(1, 2)\| = 1$.

The converse needs **there exist** appropriate light spectral projections and unitaries W_n for which $\Delta_n \to 0$, with g_n chosen as the actual exterior gap or a lower bound tending to infinity. This is the quantifier stated in lemma 15, Theorem 15.7.

Part 2. Note compared with the corrected lemma

We compared chiral_limit_note.tex against lemma 17 and its referee report.

51. Sign of j_- — Correct. With the standard Pauli matrix,

$$i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The note's $j_- = i\sigma_2 K$ is exactly the matrix in lemma 17. There is no sign discrepancy.

52. Reduced mass — Correct. In the note's basis $H_0 = (q_L, \bar{q}_R)$,

$$d_{\Sigma,s} = (\Sigma M^* \otimes 1_3) \oplus (s\,\overline{\Sigma^* M} \otimes 1_3).$$

On q_L, UD_F gives ΣM^*. On $\bar{q}_R$, the antiparticle block of U contributes the factor s, yielding $s\overline{\Sigma^* M}$. This agrees with the corrected lemma and report.

53. Reduced action and trace — Correct. The theorem retains the trace-class hypothesis, the sign-dependent real structure, the change of independent Grassmann variables in the paired sector, and the warning that the formal chiral rotation has a quantum Jacobian. The Majorana-type unpaired example prevents the Dirac conclusion from being applied to every sector.

54. Commuting subalgebra — Correct with amendment. The note states the no-scalar-fluctuation proposition and the restriction needed before calling the fields photon and gluons. Add the surviving electron pair to the explicit representation check, and state that the conclusion concerns the kept mass blocks and admissible involution. The first paragraph of the abstract should keep the same restriction.

55. Electric charge and flavour — Correct with amendment. The $U(1)^4$ statement needs **two differently charged flavours**, such as u, d. Two flavours of the same electric charge retain a larger flavour symmetry. The note also omits the corrected lemma's corresponding statement for admissible reductions: with charge included, intersect the colour-only admissible set with the charge commutant **and retain** $\Sigma^* M = (\Sigma^* M)^*$. At zero mass for u, d this leaves diagonal phases; positive nonzero diagonal masses leave signs.

56. Stationarity, reduced mass and polar part — Correct. The condition $\Sigma^* M$ Hermitian is equivalent to stationarity of $-\operatorname{Re}\operatorname{tr}(M\Sigma^*)$. The homogeneous-space statement at $M = 0$, the Hermitian involutions for $M = mI > 0$, and the unique polar minimum for invertible M agree with the corrected lemma. "Sign matrices" for positive distinct masses are to be read in a mass eigenbasis.

57. Vacuum alignment and mass signs — Correct. The note says that the stationary points include saddles and maxima, that no physical alignment energy or map to a condensate has been supplied, and that changing a reduction differs from a physical mass-sign change at fixed measure. These preserve the corrections to Remark 17.7.

58. Spacetime-dependent reduction — Correct with amendment. The displayed reduced operator and the statement that it is generally non-self-adjoint agree with Proposition 17.10. The kinetic term belongs to the separately chosen operator $D_M + \gamma_5 d_\Sigma(x)$, with the coefficient tracing both reduced blocks; it vanishes at $M = 0$. State these normalizations beside the formula and emphasize that this does not construct the massless pion effective action.

59. Fixed-background condensate expansion — Error. "A subtraction regular at $m = 0$" does not alone imply $c_V = n_0/(Vm) + O(m)$. For a finite pair-preserving spectral cutoff, subtracting a nonzero constant C is regular at zero and gives

$$c_V^{\mathrm{reg}}(m; A) = \frac{n_0}{Vm} - C + O(m).$$

Require a compatible subtraction whose finite remainder vanishes at zero with $O(m)$, for example an odd analytic subtraction in m, together with the spectral pairing. This issue is also present in the corresponding wording of the corrected lemma.

60. Banks–Casher form and determinant — Correct with amendment. The note retains the added assumptions $\rho_m(0) \to \rho_0(0)$, equicontinuity, tail and subtraction control, and the order of limits. Add $m > 0$, state the spectral-density and condensate normalization, and specify a regulator compatible with the $\pm\lambda$ pairing. The positive determinant formula is for $-iD_M + m$, as written.

The original Banks–Casher full text was inaccessible here, so the precise hypotheses should remain explicitly those of the note rather than be attributed to that paper.

61. Interval routes — Correct with amendment. For the localized branch state $\mu > 0$, $\mu l > 1$, and define $\kappa_0 > 0$ by

$$\tanh(\kappa_0 l) = \kappa_0/\mu.$$

The asymptotic $m \sim 2\mu e^{-\mu l}$ is for $\mu l \to \infty$, and the absolute limit quoted in the note follows at fixed μ. The boundary-Yukawa operator is a distinct construction from the scalar-algebra tensor-product truncation; the corrected lemma explicitly warns that identifying them at finite l needs proof. The note should retain that warning.

62. Fixed-background continuity and quantum input — Correct with amendment. The operator $D_M + \gamma_5 m$ and its square vary continuously with m. The regularized condensate itself need not have a finite limit at fixed background: a zero mode gives $n_0/(Vm)$. Specify the object whose continuity is asserted. The measure, volume sequence and order of limits remain additional inputs.

63. Provenance — Correct with amendment. The note says the identities were checked with Python on finite models. The recorded numerical checks in lemma 17 cover particular identities and examples, not every claim in the note. Name those checked identities. The earlier report also records four amendments in addition to five erroneous items; "corrected five items" should not be read as the whole scope of that report.

Part 3. Readiness for a Physical Review journal

We recommend substantial revision before submission.

1. **Separate the mathematical results from the endpoint proposal.** The introduction moves from formal KO signs to "10 to 9" and "10 to 11" limits before a map from those signs to the interval geometries or measured weak parameters has been constructed. Give the physical hypothesis, the exact results, and the unresolved identification in that order.
2. **Repair the cross-surface overclaims.** Correct the collapse statement in Table I and the abstract, the localized-mode mass formula in the discussion, the flavour-texture claim, the propinquity converse, and the condensate subtraction. State heat-cutoff and Taylor-mass scope wherever the weighted rule is summarized.
3. **Make the proof sketches show the decisive steps.** Theorem B's sketch omits the index and wall/Weyl arguments; Theorem C invokes concavity and compound matrices without defining the quantities that make the bounds work; Proposition H's displayed matrices do not alone establish the generated algebra. Proposition G needs its existential matching unitary stated in the theorem itself. The supplemental references are useful, but a reader should be able to see why each main-text conclusion follows.
4. **Define the notation needed to read the results.** In particular, define Φ_Q, Φ_U, the threshold quantities, and the symbols $f_{T_p}, g_{T_p}, D_p, K_p, \epsilon_p^{F,G}$ in Theorem C. Define κ by its trace weights, distinguish $M_{W,\mathrm{Taylor}}$ from a pole mass, and state the relative-trace or density convention on $\mathbb{R}^4$.
5. **State the status of the literature comparison.** Give a reproducible, appropriately scoped account of the searches behind the "we did not find" statements. Keep the established interval, domain-wall, truncation and spectral-propinquity ingredients visibly distinct from the proposed combination and from the still-open physical construction.

Amendments and errors

- **Chirality and labels:** scope the odd-dimensional objection, the Standard Model representation, the neutral parity criterion, particle preservation, and the Hořava–Witten comparison as in checks 1–3, 7, 10 and 12. The finite-l physical interpretation and the assignment of label 11 remain gaps.
- **Wall and collapse limits:** replace the fixed-distance bulk claim and the unrestricted tuned-bulk claim as in checks 5, 34 and 41. Keep the type-(c) localization and boundary-Yukawa scalings separate.
- **Spectral action:** define the heat cutoff, Taylor mass, trace weights and κ; restrict the asserted necessity of $m_t/\Lambda \to 0$ to the family proved in check 19.

- **Algebras and limits:** distinguish complex bicommutants from quaternionic real forms, expand the interval generation argument, and insert the existential unitary in Proposition G.
- **Discussion and note:** correct the localized-mode momentum formula, add minor bounds to the texture claim, require an $O(m)$-compatible condensate subtraction, and restore the interval-branch, charge, normalization and provenance qualifications in checks 54–63.

Overall: 63 statements checked: 16 correct, 36 correct with amendment, 4 gaps, and 7 errors.

Part R6 (referee report, no-go results and lemma 19)

Referee report on the two-model no-go results and lemma 19

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_nogo_19.md. Reproduced as delivered, with links relativized. The corrections are incorporated in lemmas 05, 06 and 19.

We checked the statements below by direct mathematical reasoning. No numerical computation was run. Each numbered verdict contributes one count to the final line.

Part 1. Bare limits and odd factorizations

1. The limit $m \to 0$ in Lemma 05, Part 4 Verdict: Correct with amendment. The fixed-rank statement is correct. The phrase "exact continuum limit throughout a fixed positive energy window" should specify the mode of convergence.

At fixed N, $D_N = mT_N$, with T_N independent of m. Hence, for fixed $t > 0$,

$$K_N(t) = \operatorname{Tr} e^{-tm^2 T_N^2} \longrightarrow 2(N-1).$$

The ultraviolet heat dimension is zero for each finite N. The distances between distinct sites are proportional to $1/m$ and diverge.

For $l \to \infty$, the condition $N/l \to \infty$ gives uniform convergence of the finite-chain dispersion to the continuum dispersion on every fixed bounded energy window. It also gives convergence of normalized heat traces and counting densities. It does not by itself give convergence of the *unnormalized difference* $K_N(t) - K_l(t)$ to zero.

To see the distinction, put

$$b = \frac{l}{2N-1}, \qquad k_r = \frac{(r-\frac{1}{2})\pi}{l}, \qquad \theta = \frac{\pi}{4N-2}.$$

Then

$$\lambda_{N,r} = \frac{\theta}{\sin\theta}\frac{\sin(bk_r)}{b}.$$

For example, taking $N = \lfloor l^{5/4} \rfloor$, expansion of the Gaussian sums gives, for fixed $t > 0$,

$$K_N(t) - K_l(t) \sim \frac{l^3}{16\sqrt{\pi}\, t^{3/2} N^2},$$

which diverges, although $K_N(t)/l \to 1/\sqrt{\pi t}$.

Direction. The finite-rank obstruction concerns the stated bare chain and heat trace. A resolved decompactification requires growing rank; the rate should be chosen for the desired notion of spectral convergence.

2. The limit $m \to \infty$ in Lemma 05, Part 4 Verdict: Correct. Every eigenvalue of the massless chiral interval, and of the calibrated finite chain, has magnitude at least

$$m = \frac{\pi}{2l}.$$

Thus, at every fixed cutoff Λ, the light spectral projection vanishes once $m > \Lambda$. At fixed N,

$$\|(D_N - i)^{-1}\| = \frac{1}{\sqrt{m^2+1}} \longrightarrow 0, \qquad d_N(i,j) \longrightarrow 0.$$

The zero resolvent limit on the original nonzero Hilbert space is not the resolvent of a finite self-adjoint operator there. The metric collapse and the empty fixed-energy sector are consequently compatible.

Direction. A nonempty light sector in a collapse calls for altered data, such as a tuned bulk mass or a scaled boundary coupling. The conclusion above applies to the stated massless interval and its calibrated truncations.

3. Proposition A.1 Verdict: Correct. For the stated graded product

$$D_F = D_0 \otimes 1 + \Gamma_0 \otimes D_I, \qquad \{D_0, \Gamma_0\} = 0,$$

the cross terms cancel and

$$D_F^2 = D_0^2 \otimes 1 + 1 \otimes D_I^2 \geq \frac{\pi^2}{4l^2}.$$

This holds for any finite KO-6 factor with the stated product operator, including an l-dependent finite Dirac operator. The fixed-cutoff light sector is empty for sufficiently small l.

Direction. The proposition points to modifications of the interval operator, boundary data, product coupling, or limiting prescription. It does not exclude a light sector in those modified families.

4. Result 3: electrodynamics Verdict: Correct. With the displayed matrices, both

$$U_+ = \mathrm{diag}(\sigma_1, \sigma_1), \qquad U_- = \mathrm{diag}(\sigma_1, -\sigma_1)$$

are grading-reversing unitary involutions. They commute with the algebra, its opposite, and D, while

$$U_+ J = J U_+, \qquad U_- J = -J U_-.$$

Thus the same D, J, γ admits both reductions, including at $m = 0$. The displayed representation and Dirac matrix agree, after the stated phase choice, with the electrodynamics data in van den Dungen–van Suijlekom, Eqs. (4.1)–(4.2).

Direction. A selection of one factorization needs additional input beyond the displayed real structure and the listed doubling conditions.

5. Lemma 4: neutral-sector criterion Verdict: Correct. In bases related by J, the KO-6 relations make the off-diagonal block M complex symmetric. Takagi factorization gives $M = \Sigma$, with nonnegative diagonal entries.

For $s = +1$, $V = I$ satisfies the criterion. For $s = -1$, the descended antiunitary j has $j^2 = -1$ and commutes with $d^2 = M^* M$. Each eigenspace of d^2, including its kernel, therefore has even complex dimension. Conversely, on each pair of equal entries of Σ, $V = \sigma_2$ is a Hermitian unitary and gives

$$(V^* K)^2 = -1, \qquad V^* \Sigma = (V^* \Sigma)^*.$$

This proves the stated equivalence, including zero singular values.

Direction. A KO-5 reduction can be recovered by neutral-sector pairing or by a symmetry enforcing the required even singular-value multiplicities. The result assumes the same scalar algebra character on the whole sector.

6. Proposition 5: the strict one-generation heavy limit **Verdict: Correct.** In the specified reduced representation, the surviving electron and down-quark sectors have both involutions from Result 3. The neutral sector has basis $(\nu_L, J\nu_L)$, zero Dirac block, and

$$\gamma = \sigma_3, \qquad J = \sigma_1 K, \qquad U = \sigma_1.$$

It supplies a KO-7 reduction. The full light space has dimension 18; a KO-5 reduction would require an antiunitary squaring to -1 on a complex space of dimension 9, which is impossible. The neutral bimodule gives the same obstruction directly.

Direction. The conclusion concerns the stated projection and reduced algebra. Adding an appropriate neutral partner can change the result; imposing particle-number preservation changes the admissible reductions, as in item 9 below.

7. Result 7: three generations **Verdict: Correct with amendment.** State that "arbitrary flavor mixing" means arbitrary matrices within the displayed species blocks, with the heavy projection chosen as a projection onto the specified top Dirac sector. Mixing between inequivalent species is outside the displayed Dirac operator.

The dimension counts are

$$96 - 6 - 12 = 78, \qquad 96 - 6 - 36 = 54.$$

The surviving three left neutrinos give a zero neutral block of size 3, so Lemma 4 gives KO-7 and excludes KO-5. Charged square flavor blocks can be brought to singular-value form by bimodule unitaries.

With three right neutrinos and a purely Dirac matrix Y_ν, the symmetric neutral block

$$M_N = \begin{pmatrix} 0 & Y_\nu^T \\ Y_\nu & 0 \end{pmatrix}$$

has every singular value of Y_ν twice, including zero. For a general symmetric Majorana block, the even-multiplicity criterion of Lemma 4 applies.

Direction. The obstruction is controlled by the neutral multiplicities. Paired neutral content or protected degeneracies provide directions to investigate.

8. No-go theorem 8: the full chiral representation **Verdict: Correct.** For the full displayed representation with $g \geq 1$,

$$e_{\mathbb{H}} = (0, 1_{\mathbb{H}}, 0)$$

has range of dimension $8g$ in H_+ and zero range in H_-. An algebra-linear grading-reversing unitary would identify these spaces. No such unitary exists, independently of the admissible D and of the sign against J.

This conclusion concerns the full quaternionic algebra, the displayed representation and grading, and the doubling criterion of Result 2. A projection that removes only one member of a weak doublet is generally not invariant under the full quaternionic algebra.

Direction. Possible directions include changing the algebra or representation, retaining the even description, and studying boundary or defect realizations of chirality.

9. No-go theorem with $[U, Q] = 0$ **Verdict: Correct.** In the stated strict light representation, an algebra- and opposite-algebra-linear U preserving particle number must map the ν_L particle subspace to an opposite-grading particle subspace with the same bimodule. No such partner is present. The KO-7 involution in Proposition 5 instead maps ν_L to its conjugate and fails $[U, Q] = 0$.

Direction. A particle-preserving reduction requires a matching particle bimodule of opposite grading, or a change in the particle content or imposed conditions. The conclusion holds for either sign against J.

Part 2. Fluctuations and the parent comparison

10. Comparison with Chamseddine–Connes–van Suijlekom Verdict: Correct. The formulas in Lemma 19 agree with arXiv:1304.7583v2, §III.2, Eqs. (10)–(11). That paper takes a "self-adjoint, universal one-form" and gives

$$A_{(2)} = \sum_j \hat{a}_j[A_{(1)}, \hat{b}_j], \qquad U = uJuJ^{-1}, \qquad UD(\omega)U^* = D(u\delta(u^*) + u\omega u^*).$$

Here $\hat{x} = JxJ^{-1}$. The formula for $A_{(2)}$ uses the same universal one-form as $A_{(1)}$; replacing it by an independently chosen one-form would change the construction. The algebraic calculation in Theorem 19.1 also extends to universal forms that are not self-adjoint.

For $J = \sigma_3 K$, the alternative pointwise choice, $JDJ^{-1} = -D$. The factor $\epsilon = -1$ in

$$\widetilde{A}_{(1)} = \epsilon J A_{(1)} J^{-1}$$

then gives the same displayed two-point formulas.

11. Theorem 19.1 Verdict: Correct. Direct evaluation gives

$$\frac{A_{(2)}}{m} = \begin{cases} \begin{pmatrix} 0 & |\phi_1|^2 \\ |\phi_2|^2 & 0 \end{pmatrix}, & J_0 = K, \\ \begin{pmatrix} 0 & \phi_1\overline{\phi_2} \\ \phi_2\overline{\phi_1} & 0 \end{pmatrix}, & J_r = \sigma_1 K. \end{cases}$$

Adding the two linear terms and D gives both factorizations in the theorem.

The represented self-adjointness condition is $\phi_2 = \overline{\phi_1}$. In this two-point example the universal one-forms are spanned by $p_1\delta p_2$ and $p_2\delta p_1$, which are adjoints and are represented faithfully for $m > 0$. Thus the same condition characterizes self-adjoint universal forms. Setting $\Phi = 1 + \phi_1$ gives

$$D'_{J_0} = m|\Phi|^2\sigma_1, \qquad D'_{J_r} = m\begin{pmatrix} 0 & \Phi^2 \\ \overline{\Phi}^2 & 0 \end{pmatrix}.$$

12. Corollary 19.2 Verdict: Error. The gauge actions and full fluctuated images in items 1–2 are correct. The common linear-only formula in item 3 is incorrect for J_0.

Let $z = e^{i\psi}$. On the gauge orbit of the zero universal form,

$$A_{(1)} = m\begin{pmatrix} 0 & z-1 \\ \bar{z}-1 & 0 \end{pmatrix}.$$

Removing $A_{(2)}$ gives the upper links

	upper link	modulus near $\psi = 0$
J_0	$m(2\cos\psi - 1)$	$m(1 - \psi^2 + O(\psi^4))$
J_r	$m(2e^{i\psi} - 1)$	$m(1 + \psi^2 + O(\psi^4))$.

For example, at $\psi = \pi/3$ the J_0 linear-only operator is zero, while the stated J_r expression has link $i\sqrt{3}\,m$. Adding $A_{(2)}$ restores the stated full results: $D' = D$ for J_0, and the link is mz^2 for J_r.

Two wording amendments accompany the correction. The set $\Phi = e^{i\psi}$ is the orbit in the potential parameter; the J_0 operator orbit consists only of D. The map $\Phi \mapsto \Phi^2$ is two-to-one away from zero and one-to-one at zero. The image statement is pointwise; a globally prescribed smooth link need not have a global smooth square root.

13. Reflection real structure and compression **Verdict: Correct.** In Counterexample 08.4,

$$J_{\mathrm{ref}} = \sigma_1 R_l K.$$

For the real lowest-pair basis

$$u_L = \sqrt{2/l}\,(\cos mx, 0), \qquad u_R = \sqrt{2/l}\,(0, \sin mx), \qquad m = \pi/(2l),$$

the identities $\cos(m(l-x)) = \sin(mx)$ and $\sin(m(l-x)) = \cos(mx)$ give

$$J_{\mathrm{ref}} u_L = u_R, \qquad J_{\mathrm{ref}} u_R = u_L.$$

The subspace is J_{ref}-invariant, and its compressed antiunitary is exactly $\sigma_1 K$. Its signs are $(J^2, JD/DJ, J\gamma/\gamma J) = (+,+,-)$.

For $e = \mathrm{diag}(1,0)$, the two compressed real structures give

$$[[D,e],\hat{e}] = m\sigma_1 \quad (J_0), \qquad [[D,e],\hat{e}] = -m\sigma_1 \quad (J_r).$$

Both two-point data satisfy order zero and fail order one.

14. Proposition 19.3, item 1 **Verdict: Correct.** Without a real structure, self-adjoint finite one-forms give every upper link $m(1+\phi_1) \in \mathbb{C}$. For fixed compression of the real parent gauge coefficient A_5, the link is

$$m + iB, \qquad B = \frac{1}{l}\int_0^l \sin\frac{\pi x}{l}\, A_5(x)\, dx \in \mathbb{R}.$$

Every real B is reached, so the fixed-compression image is the stated vertical line.

15. Proposition 19.3, item 2 **Verdict: Correct with amendment.** "A neutral radial field with no parent scalar" is fair when it means: *the finite pointwise-real fluctuation has a gauge-neutral radial parameter, while the specified pointwise-real parent has no surviving self-adjoint scalar inner fluctuation.* It does not establish a dynamical scalar field, its action, or the absence of parent fields in other models.

Let $E = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. A represented parent one-form is $Eh(x)$. Self-adjointness requires $h = ia$ with a real, and

$$K(iEa)K^{-1} = -iEa.$$

Thus the parent real fluctuation cancels. By contrast, the finite gauge-invariant link is

$$t = m|\Phi|^2 \geq 0,$$

so t is a function of the radial variable $|\Phi|$, and every $t \geq 0$ occurs.

The proof at line 71 needs this self-adjointness step. An individual commutator does not generally change sign under K: for $f = x$, $K[D,f]K^{-1} = [D,f] = E$. The sign reversal holds for the self-adjoint represented sum iEa.

The parent does possess an A_5 one-form before real symmetrization; the cancellation concerns the resulting real fluctuation. Compression of the algebra is also nonmultiplicative, so the construction gives no canonical identification of all finite potentials with parent potentials.

16. Proposition 19.3, item 3 **Verdict: Correct.** The reflected opposite algebra consists of multiplication operators, so the parent satisfies order one. For real A_5,

$$J_r D_{A_5} J_r^{-1} = D_{A_5 \circ R}.$$

Consequently its real scalar fluctuation has coefficient

$$A_5^s = A_5 + A_5 \circ R.$$

Since the compression weight is reflection-even, $B^s = 2B$, where B is calculated from the unsymmetrized A_5.

17. The table in Proposition 19.3 Verdict: Correct with amendment. Label the real structure and the meaning of B in each row. The covariant projection has fixed singular value m, and the full finite J_r orbit has link modulus m. The linear-only row has the stated modulus for J_r, locally growing at second order.

The fixed-compression row

$$\sqrt{m^2 + B^2}$$

is the formula of Lemma 13 for the unsymmetrized coefficient. For the reflection-real parent of item 3, if B continues to denote the integral of the original A_5, the corresponding expression is

$$\sqrt{m^2 + (B^s)^2} = \sqrt{m^2 + 4B^2}.$$

Equivalently, the table may retain its formula after defining its B to be the integral of the *symmetrized* coefficient. The increase is quadratic along paths for which the relevant B has a nonzero linear term; a nontrivial gauge path can also have $B = 0$.

18. Wilson phase in the paragraph following the table Verdict: Error. For a constant unsymmetrized $A_5 = a$, the transported link and its expansion are

$$mW_l^* = me^{ial} = m + i\,mla + O(a^2) = m + i\frac{\pi a}{2} + O(a^2).$$

Thus the Wilson **phase** is al; $mla = \pi a/2$ is the coefficient of the first-order imaginary change in the link. The fixed-compression value is $m + i(2a/\pi)$.

For the reflection-real fluctuation starting from the same a, the symmetrized coefficient is $2a$: the transported link is me^{2ial}, and the fixed-compression link is $m + i(4a/\pi)$.

19. Reading at the end of Lemma 19 Verdict: Correct with amendment. Replace "truncation generically breaks order one" by the demonstrated statement that spectral compression *can* break it, and state the hypotheses needed to apply the cited framework: a represented algebra, a preserved real structure, and order zero.

For example, on the circle take $D = -i\,d/dx$, $J = K$, and the full multiplication algebra. The parent satisfies order zero and order one. Compression to the zero Fourier mode leaves only scalar operators and preserves both conditions. Compression instead to the modes $-1, 0, 1$ sends e^{ix} to a truncated shift S; its conjugate is S^*, and

$$[S, S^*] \neq 0.$$

Thus this compression fails order zero. It also illustrates why the compressed operator system and the algebra it generates require separate attention.

The proposed relation to a singlet remains an analogy. In the cited 1304.7583v2, Proposition 8 and §IV, the toy model uses three complex fields, and the authors state that its quadratic term "does not generate new scalar fields"; it is expressed in fields already present in $A_{(1)}$. The present $A_{(2)}$ likewise depends on the same two-point parameters, but that fact gives no identification of the models. Devastato–Lizzi–Martinetti, arXiv:1304.0415v3, §4.3, obtain their σ from a Majorana-operator fluctuation in a larger algebra. An identification would require a map of algebras, representations, fields and gauge actions, together with an action comparison.

Amendments and errors

- Specify the convergence asserted in the $m \to 0$ continuum limit; $N/l \to \infty$ controls dispersion and normalized densities.
- Restrict the three-generation mixing statement to the displayed species blocks and specified heavy projection.
- In the parent comparison, use self-adjointness to prove the pointwise-real cancellation, and distinguish the original A_5 from its reflection symmetrization.

- Correct Corollary 19.2's J_0 linear-only link and the Wilson phase following the table.
- Qualify the general reading of truncation and retain the singlet comparison as an open analogy.

Overall: 12 Correct; 5 Correct with amendment; 0 Gap; 2 Error.

Part R7 (referee report, lemmas 20 and 21, Proposition I)

Referee report on lemmas 20 and 21, the comparison in the steer note, and Proposition I

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_20_21.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemmas 20 and 21, the steer note and the paper.

The review finds errors in the marked-point reading, the radial statement, parts of the scale comparison, Proposition I, and the no-go table. The connecting-operator calculations and the three consensus checks otherwise hold in their stated scopes.

The checks below are analytic; no numerical code was run. Source references identify the arXiv versions read.

Part 1. Connecting operators **1. Correct.** In Result 1, block multiplication gives $T = cT^*c$ and $hT + Tg = 0$. The lower off-diagonal block of order one is (2); the other follows using adjoints and the involutive algebra. In c-paired bases, the first condition is complex symmetry.

The averaging proof gives $T = R + \tau(R)$, with R intertwining the left actions and $\tau(R)$ the opposite actions. Conversely, such a decomposition gives order one on $\mathcal{A}(R)$, establishing (4). The final example has one arrow preserving the right character and the other preserving the left character; it satisfies order one although B is not contained in $\mathcal{A}(T)$. No correction is needed.

2. Correct. For the displayed rank-one T_κ, with $\kappa \neq 0$, the two selected lines must reduce their respective actions and carry the same scalar. This gives (6), of real dimension $2 + 4 + 18 = 24$. The self-adjoint off-diagonal operator has complex rank 2.

Chamseddine–Connes, §4, equations (4.6)–(4.7) use an auxiliary intertwiner, also called T there. Its compression $K : \mathbb{C}^2 \to \mathbb{C}^4$ has rank one or two. Rank two gives dimension at most $4 + 4 + 8 = 16$; rank one gives dimension 24, uniquely up to automorphism. For the displayed block, $T_\kappa x = \kappa KxK$. For a general admissible connector, the paper's K is obtained from an auxiliary intertwiner and need not equal the Dirac block. No correction is needed.

3. Correct. Result 3 and Counterexample 4 have the stated domain-preservation equations. In the scalar example, the parent has signs $(+, +, -)$ and satisfies both order conditions. The two displayed chiral vectors form the window for

$$\theta/l < \Lambda_0 < (\pi - \theta)/l.$$

Their compressed algebra is the full diagonal $\mathbb{C}^2$, and for $p = \mathrm{diag}(1, 0)$,

$$[[m\sigma_1, p], p^\circ] = -m\sigma_1.$$

The negative conclusion concerns compression of the full smooth scalar gluing algebra in this window. Restricting the parent algebra, or changing the window or representation, changes the problem; Result 5 gives a positive constant-algebra construction.

4. Correct. In Result 5, (18) and (19) preserve the boundary domain. The equality $cX = X^*c^{-1}$ is the required compatibility for J. The window contains $2n - 2$ uncoupled zero modes and the pair $\pm m$, giving (20). Constant $B = \mathcal{A}(S)$ multipliers commute with the parent Dirac operator, so the stated order-one claim follows.

With the displayed conventions, $U(x) = e^{mxF}$ carries the uncoupled domain to the coupled one, and differentiation gives (21), including its sign. Its compression gives $m(S + S^*)$. The construction assumes the Hom fibres, grading, constant multipliers and coupled channel.

5. Correct with amendment. The rank bound in Result 6 follows from the factorization $T = \mathrm{ev}_2^* X \,\mathrm{ev}_1$. A point with N fibre channels can give rank N. An exact change of boundary domain does not itself establish that factorization.

The separation argument in Result 7 is valid for unchanged finite windows when independent test functions can be supported away from the gluing sets. It applies to the stated point and fibre examples, including S^4 pullbacks after fibre integration. For a general subset upstairs in the twistor space, add the hypothesis that its image in S^4 has empty interior, so that the required independent base test functions exist. A new coupled window, boundary degrees of freedom, or a changed observable representation lies outside the result. The direction is to construct and analyze such a coupled parent.

6. Correct. Hořava–Witten, §2, states $\Gamma_{11}\chi = \chi$ for the intrinsic boundary gaugino. With a common ten-dimensional chirality convention and the standard common orbifold lift, the same-chirality comparison uses two type-(a), or two type-(b), fields. Opposite localization masses can place them near opposite ends.

For unbroken $E_8 \times E_8$, the displayed Hom space vanishes, and an operator odd for the common chirality has no matrix elements between the two retained same-chirality modes. Compactification, symmetry breaking, or additional bulk fields may give different effective couplings; those require a separate construction.

7. Correct. The three checks in Results 9–11 hold:

- The assumed weighted mass relation gives $m_{\max}/(2\sqrt{2}) \le M_W \le \sqrt{N_g}\, m_{\max}$. It forces the maximum to grow, with a fixed species necessarily unbounded only along some subsequence. The relation belongs to the specified asymptotic matching action; a full finite-cutoff calculation is a direction beyond it.
- With pointwise reality and a J_0-invariant window, inherited order zero holds exactly when the algebra generated by the compressed functions is commutative. A separate opposite action, as in the coarse correspondence, changes that hypothesis.
- If $P\gamma P$ is an actual grading, its square forces $[P, \gamma] = 0$. Thus it commutes with compressed even observables, whereas Γ_5 fails to commute with $Q(x_1)$. An added finite grading and restriction to its even algebra evade that particular inheritance claim.

The cited repository reports record earlier checks by another model. This review is a further solo check of the displayed arguments.

Part 2. Readings 8. Correct with amendment. In the CCM preprint, §2.5–2.6, especially (2.34), the right-handed up lepton is ν_R, and the off-diagonal block maps it to its conjugate through a symmetric Majorana matrix. After the stated identification, and an overall grading convention if needed, the one-generation connector has that form.

The interval construction gives one mass $m = \arctan(t)/l < \pi/(2l)$, with next positive absolute level $(\pi - \arctan t)/l$. Three identical copies give three equal masses. A general three-generation CCM Majorana matrix requires additional flavour data; it is not supplied by the single rank-one construction.

9. Error. The projector identity and

$$2\Pi_p - 1 = p_a \Gamma_a = 5Q(p \cdot x)$$

are correct. The pair of projective points determines the two lines, but it does **not** determine the connector's magnitude. Rephasing w and v also changes KxK by a phase. At fixed nonzero $|\kappa|$, the line choices are conjugate under the stated groups; $|\kappa|$ remains a modulus, and other finite data can introduce relative moduli.

The exact correction is to specify the lines **and** a coupling, together with an evaluation and normalization prescription. Moreover, $5Q(p \cdot x)$ fails to commute with the other coordinate compressions. It can be chosen

as additional finite grading data after restricting to its even algebra. That evades Result 11's inherited-grading hypothesis; it does not derive the grading from the even parent or construct the proposed point interface.

10. Correct with amendment. Equation (21) gives a constant matrix connection after a unitary change of domain, with relative endpoint rotation angle $\theta = ml$. On an interval, the open-path transport depends on endpoint frames; the physical quantity is the transport together with the specified boundary conditions. A closed Wilson loop or a dynamical gauge field needs further structure.

The gauge-Higgs comparison is an analogy at this stage: the interval calculation supplies a fixed matrix connection and no Higgs fluctuation or effective potential. The reference is Yutaka Hosotani, "Dynamical mass generation by compact extra dimensions," *Physics Letters B* **126** (1983), 309–313. For the dynamical Wilson-line and gauge-Higgs construction in an orbifold setting, see Hosotani, Noda and Takenaga, §1.

Part 3. Counting 11. Correct with amendment. In Proposition 21.1, $(q_L - q_R)/2 = (T_3/2)(1, -1)$, so the additional term in lemma 18's (33) is annihilated by the photon vector $(1, 1)$. The stated sums and removal entries reproduce:

removed	$\cos^2\theta_{\mathrm{NC}}$	$\kappa_{\mathrm{em}}/\kappa_s$
u-type	5/8	8/3
d-type	59/92	46/15
charged lepton	17/28	7/3

The table assumes the stated three-generation charges and that the other species have the mass assignments used in the removal calculation. At the endpoint where the neutral mass vanishes, the massive neutral direction is no longer distinguished: read its angle as the limit from finite mass, as lemma 18 states. Other weak-angle definitions need not have these values.

12. Error. The trace $84 + 12e^{-x}$ is correct for a top-only Dirac mass and no Majorana mass. With the positive heat cutoff and $m > 0$, its radial derivative is strictly negative. The text says there is "no finite stationary point"; at $m = 0$ the derivative is zero.

Replace that sentence by: **there is no finite nonzero stationary point on this free radial ray**. This concerns the literal flat-space action with no added potential or constraint. Compact support can give a plateau; sign-changing cutoffs, additional terms, constraints, and paths with an additional fixed noncommuting mass lie outside the strict conclusion. The result points to analyzing those changed actions or prescribed backgrounds.

13. Correct with amendment. Complexifying the real 30 gives

$$(\mathbf{5}, \mathbf{3}) \oplus (\bar{\mathbf{5}}, \bar{\mathbf{3}}),$$

and the displayed exterior-square branching has dimensions

$$192 + 24 + 8 + 1 + 60 + 45 + 60 + 45 = 435.$$

The charge tables and the charged scalar-partner counts follow. The charge operator includes the $U(1)$ commuting with $SU(5)$:

$$Q = \mathrm{diag}(3/5, 3/5, -2/5, -2/5, -2/5) + \tfrac{1}{15} I_5.$$

Also,

$$\mathbf{496} \to (\mathbf{435}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{1}) \oplus (\mathbf{30}, \mathbf{2})$$

under $SO(30) \times SO(2)$.

These are representation and charge counts. Identifying the entries with physical scalar partners requires a field construction, spin and chirality assignments, interactions and dynamics. Rivero's preprint, Tables 1–2, contains the stated diquark and meson counting; its proposed particle interpretation remains a proposal.

14. Correct with amendment. Both decompositions and the overlap of 72 follow by counting four internal states per Dirac species and colour. Without ν_R, the first family has 78, and the charged family

remains 84. A massless three-form in eleven dimensions and its six-form dual each have $\binom{9}{3} = \binom{9}{6} = 84$ on-shell polarizations.

The 84 in H_F is a complex finite-Hilbert-space dimension; the three-form count is a real bosonic polarization count. A physical identification needs a map of representations and degrees of freedom.

15. Correct with amendment. Seiberg, "Electric-Magnetic Duality in Supersymmetric Non-Abelian Gauge Theories," §1–3, *Nuclear Physics B* **435** (1995), 129–146, gives the four-dimensional $\mathcal{N} = 1$ supersymmetric result. For $N_c = 3, N_f = 5$, $9/2 < 5 < 9$; the magnetic group is $SU(2)$, with five flavours and a 5×5 gauge-singlet meson chiral superfield coupled by $W = Mq\tilde{q}$.

To turn readings 1–3 into statements, one needs:

- An $\mathcal{N} = 1$ supersymmetric theory with the required massless matter and dynamics, plus an operator and anomaly map. Five-flavour ordinary QCD alone does not supply Seiberg duality, and the magnetic $SU(2)$ needs a separate identification with any electroweak group.
- A representation and dynamics relating the finite fermionic states to the bosonic form or brane degrees of freedom; equality of the number 84 is insufficient.
- For the KO-label reading, the required involution on the stated finite sector, including its action on D, J, the algebra and its opposite. The vectorlike $SU(3) \times U(1)_{\text{em}}$ charges alone do not establish it, and neutral states still require treatment.

Part 4. Comparison with repository results **16. Correct with amendment.** Example 12.5 gives

$$\tan^2(kl) = |X|^2, \qquad k_{\text{light}} = \frac{\arctan|X|}{l}.$$

For small X, $k_{\text{light}} = |X|/l + O(|X|^3/l)$; the half-integer spectrum occurs as $|X| \to \infty$. At $X = 0$, zero has multiplicity two and must be omitted from the positive spectrum.

With $X(l) = \tan(ml)$, the light mass is exactly m and the remaining levels diverge as $l \to 0$. Identifying X with a physical Yukawa coupling, or $1/l$ with v, remains an extra hypothesis. The Wilson-line wording needs the qualification in item 10.

17. Error. CCM, (4.12) gives

$$\mu_0^2 = 2\frac{f_2}{f_0}\Lambda^2 - \frac{e}{a}.$$

Top dominance in the Dirac Yukawa trace alone does not force $e/a = 0$: the Majorana–neutrino term can remain relevant. Under exact top-only data, or the additional assumption $e = 0$, the stated truncated-potential relations give

$$\frac{m_t^2}{\Lambda^2} = 2\frac{f_2}{f_0}.$$

For the heat profile $F(u) = e^{-u^2}$, $f_2/f_0 = 1/2$, giving $x = 1$. This is a stationary point of the **truncated** potential; lemma 18's literal positive heat action has no finite nonzero stationary point on the free top-only radial ray. Its exact quadratic-background calculation at $x = 1$ gives the stated departure from the small-mass ratio.

18. Error. As written, "finite low-energy electromagnetic coupling" includes a coupling tending to zero. The trajectory in the transcript provides that case: take $\Lambda = \mu_* = v$, keep $\Lambda_5 > 0$ fixed through one-loop matching, and choose

$$\frac{8\pi^2}{g_2^2} = \frac{23}{3}\log\frac{v}{\Lambda_5} - \frac{1}{3}\log\log\frac{v}{\Lambda_5} + O(1).$$

Then $g_2 \to 0$, $m_t/\Lambda \to 0$, the top-only Taylor ratio tends to $\sqrt{8/3}$, and the low-energy electromagnetic coupling tends to zero. The text itself lists that endpoint as an escape.

The intended no-go can be stated with **nonzero limiting** α_{em} and positive finite limiting Λ_5, a matching scale above the relevant thresholds, top dominance at matching, the stated one-loop running, and the top-only heat-cutoff Taylor mass. In a step-threshold regime with $\mu_0 < m_t < \mu_*$ and nonincreasing $\alpha_{\mathrm{em}}^{-1}$ toward μ_*,

$$\alpha_{\mathrm{em}}^{-1}(\mu_0) \geq \frac{28}{3\pi} \log \frac{\mu_*}{\Lambda_5}.$$

This follows from $b_6 = 7$, $b_5 = 23/3$, and $\alpha_{\mathrm{em}}^{-1}(\mu_*) = \frac{8}{3}\alpha_3^{-1}(\mu_*)$. It bounds μ_*/Λ_5; matching then bounds g_2 away from zero, so $m_t \asymp g_2 v$ eventually conflicts with $m_t < \mu_*$.

The signs quoted in the note are correct in their respective active-field ranges: $\sum Q^2 = 20/3$ for five active quarks and three charged leptons, and $b_2 + b_Y = -11/3$ above electroweak thresholds. The thresholds are set by masses, rather than simply by v. The written route $m_t/\Lambda \to \infty$ cannot simultaneously use six-flavour matching at $\mu_* = \Lambda$ above the top. Extra thresholds or gauge mixing, altered matching, a different action or mass definition, and an $e \to 0$ endpoint are directions outside the amended no-go.

19. Error. The one-loop threshold relation

$$\Lambda_5 = \Lambda_6^{21/23} m_t^{2/23}$$

is correct under the transcript's matching assumptions. Its $n_f = 5$ and Seiberg's $N_f = 5$ count different theories; the equality of the numbers establishes no duality.

The transcript says pions *do not have to be* elementary KK modes or finite-algebra fluctuations. The comparison changes that into a stronger exclusion. Proposition 17.10 establishes absence of a kinetic term for the specified coordinate change of a fixed operator; it leaves a new operator or quantum effective action available. Proposition 17.8 gives the stated Banks–Casher order of limits under its measure and regularity assumptions.

For $T^3 = \mathrm{diag}(1/2, -1/2)$ and $Q = \mathrm{diag}(2/3, -1/3)$, the algebraic coefficient is $2N_c \operatorname{tr}(T^3 Q^2) = 1$. The reduced action supplies a starting point; a regulated fermion measure and anomaly calculation remain to be supplied.

Part 5. Paper 20. Error. Proposition I omits $\kappa \neq 0$ in item 2. At $\kappa = 0$, the block has rank zero and $\mathcal{A}(T_\kappa) = \mathcal{A}^{\mathrm{ev}}$, of real dimension 40. Add $\kappa \neq 0$.

Item 1 must also assume that the sheet-diagonal D_F already has the stated signs and order one. In item 3, specify a symmetric cutoff satisfying

$$m < \Lambda_0 < (\pi - \theta)/l.$$

The Majorana identification in the following paragraph needs the basis and flavour qualifications in item 8. The marked-pair paragraph needs the coupling modulus and grading qualification in item 9.

The abstract's wall-coupling sentence is supported for a **nonzero coupling on the chosen decomposable channel**, with the constant matrix observable algebra and the stated fibres and grading supplied as inputs.

21. Error. The table at lines 330–349 needs the following row-by-row corrections.

Row	Assessment and correction
Odd bulk	Say **no net chiral spectrum** under the specified standard smooth closed reduction. A periodic massless Dirac field on a circle has left- and right-handed zero modes. Boundaries and defects change the hypotheses. Baptista, §1, studies chiral interactions of massive non-Killing gauge fields; that is a distinct escape from an exact-isometry assumption.

Row	Assessment and correction
Odd factorization	State the full chiral Standard Model representation and the particular involution criterion of Theorem D. Reducing to invariant vectorlike sectors changes the algebra and representation.
Only KO-7	State the scalar-character neutral sector and the odd multiplicity, including zero singular values, in Theorem D. Even multiplicity and appropriate Dirac pairing are the stated alternatives.
Empty light sector	Specify the **massless type-(c)** interval and fixed spectral cutoff. A massless type-(a) interval has a zero mode for every l. Tuned bulk mass and scaled boundary coupling give the stated alternative families.
Fixed compression	Specify the local gauge transformations and fixed light projection of Proposition F. A projection invariant under a given transformation does not have this defect. The covariant projection changes with the fluctuated operator.
First order	Specify the two-point compression and inherited pointwise J. Fluctuations without order one relax that axiom; the doubled parent changes the data.
Order zero	Add J_0-invariance of the window and the inherited pointwise reality. A doubled or otherwise changed opposite action lies outside those hypotheses.
Bounded-fermion matching	Keep the leading spectral-action order, positive kinetic form, stated trace and common profile space. A tower-scale fermion removes the bound's obstruction; matching the full actions needs more.
$M_W \to 0$	Specify **massless complete doublet**, the stated heavy-doublet family, the Taylor definition, and the cutoff hypotheses including $f(0) > 0$. Keeping a doublet with a fixed nonzero mass can leave a positive residual mass even for the heat cutoff. Moreover, for the nonnegative $f(z) = ze^{-z}$, $f(0) = 0$, the top-only Taylor mass instead has $M_W^2/\Lambda^2 \sim x^2/3$. The heat-cutoff pole result has its own analytic-continuation hypotheses. A joint cutoff limit, altered light sector, or quantum effective action changes the problem.
Radial vacuum	Specify an allowed nonzero ray $V = cV_*$, $c > 0$, flat background, adequate integrability, and no added term or constraint. For nonnegative cutoff of unbounded essential support the derivative is negative. Compact support can give a plateau; other listed changes are possible directions.
Grading	Replace "grading not inherited" by "the proposed Γ_5 grading is not inherited" under the four hypotheses of Result 11. The Dolbeault window itself has an inherited scalar grading. A compressed height function can be added as finite data with a restricted even algebra.

Row	Assessment and correction
Scalar gluing	Add closed empty-interior gluing sets and independent functions supported away from them, with unchanged separate windows. Matrix-valued multipliers alone need not evade the same separation argument; a restricted multiplier algebra, changed representation, or coupled window can.
Finite limits	Specify finite limits of **connected scalar algebras** in the stated spectral-propinquity class. Truncation, matrix-valued parent algebras, or another convergence notion changes the question and needs a separate construction.

The vector-mass row points to exact calculations with the desired light masses and cutoff, while the grading and gluing rows point to an explicitly constructed coupled parent.

Amendments and errors

- Amend the pullback scope in **5**, the flavour and grading conventions in **8**, and the interval transport terminology in **10**.
- Amend the endpoint and degree-of-freedom interpretations in **11, 13–16**.
- Correct the missing connector data in **9**, the stationary-origin statement in **12**, and the missing Majorana term in **17**.
- Replace **18** by the explicitly nonzero-coupling, threshold-matched no-go, with the $e \to 0$ trajectory retained as an escape.
- Correct the transcript attribution in **19**, the hypotheses and marked-point summary in **20**, and the individual no-go rows in **21**.

Overall: 6 Correct; 8 Correct with amendment; 0 Gap; 7 Error.

Part R8 (referee report, lemma 22)

Referee report on lemma 22

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_22.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 22.

We find two errors in the stated claims: the assignment of zero modes at nonzero wall coupling, and the identification of fixed compression with the first derivative of transport. The main interval construction and its compression formulas hold, subject to the amendments below. The numerical collider-exclusion claim remains unsupported.

Throughout, let $\ell > 0$, and write $D = D_F$, $\Gamma = \Gamma_F$, and $J = J_F$ when no confusion can arise.

1. Setting — Correct.

In bases related by c, the condition on M says that its matrix is complex symmetric. Let its Takagi factorization be

$$M = U\Sigma U^T.$$

Then

$$|M| = \overline{U}\,\Sigma U^T, \qquad V = UU^T$$

give $M = V|M|$, with V unitary and $V = cV^*c$. This construction includes the kernel: zero entries of Σ still have corresponding columns in the unitary U. The extension of the polar factor on the kernel need not be unique.

The formula

$$X = \tan(\ell|M|)V^*$$

and the reality condition $cX = X^*c^{-1}$ follow. Conversely, for every X satisfying that condition, the principal functional calculus gives the unique operator

$$D_F = \ell^{-1} \arctan \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}, \qquad \ell\|D_F\| < \pi/2.$$

Thus the claimed bijection is valid. The expansion $X = \ell M^* + O(\ell^3)$ assumes fixed M.

2. Theorem 22.1 — Correct with amendment.

The boundary subspaces are Lagrangian for the first-order boundary form, giving self-adjointness; compactness of the H^1-to-L^2 embedding gives compact resolvent. Direct substitution verifies the stated signs and all three intertwining relations. The L- and R-components are orthogonal, while $R(x) = e^{x\Gamma_F D_F}$ is unitary, so W is an isometry.

For each singular value σ_k, with $\theta_k = \ell\sigma_k$, the signed spectrum is

$$\frac{n\pi \pm \theta_k}{\ell}, \qquad n \in \mathbb{Z},$$

with multiplicities added when values coincide. This gives the stated spectrum of $|D_X|$, the window, and the lower bound on the next level. The amendment is to state multiplicities, particularly when $\sigma_k = 0$: the zero mode then has multiplicity two per zero singular channel, and the two excited branches coincide.

For arbitrary signed momentum k, the scalar equation is

$$|\tan(k\ell)| = \sigma_k(X).$$

For the positive light mass $k = \sigma_k(D_F)$, its tangent is nonnegative and equals the corresponding singular value of X.

3. Corollary 22.2 — Correct.

On the uncoupled domain, multiplication by R maps the endpoint conditions to those of D_X, and

$$R^* D_X R = D_0 + E\Gamma_F D_F.$$

For one real channel, $\Gamma_F D_F = mF$, with F as defined in equation (21) of lemma 20. The sign and the placement of the spin matrix E agree.

4. Standard Model form and boundary types — Error.

The type assignment holds **at** $X = 0$: Q has type (a) and U has type (b). For $X \neq 0$, the wall conditions couple the fields, and the corresponding zero modes generally lift. For example, one channel with $X = t > 0$ has lowest masses $\pm\arctan(t)/\ell$ and no zero mode.

The correction is to say that the coupled modes continue from the type-(a) and type-(b) zero modes of the uncoupled problem.

One matrix X does encode the full finite operator, including Yukawa and Majorana data. Its entries need not have only the original block pattern: the tangent introduces higher powers of D_F. For example, with real symmetric

$$M = \begin{pmatrix} a & b \\ b & 0 \end{pmatrix},$$

the corresponding entry of $X = \tan(\ell M)$ is

$$X_{22} = \frac{\ell^3}{3} ab^2 + O(\ell^5).$$

Thus the original zero block can acquire an induced entry.

5. Theorem 22.3 — Correct with amendment.

Assume explicitly that the finite algebra and its faithful even representation are unital. The two endpoint intertwining equations are equivalent to

$$[\pi(f(\ell)), D_F] = 0.$$

The commutant of a self-adjoint operator is closed under products and adjoints. Hence C_F, and consequently A_ℓ, are unital $*$-algebras.

For every $f \in A_\ell$, multiplication preserves the whole H^1 domain and

$$[D_X, \pi(f)] = E\pi(f')$$

there. The right side extends to a bounded operator on all of H_ℓ. The opposite action also preserves the domain: this follows from domain preservation by g^* and $J^\wedge D_X = D_X J^\wedge$. Finite order zero then gives

$$[[D_X, \pi(f)], \pi^\circ(g)] = E[\pi(f'), \pi^\circ(g)] = 0.$$

No finite order-one hypothesis is needed.

This agrees with Counterexample 4 of lemma 20: its parent has order one, while its two-point compression fails finite order one. The statement concerns this parent and its pointwise algebra; it does not assert that arbitrary compressions preserve either order condition.

The Standard Model identification of C_F also needs the usual nonzero Yukawa data and the specified representation. For example, if all Yukawa blocks vanish and only an algebra-commuting Majorana block remains, the commutant is larger.

6. Theorem 22.4 — Correct with amendment.

Both compression formulas, including the minus sign in $Q_\ell(ET)$, and the symmetric exponential average for constant a are correct. The four items hold with the stated constants. In particular, interpreting $\|a\|$ as $\|\pi(a)\|$,

$$\|Q_\ell(a) - \pi(a)\| \le \frac{\ell^2}{6}\|[D,[D,\pi(a)]]\| \le \frac{2}{3}\ell^2\|D\|^2\|\pi(a)\|.$$

The constants in item 4 are respectively $8/3$ and $16/3$.

For item 1, compactly supported bulk functions suffice to test every smooth matrix-entry density. Finite-dimensional linear algebra then gives equality of the two images. The constant function a itself need not belong to A_ℓ; the theorem supplies a domain-preserving function with the same compression.

The general warning about generated algebras is sound. It also has a simple analytic example. Let

$$M = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & M \\ M & 0 \end{pmatrix}, \quad \pi(\mathbb{C}^3) = \{\mathrm{diag}(a,a) : a \text{ diagonal}\},$$

with $J = \left(\begin{smallmatrix} 0 & K \\ K & 0 \end{smallmatrix}\right)$. If P_1, P_2 are the first two diagonal channel projections, then on either graded block

$$[Q_\ell(P_1), Q_\ell(P_2)]_{12} = -\frac{\ell^2}{3} + O(\ell^4).$$

The original projections commute, whereas their compressed images generate a noncommutative algebra for sufficiently small nonzero ℓ.

The specific dimensions 15 and 30 remain **numerical claims**: the matrices and closure calculation were not retained, and the reported tolerance alone does not establish an exact rank. They should be accompanied by reproducible data and an exact or certified rank calculation. The general role of operator systems under spectral truncation is already developed in Connes and van Suijlekom, §1 and §3, arXiv:2004.14115v2.

7. Theorem 22.5 — Correct with amendment.

Every represented self-adjoint one-form is $E\pi(h)$, with $h^* = -h$. In fact **every smooth** such h arises, including those with nonzero boundary value: set

$$b(x) = \int_\ell^x h(t)\,dt.$$

Then $b(\ell) = 0$, so $b \in A_\ell$, and $[D_X, b] = E\pi(h)$.

Thus imposing compact support would exclude some connection profiles, although it loses no endpoint holonomy in the identity component: any smooth path from 1 to such a unitary can be made constant near both endpoints.

The transport formula and wall condition are correct. The proof should separate its two cases. For a general w, $\pi(w)^* D_F \pi(w)$ need not commute with J_F, so the non-real part uses the spectral and isometry proof of Theorem 22.1 **without** its real-structure hypothesis. For the real fluctuation, $U_w = \pi(w) J_F \pi(w) J_F^{-1}$ commutes with J_F, and the claimed real window follows.

There is a useful stronger statement. Put

$$\tilde{u}(x) = u(x) w^*.$$

Then $\tilde{u}(\ell) = 1$, so $\tilde{u} \in A_\ell$. With

$$\widetilde{U} = \pi(\tilde{u}) J_F \pi(\tilde{u}) J_F^{-1},$$

direct differentiation and the endpoint condition give

$$D_X + A + J^\wedge A (J^\wedge)^{-1} = \widetilde{U} D_X \widetilde{U}^*.$$

Thus every scalar connection in this interval model is gauge equivalent to the original parent under its allowed endpoint gauge group.

8. Corollary 22.6 — Error.

The gauge-orbit comparison itself is correct. Chamseddine–Connes–van Suijlekom, §III.2, equations (10)–(11), arXiv:1304.7583v2 give the quadratic fluctuation and its gauge transformation. Here the correct substitution is $v = w^*$, so the moving window carries

$$U_v D_F U_v^* = U_w^* D_F U_w.$$

For the two-point example, if $w = \mathrm{diag}(e^{i\alpha}, e^{i\beta})$, the upper link is $me^{-2i(\alpha-\beta)}$. It is $me^{2i\psi}$ in lemma 19 when ψ is defined using $v = w^*$.

The error is the assertion that fixed compression is the first derivative of the transported window for the **same** connection. Let h be constant and replace it by εh. At fixed D,

$$\frac{d}{d\varepsilon}\Big|_0 D_{\text{transported}} = -\ell[D, h],$$

$$Q_\ell(Eh) = -\frac{\ell}{2}[D, h] + O(\ell^3).$$

For example, take $D = m\sigma_1$ and $h = i\sigma_3$. Their leading terms differ by a factor of two. The real fluctuation doubles both expressions in this example and retains the discrepancy.

The correction is to state that the displayed fixed-compression expression is a tangent commutator with its own weighted generator. The moving spectral projection contributes to the derivative of the transported operator. The comparison with lemma 14 is a common transport principle, with its global path qualification; lemma 13 computes a distinct fixed compression. The first open point of lemma 19 is answered only on the specified gauge orbits and with the moving window.

9. No-go theorem 22.7 — Correct with amendment.

For the stated one-interval, zero-bulk-mass parent, fixed X, even scalar inner one-forms, and the **moving spectral window**, the finite operator is unitarily conjugate to D_F. Its singular values are therefore fixed. The author's argument and the direct endpoint-preserving gauge transformation in item 7 give the same conclusion in this scope. A fixed compression can show changes in singular values at second order, so the window must be named.

For the usual Standard Model representation with nonzero Yukawa data, the normalized shifted Higgs quaternion runs through unit quaternions, a three-real-dimensional gauge orbit. These are gauge-orbit directions. Calling them physical or eaten modes requires a spacetime gauge action and its boundary conditions; the interval calculation alone supplies no such mode count. Degenerate data, including vanishing Yukawas, can have a smaller orbit.

The listed changes of wall coupling, admissible odd bulk terms, bulk masses, or internal dimension lie outside the theorem's hypotheses and are directions for further constructions. Curvature in higher dimensions alone does not prove a radial mode. Likewise, a quantum effective action on the same fixed configuration space does not by itself add a radial coordinate; an additional composite or other degree of freedom would have to be identified.

10. Proposition 22.8 — Correct with amendment.

Evaluation shows that the unsymmetrized gauge one-form has boundary value in C_F. For the displayed masses, assume a flat product interval, a positive Yang–Mills kinetic form, the trivial background, unrestricted endpoint variation in the allowed directions, and no additional boundary kinetic or mass terms. Variation then gives Neumann conditions at 0, and Neumann on the wall Lie algebra and Dirichlet on its orthogonal complement at ℓ. The stated integer and half-integer towers follow.

The physical gauge Lie algebra is obtained from skew-adjoint algebra elements through the real gauge representation, with its kernel and any imposed unimodularity condition accounted for. For the usual Standard Model representation, $u(C_F)$ has dimension 10 before unimodularity, whereas photon plus gluons have dimension 9. The extra abelian direction must be addressed. The role of unimodularity and the hypercharge embedding is explicit in Chamseddine–Connes–Marcolli, §2.5, arXiv:hep-th/0610241.

The literature comparison is valid at the level of the boundary mechanism. The Neumann/Dirichlet towers appear in Csáki et al., §5.1, arXiv:hep-ph/0305237v1, and the fermion equation $\tan(m_n L) = M_D L$ is equation (5.6) of Csáki et al., arXiv:hep-ph/0310355v1. Their full gauge models have additional structure.

11. Corollary 22.9 — Correct with amendment.

Items 1, 2, 4 and 5 hold for fixed, orthogonal, J_F- and grading-compatible light and heavy channels, with $0 < \Theta < \pi/2$. For a general non-normal light block, write the upper wall block as

$$X_0 = \tan(\ell|M_0|)V_0^*,$$

or state the formula in terms of singular values. The light-channel coupling is $\ell M_0^* + O(\ell^3)$. Identifying it with a four-dimensional Yukawa coupling remains an additional hypothesis.

For item 3, let P_L reduce $D_F(\ell)$, and suppose $D_L = D_F(\ell)|_{P_L H_F}$ is fixed. Then the precise statement for a constant observable is

$$P_L Q_\ell(a) P_L = \operatorname{sinc}(\ell \operatorname{ad}_{D_L})(P_L \pi(a) P_L),$$

and

$$\|P_L Q_\ell(a) P_L - P_L \pi(a) P_L\| \le \frac{2}{3}\ell^2 \|D_L\|^2 \|\pi(a)\|.$$

This remains true when $\pi(a)$ mixes light and heavy channels. The limiting light compression need not be a representation: products can pass through the heavy subspace.

The full observable estimate does **not** follow from Theorem 22.4, since $\ell\|D_F(\ell)\| \to \Theta$. An explicit counterexample uses

$$D_F(\ell) = \sigma_1 \otimes \operatorname{diag}(m, \Theta/\ell), \qquad \pi(a) = 1 \otimes \sigma_1,$$

with the commutative algebra generated by 1 and $\pi(a)$ and the compatible real structure $J = (\sigma_1 \otimes 1)K$. Then

$$Q_\ell(a) = \operatorname{sinc}(\Theta - \ell m)\, \pi(a),$$

which tends to $\operatorname{sinc}(\Theta)\pi(a)$, rather than $\pi(a)$.

The ratio $m_t/M_W = 2\Theta/\pi < 1$ is a result for the stated lowest-level, zero-bulk-mass fermion channel and strict-wall gauge spectrum. Bulk fermion masses or a changed gauge wall condition are possible directions beyond that family.

12. Numerical comparison for the interior — Gap.

For a separate flat model with Neumann conditions at 0 and a finite Robin wall mass,

$$t \tan t = s, \qquad s = \frac{g_4^2 v^2 \ell^2}{4},$$

where $g_4^2 = g_5^2/\ell$ in the usual flat normalization. The Z equation has $s/\cos^2\theta_0$. This agrees with the form and sign convention of Csáki et al., §5.2, equations (5.13) and (5.22).

For $g'^2/g^2 = 3/5$, the lowest-mode squared mass ratio increases from 5/8 to 1. A small **Python** calculation evaluated the equations at the stated rounded values: $M_W\ell = 1.03$ is consistent with the quoted ratio to the displayed precision, and the next W root lies between $3.58/\ell$ and $3.59/\ell$, about 280 GeV when $1/\ell \approx 78$ GeV. This is a numerical check; the monotonicity follows separately from the equation.

The top constraints are strict:

$$M_W\ell < \frac{\pi}{2\sqrt{8/3}} \simeq 0.962, \qquad M_W\ell < \frac{\pi}{2(m_t/M_W)} \simeq 0.732$$

for the measured ratio. The latter estimate is consistent with the 2025 PDG W, Z values and top-mass table.

The collider-exclusion sentence is a **gap**: a mass alone does not establish a search exclusion without the model's couplings, branching fractions, and an applicable experimental limit. It should be removed or supplied with that analysis. The finite wall scalar also changes the strict wall boundary problem; it is an extended model outside Proposition 22.8 and No-go theorem 22.7. Within the stated flat toy model, the incompatible top and gauge mass requirements support the limited conclusion; bulk masses, warping, and running are directions to investigate.

13. Input and derived statements — Correct with amendment.

The finite algebra, representation, operator, grading, and real structure are inputs. The construction derives the endpoint commutant for that input, the interval operator and its window, the stated compression map, and the transport formulas. The Yang–Mills spectrum additionally depends on the action and boundary assumptions in item 10. The exact window identifies D_F, J_F, Γ_F; the represented algebra is obtained through compression and need not be an exact representation of A_F.

Several ingredients have broader precedents. For any reducing finite spectral subspace, the identity

$$[D_F, W^*TW] = W^*[D_X, T]W$$

holds whenever T preserves the operator domain. Operator-system compression is treated generally in Connes–van Suijlekom, §3. Boundary spectral geometry for the Standard Model was also studied by Chamseddine and Connes, §5, arXiv:1008.3980v1, and the interval fermion and gauge mechanisms occur in the Higgsless papers cited in item 10. The particular combination and formulas of lemma 22 can be stated explicitly; a wider novelty claim would need a fuller literature comparison.

Amendments and errors

- **Errors:** correct the boundary-type zero-mode statement at nonzero X and the first-derivative comparison between fixed compression and transport.
- **Gap:** support or remove the collider-exclusion claim.

- **Amendments:** state spectral multiplicities and unitality; qualify the Standard Model data, gauge quotient, and unimodularity; distinguish profiles from holonomies and real from non-real transport; scope the radial result to the moving window; provide reproducible data for the numerical algebra dimensions; state the light-block estimate and the additional assumptions of the Yang–Mills and finite-wall-scalar models.

Overall: 2 Correct; 8 Correct with amendment; 1 Gap; 2 Error.

Part R9 (referee report, lemma 23)

Referee report on lemma 23

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_23.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 23 and the note.

The boundary-spectrum formulas are correct under the stated Yang–Mills boundary conditions. The principal corrections concern the aligned-wall compression, the comparison with twisted spectral triples, the order of the holonomy double coset, and the bulk-mass extension of the one-loop potential.

The verdicts below count the lettered entries. The numerical checks use Python 3 with SciPy. They supplement the analytic arguments and do not replace them.

1. Theorem 23.1 (statement and proof)

1a. Item 1 — Correct. Each wall condition is a maximal isotropic subspace for the boundary form. The resulting first-order operator is self-adjoint with compact resolvent, and the spin grading preserves its domain and anticommutes with it.

1b. Item 2 — Correct. Under the equal-dimension and no-−1-eigenvalue hypotheses, the principal logarithm gives an anti-Hermitian K satisfying $\Gamma_0 K \Gamma_0 = -K$. The stated $D_F = \Gamma_0 K$, its singular values, and

$$\Gamma_1 = e^{lK} \Gamma_0 e^{-lK}$$

follow from the two-projection decomposition.

1c. Item 3 — Correct with amendment. The graph boundary condition and the converse hold on the stated branch $l\|D_F\| < \pi/2$. Define X explicitly as the $H_- \to H_+$ block of $\tan(lD_F)$. The spectral part of Theorem 22.1 applies without a real structure; its real-structure conclusion additionally needs the hypothesis in item 4. At an angle $\pi/2$, the principal logarithm and the isolated window used here cease to be unique.

1d. Item 4 — Correct. The antiunitary J_1 is an additional hypothesis. With $\widehat{J} = EJ_1$, the signs are $(+,+,-)$, and $J_F = \Gamma_0 J_1$ has the stated finite signs. In particular,

$$J_F \Gamma_1 J_F^{-1} = \Gamma_0 (J_1 \Gamma_1 J_1^{-1}) \Gamma_0 = -\Gamma_0 \Gamma_1 \Gamma_0.$$

The antilinearity of J_1 is compatible with $J_1 K J_1^{-1} = K$ on the principal branch.

2. Theorem 23.2 (statement and proof)

2a. Items 1 and 2 — Correct with amendment. Domain preservation is equivalent to commuting with the respective wall grading, and $[D, f] = Ef'$. The compression formula in item 2 requires the equal-dimension and branch hypotheses under which the isometry W of Theorem 22.1 exists. State those hypotheses at the item.

2b. Item 3, aligned walls — Error as stated for an arbitrary $*$-subalgebra P. A constant odd $a \in P$ generally fails the endpoint conditions and is an ambient multiplier, rather than an element of the parent algebra. More seriously, $E_0(P)$ need not equal $P \cap \{\Gamma_0\}'$.

Let $H = \mathbb{C}^2$, $\Gamma_0 = \sigma_z$, and let P be the unital $*$-algebra generated by $\sigma_x + \sigma_z$. Then

$$E_0(P) = \operatorname{span}\{1, \sigma_z\}, \qquad P \cap \{\Gamma_0\}' = \mathbb{C}1.$$

The first space is the compression image, while the second is the even part of P. Compactly supported scalar profiles show that the image of the actual boundary algebra is $E_0(P)$.

The correction is to assume $\Gamma_0 P \Gamma_0 = P$. Then $E_0(P) = P^{\mathrm{ev}}$, and the aligned-wall conclusion holds. Continue to distinguish constant ambient multipliers from parent-algebra elements.

2c. Item 4 — Correct. Under the stated J_1 hypothesis and $[P, P^\circ] = 0$, the parent has a real structure and satisfies order zero and order one, even when P contains elements odd for Γ_0. The parent grading is $\hat{\Gamma}$, which commutes with those elements. Pointwise,

$$[\pi(f), J\pi(g)^* J^{-1}] = 0, \qquad [[D, \pi(f)], J\pi(g)^* J^{-1}] = E[f', g^\circ] = 0.$$

2d. Comparison with Result 11 — Correct with amendment. The aligned-wall construction evades hypothesis 4 of Result 11: odd coordinate matrices are sent to zero rather than retained as $\Gamma_a/5$. Result 11 assumes an even parent, an inherited involutive grading, and retention of all those coordinate compressions simultaneously. Its stated escapes include restricting to the even algebra or changing the grading construction. The present construction points toward a matrix-valued parent with an inherited grading; it leaves the commutative-parent and retained-coordinate questions open.

3. Remark 23.3 (remark)

3a. Opposite action — Correct with amendment. For an odd **anti-Hermitian** h, the displayed signs for the unstarred conjugations $J_1 h J_1^{-1}$ and JhJ^{-1} are correct. The opposite representation used in order zero is $b^\circ = J_1 b^* J_1^{-1}$; state the star and the anti-Hermiticity assumption when giving signs. On the W-factor it incorporates $\sigma(b) = \gamma_W b \gamma_W$.

3b. Grand-symmetry twist — Error in the assertion that it is the same automorphism. The relevant references are:

- Agostino Devastato, Fedele Lizzi and Pierre Martinetti, "Grand symmetry, spectral action and the Higgs mass," *JHEP* 01 (2014) 042. The larger algebra and its reduction appear in §§3–5.
- Agostino Devastato and Pierre Martinetti, "Twisted spectral triple for the Standard Model and spontaneous breaking of the Grand Symmetry," *Mathematical Physics, Analysis and Geometry* **20** (2017), article 2. Its Eqs. (1.8)–(1.11) define the twisted commutator, first-order condition and automorphism; the publisher record confirms the publication details.

The latter authors call their ρ the "exchange of the left/right spinorial indices." It exchanges algebra summands. Conjugation by γ_W is inner on $M_2(\mathbb{H})$ and fixes its centre. Thus the full automorphisms have different algebraic actions. The correction is to describe a **structural analogy** between involutive automorphisms and compare the actual representations, opposites and first-order conditions before claiming an identification.

4. Theorem 23.4 and Example 23.5 (fluctuations and example)

4a. Connection removal — Correct with amendment. The displayed unitary identity follows from $u' = -hu$. It describes the perturbation $D + Eh$. For a fluctuation of a real triple, include the conjugate one-form:

$$D + Eh + \widehat{J}(Eh)\widehat{J}^{-1}.$$

In the example, the resulting generator has the form (L_F, R_F), as required for the stated action on $E \oplus E^*$. This also preserves the J_1 condition. The odd directions lie outside No-go theorem 22.7, whose hypotheses include an even algebra and a moving window.

4b. Holonomy and double cosets — Error in the order of the factors. With the convention $u(0) = 1$, $u(l) = w$, let $H_i = \mathrm{Stab}(\Gamma_i)$. A gauge transformation removing h has the form $u(x)c$, so it preserves both endpoints precisely when

$$c \in H_0, \qquad wc \in H_1, \qquad \text{equivalently} \qquad w \in H_1 H_0.$$

The holonomy quotient is $H_1 \backslash G / H_0$, or $H_0 \backslash G / H_1$ after replacing w by w^{-1}. The order matters for the stated w. Also restrict G to the unitary component reachable by a smooth path from the identity.

4c. Orbit, angles and polar coordinates — Correct with amendment. The orbit of quaternionic self-adjoint involutions with one-dimensional quaternionic $+1$ and -1 eigenspaces is

$$\mathbb{H}P^1 \simeq Sp(2)/(Sp(1) \times Sp(1)) \simeq S^4.$$

The wording "each grading with quaternionic eigenspaces" also includes ± 1; exclude those two cases. For $\Gamma_p = p_a \Gamma_a$, the eigenvalues of $\Gamma_q \Gamma_p$ are $e^{\pm i d(p,q)}$, so the principal angles are $d(p,q)/2$.

The formula on E^* is correct: p' is the reflection of p through p_0, because

$$\Gamma_{p'} = \gamma \Gamma_p \gamma.$$

The quaternion direction q supplies three angular coordinates away from the poles. It is undefined at both poles, and at $\theta = \pi/2$ the finite logarithm and lowest-window choice are degenerate. The unquotiented orbit is S^4; quotienting the relative orientation by the stabilizer leaves the radial interval.

4d. Bulk-mass texture — Correct with amendment. For $X = \tan\theta\, 1$, the light-mass approximation from Theorem 12.6 has the scale

$$M_{\text{light}} \simeq \frac{\tan\theta}{l}\, \Phi_Q \Phi_U$$

when the light eigenvalues are small compared with the relevant tower thresholds. In the convention of Theorem 12.7, $\widehat{Y} = X/l = (\tan\theta/l)1$; a dimensionless identity matrix can be stated after factoring out that common scale. Positive localization parameters suppress the wall profiles. General finite-angle eigenvalues require the secular equation.

4e. Minimal composite Higgs comparison — Correct with amendment. The coset and the four real Higgs coordinates agree with those in Agashe, Contino and Pomarol, "The Minimal Composite Higgs Model," *Nuclear Physics B* **719** (2005), 165–187, §§2–3. Their Eq. (6) identifies h/f_π with the sphere angle, corresponding here to 2θ. Their construction uses a warped AdS interval and additional fermion and boundary data. The shared coset alone does not identify the actions or potentials.

5. Theorem 23.6 (boundary conditions and theorem)

5a. Connector at one wall — Correct with amendment. The stated gauge boundary conditions form a consistent boundary-value problem with the connector treated as prescribed wall data. A real fermion connector can also be placed at one wall. For example, choose a unit vector e in the $+1$ eigenspace of Γ_1, put $f = J_1 e$, and set

$$K_c = |f\rangle\langle e| - |e\rangle\langle f|, \qquad \Gamma_{1,c} = e^{\eta K_c} \Gamma_1 e^{-\eta K_c}.$$

Then K_c anticommutes with Γ_1, commutes with J_1, and J_1 anticommutes with $\Gamma_{1,c}$. Restricting the wall algebra to the common stabilizer gives the connector conditions and retains the parent order conditions.

The amendment is to specify whether the connector is prescribed gauge-breaking data or a term in the fermion boundary operator. In the latter case, $\Gamma_{1,c}$ changes the affected fermion tower; the universal fermion product later in the lemma must be recalculated for that channel.

5b. Charged and neutral masses — Correct. An independent solution uses principal angles between the Neumann subspaces at the two walls. Put $s = \sin(2\theta)$. For each charged direction the squared sine of the angle is $s^2/2$. In the neutral sector the two Neumann directions are L' and $\sqrt{3/5}\, R' + \sqrt{2/5}\, B$; their projection onto the Dirichlet complement at the first wall has one nonzero squared singular value $4s^2/5$. Consequently

$$\sin^2(M_W l) = \frac{s^2}{2}, \qquad \sin^2(M_Z l) = \frac{4s^2}{5},$$

which give the stated tangent equations. This derivation also covers $\cos(2\theta) = 0$, where the divisions in the written proof need a limiting interpretation.

5c. Massless modes — Correct with amendment. For $0 < \theta < \pi/2$, the intersection has dimension nine: the photon and eight gluons. At each endpoint it has dimension twelve, with three additional electroweak zero modes. The phrase "massless for all θ" is correct; state the endpoint enhancement explicitly.

5d. Normalization — Correct. With the stated fundamental trace forms, equal five-dimensional coefficients, flat profiles and no boundary kinetic terms,

$$g_3 = g_2 = \sqrt{\frac{5}{3}}\, g_Y, \qquad \frac{1}{e^2} = \frac{8}{3g_2^2}.$$

Indeed the hypercharge generator has squared trace norm $1 + 2/3 = 5/3$, while the photon generator has squared norm $1 + 1 + 2/3 = 8/3$. Here g_2 is the normalized four-dimensional zero-mode coupling.

5e. Items 1 and 2 — Correct. Expanding the sine equations gives $M_W^2 \sim 2m^2$, $M_Z^2 \sim (16/5)m^2$ as $\theta \to 0$. At $\theta \to \pi/2$, both displayed branches tend to zero and their squared ratio tends to $5/8$. The ratio itself is undefined at the endpoints.

5f. Item 3 — Correct. Let $a = \sin(2\theta)$. Then

$$\frac{M_W^2}{M_Z^2} = \left[\frac{\arcsin(a/\sqrt{2})}{\arcsin(2a/\sqrt{5})}\right]^2.$$

For $0 < a \le 1$, this decreases with a, since $x/(\sqrt{1-x^2}\arcsin x)$ increases. Hence the midpoint is the global minimum, with exact value

$$\left(\frac{\pi/4}{\arctan 2}\right)^2 = 0.5032315\ldots.$$

The upper value $5/8$ is approached at the endpoints. This supplies the proof missing from the grid check.

5g. Item 4 — Correct. The ratio m/M_W increases from $1/\sqrt{2}$ to infinity. Python 3 with SciPy gives $\theta = 1.05888846\ldots$ for $m/M_W = \sqrt{8/3}$.

6. Comparison with lemma 18 and Proposition F

6a. Constants 2 and 16/5 — Correct with amendment. These are the coefficients in the lower bounds for the fully massive finite-action pole masses in Proposition 14 of lemma 18. They are not the small-x pole coefficients there: those are 3 and 24/5. The common 5/8 limit is a valid comparison, with different limiting behaviour: the finite-action poles grow in the lemma 18 heavy limit, while the interval vector masses tend to zero at the antipodal endpoint.

Specify the trace or Hilbert–Schmidt norm in the commutator-norm claim. With the ordinary operator norm, the charged example gives m^2, rather than $2m^2$.

6b. Factor 3/2 — Correct with amendment. Proposition F gives the factor 3/2 for the stated leading finite spectral-action mass matrix. Comparing it with the assumed five-dimensional Yang–Mills quadratic form gives $3m^2$ versus $2m^2$ in the universal small-mass example. A calculation of the full parent spectral action, with its tower and boundary conditions, remains necessary for a direct matching claim.

7. Proposition 23.7 (heat trace)

7a. Heat trace and monotonicity — Correct. With $s = l\Lambda$,

$$\mathrm{Tr}_{\mathrm{channel}}\, e^{-D^2/\Lambda^2} = \frac{2s}{\sqrt{\pi}}\,\vartheta_3(\theta, e^{-s^2}).$$

The Jacobi product proves strict decrease on the interval, with zero derivative at the endpoints. At fixed $x = \theta^2/s^2$, its towerless limit is $2e^{-x}$. At $\theta = \pi/2$, the full lowest eigenspace is degenerate even though the heat trace is well defined.

7b. Escape from No-go theorem 9 — Correct with amendment. No-go theorem 9 assumes finite internal data, the positive-Laplace cutoff class, the literal flat-space spectral action, $A = 0$, and an allowed unconstrained ray $V = cV_*$, $c > 0$, without an added scalar potential. Its conclusion is strict decrease along that ray. The interval family changes the operator family and replaces the free ray by a compact relative-grading orbit with a tower. It therefore lies outside those hypotheses.

For the stated heat action, the antipodal point is a minimum of this family. Added potentials, constraints, other cutoff classes and quantum effects remain separate possibilities. The result points toward analysing the full tower and any additional scalar terms.

8. One-loop potential (products and potential)

8a. Three products — Correct. The product for a principal angle α, normalized at $\alpha = 0$, is

$$1 + \frac{\sin^2 \alpha}{\sinh^2 x}.$$

Using the fermion angle θ and the two gauge angles from 5b gives all three displayed products.

8b. Fermion coefficient — Correct with amendment. With the conventional four-dimensional Dirac determinant and each physical species counted once, $n_f = 2N_D$; one colour-triplet Dirac top contributes six. The universal example gives masses to both members of each weak doublet. For one complete generation it contains eight physical Dirac species, hence $n_f = 16$, before any connector modification; three identical generations give $n_f = 48$. A top-only $n_f = 6$ potential therefore needs a separately specified fermion sector or a complete accounting of spectator channels.

8c. Potential and endpoint curvatures — Correct with amendment. The logarithmic integrand and the vector multiplicities are correct under the stated one-loop assumptions. The curvature sentence omits the prefactor on its left-hand side. Let

$$I = \int_0^\infty \frac{x^3\,dx}{\sinh^2 x} = \frac{3}{2}\zeta(3), \qquad J = \int_0^\infty \frac{x^3\,dx}{\cosh^2 x} = \frac{9}{8}\zeta(3).$$

The precise expansions are

$$8\pi^2 l^4[V(\theta) - V(0)] = (10.8 - n_f) I\,\theta^2 + o(\theta^2),$$

and, with $\epsilon = \pi/2 - \theta$,

$$8\pi^2 l^4[V(\tfrac{\pi}{2} - \epsilon) - V(\tfrac{\pi}{2})] = (10.8 I + n_f J)\epsilon^2 + o(\epsilon^2).$$

For $n_f \geq 0$, the latter curvature is positive.

8d. Numerical absence of an interior minimum — Gap in any conclusion beyond the sampled grid. The recorded computation is explicitly a grid statement, and its original code was not retained. An independent Python 3/SciPy calculation of the derivative found one sign change for each tested $n_f = 1, 2, 3, 4, 6, 8$; for $n_f = 8$, the stationary angle is approximately 0.3382. The known integral $I = 3\zeta(3)/2$ agreed to 5×10^{-16} in absolute error, and a 1% perturbation failed the same check. A grid does not exclude additional stationary points between samples, so the claim about absence of an interior **local** minimum for all parameters in the displayed massless family remains open in the range $0 < n_f < 10.8$.

Two stronger statements follow analytically from the displayed potential:

- For every $n_f > 0$, $\theta = \pi/2$ is the unique global minimum on $[0, \pi/2]$. The gauge logarithms are nonnegative, and the negative fermion logarithm is minimized when $\sin^2 \theta = 1$.
- For $n_f \geq 10.8$, the potential is strictly decreasing throughout the interior. To see this, let $t = \sin^2 \theta$ and

  $$\mathcal{J}(z) = \int_0^\infty \frac{x^3\,dx}{\sinh^2 x + z}.$$

  The derivative, apart from the positive overall factor, is

  $$(1 - 2t)\left[6\mathcal{J}(2t(1-t)) + \frac{24}{5}\mathcal{J}\left(\frac{16}{5}t(1-t)\right)\right] - n_f \mathcal{J}(t).$$

  For $t \leq 1/2$, both arguments of $\mathcal{J}$ in the bracket are at least t, and $\mathcal{J}$ decreases. For $t > 1/2$, both contributions have the required sign.

These conclusions concern the displayed massless flat products, the stated gauge content and boundary conditions, and one-loop order. They direct attention to changed form factors or additional terms for an interior vacuum.

8e. Leading-logarithm form — Correct with amendment. The formal first term is $B\sin^2(2\theta) - A\sin^2\theta$, with $A, B > 0$, and any interior stationary point of that expression is a maximum. The logarithm expansion is not uniform at $x = 0$; the next term has an infrared-divergent integral. Present the expression as a formal leading approximation and use the full integral for conclusions about its global shape.

8f. Bulk-mass scan — Error as a finite one-loop potential for general scanned masses. Let $a = c_Q + c_U$. The stated fermion factor satisfies

$$G(x) = 1 - \frac{a}{x} + \frac{a^2}{2x^2} + O(x^{-3}),$$

so, for $t = \sin^2\theta$,

$$\log(1 + t(G-1)) = -\frac{ta}{x} + \frac{t(1-t)a^2}{2x^2} + O(x^{-3}).$$

Multiplication by x^3 gives angle-dependent ultraviolet divergences whenever $a \neq 0$. A finite quadrature value then depends on a regulator and subtraction prescription; the reported scan does not establish a regulator-independent potential.

The exceptional case $c_U = -c_Q$ has a common bulk mass on the two mixed components and

$$G(x) = 1 + \frac{w^2}{x^2\sinh^2 w},$$

which has the required exponential decay. Keeping the flat gauge products is consistent as a **separate tree-level gauge-sector assumption**. For a coupled $Sp(2)$ gauge theory, unequal masses on components mixed by the gauge group require gauge-invariant completion or explicit breaking data. The correction is to specify that completion, the regulator and allowed wall counterterms, then recompute the potential.

8g. Escapes and spinor comparison — Correct with amendment. A frequency-two fermion contribution, modified gauge form factors, wall potentials and wall terms are possible changes to the stated flat calculation; each requires its own model and coefficients. For the comparison with Agashe–Contino–Pomarol, their §2.1, Eqs. (19)–(23), uses spinorial $SO(5)$ fermions and permits an approximate potential

$$\alpha\cos(h/f_\pi) - \beta\sin^2(h/f_\pi).$$

With $h/f_\pi = 2\theta$, the displayed leading flat expression has this angular form with $\alpha = A/2$ and $\beta = -B$, up to a constant. Their fermion form factors can also give the sign and competition needed for an interior minimum. The flat scan therefore has its stated scope; it does not exclude the spinorial composite-Higgs construction.

9. Measured values and open questions (comparison)

9a. Numerical comparison — Correct with amendment. Using the 2026 Particle Data Group values $m_t = 172.60$ GeV, $M_W = 80.3625$ GeV and $M_Z = 91.1879$ GeV gives $m_t/M_W \simeq 2.1478$ and $M_W^2/M_Z^2 \simeq 0.7767$. The top entry is a direct-measurement mass and its definition should be kept distinct from a specified pole-mass scheme. PDG quark table, p. 2; PDG gauge-boson table, pp. 1–2.

Python 3 with SciPy gives $\theta \simeq 1.1626$, $M_W l \simeq 0.5413$, $1/l \simeq 148.5$ GeV and the model ratio $M_W^2/M_Z^2 \simeq 0.5814$. For the separate wall-scalar model of lemma 22, fitting the measured vector ratio gives $M_W l \simeq 1.0282$; its lowest no-bulk-mass fermion channel then has $m/M_W < 1.528$. Thus the stated mismatch holds for those flat, tree-level families with the specified lowest-level and coupling assumptions. Bulk masses, warping, running and a combined wall/Wilson-line model remain directions to examine.

9b. Open questions — Correct with amendment. The listed questions remain appropriate. Add the real connector's contribution to the fermion tower, a gauge-invariant and renormalized bulk-mass potential,

the degenerate antipodal window, and the scalar kinetic normalization needed to identify a canonically normalized Higgs field.

Amendments and errors

- Assume $\Gamma_0 P \Gamma_0 = P$ for the aligned-wall even-algebra claim, and distinguish ambient constant multipliers from elements of the boundary algebra.
- Replace the claimed equality with the grand-symmetry twist by a comparison of the involutions and their first-order conditions.
- Reverse the holonomy factors for the convention used: $w \in H_1 H_0$, with quotient $H_1 \backslash G / H_0$.
- Specify the real fluctuation, the connector's role, the altered fermion tower, endpoint degeneracies and trace norms.
- Add the missing $1/l$ scale and light-mode approximation to the bulk-mass texture; add $1/(8\pi^2 l^4)$ to the potential curvatures.
- Restrict the unproved absence of local minima to the sampled grid, and replace the generic bulk-mass scan by a calculation with a stated regulator, counterterms and gauge completion.

Overall: 11 Correct; 19 Correct with amendment; 1 Gap; 4 Error.

Part R10 (referee report, lemma 25)

Referee report on lemma 25

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_25.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 25 and the note.

The leading constant-profile mass formula is supported after the amendments below. The principal mathematical error is the extra factor of s in the displayed heat-trace derivative. The identification with a physical lowest gauge mode requires additional boundary and spectral assumptions.

The calculations in this report are analytical; no numerical code was run. Numerical values below are taken from the cited records.

1. **Proposition 25.1 — Correct with amendment.**

   The unfolding in lemma 25 is correct after multiplying it by $1/\sqrt{2}$. It maps the H^1 domain of D_0 onto the periodic H^1 domain on the circle. With

   $$D_F = \begin{pmatrix} 0 & M^* \\ M & 0 \end{pmatrix},$$

   the connection term satisfies

   $$E\Gamma_F D_F : \quad \Psi_Q \mapsto M^* \Psi_U, \qquad \Psi_U \mapsto M \Psi_Q.$$

   Thus the unfolded operator is $-i\partial_x + D_F$, with signed spectrum $n\pi/l + \mu_i$.

   The amendment is to name the frame. Corollary 22.2 states

   $$D_X = R(D_0 + E\Gamma_F D_F)R^*, \qquad R(x) = e^{x\Gamma_F D_F}.$$

   An even T that is constant **after transport** unfolds to $1 \otimes T$. In the original frame it is $R(x)TR(x)^*$, which generally varies with x. Conversely, a constant original-frame T becomes R^*TR after transport and generally does not unfold to $1 \otimes T$. For broken T, it also need not preserve the original coupled wall domain. The transported field generally has an odd bulk component and therefore requires an admissible enlarged gauge algebra; it is not automatically a field of the even algebra in lemma 22.

   There is a separate error in the holonomy sentence at line 31. With the stated convention, put $U = e^{-i\pi\Gamma_0/4}$. The correct relation is

   $$\Gamma_1 \Gamma_0 = e^{2l\Gamma_0 D_F} = U e^{2ilD_F} U^*.$$

Conjugation by Γ_0 instead sends e^{2ilD_F} to its inverse. The circle holonomy can acquire the inverse sign under the opposite parallel-transport convention.

2. **Theorem 25.2 — Correct with amendment.**

   The exact kernel at line 39 and its quadratic coefficient have the stated signs. Lemma 18 states its theorem for a finite internal trace; an extension to this infinite tower must be supplied. Here it is available: V and T split into finite-dimensional blocks indexed by n. Applying the finite theorem to a symmetric finite set of blocks and passing to the limit is justified by Gaussian bounds, uniform for D_F in a bounded neighborhood of zero. The same bounds justify differentiation and Taylor expansion under the sum.

   Let $t = 1$, $d = \mu_i - \mu_j$, $b = u(1-u)$, and

$$E(u) = e^{-[(1-u)(k+\mu_i)^2 + u(k+\mu_j)^2]}.$$

   An exact integration-by-parts form of the $q = 0$ bracket is

$$d^2 \int_0^1 E(u)\big[1 - b(2k + \mu_i + \mu_j)^2\big]\, du.$$

   Its expansion begins

$$d^2 e^{-k^2}\left[1 - \frac{2}{3}k^2 + \frac{2k^3 - 5k}{3}(\mu_i + \mu_j) + O(\|\mu\|^2)\right].$$

   The displayed cubic term is odd in k and cancels between n and $-n$. Similarly, the q^2 bracket begins

$$e^{-k^2}\left[\frac{2}{3} - \frac{2k}{3}(\mu_i + \mu_j) + O(\|\mu\|^2)\right],$$

   and its linear term cancels. The grading symmetry gives the same evenness for the summed expression.

   The proof should include these cancellations and remainder terms. At fixed $s = l\Lambda$, for fixed nonzero Hermitian T, the stated $O(\|D_F\|^3/\Lambda)$ mass remainder is valid but can be sharpened to

$$M^2_{\text{Taylor}} = \left(\frac{3}{2} - \langle k^2\rangle\right)\frac{\|[D_F, T]\|^2_{\text{HS}}}{\|T\|^2_{\text{HS}}} + O\left(\frac{\|D_F\|^4}{\Lambda^2}\right).$$

   Here $k_n = n\pi/(l\Lambda)$ is dimensionless. Equations presented later without a remainder must be labeled leading-order formulas.

3. **Spectral dimension — Error.**

   The factor at line 60 is wrong. Direct differentiation gives

$$\boxed{\langle k^2\rangle = \frac{1}{2}\frac{d\log S}{d\log s} = \frac{s}{2}\frac{d\log S}{ds}.}$$

   The extra s multiplying the logarithmic derivative must be deleted. The subsequent coefficient $(3 - d_{\text{eff}})/2$ uses the corrected identity.

   The spectral-dimension interpretation is correct for the **unshifted unfolded circle**, equivalently the paired Neumann-plus-Dirichlet interval trace. It should not be called the spectral dimension of an arbitrary interval boundary problem. Indeed, their separate heat traces are

$$S_N = \frac{S+1}{2}, \qquad S_D = \frac{S-1}{2},$$

   and have different logarithmic derivatives. For nonzero D_F, the actual parent heat trace contains $e^{-t(k_n+\mu_i)^2}$; S is the unperturbed trace used in the leading expansion.

4. **The two limits and covariance — Correct with amendment.**

   The limiting factors 3/2 and 1 are correct. The exponential corrections have polynomial prefactors. More precisely,

$$d_{\text{eff}}(s) \sim \frac{4\pi^2}{s^2} e^{-\pi^2/s^2} \quad (s \to 0), \qquad d_{\text{eff}}(s) = 1 - 4s^2 e^{-s^2} + O(s^2 e^{-2s^2}) \quad (s \to \infty).$$

   State the order of limits: first take the small-D_F/Λ coefficient at fixed s, then take either limit.

   The five-dimensional covariance argument is valid on the unfolded circle for this operator and these insertions. With Hermitian connections the curvature convention gives $F_{5\mu} = i[D_F, A_\mu]$, up to the sign convention for the covariant derivative; its norm gives the quoted commutator quotient.

   On a general space with boundary, the bulk local coefficient alone does not rule out boundary contributions. In the present calculation the full operator with the specified transported fields is unitarily equivalent to the circle problem, so its trace has the required cancellation. State that specific reason rather than a general boundary assertion.

5. **Corollary 25.3 — Correct with amendment.**

   The coefficients agree with Remark 16 of lemma 18 and with the CCM mass relation, Eq. (4.16) at leading order. With the literal particle-and-antiparticle trace,

$$Q_{11} = 12, \qquad K_{11} = \sum_f N_f m_f^2,$$

   which gives $(3 - d_{\text{eff}})/24$. For the top alone, $K_{11} = 3m_t^2$, giving the displayed ratio and

$$y_t = \frac{2g}{\sqrt{3}} \quad (d_{\text{eff}} = 0), \qquad y_t = \sqrt{2}\, g \quad (d_{\text{eff}} = 1).$$

   The first is also CCM's top-dominance approximation, Eq. (5.27).

   The neutral result $M_W^2/M_Z^2 = 5/8$ follows **to this order**, with the common trace and normalization. "Massless directions are unaffected" should say that their mass coefficient remains zero; their kinetic coefficients can change.

   The twelve doublets have the same tower factor S in the stipulated flat, bulk-mass-free model, including when only the top has nonzero Dirac mass. With bulk masses used to localize light fields, their spectra and overlap matrix elements change. The kinetic coefficient must then be recomputed from all eigenstates and overlaps; it cannot generally be obtained by assigning the same S to every doublet. Light zero modes can still contribute even when their excited modes are suppressed. For an explicit localization mechanism and profiles, see Adachi, Kurahashi and Maru, arXiv:1404.4281v1, Eqs. (2.10)–(2.11).

6. **General cutoffs — Correct with amendment.**

   The weighted formula is correct for the positive Laplace class and moment hypotheses in lemma 18, Eq. (4). It is a leading small-D_F formula. A precise version is

$$S_\nu(s) = \int S(s/\sqrt{\tau})\, d\nu(\tau), \qquad d_\nu = \frac{d \log S_\nu}{d \log s} = 2\langle \tau k^2 \rangle_\nu,$$

   so that the coefficient is $(3 - d_\nu)/2$. Under those moment assumptions, dominated convergence gives the stated endpoints. The crossover and its corrections depend on ν.

   Replace "general cutoffs" by "positive Laplace cutoffs satisfying (4)." A general smooth spectral cutoff, including a nonconstant cutoff flat near zero, need not belong to that class.

7. **Scope and the lowest mode — Gap.**

   The small-angle comparison with Theorem 23.6 is correct. For its *universal* fermion mass $m = \theta/l$, the $d_{\text{eff}} = 1$ formula gives

$$M_W^2 = 2m^2, \qquad M_Z^2 = \frac{16}{5}m^2$$

   at leading order. Expanding the exact Yang–Mills formulas gives, respectively,

$$2m^2 - \frac{4}{3}m^4l^2 + \cdots, \qquad \frac{16}{5}m^2 - \frac{64}{75}m^4l^2 + \cdots.$$

   Lemma 25 directly computes a Taylor quotient for a specified constant profile. For the unrestricted unfolded circle, translation invariance separates its zero Fourier sector from nonzero Fourier sectors; one must still diagonalize the generator matrix within that sector. Calling the result the first nonzero term of a **lowest-mode mass** also requires an admissible gauge field, a positive kinetic form and an isolated lowest branch. Boundary constraints or additional terms can alter the profile problem. Theorem 23.6 establishes the comparison for its own Yang–Mills boundary problem; it does not establish the full finite-cutoff spectral-action gauge spectrum.

   Add these hypotheses to Scope, together with the transported-frame qualification, the distinction between Taylor and pole masses, and the fact that the calculation is about a prescribed background rather than a demonstrated vacuum. "First order in D_F" refers to the mass; the displayed mass **squared** starts at quadratic order.

8. **Reading — Correct with amendment.**

   The factor 3/2 in Proposition F is the finite-action coefficient relative to the Yang–Mills commutator quotient in the stated trace. The crossover explains that coefficient's two limits in the present constant-profile calculation. Proposition F also contains a bounded-fermion versus tower-scale obstruction and overlap conditions; those do not follow solely from the factor 3/2. The reading should retain their separate hypotheses.

   The direct-measurement figure uses the PDG 2026 top listing, p. 1, 172.60 ± 0.27 GeV, and the PDG 2026 *W* listing, 80.3625 ± 0.0077 GeV. The top value is a direct-measurement Monte Carlo mass, with a scheme interpretation discussed by PDG.

   The remembered 163 GeV is supportable with a scheme and scale: Yoon and Peskin, arXiv:1810.12352v3, §3.3 and Eq. (23) use approximately $\overline{m}_t(\overline{m}_t) = 163$ GeV for a 173 GeV pole mass. They also quote about 147 GeV at 2 TeV. Dividing either running mass by an on-shell W mass is only a formal orientation, rather than a common-scale matching test.

   The proposed matching scale of order $30M_{\text{KK}}$ and its running are outside lemma 25. The cited steer note labels its calculation unrefereed, uses remembered tower coefficients, and reports that the three couplings do not meet at one point. Retain the figure only as a conditional estimate from that separate model.

9. **Literature — Gap.**

   The endpoint $m_t = 2M_W$ is already present in flat five-dimensional gauge–Higgs literature. Cacciapaglia, Csáki and Park, *Fully Radiative Electroweak Symmetry Breaking*, arXiv:hep-ph/0510366v2, §5 explicitly state the tree-level relation for a suitable rank-four representation and discuss the assumptions on the top zero mode. Adachi and Maru, arXiv:1804.06012v2, Eq. (3.23) obtain a limit approaching $2M_W$ with their specified brane mixings. These are representation-dependent mechanisms; they do not establish the lemma's spectral-action crossover.

   Ordinary universal-extra-dimension models need separate specification. For example, Cordero-Cid et al., arXiv:1108.2926v4, §2 formulate the five-dimensional Standard Model with separate Higgs and Yukawa sectors; universal propagation alone does not impose $m_t = 2M_W$.

The inspected primary records do not exhibit the particular $(3-d_{\text{eff}})/24$ coefficient or its theta-function crossover. We cannot claim novelty on that basis. The search used web-index queries for spectral action, heat kernels, compactification and gauge–Higgs mass relations, including a 2026 query; it followed the 2014 Adachi–Kurahashi–Maru reference to the 2006 paper. A systematic forward-citation search, author recency sweep and broader database search remain open.

10. **Paper's final sections — Error.**

    The sum-rule section of boundary_condition_note.tex carries the leading result into an unqualified displayed equality. Add the small-mass remainder and the transported-frame, trace and constant-profile assumptions. In particular, the towerless top-only Taylor expansion already contains the nonzero correction

    $$M_W^2 = \frac{3}{8} m_t^2 - \frac{23}{128} \frac{m_t^4}{\Lambda^2} + \cdots$$

    in lemma 18, Eq. (19). Thus the leading equality must not be stated as an exact finite-mass result. Apply the same qualification to the abstract and to "throughout" for the neutral ratio.

    The statement that running "would be a power law" also requires a specified effective theory, thresholds and renormalization prescription. The subsequent **Status** section contains separate claims based on lemma 24 and its literature search; those lie outside lemma 25 and were not audited here. Provenance should be updated if this report is recorded.

Amendments and errors

- Correct the factor in the heat-trace derivative and identify the unperturbed paired interval or circle trace.
- Name the transported frame for T, correct the holonomy conjugation, and state the admissible gauge algebra and wall conditions.
- Add the finite-block convergence argument, odd-k cancellations and leading-order remainders.
- Qualify the covariance, cutoff, neutral-ratio and lowest-mode statements by their hypotheses.
- Cite the top-mass schemes and prior gauge–Higgs literature; treat the crossover's priority as open pending a wider search.
- Amend the paper and abstract to state leading-order results, and keep the separate running and Status claims within their own scopes.

Overall: 0 Correct; 6 Correct with amendment; 2 Gap; 2 Error.

Part R11 (referee report, lemma 24)

Referee report on lemma 24

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_24.md. Reproduced as delivered, with links made relative. The amendments are collected at the end of lemma 24.

The main operator-window construction and the new one-loop argument are correct in their stated families. The principal gap is the statement of convergence of compression maps for general observables as the interval shrinks: the varying source spaces need an identification, and different interval profiles have different limits.

This report gives one verdict for each of the ten requested items. The checks are analytic; no numerical computation was used.

1. **Result 1 — Correct.** In Result 1, $C_i^2 = \Pi$, $(\operatorname{sgn} A)^2 = 1 - \Pi$, and the cross terms vanish. Thus both χ_i are self-adjoint involutions and anticommute with E. They commute with $\hat{\gamma}$, so the boundary domain is preserved by the grading.

   Let $B = SA/r$. At 0, the complementary boundary condition selects $B < 0$; at ℓ, it selects $B > 0$. The reversal makes the boundary term in (3) nonnegative and gives (2). The finite kernel conditions are maximal isotropic and give the interval operator of lemma 23.

The H^1 realization is correct. Decomposition into eigenspaces of A gives self-adjoint finite-dimensional interval problems. On their direct sum, (3) controls both q' and Aq; ellipticity of A identifies the graph domain with the stated H^1 domain. Rellich compactness then gives compact resolvent. The projection Π reduces both the expression and the domain.

Differentiation verifies (4). The principal-logarithm hypothesis gives $\ell\|D_F\| < \pi/2$, and the interval spectrum together with (2) gives (5). Thus, when $\Delta_K/r > \|D_F\|$, the displayed embedding is the isolated finite window.

2. **Result 2 — Correct with amendment.** With the endpoint convention made explicit, condition (6) gives the claimed local self-adjoint elliptic realization. Write G_0 at $x = 0$ and G_ℓ at $x = \ell$, or write G_i at $x = i\ell$ for $i \in \{0, 1\}$. The present "at $x = i$" leaves the index convention implicit.

   For $\chi = ZG$, the adapted tangential principal symbol exchanges the two χ-eigenspaces because $[G, \sigma_A] = 0$. The normal symbol exchanges them as well. Bär–Ballmann, Corollary 4.18 and Theorem 4.11 give the relevant ellipticity and self-adjointness criteria. Their Example 4.20 assumes anticommutation with the full adapted operator; (6) supplies the principal-symbol condition needed for Corollary 4.18.

   Equation (7) is correct for the stated maximal smooth scalar $*$**-algebra**. Preservation by both f and $\bar{f}$ makes the boundary subspace reducing, hence gives commutation with χ_i. Commutation with $C_i^2 = \Pi$ then forces multiplication by f_i to preserve $H^0(\mathbb{CP}^3, \mathcal{O}(1))$. Those sections have no common zero, so f_i is holomorphic and therefore constant.

   The local hypothesis is restrictive. On the unaugmented Dolbeault spinor bundle, the six-dimensional complex Clifford representation is irreducible. A fibre endomorphism commuting with every $\sigma_A(y, \xi)$ is scalar; a smooth self-adjoint involution of this form is consequently $+1$ or -1 on connected K. Additional fibres and other local elliptic boundary projectors are directions outside this sufficient criterion.

3. **Result 3 — Correct with amendment.** Equations (9)–(11) follow as stated. Equation (9) proves that $g^2 = 1$ is equivalent to $[G, \Pi] = 0$. Scalar multiplication commutes with G, giving (10); the complex-linear commutant of the generated $M_4(\mathbb{C})$, or of $M_2(\mathbb{H})$ acting on $\mathbb{C}^4$, is $\mathbb{C}1$. Hence the self-adjoint involution g is ± 1.

   The auxiliary example also checks: $Q_K(y_a) = \Gamma_a^{\text{section}}/5$, and the Clifford traces give $\operatorname{tr} g^2 = 16/5$. Applying (9) and taking the largest eigenvalue gives the displayed $2/\sqrt{5}$ lower bound on boundary-trace mixing.

   Amend the marked-point sentence at lines 236–244 to say that the **natural field displayed there** supplies no separately marked point. A local field can encode a chosen point: with the auxiliary bundle, $G_p(y) = p_a \Gamma_a^{\text{aux}}$ is a local constant field whose coefficients contain p. The restriction (11) concerns the unchanged four-section window and an actual involution obtained by compression. A coupled window, extra fibres, or spectral walls remain directions to examine.

4. **Result 4 — Correct with amendment.** Inserting W_ℓ into the scalar multiplier gives (12), including its independence of r. For real S^4 functions, (15) follows from quaternionic Hermiticity and the six independent matrices $1, \Gamma_a$. When the walls are aligned,

$$\mathcal{E}_p(\Gamma_a) = p_a \Gamma_p,$$

   so direct scalar compression gives (13). Generating $M_2(\mathbb{H})$ first and then taking its Γ_p-centralizer gives (14).

   For (16), state explicitly that the two Hom spaces and the two product-factor scalar algebras of Example 16.9 have been supplied. The V-side left action on the second Hom space commutes with its right grading, giving

$$\mathbb{R} \oplus \mathbb{R} \oplus M_4(\mathbb{C}).$$

   The endpoint argument proves equality of the **finite-dimensional linear images**: interior-supported scalar functions separate the same matrix-entry densities. It does not by itself identify the parent

algebras, their norms, gauge groups, or one-forms. The restriction $\mathbb{R} \oplus \mathbb{R}$ applies to real S^4 scalars, aligned walls, and direct compression. Intermediate algebra generation and other windows remain available constructions.

5. **Result 5 — Gap.** Equations (17)–(20) are correct under their displayed hypotheses. In particular, the complementary resolvent in (17) has norm at most r/Δ_K; at fixed D_F, the interval tower also decouples as $\ell \to 0$. The Schur estimates (18) hold with the stated positive g_r. Equation (19) holds for an x-independent ambient multiplier, and (20) follows from the even part of the conjugation series.

   The final assertion about convergence of **the compression maps** needs a common domain and a specified collapse map. For example, take aligned walls, $D_F = 0$, and the admissible scalar functions

   $$f_\ell(y, x) = 6(x/\ell)(1 - x/\ell).$$

   Both endpoint values are zero, but $\Phi_\ell(f_\ell) = 1$. Thus a general family of observables cannot be replaced by its value at the collapsing endpoint.

   One exact correction is to fix $F(y, s)$ on $K \times [0, 1]$, set $f_\ell(y, x) = F(y, x/\ell)$, and state

   $$\Phi_\ell(f_\ell) \longrightarrow \mathcal{E}_0 Q_K \left(\int_0^1 F(\,\cdot\,, s)\, ds \right).$$

   For x-independent ambient multipliers, the limit is $\mathcal{E}_0 Q_K$, as follows from (19). Nonconstant x-independent functions on K are ambient multipliers; they do not themselves obey the spectral-wall endpoint restriction. Result 4's equality of finite images supplies representatives with the required endpoint values.

   The rate discussion after (18) also needs a proof or a strengthened assumption: the two displayed rv_r conditions alone do not control the denominator g_r. A simple sufficient addition is

   $$r(b_r + L) \le (1 - \varepsilon)\Delta_K$$

   for some fixed $\varepsilon > 0$. The stated $b_r = O(1/\ell)$ case can instead be justified from the stronger complementary estimate

   $$|\mathcal{D}_{r,Q}| \ge \left(\frac{\Delta_K^2}{r^2} + \frac{\pi^2}{4\ell^2} \right)^{1/2},$$

   obtained by applying the one-ended Poincaré inequality to the positive and negative B-spectral components. Together with $g_r > 0$ and fixed L, this supplies a complementary gap of order $1/r$.

6. **Result 6 — Correct.** The table in Result 6 separates the chosen data from the algebraic consequences. The proposed $\widehat{J} = E\mathcal{J}_1$ has the stated signs and induces $J_F = \Gamma_0 J_1$ when all the displayed commutation and anticommutation hypotheses hold. Those are sufficient conditions; the note does not construct the required tangential extension or establish its order-zero property.

   The quaternionic j commutes with the Γ_a, so j alone fails the required anticommutation with Γ_p. At the finite level, if the two gradings are Γ_p, Γ_q, a unit $n \perp p, q$ gives $J_1 = \Gamma_n j$ with the required square and anticommutation. Extending that choice to the stated parent remains a separate task.

   The connector calculation gives the displayed algebra after choosing the two complex lines and nonzero coupling. A marked quaternionic line supplies neither of those complex lines by itself.

7. **Result 7 — Correct.** For No-go theorem 22.7, the endpoint-preserving unitary $v = u\, u(\ell)^*$ implements the fluctuation and preserves the original domain. The corresponding real unitary works under the stated order-zero condition. Consequently the moving spectral window is unitarily equivalent to the original one, with the same singular values. The displayed fixed-compression example also has the stated sign and can change singular values at second order.

   The conclusion concerns one interval, fixed wall coupling X, zero bulk mass, scalar inner fluctuations from the even algebra, and the moving spectral window. It is an exact spectral statement without

an action or loop expansion. Variable wall coupling, odd connection directions, suitable bulk masses, further internal connection data, and additional fields lie outside those hypotheses. The result directs the search for a radial variable toward those constructions; it does not decide the full fluctuation space of the $K \times I_\ell$ parent.

8. **Lemma 8 and Theorem 9 — Correct with amendment.** The proof of Lemma 8 checks. The displayed formula for b'' is positive for $x > 0$, since $\sinh x \cosh x > x$. Log-sum-exp is convex and increasing in each argument, so the integrand in (21) is jointly log-concave. Its integral is finite for every u, and Prékopa's Theorem 6, p. 342 applies. Differentiation under the integral gives $0 < e < 1$ and nondecreasing e.

   There is also a separate analytic check of the key monotonicity, using a **covariance argument**. Let

$$\rho(s) = \frac{(\operatorname{arsinh}\sqrt{s})^3}{2\sqrt{s(1+s)}}, \qquad L(v) = \log \rho(e^v).$$

   The substitution $s = \sinh^2 x$ writes $\mathcal{J}(z) = \int_0^\infty \rho(s)/(s+z)\, ds$. For $v = 2 \log \sinh x$,

$$L'(v) = \frac{3 \tanh x}{2x} - \frac{1 + \tanh^2 x}{2},$$

   which is strictly decreasing: $x \operatorname{sech}^2 x - \tanh x < 0$. With $z = e^u$, let the probability density in y be proportional to $e^y \rho(e^{u+y})/(1+e^y)$. Then

$$e(e^u) = \mathbb{E}\left[\frac{1}{1+e^y}\right], \qquad \frac{d}{du} e(e^u) = \operatorname{Cov}\left(\frac{1}{1+e^y}, L'(u+y)\right) > 0.$$

   Both functions in the covariance are strictly decreasing. This check uses the same integral but does not use the log-concavity or Prékopa argument.

   For Theorem 9, the differential inequality for R_c follows from $a = ct(1-t) \geq t$, $e(a) \geq e(t)$, and $e(t) < 1$. Dominated convergence gives

$$H(0+) = 54/5, \qquad H(1/2-) = 0.$$

   Hence H is strictly decreasing and all three conclusions follow. The independent covariance check above verifies the essential monotonicity; the final classification uses the derivative calculation (23).

   Amend the theorem to state $\theta \in [0, \pi/2]$ explicitly. At $\theta = \pi/2$, the determinant has the stated continuous endpoint value, while the isolated principal-logarithm window is degenerate. The no-interior-minimum conclusion assumes the displayed flat, massless, one-loop determinant, its gauge multiplicities and connector wall condition, frequency-one fermions, no fermion-connector modification, and no additional angle-dependent term. Modified form factors, fermion sectors, bulk masses, wall terms, and the uncomputed K-tower effective action remain directions for an interior vacuum.

9. **Result 10 — Correct with amendment.** For real finite opposite masses, identity (17) of this result's discussion—namely equation (10)'s displayed G-identity at lines 658–663—makes the fermion contribution strictly decreasing in t. With the separately assumed flat gauge terms and $n_f > 0$, $t = 1$ is the unique global minimum.

   For $a = c_Q + c_U \neq 0$, the displayed expansion of G and of the logarithm is correct. State that the divergent contribution has nonzero fermion multiplicity and is considered without cancellation by additional sectors. A regulator, subtraction and specified physical completion are needed to draw a conclusion from the general-mass potential.

   The opposite-mass Python scan in lemma 23 covers the listed finite sets of masses and multiplicities and 721 sampled angles. It gives a numerical result for those samples; it does not exclude a local minimum between samples or at unsampled parameters. General masses with a specified renormalized completion and changed gauge form factors remain open directions.

10. **Consequence for T0 and T0b, and the note — Correct with amendment.** The consequence keeps the central questions open: the construction supplies an operator and grading window with prescribed spectral walls, while the coupled local scalar window, compatible reality, second sheet and connector require further data. The no-go statements do not close either T0 or T0b.

In the note, replace "higher modes" in the radius-collapse sentence by **higher K-modes**. At fixed ℓ, the interval tower remains; its decoupling uses the subsequent $\ell \to 0$ limit. Also identify the exact window there as the operator-and-grading window, since the required parent reality structure is conditional.

At the one-loop statement, state $0 < n_f < 54/5$ explicitly for the unique interior maximum and retain the determinant assumptions listed in item 8. When this review is recorded in the repository, update the provenance sentence at line 185 to identify the review and Astra 6's one-loop proof.

Amendments and errors

- Make the local-wall endpoint indices explicit and distinguish Bär–Ballmann's principal-symbol criterion from its stronger full-operator example.
- Qualify the marked-point statement to apply to the displayed natural field.
- State the product-factor inputs for (16) and the scope of equality of compression images.
- Define the source identification and averaging map for interval-collapse convergence; add a denominator estimate to the mixing-rate proof.
- State the angle interval, endpoint degeneracy, and complete determinant hypotheses for Theorem 9 and the paper note.
- Scope the bulk-mass scan to its sampled parameters and state the nonzero-multiplicity assumption for the ultraviolet divergence.
- Update the note's radius-collapse and provenance wording when the review is recorded.

No error was established in the displayed equations. Item 5 has the stated gap in the general compression-map conclusion.

Overall: 3 Correct; 6 Correct with amendment; 1 Gap; 0 Error.